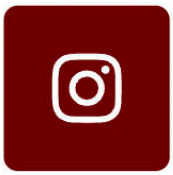


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Surg Cardio

Important basic heart failure points for Surgery	
Left heart failure	- Presents as pulmonary findings (i.e., dyspnea, orthopnea, paroxysmal nocturnal dyspnea). - This is because left-heart problems cause a backup of pressure onto the pulmonary circulation, leading to increased pulmonary capillary hydrostatic pressure → transudation of fluid into the alveolar spaces (pulmonary edema). Sometimes this can also cause pleural effusion. - Left atrial pressure (LAP) = pulmonary capillary wedge pressure (PCWP). - Therefore, if there is left heart pathology, PCWP is high (exceedingly HY). - Conversely, if a Q gives you normal PCWP, you know there's nothing wrong with the left heart. Qs will often give high PCWP and low BP, where you need to know immediately that means cardiogenic shock. - What USMLE will do is give you some sort of left-heart pathology + dyspnea, and then ask for the cause of the dyspnea → answer = "increased pulmonary capillary hydrostatic pressure." Another answer in this case is "increased alveolar-arteriolar (A-a) oxygen gradient."
Right heart failure	- Presents as systemic findings – i.e., jugular venous distension (JVD) and peripheral edema. - Since blood cannot enter the right heart as easily, it backs up to the neck veins (JVD) and venous circulation (increased hydrostatic pressure in veins → transudation of fluid into legs). The Q might mention that central venous pressure is high. - Hepatosplenomegaly can also be seen in RHF but is very rare on USMLE. - Normal jugular venous pressure (JVP) is 3cm above the sternal angle. JVD would be higher than this. Sometimes questions can write that jugular venous pulsations are seen 3cm above the sternal angle and the student erroneously thinks this means JVD, but this is not the case.
Congestive heart failure	- Congestive heart failure = left heart failure + right heart failure. - The most common cause of right heart failure is left heart failure. Simply adding the two together, we now call that congestive heart failure. - In congestive heart failure, we'll see both left- and right-heart failure findings – i.e., patient will have dyspnea, JVD, and peripheral edema. - PCWP is elevated in these patients, since the left heart has pathology.
Cor pulmonale	- Cor pulmonale is defined as right-heart failure due to a pulmonary cause. In other words, the left heart is completely normal in cor pulmonale and PCWP is normal. - Cor pulmonale will be a patient who has JVD and peripheral edema in the setting of obvious and overt lung disease, such as 100-pack-year smoking history, cystic fibrosis, or pulmonary fibrosis. These can present with lung findings such as wheezes, where you as the student need to say, "It just doesn't seem like they're focusing on left-heart failure as the cause of the right-heart failure here. It seems the 100-pack-year smoking Hx causing COPD is why the right heart is failing." - The patient can have a "boot-shaped" heart colloquially, which refers to right ventricular hypertrophy without left ventricular hypertrophy. - If the patient has COPD, the massively hyperinflated lungs will push the heart to the midline, causing a long, narrow cardiac silhouette, with a point of maximal impulse in the sub-xiphoid space. - You must know that *pulmonary hypertension* is the reason the right heart decompensates. In both cor pulmonale and congestive heart failure, the right heart experiences increased afterload because of *pulmonary hypertension*.

	- Endothelin 1 is vasoconstrictor and key mediator in pulmonary hypertension. USMLE wants you to know this is increased in both cor pulmonale and left heart failure. Bosentin is an endothelin 1 receptor antagonist. - Nitric oxide synthase, in contrast, USMLE wants you to know is decreased in pulmonary hypertension (makes sense, since NO dilates). - A loud P2 and tricuspid regurgitation are HY findings in cor pulmonale. I discuss these in more detail in the tables below.

Bilateral pitting peripheral edema	
Cardiac	- Right heart failure (either due to cor pulmonale or congestive) → ↓ ability to fill right heart → ↑ central venous pressure → ↑ **systemic venous hydrostatic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Hepatic	- Cirrhosis → ↓ hepatic production of albumin → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Nephrogenic	- Proteinuria → hypoalbuminemia → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Drugs	- Dihydropyridine calcium channel blockers (i.e., amlodipine, nifedipine).
Dietary	- Strict vegetarianism or veganism → ↓ dietary protein consumption → ↓ **intravascular oncotic pressure** → transudation of fluid from systemic veins/venules into interstitium of legs.
Pregnancy	- A little bit of peripheral edema is normal in pregnancy due to compression of IVC.

Unilateral non-pitting edema	
Lymphatic insufficiency	- Malignancy (e.g., peau d'orange of breast), Hx of surgery (e.g., mastectomy), *Wuchereria bancrofti* (elephantiasis). - Stewart-Treves syndrome is lymphangiosarcoma in the setting of chronic lymphatic insufficiency; shows up on Surg form following mastectomy – i.e., violaceous lesions on elbow in patient with lymphatic drainage issue ever since a mastectomy many years ago.
Thyroid	- Pretibial myxedema (Graves) → mucopolysaccharide deposition in skin + surrounding edema. - Myxedma (severe hypothyroidism) → despite the name, it refers to general "severe hypothyroidism," not just skin changes; can cause carpal tunnel syndrome. - "Pretibial myxedema" is only seen in Graves. Paradoxical hyperthyroidism seen in Hashimoto causing pretibial myxedema is astronomically rare and will get you questions wrong on USMLE.

HY Valvular / flow abnormalities on USMLE	
Atrial septal defect	- **Fixed splitting of S2.** - Can sometimes be associated with a systolic flow murmur, since more blood L → R from the LA → RA means more blood flow across the pulmonic valve. So Q might say "fixed splitting of S2 and a systolic murmur." - Sometimes can be seen in Qs as "wide, fixed splitting." I only mention this because some students get pedantic / ask about this. "Wide splitting" just means right ventricular hypertrophy. So if the Q says "wide, fixed splitting," they're saying the patient has RVH due to an ASD.

	- Patent foramen ovale = ASD on USMLE. Don't confuse with patent ductus arteriosus (discussed below). - ASDs do not need to be repaired unless patient has evidence of pulmonary hypertension, RVH, or develops an arrhythmia (i.e., usually AF). - If adult with ASD has paradoxical embolus (DVT going through the ASD to the brain/arterial circulation), this is also indication for repair.
Ventricular septal defect	- Holosystolic (aka pan-systolic) murmur at lower left sternal border. - Can be associated with a diastolic rumble or enlarged left atrium (if more blood going L → R across VSD, then more blood is returning to the LA from the lungs → LA dilatation). - Seen as part of tetralogy of Fallot (VSD, RVH, overriding aorta, pulmonic stenosis). - If a VSD is repaired, USMLE wants ↑ LV pressure, ↓ RV pressure, and ↓ LA pressure as the changes now seen in the heart. - LA pressure ↓ following repair because less blood is circulating through the pulmonary circulation back to the LA. - VSD does *not* cause cyanosis at birth. Only years later after the higher blood flow to the lungs results in pulmonary hypertension, followed by right ventricular hypertrophy and reversal R → L (Eisenmenger) does the patient become cyanotic. - Murmur can be silent or soft at birth, followed by loud at 7 days of life. The USMLE will ask why the murmur is louder now → answer = decreased pulmonary vascular resistance – i.e., the lungs open up during the first week of life, resulting in decreased RV pressure and an increase in the L → R pressure gradient (louder murmur). - Conversely, if they ask why the murmur was softer at birth compared to now, the answer is "increased pulmonary vascular resistance," where the lungs were still closed at the time, so there was a lesser gradient L → R (softer murmur). - VSDs are repaired if patient develops pulmonary hypertension, RVH, arrhythmia, Eisenmenger syndrome, recurrent endocarditis (turbulence of blood due to VSD can ↑ risk of valve infections), or aortic regurgitation (if VSD located near the aortic valve).
Atrioventricular septal defect	- Seen in Down syndrome. - Between the atrium and ventricle, aka "endocardial cushion defect," although this latter term can also apply to ASD and VSD in Downs.
Mitral regurgitation	- Holosystolic (pan-systolic) or just regular "systolic," 29 times out of 30. - Q on old NBME has MR as "mid-systolic," but I contend this is erratum. - Most USMLE questions will not mention it radiating to the axilla. - Highest yield cause of MR on USMLE is post-MI papillary muscle rupture. USMLE is obsessed with this. They'll say hours to days after an MI, patient has new-onset systolic murmur → answer = MR. - Seen **acutely** in rheumatic heart disease (valve scars over years later and becomes mitral stenosis). - Can be caused by general ischemia / dilated cardiomyopathy. - Can cause **JVD** (i.e., back up all the way to the right heart); this is asked multiple times on the new NBMEs. - Surgery forms want you to know that you do not do preoperative stress tests to determine perioperative MI risk if the patient has mere mitral regurg without other risk factors. For example, one of the Surg forms gives a Q where smoker with MR has no shortness of breath or chest pain with exertion, and the answer is "no further management indicated," where exercise stress test is wrong. - If the patient has Sx of heart failure or ischemia, then we do pre-op stress test to determine MI risk. I discuss stress tests later in this chapter.

	- The mitral valve is replaced if the patient develops severe pulmonary symptoms (i.e., shortness of breath / reduced exercise tolerance), reduced ejection fraction, arrhythmia, or endocarditis if valve function is destroyed.
Mitral stenosis	- Described as “rumbling diastolic murmur with an opening snap”; can also be described as “decrescendo mid-late diastolic murmur” (i.e., following the opening snap). - Can cause a right-sided S4 if the pressure backs up all the way to the right heart (seen on NBMEs sometimes; this confuses students because they think S4 must be LV, but it’s not the case). An S4 is a diastolic sound heard in either the LV or RV when there is diastolic stiffening due to high afterload. - 99% of mitral stenoses are due to Hx of rheumatic heart disease (i.e., the patient had rheumatic fever as a child, where at the time it was mitral regurg, but years later it has now become mitral stenosis). - One 2CK NBME Q mentions patient with history of rheumatic heart disease who, years later, now has 4/6 rumbling diastolic murmur **without** an opening snap; this is still mitral stenosis. Although opening snap is buzzy for MS, just be aware it’s not mandatory and that this Q exists on NBME. - Other HY presentation on USMLE is pregnant women with new-onset dyspnea in 2nd trimester and a diastolic murmur. This is because 50% increase in plasma volume by 2nd trimester causes the underlying subclinical MS to become symptomatic. Don’t confuse this with severe dyspnea and peripheral edema in late third-trimester, which is instead peripartum cardiomyopathy (antibody-mediated). - The 1% of MS that’s not due to Hx of RF can be marantic (non-bacterial thrombotic endocarditis; NBTE) → endocarditis seen due to hypercoagulable state in the setting of malignancy, where the vegetations are small and verrucous, on both sides of the valve. This is in contrast to bacterial endocarditis, which causes large, floppy vegetations that lead to MR, not MS. - Libman-Sacks endocarditis seen in SLE is due to antiphospholipid antibodies and is a type of NBTE. - Balloon valvuloplasty is the 1st-line Tx for mitral stenosis. This is done if patient has minimal calcification of the valve + has pulmonary HTN. - Mitral valve replacement is done if balloon valvuloplasty fails, if patient has severe MS with dyspnea, arrhythmia, or calcification of the valve.
Mitral valve prolapse	- Most common murmur. - Described as mid-systolic click. - “Myxomatous degeneration” is buzzy term that refers to connective tissue degeneration causing MVP in Marfan and Ehlers-Danlos. - Almost always asymptomatic. On 2CK forms, they want you to know about “mitral valve prolapse syndrome,” which is symptomatic MVP that presents as repeated episodes of “fleeting chest pain” on the left side in an otherwise healthy patient 20s-30s. They might say there is Hx of MI in the family, but this is MVPS, not MI. Answer on surgery form is “no treatment necessary." Beta-blockers can be used in theory to help with MVP, but this is a wrong answer on that Q. - USMLE loves using MVP as a distractor in panic disorder questions. They will give long paragraph about panic attack/disorder + also mention there’s a mid-systolic click; they’ll ask for cause of patient’s presentation → answer = panic disorder, not MVP → student is confused because they say mid-systolic click, but the MVP isn’t the cause of the patient’s presentation; the panic disorder is; MVP’s are usually incidental, benign, and asymptomatic. - MVP does not progress to mitral regurg almost always. So don’t think that MVP and MR are the same.

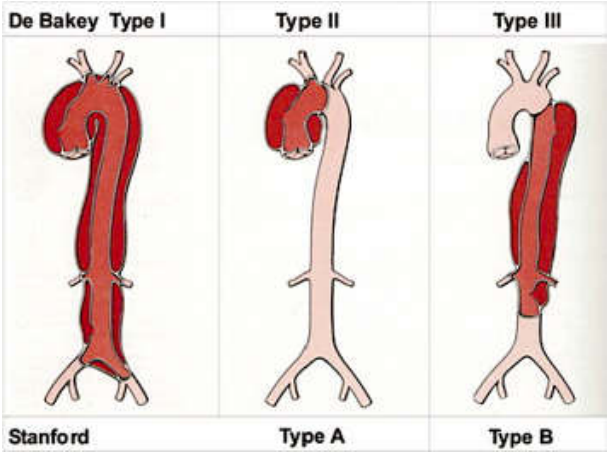
Aortic regurgitation	- Decrescendo holo-diastolic (pan-diastolic) murmur; can also be described as "early diastolic murmur," or "diastolic murmur loudest after S2." - Causes wide pulse pressure (i.e., big difference between systolic and diastolic pressures, e.g., 160/50, or 120/40) → results in head-bobbing and bounding pulses (don't confuse with slow-rising pulses of aortic stenosis). - The bounding pulses can be described on NBME as "brisk upstroke with precipitous downstroke." In turn, they can just simply say, "the pulses are brisk," meaning the systolic component is strong. - I would say 4/5 times bounding pulses means AR. The other 1/5 will be PDA and AV fistulae (discussed below). Bounding pulses occur when blood quickly leaves the arterial circulation. In AR, the blood quickly collapses out of the aorta back into the LV. In PDA, it leaves the aorta and enters the ductus arteriosus; in AV fistulae, it leaves for a vein. - Highest yield cause on USMLE is aortic dissection → can retrograde propagate toward the aortic root causing aortic root dilatation and AR. - Even though MVP is most common in Marfan and Ehlers-Danlos, AR is second most common in these patients, since if they get aortic dissection, this can lead to AR. - Can lead to volume overload on the LV and eccentric hypertrophy. - Aortic valve is replaced if patient has EF <50%, there is significant left ventricular dilatation, or if severe endocarditis has obliterated the valve.
Aortic stenosis	- Mid-systolic murmur, or just "systolic" murmur; can also be described as "late-peaking systolic murmur with an ejection click." - Radiates to the carotids. This descriptor shows up quite frequently on NBME (way more than radiation to the axilla for MR). - Causes slow-rising pulses, aka "pulsus parvus et tardus" (don't confuse with bounding pulses of AR). - SAD → Syncope, Angina, Dyspnea; classic combination seen in AS, albeit not mandatory. If you get a question where they say systolic murmur but you're not sure of the diagnosis, if they say chest pain or fainting, you know it's AS. - Often caused by bicuspid aortic valve. The patient need not have Turner syndrome and often won't. Bicuspid valve is usually inherited as an autosomal dominant familial condition. - The bicuspid valve *need not calcify in middle-age prior to the AS forming.* I've seen it cause AS in high schooler **and in a baby** on NBME. - Do aortic valve replacement on 2CK if 1) cross-section of valve is <1.0 cm^2, or 2) there is SAD. They ask both of these as separate Qs where they want valve replacement.
Tricuspid regurgitation	- Will be described on USMLE as a holosystolic murmur that increases with inspiration. - Right-sided heart murmurs get worse with inspiration → diaphragm moves down → decreased intra-thoracic pressure → increased right-heart filling. - Can cause pulsatile liver. - Highest yield cause of TR on USMLE is **pulmonary hypertension / cor pulmonale.** I see this all over the NBME exams. For whatever reason, these conditions do not cause pulmonic regurg; they cause tricuspid regurg. In other words, if you see tricuspid regurg in a Q, your first thought should be pulmonary hypertension or cor pulmonale (right heart failure due to a pulmonary cause). - IV drug user endocarditis is obvious risk factor for TR, but weirdly nonexistent on USMLE. - Carcinoid syndrome is theoretical cause but lower yield.
Tricuspid stenosis	- Nonexistent murmur on USMLE. I don't think I've ever seen this assessed once on any NBME exam for Steps 1 and 2 combined.

	- In theory, would be a rumbling diastolic murmur similar to mitral stenosis, but would increase with inspiration since it's on the right side of the heart.
Pulmonic regurgitation	- Same as with tricuspid stenosis, this is a nonexistent murmur on USMLE. I've never seen it assessed. In theory it would be the same as aortic regurg but on the right (i.e., holo-diastolic murmur), but increases with inspiration.
Pulmonic stenosis	- Seen in tetralogy of Fallot. - Described as mid-systolic murmur, or just regular "systolic" murmur, that increases with inspiration, at the left sternal border, 2nd intercostal space. This is the theoretical location, whereas AS is the 2nd intercostal space on the right, not left. But the USMLE often isn't strict about murmur locations this way.
Patent ductus arteriosus (PDA)	- Murmur described three ways on USMLE: 1) continuous, machinery-like murmur; 2) pan-systolic pan-diastolic murmur (meaning it's continuous throughout both systolic and diastole); and 3) to-and-fro. The latter shows up on offline 2CK NBME. - Classically associated with congenital rubella (HY). They'll give a kid born with a PDA and then ask what the mom experienced while pregnant; answer = arthritis and/or rash (rubella often presents as arthritis in adults). - Indomethacin (NSAID) will close the PDA. - Prostaglandin E1 is used to keep a PDA open (if a kid with congenital heart malformations is born cyanotic and we need to buy time until surgery). - An open PDA can mask cyanosis in a newborn in a variety of conditions (i.e.,., hypoplastic left heart syndrome or pre-ductal coarctation). If they tell you a kid is born with normal APGAR scores but a week later becomes cyanotic and they ask why, the answer is "closure of ductus arteriosus."
Tetralogy of Fallot	- 1) Pulmonic stenosis; 2) RVH; 3) overriding aorta; 4) VSD. - If you're asked which component most determines prognosis, the answer is the degree of pulmonic stenosis. - The child will not be cyanotic at birth, but then years later, will develop Eisenmenger syndrome (i.e., a reversal of the L→R shunt over the VSD to be R→L) and cyanosis, where the stem gives a school-age kid who squats on the playground to relieve symptoms. - Squatting ↑ afterload, which ↑ LV pressure, which ↓ the pressure gradient of the R→L shunt, thereby mitigating cyanosis. - Squatting also ↑ preload by ↑ venous return back to the right heart. But it is the effect of ↑ afterload that is most related to the ↓ in symptoms. - Tx is surgical correction in infancy or early childhood.
Functional (flow) murmur	- Systolic murmur seen in the setting of higher heart rate caused by infection, anemia, or pregnancy. Caused by increased flow across the pulmonic and/or aortic valves. - Known as a functional murmur because this means it goes away once the heart rate comes back down. - Seen all over 2CK Peds forms, where they try to trick you into thinking the kid has a valvular pathology of some kind, but there isn't; there will merely be an infection or simple viral infection. - Can be seen sometimes with ASD, where the patient will have fixed splitting of S2 "plus a systolic murmur" → merely higher right-sided volume, so more flow across the pulmonic valve. - No Tx necessary.
Venous hum	- On 2CK Peds form; described as a murmur in the neck that abates when the kid is laid supine + the neck rotated. - Benign + don't treat.

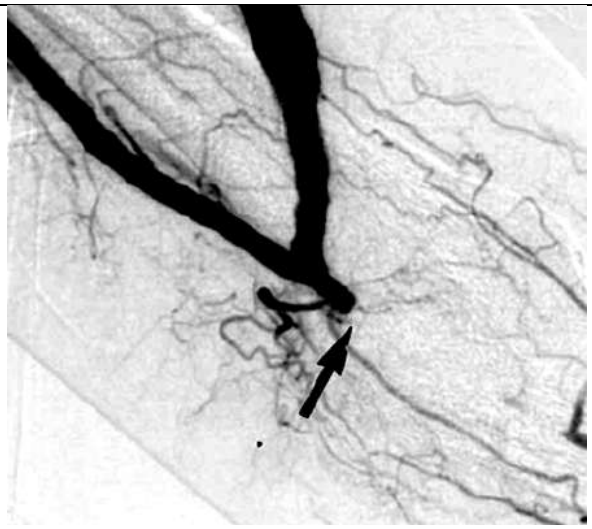
"Ball-in-valve" murmur	- Associated with cardiac tumors (i.e., myxoma in adult, or rhabdomyoma in kids for tuberous sclerosis). - Described as a diastolic rumbling murmur that abates when the patient is re-positioned unconventionally (e.g., onto his or her right side).
Persistent fetal hypertension	- Aka persistent fetal circulation. - Q will give a post-term birth at 42 or 43 weeks + meconium-stained fluid + echo of the neonate shows a R→L shunt across the foramen ovale. Student says, "Wait, but isn't the foramen ovale between the atria, and that's only open in the fetus but is supposed to close after birth?" Correct. Hence we have *persistent* fetal circulation. - Answer on USMLE will be "failure of pulmonary vasodilation." Meconium aspiration syndrome can ↓ opening of the lung vasculature, leading to ↑ right heart pressure and ↑ risk of persistent fetal circulation.
S3 versus S4	- Both are diastolic sounds. - S3 is due to high volume/preload in the left ventricle, causing a reverberation against the wall. - S3 can sometimes be physiologic (i.e., normal / no problem) in pregnancy and high-endurance athletes. Patient will have eccentric hypertrophy (sarcomeres laid in linear sequence). If pathologic, it is due to dilated cardiomyopathy with reduced ejection fraction (<55%) or high-output cardiac failure (EF >70%). - There is one question on IM CMS form 7 where they give an S3 in diastolic dysfunction. I'm convinced this is an erratum, but I need to mention it because it exists on the NBME form. - S4 is due to high pressure/afterload on the left (but sometimes right) ventricle, causing a stiffened ventricle with diastolic dysfunction and concentric hypertrophy (sarcomeres laid in parallel). It is always pathologic. It is usually caused by systemic hypertension causing afterload on the LV, or aortic stenosis. - S4 can sometimes be right-sided on USMLE. There is a 2CK Q where they give severe mitral stenosis and say there's an S4, but it's for the RV not LV. Some weird/annoying points: - The combo of S3 and S4, seen together in the same vignette, can be seen in high-output cardiac failure. For example, they will say a patient as an AV fistula/conduit, or has Paget disease, and they will say there's S3 and S4 and ask for diagnosis → answer = high-output cardiac failure. The take-home point is that high-output failure can present with either an isolated S3 or the combo of S3 and S4 together, but never S4 alone on USMLE.
Loud P2	- One of the highest yield cardiac sounds on USMLE, almost always overlooked by students. - Means pulmonary hypertension or cor pulmonale on USMLE. - The pulmonic valve slams shut due to high pressure distal to it. - For example, they'll give a smoker who simply has a loud P2 → this just means patient has pulmonary hypertension. Not complicated. - Also recall that I said above that highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale. So both what I want you to remember is both TR and loud P2 for this. - Sometimes the UMSLE will just say "loud pulmonic component of S2," or "loud S2," rather than saying "loud P2." I've never seen "loud A2" on USMLE, but in theory this means systemic hypertension. - A soft P2 refers to pulmonic stenosis, but is LY.
Wide splitting of S2	- Means right ventricular hypertrophy on USMLE. - A2 and P2 are far apart. - You don't have to worry about the mechanism. But in short, the more pressure you have in a ventricle, the more delayed the semilunar valve will close. So if we have RVH, P2 occurs later, widening the split.

	- Wide splitting of S2, right-axis deviation on ECG, and right bundle branch block (RBBB) all = right ventricular hypertrophy on USMLE.
Paradoxical splitting of S2	- Means left ventricular hypertrophy on USMLE. - A2 occurs after P2 (normally we have A2 before P2). - Left ventricular pressure is high and A2 delayed to the point that it actually occurs on the opposite side of P2. - Paradoxical splitting of S2, left-axis deviation on ECG, and left bundle branch block (LBBB) all = left ventricular hypertrophy on USMLE.
Coarctation of the aorta	- Refers to narrowing of the aortic arch (this is referred to as coarctation; do not use the word stenosis to describe this). - Classically seen in Turner syndrome, but absolutely not mandatory. Shows up idiopathically in plenty of NBME Qs. I point this out because students often think the patient must have Turner syndrome. - Presents as upper extremities that have higher BP, brisk pulses, and are warmer; the lower extremities have lower BP, weak pulses, and are cooler. - Sometimes the Q can just say, "the radial pulses are brisk." → The implication is, "Well if they're saying specifically that the radial pulses are brisk, that must mean the pulses in the legs aren't." - Murmur sound not important for USMLE. Can sometimes be described as a systolic murmur heart in the infrascapular region. - Can cause LVH with left-axis deviation ECG (on NBME). - USMLE doesn't give a fuck about pre- vs post-ductal. Pre-ductal in theory will be a very sick neonate. Post-ductal will be an adult (most cases).
Subclavian steal syndrome	- Confusing condition when you're first learning things. - The vertebral artery (goes to brain) is the first branch of the subclavian artery (goes to arm). - If there is a narrowing/stenosis of the proximal subclavian prior to the branch point of the vertebral artery, this can lead to lower pressure in the vertebral artery. - This can cause a backflow of blood in the vertebral artery, producing miscellaneous neuro findings such as dizziness. - **Blood pressure is different between the two arms.** - USMLE will ask the Q one of two ways: 1) they'll give you dizziness in someone who has BP different between the arms and then ask for merely "subclavian steal syndrome," or "backflow in a vertebral artery" as the answer. Or 2) they'll give you BP in one of the arms + give you dizziness, then the answer will be, "Check blood pressure in other arm." - Next best step in Dx is CT or MR angiography (asked on 2CK NBME). - I should point out that probably 3/4 questions on USMLE where blood pressure is different between the arms, this refers to aortic dissection. But 1/4 is subclavian steal syndrome. As per my observation.
Vertebral artery stenosis	- Presents same as subclavian steal syndrome with otherwise unexplained dizziness, but blood pressure is not different between the arms because the subclavian is not affected. - Caused by atherosclerosis. CT or MR angiography can diagnose. - "Vertebrobasilar insufficiency" is a broader term that refers to patients who have either subclavian steal syndrome or vertebral artery stenosis.
Vertebral artery dissection	- 2CK Neuro forms assess vertebral artery dissection, where they want you to know a false lumen created by dissection in a vertebral artery can lead to stasis and clot formation, which in turn can embolize to the brain and cause stroke. - NBME can mention recent visit to a chiropractor (neck manipulation is known cause). - The answer on the NBME is **heparin** for patients who have experienced posterior stroke due to vertebral artery dissection. Sounds weird because

	it's arterial, but it's what USMLE wants. Take it up with them if you think it's weird.
Carotid artery dissection	- Shows up on 2CK form as patient with stroke-like presentation + who simultaneously has **ipsilateral facial/neck pain.** - The pain is due to stretching of nociceptors secondary to vascular dilation. - Stasis within false lumen can lead to embolus to brain/eye.
Carotid artery stenosis	- Caused by atherosclerosis. - HTN biggest risk factor for atherosclerosis specifically of the carotids (strong systolic impulse pounds the carotids → endothelial damage → atheromatous plaque formation). - Carotid bruit only seen in about 25% of Qs. Don't rely on this as crutch. - Vignette will give a stroke, TIA, or retinal artery occlusion in the setting of a patient with HTN. → You have to be able to make the association that a plaque from one of the carotids has launched off, since HTN = ↑ risk. - USMLE will then ask for management (2CK only): - Do carotid duplex ultrasonography as next best step in diagnosis to look for degree of occlusion. I've never seen carotid angiography as a correct answer on NBME exams. - If occlusion >70% symptomatic, or >80% asymptomatic, then do endarterectomy. "Symptomatic" = stroke, TIA, or retinal artery occlusion. A mere bruit is not a symptom; that is a sign. - If under these thresholds, do medical management only, which requires a triad of: 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy. - The USMLE will not force you to choose between low- and high-potency statins. - USMLE tends to list lisinopril as their favorite ACEi for HTN control. - It's to my observation aspirin alone is sufficient on NBME exams for anti-platelet therapy, even though in real life patient can receive either aspirin alone; the combo of aspirin + dipyridamole; or clopidogrel alone. - USMLE will not give borderline carotid occlusion thresholds – i.e., they'll say either 30% or 90%. If they list the % as low, look at the vignette for the drugs they list the patient on. Sometimes they'll show the patient is already on statin, lisinopril, and aspirin, and then the answer is just "continue current regimen." I have once seen "add clopidogrel" as a wrong answer in this setting, which makes sense, since the combo of aspirin + clopidogrel is never given anyway. - Sometimes they will give you a low carotid occlusion % + say the patient is on 2 of 3 drugs in the triad, and then the answer is just "add aspirin," or "add statin," or "add lisinopril." - If the vignette doesn't mention elevated BP but says you have some random dude over 50 with a stroke, TIA, or retinal artery occlusion, the next best step is carotid ultrasonography to look for carotid stenosis. In other words, it is assumed the patient has a carotid plaque in this setting. - If the vignette gives patient with episodes of unexplained *syncope* or light-headedness, but not stroke, TIA, or retinal artery occlusion, then the next best step is ECG, followed by Holter monitor, looking for atrial fibrillation (AF causes LA mural thrombus that launched off to brain/eye). - The triad of 1) statin; 2) ACEi or ARB; and 3) anti-platelet therapy is also done for general peripheral vascular disease unrelated to carotid stenosis (i.e., if a patient has intermittent claudication). - Stroke, TIA, or retinal artery occlusion, if they don't mention HTN, but they mention an *abdominal* bruit, you will still do a carotid duplex ultrasound. The implication is that the bruit in the abdomen could be a AAA or RAS, where atherosclerosis in one location means atherosclerosis everywhere, so the patient likely has carotid stenosis by extension. They

	once again need not mention carotid bruit; apparently it is not a sensitive finding (i.e., we cannot rule-out ↑ occlusion just because we don't hear it).
Aortic dissection	- As discussed above in the aortic regurg section, USMLE loves this as most common cause of AR due to retrograde propagation toward the aortic root. For example, patient with Hx of HTN, cocaine use, or a connective tissue disorder (i.e., Marfan, Ehlers-Danlos) who has a diastolic murmur, you should be thinking immediately that this is dissection. - "Medial necrosis" is a term that is used on NBME exams to describe changes to the aorta in dissection. In the past, "cystic medial necrosis" used to be buzzy for dissection due to Marfan syndrome, but I haven't seen USMLE care about this. I have, however, seen a dissection Q on NBME where it is due to hypertension, and simply "medial necrosis" is the answer. - As mentioned above, 3/4 Qs where BP is different between the arms refers to aortic dissection. A Q on 2CK IM form 7 has "thoracic aortic dissection" where not only is the BP different between the arms, but it's also different between the L and R *legs* (i.e., L-leg BP is different from R-leg BP) → sometimes thoracic aortic dissections can anterograde propagate all the way down to the abdominal aorta.  - You do **not** need to memorize these aortic aneurysm types. I'm just showing you that if the common iliacs are involved (as with left image), BP can differ as well *between the legs*. - Tx for ascending aortic aneurysm (type A) = labetalol + surgery. - Tx for descending aortic aneurysm (type B) = labetalol alone initially.
Traumatic rupture of the aorta	- Caused by deceleration injury. Most common cause of death due to car accident or fall. Exceedingly HY on 2CK. - Will be described as patient following an MVA who has "widening of the mediastinum." They'll then ask for the next best step → answer = aortic angiography (aka aortography), OR CT angiography. - New 2CK form has "CT scan of the chest" straight up as the answer, which refers to CT angiography. NBME/USMLE will not force you to choose between aortography or CT angiography; they'll just list one. - Labetalol used first-line in patients who have aortic dissection and traumatic rupture of the aorta. Nitroprusside comes after. - Labetalol is answer on NBME even in patient who has *low BP* due to rupture or dissection due to the drug ↓ shearing forces. I've seen students get this wrong saying, "But patient has low BP though." My response is, file a complaint with the exam not with me. - 2CK Q gives "esmolol + nitroprusside" as answer to a traumatic rupture Q, but almost always, they will just want "labetalol." - Emergency surgical repair is indicated following IV drug administration.
Aortic aneurysm	- Can present as "visible pulsation" on USMLE.

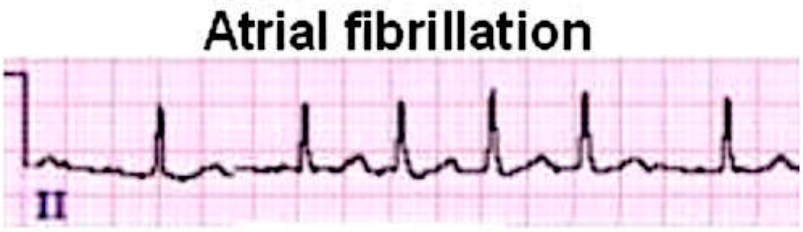
	- For aortic aneurysm, they can say "visible pulsation above the manubrium," or "pulsatile mass above the manubrium." **There can also be a tracheal shift. I've seen students select pneumothorax here.** But for whatever reason you can get tracheal shift in thoracic aortic aneurysm. For AAA, there can be "visible pulsation in the epigastrium." - Biggest risk factor for AAA is smoking. - Do a one-off abdominal ultrasound in men 65+ who are ever-smokers. - For Surgery, AAA repair is indicated if the aneurysm is >5.5 cm or the rate of change of size increase is >0.5cm over a 6-month period. This is on Surgery form, where they give a patient with a 4-cm AAA and ask why serial ultrasounds are indicated → answer = "size of aneurysm." - In general, perioperative MI risk is assessed using a pre-op stress test. 2CK NBME Q has dipyridamole and thallium pharmacologic stress test as answer in patient with 6-cm AAA prior to surgery. - Diabetes is protective against aneurysm. Non-enzymatic glycosylation of endothelium causes stiffening of the vascular wall. - Don't do AAA repair on USMLE in patient who has advanced comorbidities or terminal disease, e.g., stage 4 lung cancer. - 2CK Surg loves "pulsatile hematoma" in the neck in trauma patients, where the answer is "endotracheal intubation." Sounds nitpicky, but shows up repeatedly.
Arteriovenous fistula	- Can be idiopathic, iatrogenic (i.e., dialysis), from injury (i.e., stab wound), or caused by other disease (i.e., hereditary hemorrhagic telangiectasia or Paget disease of bone). - Similar to aortic aneurysms, AV fistulae can sometimes present with pulsatile mass, *but in a weird location*, e.g., around the left ear in patient with tinnitus (on NBME exam). Student says, "Why is it at the left ear though?" → No fucking idea. Take it up with NBME. - Highest yield point is they can cause **high-output cardiac failure.** This is because blood quickly enters the venous circulation from the arterial circulation → combo of ↑ preload back to right heart + poorer arterial perfusion distal to the fistula → compensatory ↑ CO. - AV fistulae can sometimes present with a **continuous machinery murmur similar to a PDA,** since blood is continuously flowing through it. They might say a continuous machinery-like murmur is auscultated in the leg at site of prior stab wound. - As discussed earlier, they can present with bounding pulses similar to AR. - Student says, "Well how am I supposed to know if it's AV fistula then if it sounds like other conditions too?" → by paying attention to HY points like, "Is there lone S3 or S3/4 combo or EF >70%? Is there Hx of penetrating trauma? Or does the patient have Paget? Etc." - 2CK Surg Q shows you obscure angiogram of a fistula in the leg + tells you there's a continuous machinery murmur; they ask what most likely determines prognosis in this patient → answer = "size of lesion."

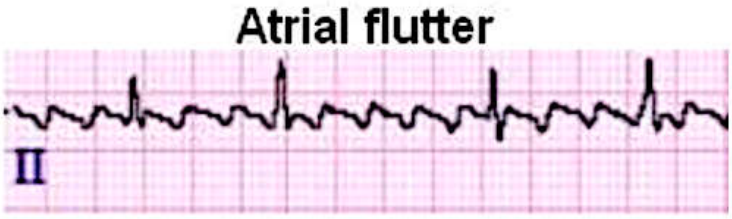


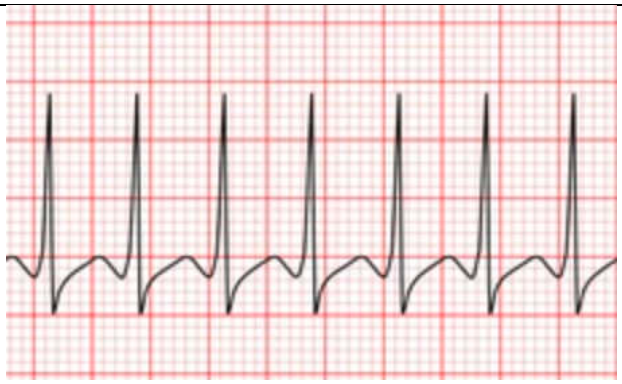
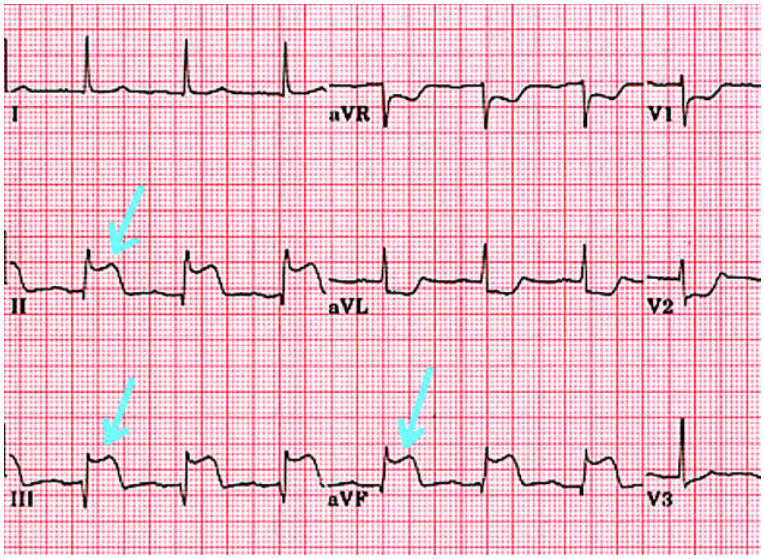
- Surg form shows obscure image similar to above (without the arrow) + they tell you there's continuous murmur → answer = "size of lesion."
- Another NBME Q gives 45-year-old male will nosebleeds since adolescence + S3 heart sound + dyspnea + they show you pic of tongue; they ask for the cause of dyspnea.



- Answer = "Pulmonary arteriovenous fistula" (leading to high-output failure); diagnosis is hereditary hemorrhagic telangiectasia. USMLE will basically always show you a pic of red dots on the tongue/mouth or finger in a patient with nosebleeds.
- Likewise, be aware intraosseous AV fistulae can occur in Paget, as mentioned before.

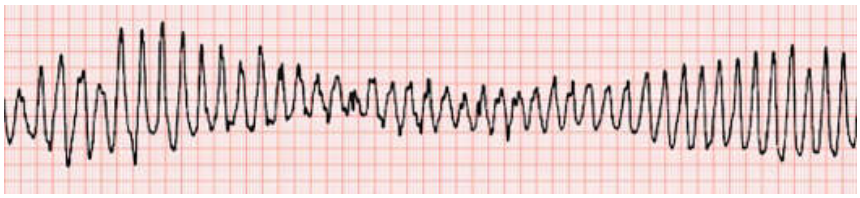
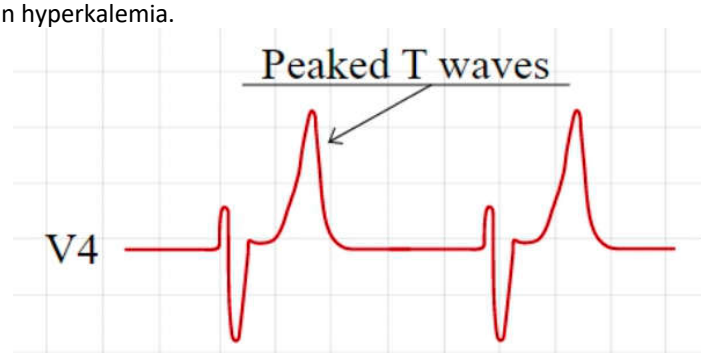
HY Murmur / ECG points for Surg	
Atrial fibrillation (AF)	- Described as “irregularly irregular” rhythm.  - Notice how the QRS complexes are at random and irregular distances from one another. This is the “irregularly irregular” pattern. - AF is hugely important because it can cause turbulence/stasis within the left atrium that leads to a LA mural thrombus formation. This thrombus can launch off (i.e., become an embolus) and go to brain (stroke, TIA, retinal artery occlusion), SMA/IMA (acute mesenteric ischemia), and legs (acute limb ischemia). - AF HY in older patients, especially over 75. Vignette will usually be an older patient with a stroke, TIA, or retinal artery occlusion, who has *normal* blood pressure (this implies carotid stenosis is not the etiology for the embolus). - AF usually is paroxysmal, which means it comes and goes. The vignette might say the patient is 75 + had a TIA + BP normal + ECG shows sinus rhythm with no abnormalities → **next best step is Holter monitor (24-hour ambulatory ECG monitor)** to pick up the paroxysmal AF (e.g., when the patient goes home and has dinner). - After AF is diagnosed with regular ECG or Holter, 2CK wants **echocardiography** as the next best step to visualize the LA mural thrombus. - Patient who has severe abdominal pain in setting of AF or hyperthyroidism (which can cause AF), diagnosis is acute mesenteric ischemia; next best step is mesenteric angiography; Tx is laparotomy if unstable (answer on NBME). - Severe pain in a leg + absent pulses in patient with irregularly irregular rhythm = acute limb ischemia; USMLE wants “embolectomy” as answer. - Any structural abnormality of the heart, either due to LV hypertrophy, ischemia, growth hormone/anabolic steroid use, prior MI, etc., can lead to AF. - For 2CK, you need to know AF patient will get either NOAC or warfarin. Aspirin is no longer indicated for AF management. This is determined by the CHADS2 score. There are variations of the score, but the simple **CHADS2** suffices for USMLE → **C**HF, **H**TN, **A**ge 75+, **D**iabetes, **S**troke/TIA/emboli. Each component is 1 point, but stroke/TIA/emboli is 2 points. More points favors warfarin. - “Emboli” refers to Hx of AF leading to stroke, TIA, acute, mesenteric ischemia, or acute limb ischemia – i.e., any Hx of embolic event. 2CK IM form 7 gives short vignette of 67F with chronic AF + Hx of acute limb ischemia + no other info relating to CHADS, and answer is warfarin to prevent recurrence; aspirin is wrong. - Some students will ask about NOACs, e.g., apixaban, etc., for non-valvular AF → I’ve never seen NBME care about this stuff. They seem to be pretty old-school and just have warfarin as the answer, probably because there isn’t debate around whether it can be used; use of NOACs is less textbook. - AF patient should also be on rate control before rhythm control. The USMLE actually doesn’t give a fuck about this component of management, although in theory metoprolol or verapamil is standard. You could be aware for Step 3 that flecainide is first-line for rhythm control if patients fail rate-control and have a structurally normal heart and no coronary artery disease. - 2CK form has “electrical cardioversion” as the answer for patient with AF who has hemodynamic instability (i.e., low BP). What you need to know is:

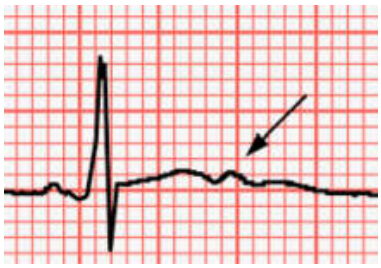
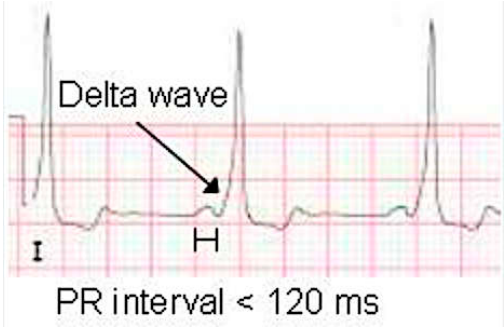
	sometimes AF can trigger “rapid ventricular response,” where HR goes >150 and low BP can occur.
Atrial flutter	- Has classic sawtooth appearance.  - Low yield for USMLE. I think it’s asked once on a 2CK NBME. But as student you should know it exists / the basic ECG above.
Ventricular tachycardia (VT)	- Causes wide-complex QRS complexes (>120 ms; normal is 80-120 ms).  - Exceedingly HY for 2CK that you know VT is wide-complex, whereas SVT is narrow-complex. If you look at above ECG, even if you say, “No idea what I’m looking at.” You can tell the complexes look wide like mountains in comparison to a typical ECG. - VT is treated with anti-arrhythmics – i.e., amiodarone. If patient has coma or hemodynamic instability (low BP), the NBME answer is direct current countershock or cardioversion (same thing). - Premature ventricular complex (PVC) is asked on 2CK.  - Note on the above strip, we have a wide complex (meaning ventricular in origin) that occurs earlier (hence premature). What they do on the NBME is show you this strip and ask where this abnormality originates from, then the answer is just “ventricle.” - Don’t treat PVCs on USMLE.
Supraventricular tachycardia (SVT)	- Causes narrow / needle-shaped complexes. Make sure you’re able to contrast this with VT above, which is wide-complex.

	 - Notice the complexes are narrow / look like needles. This means the tachy originates above the ventricles (hence SVT). - Treatment of SVT exceedingly HY on 2CK. - First step is carotid massage (aka vagal maneuvers). In pediatrics, they can do icepack to the face. - If the above doesn't work, the next step is give adenosine (not amiodarone). - Same as with VT, if the patient has coma or low BP, shocking the patient is the first step. In other words, for both SVT and VT, you must shock first in the setting of coma or hemodynamic instability. It's for stable SVT and VT that the treatments differ on USMLE.
Acute MI (STEMI)	- Will present as ST-elevations in 3-4 contiguous leads.  - The above is an inferior MI, as evidenced by ST-elevations in leads II, III, and aVF. The answer for the affected vessel is the posterior descending artery (PDA supplies the diaphragmatic surface of the heart); since >85% of people have right-dominant circulation (meaning the PDA comes of the right main coronary), sometimes the answer for inferior MI can just be "right coronary artery." - If the Q says left-dominant circulation, the sequence USMLE wants is: left main coronary → left circumflex → PDA. - The apex of the heart is supplied by the left anterior descending artery (LAD). If there are ST-elevations in leads V1-V3, choose LAD as the answer. - The left-lateral heart is supplied by the left circumflex artery. If there are ST-elevations in leads V4-V6 for lateral MI, choose left circumflex. - Reciprocal ST-depressions in the anterior leads V1-V3 can reflex posterior wall MI (i.e., we have "elevations" out the back of the heart, so they look like depressions on the anterior wall leads).

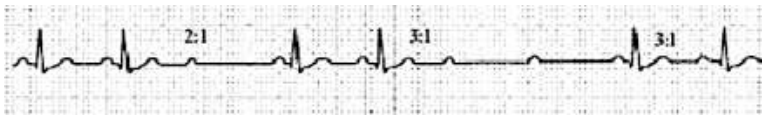
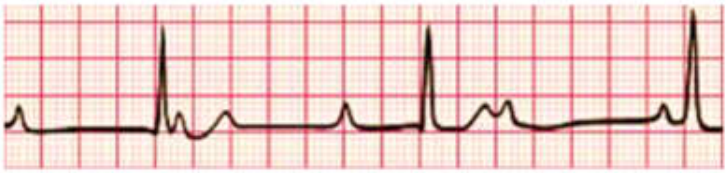
	- Post-MI papillary muscle rupture resulting in mitral regurg is exceedingly HY. As discussed earlier, if patient has MI followed by new-onset systolic murmur hours to days later, with or without dyspnea, that's mitral regurg. - Stroke-like presentation in patient who had MI weeks ago → "embolus from ventricular septal aneurysm" (on 2CK Neuro form). - Most common cause of death due to MI is ventricular fibrillation (VF). - Fibrosis of myocardium in the months-years post-MI increases risk of arrhythmias such as AF, SVT, VT, etc. There's no specific arrhythmia you need to memorize. Just know the risk is there in the future. - Q waves on an ECG mean old MI / history of MI. The vignette might give you patient who has light-headedness / fainting + they say patient has Q waves in II, III, aVF, and the answer will be something like "paroxysmal supraventricular tachycardia." Student thinks this specific arrhythmia matters, but it doesn't. The point is that Hx of MI means patient is at risk for nearly any arrhythmia now. - MI classically causes coagulative necrosis of the myocardium. - With cardiogenic shock as a result of MI, the arrows USMLE wants are: ↓ cardiac output, ↑ peripheral vascular resistance, ↑ PCWP. - MI can lead to acute tubular necrosis from cardiogenic shock → acute drop in renal perfusion. This is not pre-renal. I discuss this in detail in the renal section. - First treatment for MI is aspirin. After aspirin is given, the next drug to give is clopidogrel (an ADP P2Y12 blocker) as dual anti-platelet therapy. - USMLE wants you to know anyone with acute coronary syndrome (i.e., MI or unstable angina) gets coronary catheterization. This is answer on new 2CK NBME exam. - It's to my observation that more extensive management of MI on USMLE, such as use of beta-blockers, nitrates, morphine, oxygen, statin, percutaneous coronary intervention, etc., isn't assessed in detail. I can comment, however, that one 2CK Q wants you to know nitrates are contraindicated in right-heart MIs, which includes inferior MI in most people due to the right coronary supplying the PDA. This is because right-sided MIs are preload-dependent, which means they need sufficient preload to maintain BP. - Percutaneous coronary intervention (PCI) is done in patients with STEMIs within 90 minutes of reaching hospital.
Pericarditis	- Shows up on ECG as diffuse ST-elevations (i.e., in all leads rather than 3-4 contiguous leads as with MI). PR depressions can also be seen, but I've never seen the USMLE give a fuck about the latter. - Patient will have pain that's worse when lying back, better when leaning forward. In turn, the patient can present walking through the door bent over at the waist. - Serous pericarditis will be post-viral, secondary to autoimmune disease, or due to cocaine use. - NBME Q gives pericarditis + a bunch of different organism types (i.e., fungus, bacterium, etc.), and answer is "virus." - Patient with rheumatoid arthritis or SLE notably at risk for pericarditis. In other words, don't get confused if they mention pericardial friction rub in vignette of RA or SLE; this is common. - For cocaine use, they'll say a 22-year-old male has chest pain after a night of heavy partying + ECG shows diffuse ST-elevations → Dx = pericarditis. - Uremic pericarditis is HY for 2CK. Q will give ultra-high creatinine and BUN and say there's a friction rub → treatment = hemodialysis. - Treatment for pericarditis is same as acute gout → NSAIDs, colchicine, steroids. - Fibrinous pericarditis is post-MI and occurs as two types: 1) literally "post-MI fibrinous pericarditis," which will simply be friction rub within days of an MI; 2) Dressler syndrome (antibody-mediated fibrinous pericarditis occurring 2-6 weeks post-MI).

	- ECG is first step in Dx of pericarditis, but USMLE wants echocardiography as next best step in order to visualize a concomitant effusion that can occur sometimes. Vignette will give you stereotypical pericarditis + will ask for next best step in diagnosis; ECG might not be listed and you're like huh? → Answer is echocardiography to look for potential effusion concomitant to the pericarditis.
Chronic constrictive pericarditis	- I should make note that chronic constrictive pericarditis is a separate condition that doesn't present with the standard pericarditis findings as described above. - This is low-yield for USMLE, but students ask about it because it can be confused with tamponade. - There's two ways this can show up: 1) **Tuberculosis** is a classic cause; there may or may not be calcification around the heart on imaging. So if you get a Q where patient has TB + some sort of heart-filling impairment → answer = chronic constrictive pericarditis. 2) **Kussmaul sign** will be seen in the Q, where JVD occurs with *inspiration* rather than expiration. - Normally, inspiration facilitates RA filling (↓ intrathoracic pressure → ↑ pulmonary vascular compliance/stretching → ↑ high-low pressure gradient from right heart to the lungs → ↓ in afterload on RV from the lungs → blood moves easier from right heart to the lungs → blood is pulled easier from SVC/IVC to the RA). - However, if there is ↑ compressive force on the heart, the ↑ in negative intrathoracic pressure during inspiration is not transmitted to the right side of the heart, so JVP does not ↓ (and can even paradoxically can ↑). - In tamponade, however, as discussed below, the ↑ in negative intrathoracic pressure during inspiration *is able* to be transmitted to the right side of the heart, so Kussmaul sign does not occur. This is likely because in constrictive pericarditis, the rigid pericardium prevents expansion of the right heart altogether, whereas in tamponade, the pericardium isn't rigid per se, but is just filled with blood that can move/shift during the respiratory cycle, thereby allowing right heart expansion during inspiration.
Pericardial effusion / Cardiac tamponade	- Cardiac tamponade = pericardial effusion + low blood pressure. - What determines whether we have a tamponade or not is the *rate* of accumulation of the fluid, not the volume of the fluid – i.e., a stab wound or post-MI LV free-wall rupture resulting in fast blood accumulation, even if smaller volume, might cause tamponade, but cancer resulting in slow, but large, accumulation might not cause tamponade. - Tamponade presents as **Beck triad: 1) hypotension, 2) JVD, 3) muffled/distant heart sounds.** The question will basically always give hypotension and JVD. Occasionally they might not mention the heart sounds. But you need to memorize Beck triad as HY for tamponade. - Pulsus paradoxus (i.e., drop in systolic BP >10 mm Hg with inspiration) is classically associated with tamponade, although not frequently mentioned in vignettes. I've seen a 2CK NBME Q where they say "the pulsus paradoxus is <10 mm Hg," which is their way of saying the Dx is not tamponade. I consider that wording odd, but it's what the vignette says. - ECG will show electrical alternans / low-voltage QRS complexes.

	 - You can see the amplitudes (i.e., heights) of the complexes are short. This refers to "low-voltage." You can also see the heights ever so slightly oscillate up and down. This refers to electrical alternans. They show this ECG twice on 2CK NBMEs. - If the Q asks for next best step in *diagnosis,* choose ECG as first step if listed If not listed, then choose echocardiography, which confirms fluid over the heart. - If the Q asks for next best step in management for tamponade when the vignette is obvious, choose pericardiocentesis or pericardial window. USMLE will not list both; it will be one or the other. NBME 8 offline for 2CK has pericardial window as answer, where pericardiocentesis isn't listed.
Torsades de pointes (TdP)	- HY type of VT that has sinusoidal pattern on ECG.  - USMLE wants you to know this can be caused by some anti-arrhythmic agents, such as the sodium- and potassium-channel blockers, such as quinidine and ibutilide, respectively. They ask this directly on the NBME exam, where Q will say patient is given ibutilide + what is he now at increased risk of → answer = torsades. - QT prolongation is risk factor for development of TdP. Agents such as anti-psychotics, macrolides, and metoclopramide prolong the QT. - Tx USMLE wants is magnesium (asked directly on new 2CK form), which stabilizes the myocardium in TdP.
Peaked T wave	- Seen in hyperkalemia.  - Asked once on one of the 2CK forms, where they show the ECG. - Highest yield point is that if a patient has hyperkalemia and ECG changes, the Tx USMLE wants is IV calcium gluconate or calcium chloride, which stabilizes the

	myocardium. Calcium gluconate is classic, but calcium chloride shows up as an answer on a 2CK NBME.
U-wave	- Means hypokalemia.  - Shows up on NBME 12 for 2CK in anorexia patient. First time I've ever seen it show up anywhere on NBME material. But Q doesn't ride on you knowing it means hypokalemia to get it right. It's HY and pass-level to know that purging (anorexia or bulimia) causes hypokalemia anyway.
Delta wave	- Seen in Wolff-Parkinson-White syndrome (WPW; accessory conduction pathway in heart that bypasses the AV node, resulting in reentrant SVT). - Classically described as a "slurred upstroke" of the QRS, where the PR interval is shortened.  - Both the delta-wave and WPW have basically nonexistent yieldness on USMLE, but I mention them here so you are minimally aware.
J waves	- They mean hypothermia. You don't need to be able to identify on ECG. Just know they exist, as they show up in a 2CK vignette where patient has body temperature of 89.6 F (not 98.6).

HY Heart block points for Surg	
First degree	- Prolonged PR interval (>200 ms). Should normally be 80-120 ms.  - Note that above on the ECG, the PR-segment in particular (just prior to the QRS complex) is extra-long. - Not really assessed on USMLE. Just know the definition. - Don't treat on USMLE.
Second degree Mobitz type I (aka Wenckebach)	- Gradually prolonging PR interval until QRS drops. Then cycle repeats.

	 - Don't treat on USMLE.
Second degree Mobitz type II	- *No* gradual prolongation of PR interval, followed by a random dropping of the QRS.  - Can also sometimes occur as patterns of 2:1, 3:1, etc., where there will be a P to QRS ratio of 2:1 or 3:1, etc.  - Regardless as to whether the dropped QRS is random or in a numerical pattern, there is no gradual prolongation of the QRS before the dropped complex. - More dangerous than Mobitz I. This is because Mobitz II has higher chance of progression into type III heart block. - Treatment on USMLE is insertion of pacemaker. This is asked on a new 2CK NBME exam.
Third degree	- Two things you want to look for on ECG: - 1) Ultra-slow HR (i.e., 30-40). You'll see the QRS's are super far apart. This is the ventricular escape rhythm. - 2) No relationship between the P-waves and QRS complexes. Third degree AV block  - Treatment on USMLE is insertion of pacemaker. - So what you want to remember is that Mobitz II and 3rd-degree are the ones where we insert pacemaker; 1st-degree and Mobitz I we don't.

HY Cardiomyopathy points for USMLE	
Dilated (DCM)	- Can be isolated ventricular or diffuse 4-chamber dilation. Causes are multifarious, but a key feature is **systolic dysfunction,** where ejection fraction is reduced (i.e., <55%, where normal range is 55-70). - CXR shows enlarged cardiac silhouette.

	 - An S3 heart sound can sometimes be heard. - Cardiac exam shows lateralized apex beat. Can be described as the point of maximal impulse being in the anterior axillary line. Should be noted that this lateralization just means an enlarged LV, so it is non-specific, and can also be seen in LV hypertrophy from any cause. But many vignettes will mention it. - The arrows USMLE wants are: ↓ EF; ↑ LVEDV; ↑ LVEDP. I discuss this stuff in more detail in my HY Arrows PDF. - Causes of DCM are **ABCD**: - **A**: Alcohol. - **B**: Wet Beriberi (thiamine deficiency). - **C**: Coxsackie B virus, Cocaine, Chagas disease. - **D**: Drugs → doxorubicin (aka Adriamycin). - Other notable causes are: - Pregnancy (peripartum cardiomyopathy); hemochromatosis; rheumatic heart disease (myocarditis leading to DCM).
Hypertrophic (HCM)	- Heart failure due to systemic hypertension. - Characterized by diastolic dysfunction (the heart can pump just fine but cannot expand as easily). - Can be associated with S4 heart sound. - The arrows USMLE wants are: ↔ EF; ↔ LVEDV; ↑ LVEDP. - Ejection fraction is normal because the heart can pump perfectly fine. - Students get confused about LVEDV, thinking it should be low, if the heart cannot expand as easily. But this is not the case for USMLE. They want you to know normal volume can be achieved; *it just merely requires more force/pressure to get there.* - As with DCM, the apex beat / point of maximal impulse can be lateralized, which merely reflects LV hypertrophy. - As described earlier, paradoxical splitting of S2, left-axis deviation on ECG, and LBBB can all be seen due to LVH. When you see these findings in vignettes, don't get confused. They just mean LVH. - Often associated with hypertensive retinopathy (fundoscopy shows narrowing of retinal vessels, flame hemorrhages, and "AV-nicking"), hypertensive nephropathy (hyperplastic arteriolosclerosis with increased creatinine).

Hypertrophic obstructive (HOCM)	- Caused by mutations in β-myosin heavy-chain gene; autosomal dominant; results in disordered/disarrayed myocardial fibers. - HOCM causes asymmetric septal hypertrophy that results in the anterior *mitral* valve leaflet obstructing the LV outflow tract (so it can sound similar to aortic stenosis). - Classically sudden death in young athlete; cause of death is **ventricular fibrillation** due to acute left heart strain in the setting of fast heart rate. - Presents with systolic murmur that worsens with Valsalva or standing (aortic stenosis, in contrast, gets softer or experiences no change with Valsalva) or standing. - HOCM and MVP are the only two murmurs that get worse with less volume in the heart. All other murmurs get worse with more volume. Valsalva increases intra-thoracic pressure and decreases venous return, so there's less volume in the heart. Standing simply decreases venous return. - NBME wants you to know that ↑ HR from exercise or stress means **"diastole is shortened more than systole."** - Beta-blockers (metoprolol or propranolol are both answers on NBME) are given to slow heart rate, which maximizes diastolic filling and decreasing symptoms / risk of death. - Implantable cardioverter-defibrillator (ICD) is indicated if patient develops any type of arrythmia, syncope, low blood pressure, or LV wall thickness >30mm. - Septal myectomy is done if the left ventricular outflow tract pressure gradient is >50mm Hg. But it should be noted ICD is usually first-line.
Restrictive (RCM)	- Heart failure due to diastolic dysfunction, where HTN is not the cause. - **JVD** is HY for RCM. An S4 can also be seen. The heart will not be dilated. - HY causes are Hx of radiation (leads to fibrosis), amyloidosis, and hemochromatosis. - Student might say, "I thought you said hemochromatosis was DCM. So if we have to choose on the exam, which one is it?" The answer is, whichever the vignette gives you. If they say a large cardiac silhouette with an S3 and lateralized apex beat, that's DCM. If they say JVD + S4 + nothing about a lateralized apex beat, you know it's RCM. - Amyloidosis is protein depositing where it shouldn't be depositing. Highest yield cause of amyloidosis on USMLE is multiple myeloma, which will lead to RCM. - Since RCM is diastolic dysfunction, the arrows are the same as HCM, which are: ↔ EF; ↔ LVEDV; ↑ LVEDP.

HY Angina points for USMLE	
Stable angina	- Chest pain that occurs predictably with exercise. - Due to atherosclerotic plaques causing >70% occlusion; can be calcific. - Classically causes ST depressions on ECG. - Nitrates (e.g., sublingual isosorbide dinitrate) used as Tx → nitrates "donate" nitric oxide (NO) that upregulates guanylyl cyclase within **venous** smooth muscle → increased cGMP → relaxation of venous smooth muscle → increased venous pooling of blood → decreased venous return → decreased myocardial oxygen demand → mitigation of chest pain. - Nitrates are contraindicated with PDE-5 inhibitors (e.g., Viagra) due to risk of low blood pressure. - Sodium nitroprusside used for hypertensive emergencies dilates **arterioles** in addition to the veins. If USMLE asks you where this drug acts, choose arterioles.
Unstable angina	- Chest pain that is unpredictable and can occur at rest.

	- Due to partial rupture of atherosclerotic plaque leading to partial occlusion. - ST depressions on ECG. - Diltiazem is answer on new 2CK NBME for patient with unstable angina. - Patients need cardiac catheterization.
Prinzmetal angina (variant angina pectoris)	- Vasospastic angina that occurs at rest (i.e., watching TV or while sleeping) in younger adults; it is not caused by atherosclerosis. - ST elevations are seen on ECG. - You must know that Prinzmetal is also known as variant angina pectoris. There is an NBME Q that gives vignette of Prinzmetal, but answer is "variant angina pectoris." - Treatment is nitrates (can cause coronary artery dilation unrelated to the venous pooling effects) or dihydropyridine calcium channel blockers (e.g., nifedipine). Avoid α1-agonists in these patients (cause vasoconstriction), as well as non-selective β-blockers like propranolol (can cause unopposed α effects).

Hypertensive Emergency + urgency	
Emergency	- HTN >180/120 + signs of end-organ damage. - The latter can be hypertensive encephalopathy (confusion), nephropathy (poor renal function tests), retinopathy, acute heart failure, etc. - BP should be ↓ by no more than 20-25% in the first hour, as drastic ↓ can compromise perfusion to the brain and vital organs. - Blood pressure should be brought under 160/100 by 24-48 hours. - Drugs used are IV sodium nitroprusside, IV nicardipine, IV labetalol, and oral captopril.
Urgency	- HTN >180/120 + no signs of end-organ damage. - Blood pressure should be brought under 160/100 by 24-48 hours. - Drugs used are IV sodium nitroprusside, IV nicardipine, IV labetalol, and oral captopril.

HY Endocarditis points	
Acute endocarditis	- Bacterial infection of valve in patient with no previous heart valve problem. - Caused by *Staph aureus* on USMLE. - Left-sided valves (i.e., aortic and mitral) most commonly affected because of greater pressure changes (i.e., from high to low) within left heart, resulting in turbulence that enables seeding. - IV drug users → venous blood inoculated with *S. aureus* → travels to heart and causes vegetation of tricuspid valve. - *Staph aureus* is coagulase positive.
Subacute endocarditis	- Bacterial infection of valve in patient *with history* of valve abnormality (i.e., congenital bicuspid aortic valve, Hx of rheumatic heart disease). - Caused by *Strep viridans* on USMLE. You need to know *S. viridans* is can be further broken down into: *S. sanguinis, S. mutans,* and *S. mitis*. - Hx of dental procedure is HY precipitating event, where inoculation of blood occurs via oral cavity → previously abnormal valve gets seeded.
Random points	- New-onset murmur + fever = endocarditis till proven otherwise on USMLE. - Reactive thrombocytosis (i.e., high platelets) can occur due to infection. This is not unique to endocarditis, but it is to my observation USMLE likes endocarditis as a notable etiology for it. In other words, if you get an endocarditis question and you're like, "Why the fuck are platelets 900,000?" (NR 150-450,000), don't be confused. - Hematuria can occur from vegetations that launch off to the kidney. - Endocarditis + stroke-like episode (i.e., focal neurologic signs) = septic embolus, where a vegetation has launched off to the brain.

	- Janeway lesions, Osler nodes, splinter hemorrhages, etc., are low-yield for USMLE and mainly just school of medicine talking points. - HACEK organisms nonexistent on USMLE.
Management	- Blood cultures *before* antibiotics is important for 2CK. - Transesophageal echocardiography (TEE) confirms diagnosis after blood cultures. Transthoracic echocardiography (TTE) is not done for endocarditis. - For 2CK, empiric treatment for endocarditis is vancomycin, PLUS either gentamicin or ampicillin/sulbactam. - Vancomycin targets gram-positives (including MRSA). Gentamicin targets gram-negatives. - Endocarditis prophylaxis given prior to a dental procedure is usually ampicillin or a second-generation cephalosporin, such as cefoxitin. - Indications for endocarditis prophylaxis are: 1) Hx of endocarditis (obvious); 2) If there is any prosthetic material in the heart whatsoever; 3) If there is any congenital cyanotic heart disease that has not been completely repaired (if it's been completely repaired with prosthetics, give prophylaxis); 4) Hx of heart transplant with valvular regurgitation of any kind. - Highest yield point for USMLE about endocarditis prophylaxis is that mitral valve prolapse (MVP) and valve regurgitations or stenoses are ***not*** an indication. In other words, do not give prophylaxis if the patient has MVP, MR, AS, etc. In addition, bicuspid aortic valve is ***not*** an indication.

Conditions confused for cardiac path	
Panic attack	- Psych forms love trying to make you think this is an MI. - They'll give you young, healthy patient who feels doom / like he or she is going to die. - Sometimes they mention in stem Hx of MI in family as distraction. - They can say patient has mid-systolic click, as discussed earlier, and then they ask for cause of patient's symptoms → answer = panic disorder, not MVP. Student gets confused, but MVP is almost always asymptomatic, where panic attack is clearly cause of the patient effusively hyperventilating. - Treat with benzo.
Orthostasis	- Orthostatic hypotension is defined as intravascular fluid depletion causing a drop of systolic BP >20 mmHg and diastolic BP >10 mmHg when going from supine to standing. - Shows up on 2CK IM form as exactly a drop of 20 and 10, respectively, for systolic and diastolic BPs in a patient with fainting → answer = "intravascular fluid depletion." - Diuretic use is big risk factor.
Vasovagal syncope	- Fainting in response to stressor (e.g., emotional trigger). - Stress triggers an initial sympathetic response, which in turn triggers a compensatory parasympathetic response. This latter response is excessive in some people, where the peripheral arterioles dilate and the heart slows too much → decreased cerebral perfusion → lightheadedness/fainting. - 2CK wants you to know a tilt-table test can be used to diagnose, where a reproduction of symptoms can occur.
Carotid sinus hypersensitivity	- USMLE likes this for both Steps 1 and 2. - They'll say dude was shaving then got lightheadedness or fainted. Mechanism is ↑ stretch of carotid sinus baroreceptors → ↑ afferent CN IX firing to solitary nucleus of the medulla → ↑ efferent CN X parasympathetic firing down to cardiac nodal tissue → ↓ HR → ↓ CO → ↓ cerebral perfusion.
Costochondritis	- Inflammation of cartilage at rib joints.

	- Will present as chest pain that worsens with palpation or when patient reaches over the head or behind the back. These two findings are clear indicators we have an MSK condition, not cardiac. - Can be idiopathic, caused by strain (e.g., at the gym), or even post-viral.
Pleurodynia	- MSK condition asked twice on 2CK material that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. Sometimes students choose pericarditis, etc., even though the presentations are completely disparate. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Viral pleurisy	- Viral infection causing inflammation of the pleura (layers covering the lungs), leading to sharp chest pain. - If this is the answer, CK will be normal (unlike pleurodynia, because it's not MSK).
Diffuse esophageal spasm	- Can cause angina-like pain in patient without cardiovascular disease. - I discuss DES more in Gastro chapter.
Gastroesophageal reflux	- Can present as chest pain confused for MI. ECG will be normal, clearly. - I discuss GERD more in Gastro chapter.

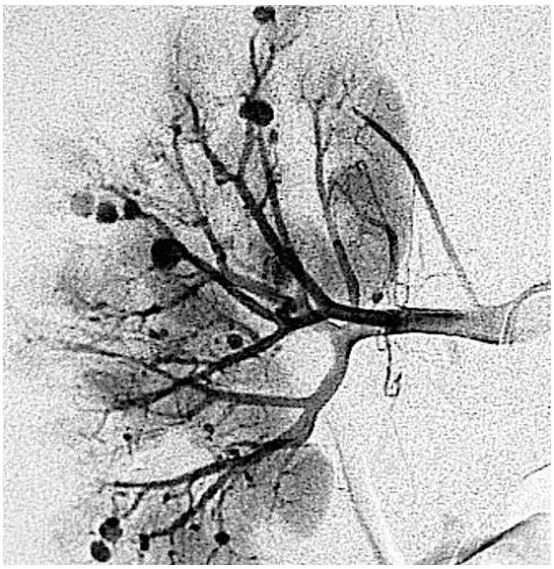
Arterial vs venous disease	
Arterial disease	- Caused by atherosclerotic disease; presents as **diminished peripheral pulses** in patient over 50 who has risk factors, e.g., diabetes, smoking, HTN. - Lower legs can be shiny and glabrous (trophic changes). - Arterial ulcers are small and punched-out; located on tops/bottoms of feet and toes.  - Ankle-brachial indices (ABIs) are first step in diagnosis (exceedingly HY on 2CK), which compare BP in ankle to the arm; if <0.9, this reflects ↓ peripheral blood flow due to atherosclerosis. - If ABIs are not listed as first step in diagnosis for whatever reason, choose Doppler ultrasound. There is one 2CK NBME Q where this is the case. - After ABIs, next step is exercise stress test (if listed) in order to determine exercise tolerance. If not listed, go straight to "recommend an exercise / walking program." Do not choose cilostazol first or arteriography as answers. - 2CK CMS Surg form 5 has "prescription for an exercise program" as the answer. Students say, "Why does it say 'prescription'?" No fucking idea. Ok? - Surgery is indicated in the event of critical limb ischemia, which is when there is chronic ischemic rest pain, ulcers, or gangrene. - First surgical intervention is usually angioplasty +/- stenting. Endarterectomy and bypass surgery are indicated for more severe blockages.

	- All patients with arterial disease should be on triad of 1) ACEi/ARB, 2) statin, 3) anti-platelet therapy (same as carotid stenosis). This is unrelated to the management sequence of exercise program → cilostazol → surgery.
Venous disease	- Congestion of venous system usually from valvular incompetence; idiopathic / familial; varicose veins are one type of venous disease and are not synonymous; patients can have venous disease without varicosities. - **Peripheral pulses are normal** (those reflect arteries, not veins). - Lower legs demonstrate "brawny edema," which is a brown, hemosiderin-laden edema due to ↑ pressure / micro-extravasations; hyperpigmentatory changes resulting in brown/red skin is known as stasis dermatitis, aka post-phlebitic syndrome; the latter is a term is asked on 2CK, so know the annoying vocab. - Venous ulcers are large and sloughy, and located at the malleoli.  - Diagnose with venous duplex ultrasonography of the legs; first treatment is compression stockings. Never choose answers such as venous stripping or glue agents, etc. - Venous disease ↑ risk for DVT and superficial thrombophlebitis (STP). If patient has active DVT or STP, answer = subcutaneous enoxaparin (heparin) over compression stockings. - Surgery is indicated if there are significant skin changes, venous ulcers, or for symptomatic varicose veins (i.e., pain, itching, swelling, cramps). - Sclerotherapy is often chosen as the first surgery, where a sclerosing (scarring) agent is injected into the vein, causing it to close where the blood is re-routed to other veins. I mention this for the sake of our Surgery PDF here, but on NBME Qs, I've never seen surgery as correct. The answer is basically always just compression stockings, or rarely heparin for DVT or STP as I mentioned above.

Cardiac stress test points	
- Most 2CK Qs that ask about stress tests are in the context of evaluating patients for perioperative MI risk. - It is rare the Q will force you to choose between different types of stress tests. 4/5 Qs will just list one stress test, where it is simply assessing, "Do you know a stress test should be done, period, in this scenario." - Stress tests are also done for peripheral arterial disease prior to recommending an exercise/walking program (as mentioned above).	
Exercise ECG	- Most common stress test. - The answer on USMLE for patients who have stable angina, where you're looking for ST depressions (i.e., evidence of ischemia) with exertion. - **Requires a patient has a normal baseline ECG in order to perform.** - In other words, the Q will give you a big 15-line paragraph + mention in the last line that the patient's baseline ECG shows, e.g., a LBBB from a year ago that's unchanged. This means ECG stress test is wrong in this situation, since you need to have a normal

	ECG to do it. The 1/5 Qs that force you to choose between stress tests want you to know this detail, basically, where you just choose the non-ECG stress test instead.
Exercise echo	- Used to look for heart failure (i.e., ↓ EF) with exertion, not overt ischemia. - In other words, the answer on USMLE for patients who don't get chest pain with exertion (i.e., don't have stable angina), but who get shortness of breath with exertion. This reflects, at a minimum, left heart decompensation with possible ↓ EF. - Also the answer for patients who have abnormal baseline ECG.
Pharmacologic	- Refers to numerous answer choices on USMLE – i.e., dobutamine-echo, dipyridamole-thallium. - The answer on USMLE for patients who cannot exercise, such as in the setting of angina when merely walking up a single flight of steps, or in patients imminently undergoing major surgery (e.g., AAA repair), where perioperative MI risk needs to be assessed. I have seen both of these scenarios on 2CK forms. - The USMLE will typically not force you to choose between stress tests. As I mentioned at the top of this table, they will usually just have the pharmacologic stress test as the only one listed. - Dobutamine is a β1-agonist that stimulates the heart (i.e., ↑ oxygen demand). Echo can then be done to look for ↓ EF (i.e., heart failure). - Dipyridamole is a phosphodiesterase inhibitor that dilates arterioles. HR goes up to compensate, thereby ↑ myocardial oxygen demand. Thallium is then used to look at perfusion of the myocardium.
Cardiac scintigraphy	- "Cardiac scintigraphy" is a broad term that refers to any evaluation of the heart in which some form of radiotracer is used (i.e., thallium, technetium, sestamibi). - This is the same as pharmacologic stress test for all intents and purposes on USMLE, even though technically it need not require myocardium is stimulated and can just be used to look at blood flow to the heart in the resting state. - The point is: This is an answer on 2CK sometimes as just another way of them writing "pharmacologic stress test." Choose it if the patient cannot exercise.
Myocardial perfusion scan	- "Myocardial perfusion scan" is one type of cardiac scintigraphy that evaluates blood flow to myocardium. It is non-invasive, whereas coronary angiography is invasive and evaluates coronary blood flow via the use of a catheter. - This is interchangeable with cardiac scintigraphy and pharmacologic stress test on USMLE for all intents and purposes.

Knee trauma causing popliteal arterial injury	
Knee dislocation	- If Q gives you MVA where the knee is injured + absent pulses distally, the sequence of answers they want is knee relocation first, followed by arteriography to look for popliteal artery injury. - In one of the 2CK Qs, they already tell you the knee is relocated, then the answer is "arteriography with runoff." Students say, "what's the runoff part?" No fucking idea, it's just what they want.
Penetrating trauma	- If Q gives gunshot wound to the knee + absent distal pulses, go straight to "surgical exploration" as the answer. - This could be thought of as the knee-equivalent of a gunshot wound to the abdomen, where straight to laparotomy (even if patient is stable) is the answer.

Vasculitides (pleural for vasculitis) for Surg	
Polyarteritis nodosa (PAN)	- Medium-vessel vasculitis that causes a "string of pearls" appearance of the renal vessels.  - Causes fibrinoid necrosis, which means it looks like fibrin but it ain't fibrin. - Offline NBME has "segmental ischemic necrosis" as the answer. - **Can be caused by hepatitis B.** - For whatever reason, USMLE wants you to know PAN spares the lungs – i.e., it does not affect the pulmonary vessels.
Takayasu arteritis	- Aka "pulseless disease." Classically affects Asian women 40s or younger. - Inflammation of large vessels, including the aorta. - Always affects the subclavian arteries (which supply the arms), which is why it can cause weakly, or non-palpable, pulse in the upper extremities.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many 2CK NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another 2CK Neuro CMS Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.
Thromboangiitis obliterans	- Aka Buerger disease; technically a vasculitis.

	- Dry gangrene of the fingers or toes seen generally in male over 30 who's a heavy smoker. - Treatment is **smoking cessation.** - Don't confuse with Berger disease, which is IgA nephropathy.
Ascending aortitis	- Tertiary syphilis can cause ascending aortitis + aortic aneurysm. - Causes "tree-barking" of the aorta.

Thrombophlebitis	
Deep vein thrombosis	- DVT will be unilateral thigh or lower leg swelling in patient with risk factors such as: post-surgery, prolonged sedentation, OCP use, Hx of thrombotic disorders (e.g., Factor V Leiden, prothrombin mutation). - Virchow triad for ↑ DVT risk: 1) venous stasis (e.g., post-surgery sedentation), 2) hypercoagulable state (e.g., estrogen use, underlying malignancy), 3) endothelial damage (i.e., smoking). - OCPs contraindicated in smokers over 35 because estrogen causes hypercoagulable state for two reasons: 1) estrogen upregulates fibrinogen; 2) estrogen upregulates factors Va and VIIIa. - USMLE loves nephrotic syndrome as cause of DVT (loss of antithrombin III in the urine → hypercoagulable state). - Antiphospholipid syndrome → DVTs despite paradoxical ↑ PTT (i.e., if PTT is high, you'd think you have bleeding diathesis, not thromboses); may or may not be due to SLE. Antibodies against phospholipids cause *in vivo* clumping of platelets + ↑ clot initiation, but disruption of *in vitro* PTT assay means ↑ PTT. - Major danger is DVT can embolize to lungs causing PE → acute-onset shortness of breath and tachycardia + death if saddle embolus. - Homan sign can mean DVT, which is pain in the calf with dorsiflexion of foot. - Diagnose DVT with duplex venous ultrasound of the leg/calf. - Treatment is heparin. - Harder surgery stuff for 2CK is that they care about prophylactic vs therapeutic doses of heparin. Prophylactic dose is lower-dose and is used perioperatively in patients with venous disease/stasis or who are high risk. If a patient has an actual full-blown DVT, however, give therapeutic dose, which is higher-dose. - There are two 2CK Qs on this stuff. One just mentions a guy going into surgery who has Hx of venous stasis → answer = "prophylactic heparin dose"; "therapeutic heparin dose" is wrong answer. - The second question gives a guy *who's already on prophylactic heparin* but gets a DVT anyway. The answer is then "heparin." It's weird because students are like, "Wait what? He's already on heparin though." And I'm like, yeah, but what they mean is, we have to give therapeutic dose now for the active DVT, which is higher dose. - DVT can rarely cause stroke if an ASD is present (paradoxical embolus). Dumb and low-yield, but it shows up, and students get fanatical over it.
Post-op migratory	- Thrombophlebitis means inflammation of a vein. - Post-surgery, this is usually due to changes in hemostasis and coagulability. - Will present as pink/red painful lesions appearing asymmetrically on the limbs within days of surgery. You just need to be able to diagnose this.
Trousseau sign of malignancy	- Migratory thrombophlebitis classically due to head of pancreas adenocarcinoma. But this can also be seen with adenocarcinomas in general, e.g., pulmonary.
Catheter-associated septic thrombophlebitis (CAST)	- Shows up on 2CK Surg form as patient who had a catheter in and then develops a 4-cm indurated, painful, fluctuant cord in his arm (refers to vein). - Answer = "excision of vein." Obscure question, but not my opinion. Take it up with NBME if you think it's weird.

Pelvic septic thrombophlebitis (PST)	- Important for 2CK Obgyn Qs. - The answer on USMLE in a woman who has post-partum endometritis (fever + tender lower abdomen) with persistent fever >48 hours despite antibiotics." - Endometritis can lead to ↑ risk of local infective clots in the ovarian veins. - If they give you a post-partum woman with sepsis (i.e., SIRS + infection), but the vignette doesn't fit PST as described above, the answer is "puerperal sepsis" on the Obgyn form. The latter is a more general term and can refer to many causes of post-partum sepsis (including PST confusingly enough).
Superficial thrombophlebitis	- Painful palpable cord in the ankle that may or may not track up to the knee. - Seen in patients with venous insufficiency. - Answer is "subcutaneous enoxaparin." Compression stockings are typically the answer for first step in venous insufficiency, but if you have an active ST or DVT, heparin must be given as first step. - There will occasionally be some intentional redundancy on my end with things I write in this doc if I believe they're HY enough (as with this).

Random HY cardio pharm points	
Verapamil	- Non-dihydropyridine calcium channel blocker (acts on nodal calcium channels). - **Causes constipation.** - Used for AF for rate control sometimes in place of metoprolol (don't worry about the use-case; USMLE doesn't give a fuck; just know MOA and side-effect).
Nifedipine	- Dihydropyridine calcium channel blocker (acts on arteriolar calcium channels → dilates arterioles → ↓ peripheral vascular resistance → ↓ BP). - **Causes peripheral edema / fluid retention.** - Used for essential HTN in patients without diabetes, atherosclerotic disease, or renal disease (if that sounds confusing, I talk about this in extensive detail in my HY Risk Factors PDF).
β-blockers	- Can cause depression and sexual dysfunction. - Avoid in patients with lung disease or history of severe or psychotic depression. No longer contraindicated in peripheral vascular disease in most cases. - Metoprolol used for rate-control in AF. - Labetalol used first-line in patients who have aortic dissection and traumatic rupture of the aorta. Nitroprusside comes after. - 2CK Q gives "esmolol + nitroprusside" as answer to a traumatic rupture Q, but almost always, they will just want "labetalol."
Amiodarone	- Potassium channel blocker. - Can cause TdP, greyish skin discoloration, and thyroiditis. - Used for VT in patients without coma or low BP.
Quinidine	- Sodium channel blocker. - Can cause TdP and cinchonism (headache + tinnitus).
Digoxin	- Directly blocks myocardial Na^+/K^+ ATPase pump → causes indirect inactivation of myocardial Na^+/Ca^{2+} ATPase → more Ca^{2+} remains in myocardial cell → increased contractility. - Also has parasympathomimetic effect at nodal tissue that slows HR. - In other words, digoxin both slows HR + increases contractility. - Hypokalemia can cause toxicity. This is because digoxin binds to extracellular K^+ binding site, so if less K^+ is around to compete, lower dose is needed to induce effect. - Toxicity presents classically as yellow/wavy "Vincent van Gogh" vision.
Lisinopril	- USMLE-favorite ACE inhibitor. - Can cause dry cough; also can ↑ serum K^+; avoid in hereditary angioedema. - Used for HTN in patients with pre-diabetes, diabetes, atherosclerotic disease, or renal disease (I talk about this in HY Risk Factors PDF in more detail).
Valsartan	- Angiotensin II receptor blocker (ARB).

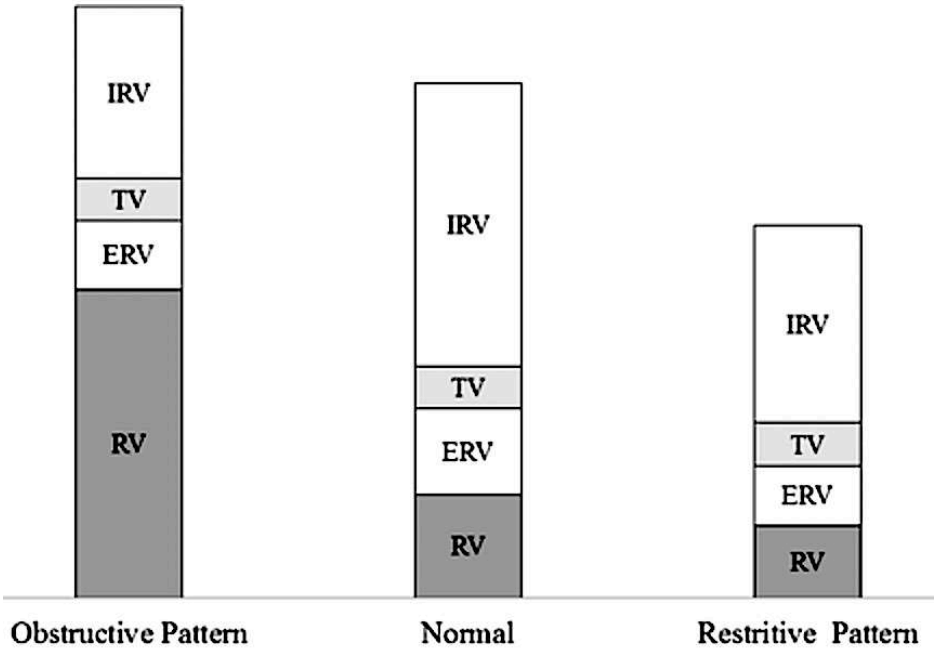
	- Use-cases are identical on USMLE to ACEi (i.e., if you see both as answer choices to a question, they're usually both wrong because they're the "same"). - Doesn't cause dry cough the way ACEi do.
Isoproterenol	- β1/2 agonist → increases HR and decreases peripheral vascular resistance.
Oxymetazoline, Phenylephrine	- α1 agonists → constrict arterioles → ↑ BP → HR ↓ due to baroreceptor reflex. - Highest yield uses on USMLE are for **nasal decongestion** → constrict capillaries within nasal mucosa → ↓ inflammation → relief of congestion. - Can cause rhinitis medicamentosa, which means rebound nasal congestion upon withdrawal if used non-stop for ~5 days. In other words, patients should use only as needed for a maximum of about ~3-4 days while sick. - Can theoretically be used in the acute setting of nasal bleeding. For surgery, choose nasal packing first for management. Then answers like α1 agonists and cauterization can be considered.
Hydralazine	- Dilates arterioles → ↓ BP. - Used for hypertensive emergencies in pregnancy. - Affects calcium currents (but not a calcium channel blocker).
Nitrates	- As discussed earlier, they liberate NO which ↑ guanylyl cyclase → relaxation of venous smooth muscle → venous dilation/pooling → ↓ preload on heart → ↓ oxygen demand → relief of anginal pain. - For sodium nitroprusside, choose arterioles as site of action.
Statins	- Statins have 2 HY MOAs on USMLE: 1) inhibit HMG-CoA reductase; 2) upregulate LDL receptors on hepatocytes. - Can cause myopathy and toxic hepatitis. An offline NBME has myopathy as correct over toxic hepatitis. - Indications for statins on 2CK vary depending on the source (i.e., whether to use the 70 vs 100 mg/dL cutoff in certain scenarios), but for USMLE, give if: - Age 20-39 if LDL > 190 mg/dL. - Age 40-75 if LDL > 100 mg/dL. - Age 20-75 in diabetic if LDL >100 mg/dL. - Some sources use 70, rather than 100, for the latter two cutoffs. What I can say is that I've seen 2CK NBME Qs where they use 100 as the cutoff (i.e., they give an LDL of 95 and statin is wrong, implying LDL is satisfactory). - Some sources incorporate a CVD risk % of >7.5%, which is more of a moot / pedantic talking point. I've seen one 2CK NBME Q where a CVD risk % shows up in the stem, but it doesn't rely on you knowing that point to get it right.
Fibrates	- Fibrates (e.g., fenofibrate) upregulate PPAR-α and lipoprotein lipase; best drugs to decrease triglycerides. - Give if triglycerides >300 mg/dL. - Can cause myopathy and hepatotoxicity, as well as cholesterol gall stones. - If the USMLE asks you why statins + fibrates combined have ↑ chance of myopathy, the answer is "P-450 interaction."
Ezetimibe	- Blocks cholesterol absorption in the small bowel.
Cholestyramine	- Bile acid sequestrant. Causes reduced enterohepatic circulation of bile acids at terminal ileum → liver must now convert more cholesterol into bile acids in order to replenish them → liver pulls cholesterol out of the blood to accomplish this.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.
Niacin	- Vitamin B3. Two MOAs USMLE wants you to know: 1) ↓ VLDL export by the liver; 2) ↑ HDL more than any other medication. - Deficiency → pellagra → 3Ds → dementia, dermatitis, diarrhea. - Can present as delirium instead of overt dementia (biggest risk factor for delirium is underlying dementia, so old patient with delirium often has underlying cognitive decline); dermatitis will present on USMLE as either hyperpigmentation of the skin of the forearms or Casal necklace. - Administration can cause flushing (caused by prostaglandin, so mitigated by asprin), gout, and insulin resistance.
Procainamide	- Sodium channel blocker that can be used in Wolff-Parkinson-White.

	- Can cause drug-induced lupus with anti-histone antibodies.
Flecainide	- Sodium channel blocker used as first-line rhythm control in patients fail rate-control for AF. - Patient should have no structural or coronary artery disease. Otherwise use a potassium channel blocker like amiodarone or dofetilide.
Dofetilide, Ibutilide	- Potassium channel blockers. - Can cause torsades (asked directly on NBME).
Omega/fish oils	- Decrease triglycerides.
Morphine	- 2CK Surgery form gives patient who is in pain from surgery despite being on max doses of morphine + they say blood pressure is elevated + ask what is next best step → answer = "increase bolus of morphine." Unusual, since they say max dose of morphine already, but 1) we always full treat pain, and 2) explanation for the Q talks about how pain can lead to high BP. - My point here is that just be aware HTN can be caused by pain → treating pain can bring down the BP. - Separately, treatment of MI, acute limb ischemia, and sickle crisis usually includes oxygen + morphine. - Answer on NBME exam for how a patient using a self-administering pump develops morphine overdose = "morphine is converted into active metabolites that accumulate." - Otherwise morphine is odd drug for me to put in this table; didn't know where else to put it though.

Surg Pulmonary

Obstructive vs restrictive lung disease		
	Obstructive	**Restrictive**
HY conditions	- Asthma. - COPD (chronic bronchitis + emphysema). - **Old age (i.e., healthy person age 70 vs 20)**	- Pulmonary fibrosis due to any cause (i.e., idiopathic; radiation; drugs; auto immune disease). - Pneumoconioses.
FEV1	↓	↓/↔
FVC	↓	↓
FEV1/FVC	↓ (<70%; normal is 70%)	↑/↔ (>70-80%)
TLC	↑	↓

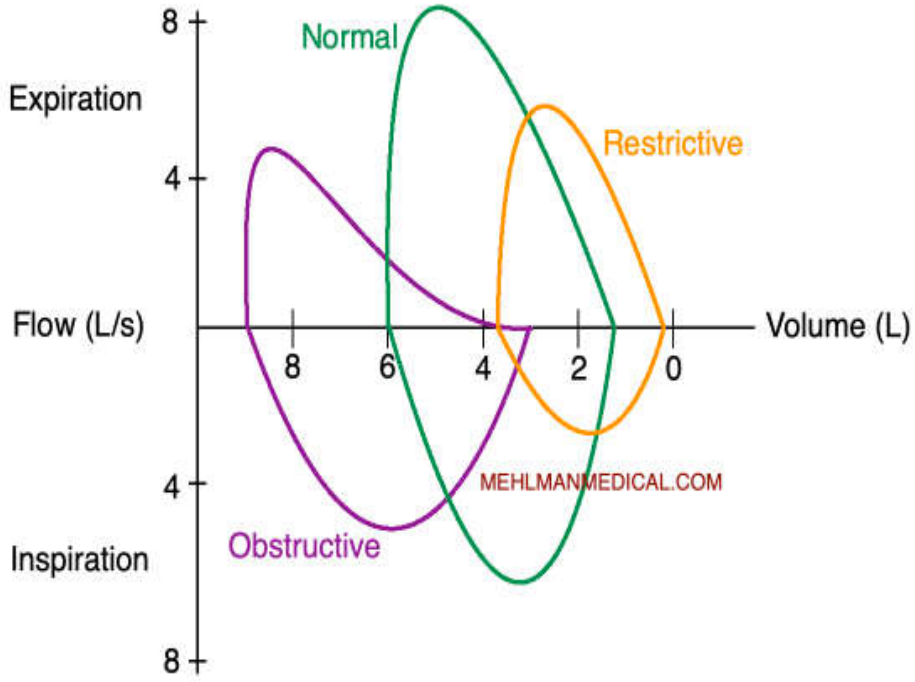
- The above lung volumes are the HY ones you need to know for USMLE.
- FEV1 = Forced Expiratory Volume in 1 second = amount of air patient can forcefully blow out in 1 second, starting from maximal inhalation.
- FVC = Forced Vital Capacity = amount of air patient can blow out starting from maximal inspiration all the way until maximal expiration.
- The reason FEV1/FVC is ↑/↔ in restrictive, as compared to ↓ in obstructive, is because of **radial traction,** which is a "stickiness" on the outside of the airways (such as from fibrosis) that prevents them from closing as rapidly as obstructive.
- TLC = Total lung capacity → increased in obstructive due to air-trapping (hence "obstructive").



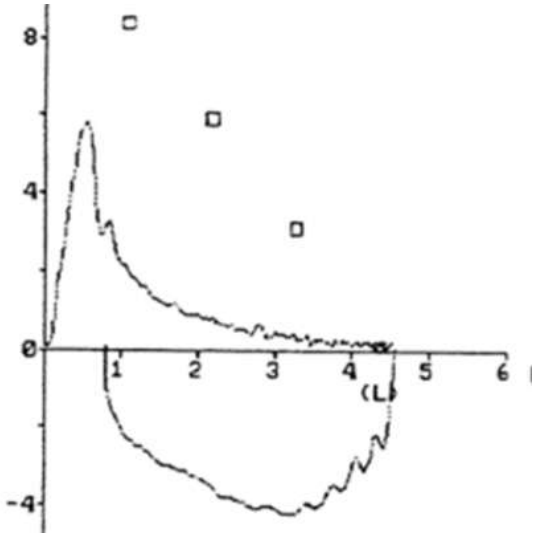
TV	↔	↔
ERV	↓	↓
IRV	↓	↓
RV	↑↑	↓
FRC (ERV + RV)	↑	↓
DLCO	↓ (but ↑ in asthma on USMLE)	↓ (↔ if neuromuscular)

- TV = Tidal Volume; generally preserved in lung diseases.
- ERV = Expiratory Reserve Volume = additional amount that can be forcibly exhaled following the end of a normal exhalation.

- IRV = Inspiratory Reserve Volume = additional amount that can be forcibly inhaled following the end of a normal inhalation.
- RV = Residual Volume = amount of air remaining in the lungs after the end of maximal expiration.
- FRC = Functional Residual Capacity = total amount of air remaining in lungs after end of normal expiration.



- Don't freak out about the above flow-loops. For USMLE (all Steps), you just need to know their general shape. I'd say the most important point is that the expiratory component of the obstructive curve (purple) tends to have a scooped-out, or concave, or L-shape.



- What USMLE will do is give you a real bootleg Windows 95-type spirometry curve as per above in a 34-year-old + no family Hx + 5-year smoking Hx. You look at curve and say, "No idea what I'm looking at, but top of curve is concave, and I know that's obstructive. So between COPD or asthma here, which are both concave and hence possible answers, I'll choose asthma as most likely in this particular patient."

HY respiratory cancers for Surg + Screening	
Screening	- Lung cancer screening with annual low-dose chest CT is done in patients who meet all of the following: 1) age 50-80; 2) have 20-pack-year Hx of smoking; and 3) smoked within the past 15 years.
Small cell bronchogenic carcinoma	- Small basophilic (purple) cells. NBME can describe these cancer cells as "twice the size of lymphocytes." - Occurs centrally in the lung (i.e., hilar / medially). Smoking biggest risk factor. - Paraneoplastic syndromes exceedingly HY: 1) SIADH (ADH secretion); 2) Cushing syndrome due to ACTH secretion; 3) Lambert-Eaton syndrome (production of antibodies against presynaptic voltage-gated calcium channels); 4) Cerebellar dysfunction / ataxia (anti-Hu/-Yo antibodies). - Treatment on 2CK is "chemotherapy." Do not do surgery. I believe this is the only time I've seen straight-up "chemotherapy" as correct answer on NBME.
Squamous cell carcinoma	- Stains positive for keratin; may have keratin pearls on histo . - Similar to small cell, occurs centrally in the lung (i.e., hilar / medially). Smoking biggest risk factor. In other words, the two cancers that start with an "**ssss**" sound for "**c**entral" (i.e., **S**mall cell and **S**quamous cell) are **C**entral. - Can cavitate → if USMLE gives you lung cancer with a cavitation (i.e., cavity/hole), the answer is squamous cell. - Highest yield point is that it secretes PTHrp (parathyroid hormone-related peptide). This acts like PTH and increases calcium / decreases phosphate, but this is not the same as PTH. Endogenous PTH is *suppressed* due to negative feedback from high calcium. In other words, choose a down arrow for PTH in squamous cell carcinoma of the lung.
Adenocarcinoma of the lung	- Adenocarcinoma means cancer of glands. If the Q says something about biopsy showing glandular morphology, you know they're talking about adenocarcinoma. - The answer on USMLE for lung cancer in a non-smoker; classically "female non-smokers," but I've seen NBME Qs with this in men. - Normal ground/Earth radiation due to radon is accepted cause of lung cancer in non-smokers. There is NBME Q where they mention non-smoker living in a basement and develops lung cancer. The correlation is probably nonsense in real life, but it's in an NBME question somewhere. - Does not occur centrally on NBME exam, unlike small cell and squamous, and hence will be described as apical or peripheral lung lesion in non-smokers. - Apical tumors can are known as Pancoast tumors and cause Horner syndrome (miosis, ptosis, anhidrosis) due to impingement on C8 superior cervical ganglia (sympathetic nerves). They can also cause SVC syndrome (flushing of the face + congestion of neck veins) or brachiocephalic syndrome (only right side of face/neck) due to impingement on venous return. (+) Pemberton sign is worsening of flushing + neck vein congestion when raising the arms above the head. - Can be associated with migratory thrombophlebitis (Trousseau sign of malignancy). The latter is not limited to head of pancreas cancer. Adenocarcinomas in general are known to be associated with hypercoagulable state due to malignancy. USMLE won't ask specific mechanism, but liberation of tissue factor (factor III) by these cancers is a proposed etiology. - Associated with hypertrophic osteoarthropathy (clubbing + hand pain due to lung cancer); mechanism is fibrovascular proliferation. Literature says adenocarcinoma of lung is most common cause, although the association isn't 100% specific.
Large cell carcinoma	- Nonexistent lung cancer on USMLE. Bogus/garbage diagnosis. I don't think I've ever seen this assessed.
Bronchogenic carcinoid tumor	- Pulmonary nodule that secretes serotonin or serotonin-like derivatives, resulting in carcinoid syndrome (i.e., tachycardia, flushing, diarrhea). - A type of neuroendocrine tumor.

	- Carcinoid tumors are classically appendiceal and of the small bowel, but you should be aware that bronchogenic carcinoids exist. - USMLE wants urinary 5-hydroxy indole acetic acid (5-HIAA) for initial step in Dx.
Mesothelioma	- Cancer of "mesothelial cells" (answer on NBME) seen in patients with prior occupational exposure to asbestos. - Asbestosis occurs first, which then gives rise to mesothelioma years later. - Shipyard and construction workers are buzzy for prior occupational exposure. - Asbestosis will be described as pleural or supradiaphragmatic plaques ("soft tissue plaques seen on CXR"). Pulmonary biopsy shows ferruginous bodies. - Calretinin (+); a protein that is highly indicative of mesothelioma on staining. - Mesothelioma appears as a whiteish cancer and is described as an "encasing rind of pleural-based tumor" (i.e., circumferentially surrounds/wraps around the lungs).
Nasopharyngeal carcinoma	- Can be caused by EBV. - A type of squamous cell carcinoma of the airway.
Laryngeal cancer	- Squamous cell carcinoma of vocal cords. - Smoking is major risk factor. - New NBME Q wants you to know this spreads to **cervical lymph nodes.**
Laryngeal papillomatosis	- Pediatric condition characterized by warts of the vocal cords. - Lesions will have papillary structures on biopsy. - Due to HPV 6/11 exposure from maternal vaginal canal.

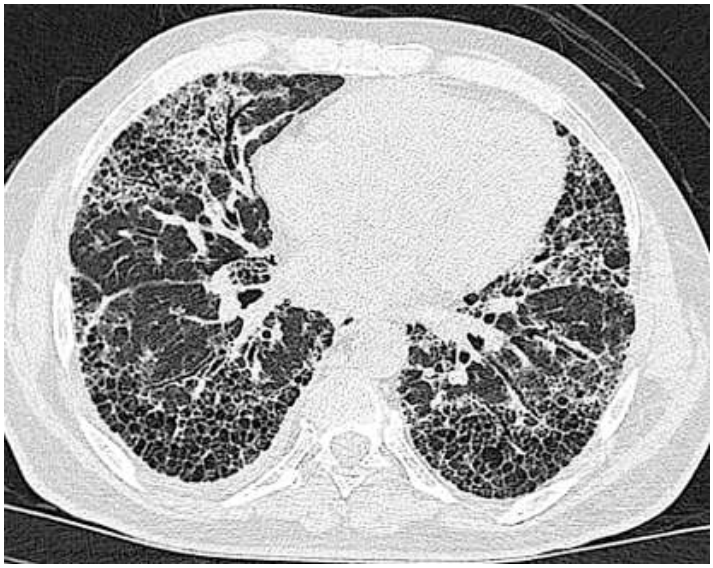
Pneumoconioses for Surg	
Asbestosis	- As already mentioned above in the mesothelioma section, this is associated with shipyard workers, construction workers, and electricians. It can give rise to mesothelioma later in life. - Ferruginous bodies (dumbbell-shaped structures) on lung biopsy. - Causes restrictive lung pattern.
Berylliosis	- Occupational exposure to beryllium in the aerospace / aeronautical industry. - Causes restrictive lung pattern. - Can cause granulomas.
Silicosis	- Occupational exposure to silicon (sand) in foundry or stone quarry workers. - Can cause egg-shelf calcifications in upper lobes. - Increases risk of tuberculosis infections. - Avoid anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) in these patients due to increased TB risk (TNF-α needed to suppress/fight TB).
Anthracosis	- Aka "coal miner's lung." - Black discoloration of the lung. - Can be either obstructive or restrictive.
Caplan syndrome	- Rheumatoid arthritis + any pneumonociosis, presenting as pulmonary nodules. - Clinical relevance is that patients with RA are at increased risk for developing pneumoconioses if they have a workplace exposure.

HY General lung conditions for Surg	
Idiopathic pulmonary fibrosis (Usual interstitial pneumonitis)	- The answer on USMLE for a patient over the age of 50 who has 6-12+ months of unexplained dry cough. This is how it shows up 4/5 times. - Textbook restrictive lung disease, with $\leftrightarrow$ or $\uparrow$ FEV1/FVC. The reason for the FEV1/FVC being greater than in obstructive lung disease is radial traction, as mentioned earlier. - CXR and CT scan show "reticular" or "reticulonodular" pattern. These descriptors are exceedingly HY on USMLE, where students will overlook them in the vignette, but they are hugely buzzy for restrictive lung disease. They are colloquially known as "honeycombing," but I have not

seen the USMLE give a fuck about the latter colloquialism. They frequently just say “reticular” and “reticulonodular,” and then you know right away, “Boom. Restrictive lung disease,” i.e., fibrosis, etc. Tangentially, it’s to my observation that NBME will say “reticulogranular” frequently for NRDS, but the two vignettes are clearly disparate anyway.

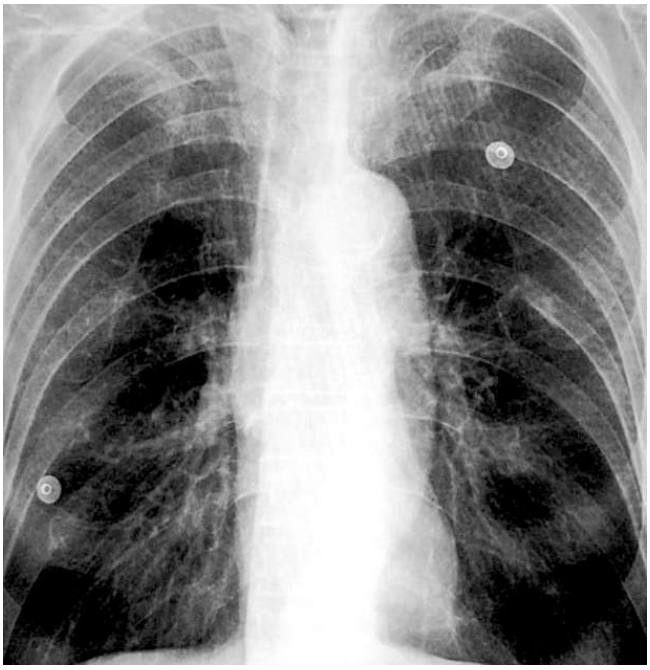


- After the CXR and spirometry are performed, 2CK wants “high-resolution CT of chest” as answer for next best step.



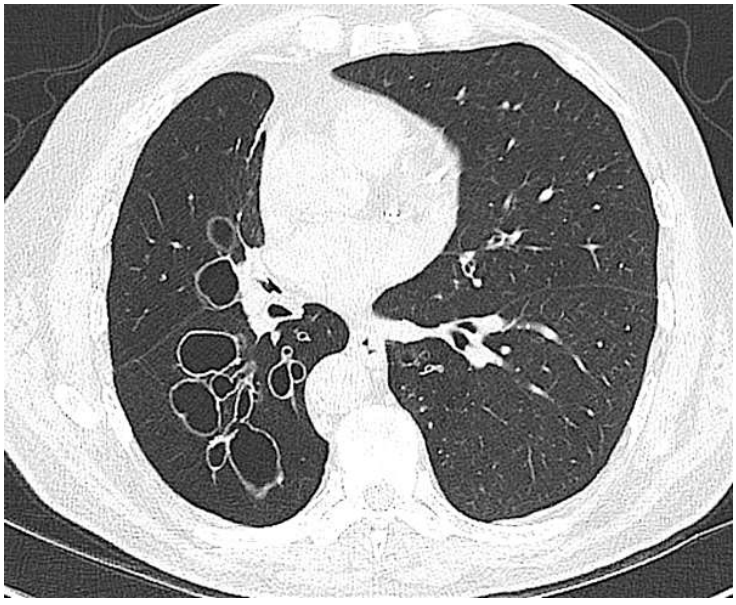
“Honeycombing” = **reticular / reticulonodular pattern**.

- New 2CK NBME wants “lung biopsy” as answer to confirm diagnosis of interstitial lung disease (i.e., idiopathic pulmonary fibrosis) after imaging.
- Vignette can also mention loud P2 (means pulmonary hypertension) with “dry inspiratory crackles heard bilaterally.”
- 1/10 times, the Q will be patient over 50 with increasing fatigue and shortness of breath over 6-12 months, with only 1 month of cough, where it initially sounds like heart failure, and they’ll say CXR shows “interstitial markings” instead of reticular/reticulonodular patterning. However, they say patient has “↑ FEV1/FVC showing restrictive pattern” in the stem, which gives it away.
- You need to know that **“usual interstitial pneumonitis” (UIP)** is another name for idiopathic pulmonary fibrosis. Yes, the name is weird, but it’s not my opinion and it’s asked on NBMEs, where instead of writing “idiopathic pulmonary fibrosis” as the answer, they write “usual interstitial pneumonitis.” UIP is technically a broad term that can refer to many restrictive lung conditions, but as I said, on NBME they use this synonymously with idiopathic pulmonary fibrosis.

	- **Tx on 2CK = pirfenidone** → anti-fibrotic agent that inhibits TGF-β-mediated synthesis of collagen.
Chronic obstructive pulmonary disease (COPD)	- COPD = chronic bronchitis + emphysema. - Smokers will have combination of the two. When we say a smoker has COPD, we are saying he/she has chronic bronchitis + emphysema at the same time. - The term COPD can in theory apply to any obstructive disease of the lung that is chronic (e.g., asthma, Kartagener, etc.) but when the term is used without any specific condition attached, it refers to the combo of chronic bronchitis and emphysema.  - Hyperinflated lungs in COPD (due to air trapping) can push the heart to the midline. NBME will say there's a "long, narrow cardiac silhouette," or a "point of maximal impulse palpated in the sub-xiphoid space." In left ventricular hypertrophy, in contrast, there will be a lateralized apex beat, or a point of maximal impulse in the anterior axillary line. - Home oxygen is indicated on 2CK if O2 sats are: - <88% saturation (55 mm Hg), or - <89% saturation (60 mm Hg) if the patient has cor pulmonale. - First-line Tx is now considered to be a long-acting muscarinic receptor antagonist (i.e., LAMA such as tiotropium) or a long-acting β2 agonist (i.e., LABA such as olodaterol), either alone or in combination. - If insufficient, add an inhaled corticosteroid (e.g., fluticasone). - Should be noted that a question on an earlier NBME form that has ipratropium (SAMA) as the answer for 1st-line in COPD, but there aren't any other anti-muscarinic or β2 agonists listed. So the point is that if you are forced to choose a SAMA or SABA (i.e., albuterol) on USMLE, these are OK, but newer guidelines say start with a LAMA or LABA. - "Exacerbation of COPD" as a diagnosis on USMLE will always give a patient with *high CO2*. This is really important. In other words, if they give you a big paragraph and you're not sure of the Dx, if you see CO2 is not elevated, the answer is not COPD exacerbation. - COPD exacerbations can be triggered by minor chest infections, so always **give antibiotics** on 2CK, even though etiology is usually viral.

	- Patients with COPD are chronic CO2 retainers, so they will have a chronic respiratory acidosis (i.e., ↑ CO2 and ↑ bicarb; pH can either be ↓ or compensated back into normal range). - As mentioned earlier, for 2CK Surg, lung cancer screening with annual low-dose chest CT is done in patients who meet all of the following: 1) age 50-80; 2) have 20-pack-year Hx of smoking; and 3) smoked within the past 15 years.
Chronic bronchitis	- Chronic bronchitis = productive cough for at least 3 months in a year, for 2+ years. - Reid index >0.5 (ratio of the thickness of bronchial mucous-secreting glands to the bronchial wall itself). <0.4 is considered normal. - Chronic bronchitis is known as "blue bloater" because the mucous sits in the alveolar spaces and impairs gas exchange. This results in a shitload of hypoxic vasoconstriction → patient can become acutely blue and hypoxic during exacerbations. - The hypoxic vasoconstriction of pulmonary vessels causes pulmonary hypertension (i.e., if the vessels constrict, then pressure in the more proximal pulmonary arterioles increases). This increased afterload on the right heart can lead to right ventricular hypertrophy and right heart decompensation. When right heart failure (i.e., evidence of JVD or peripheral edema) occurs due to a pulmonary cause, we now call that cor pulmonale, as discussed in the Cardio PDF. - The pulmonary hypertension can cause a loud P2 and tricuspid regurg prior to cor pulmonale occurring. - Don't confuse chronic bronchitis with acute bronchitis, which can present as a worsening cough in patient (with or without COPD) following a viral infection. This is a temporary irritation/inflammation of the airways that is self-resolving.
Emphysema	- Emphysema = loss/destruction of alveolar surface area. - When alveolar surface area is reduced, so is alveolar capillary surface area, since the capillaries are within the alveolar walls → ↓ gas exchange. - Known as "pink puffer," since although gas exchange is impaired, sudden and acute increases in hypoxic vasoconstriction, as with chronic bronchitis, are not a feature. - "Bullous changes" on CXR are synonymous with emphysema on USMLE. - Smokers can get centri-acinar emphysema (proximal alveolar structure is destroyed). - α1-antitrypsin deficiency patients get pan-acinar emphysema (entire alveolus destroyed).
α1-antitrypsin deficiency	- Codominant genetic condition resulting in emphysema and hepatic cirrhosis. - *ZZ* allele combo is worst and results in disease (asked on USMLE). - α1-antitrypsin is an enzyme produced by the liver that travels to the lungs and breaks down neutrophilic elastase. Elastase is an enzyme that normally causes damage to the alveoli. Homeostatically, elastase is required in small amounts for normal pulmonary function and cell turnover, but in high amounts it destroys the alveoli, resulting in emphysema. - Vignette will give young adult who has sibling or parent who's had early-onset emphysema or cirrhosis. - The USMLE Q can absolutely say the patient is a smoker or drinks alcohol. For instance, they might say the patient is 34 and has been smoking for 5 years and has bullous changes on CXR, or that the father died from alcoholic cirrhosis, and the student thinks, "Oh that can't be α1-antitrypsin deficiency though because they said smoking/alcohol."

	But the key point is that this condition *increases the risk* for early-onset emphysema and cirrhosis. Normally, COPD should take 20+ years of smoking to develop, not 5.
Asthma	- Bronchospasm that occurs either idiopathically/hereditarily, or in response to certain allergens or cold air. - One-third of patients with asthma only present with a dry cough and no problems breathing. This is called cough-variant asthma. It will often present as a patient with a dry cough that's worse in the winter. - Can also present as part of atopy constellation – i.e., dry cough in winter, seasonal allergies / rhinoconjunctivitis / urticaria in spring, and eczema in summer. - Aspirin-induced asthma mechanism HY for USMLE → inhibition of COX by aspirin → shunting of arachidonic acid down lipoxygenase pathway → increased leukotrienes → increased bronchoconstriction. - Samter triad = aspirin allergy, asthma (due to aspirin), nasal polyps. - "Increased expiratory phase" is a buzzy phrase that will be thrown into quite a few asthma vignettes. It is not specific for asthma and can refer to any obstructive pathology, but I just make note of it here because you'll see it quite a bit for asthma and say, "What's that mean?" → in obstructive conditions, it takes us a lot longer to exhale (FEV1 is ↓↓). - Acutely, asthma causes ↓ CO2, ↑ pH, ↔ bicarb. This is an acute respiratory alkalosis. Even though the patient is having difficulty breathing, CO2 diffuses quickly, whereas O2 diffuses slowly, so insofar as the patient's respiratory rate is ↑, CO2 will be ↓. Bicarb is unchanged because it takes the kidney about a day to alter excretion. In contrast, CO2 is ↑ in COPD because there is large amounts of hypoxic vasoconstriction due to excessive mucous (chronic bronchitis) or ↓ alveolar surface area, so CO2 can't get out, even with faster breathing. In asthma, alveolar surface area is intact, and the degree of mucous production and hypoxic vasoconstriction is not nearly as bad as chronic bronchitis. - The combination of ↓ CO2 and ↓ O2 in acute asthma attack is known as type I respiratory failure. Eventually, the patient will begin to fatigue and breathing rate will slow. This will be observed as CO2 and pH rebounding to normal levels *despite* O2 remaining low. This means the patient is in transition to a type II respiratory failure (i.e., hypoventilation), where we have ↑ CO2 and ↓ O2. USMLE wants intubation as the intervention. - 2CK Q gives vignette of asthma and then asks for best initial step in diagnosis → answer = "spirometry." The expiratory component of the curve, as discussed earlier, will appear concave. - Should be aware methacholine challenge can also be done to diagnose. This is a muscarinic agonist that bronchoconstricts and can induce symptoms. Never give during acute episode. This can be tried between episodes. But if you're forced to choose diagnostic modality, go with simple spirometry. - USMLE cares about both outpatient management of asthma as well as acute attacks. - For outpatient management: 1) First-line Tx is β2-agonist (albuterol) for acute attacks. 2) If the Q says the patient has weekly episodes, or they ask for what could decrease risk of future episodes if patient is already on albuterol, the answer is inhaled corticosteroid (ICS; i.e., fluticasone). 3) If the combo of albuterol + ICS is insufficient, the next step is increasing the dose of the ICS.

	4) If that doesn't work, adding a LABA (i.e., salmeterol) is the next step. 5) If insufficient, agents such as mast cell stabilizers, leukotriene blockers, etc., can be tried. If patient has Hx of aspirin allergy, the latter in particular can be effective. 6) Last resort for outpatient asthma is oral prednisone. It is most effective at decreasing recurrence of episodes, but we want to avoid it if at all possible because of the risk of Cushing syndrome and ↓ linear bone growth in Peds. - Acute management of severe asthma attack (i.e., not outpatient) is nebulized albuterol, oxygen, and IV methylprednisolone. Inhaled corticosteroids have no role in acute asthma management. - NBME for 2CK wants you to know that any asthma patient who requires hospital management for an acute episode must automatically be given ICS (i.e., fluticasone) on discharge. In other words, if patient isn't currently on an ICS, it must be commenced at discharge. I'm explicit here because this particular NBME Q doesn't mention albuterol anywhere in the stem, but inhaled fluticasone is the answer. - Another 2CK NBME Q gives a patient who is currently not receiving any asthma medications but who gets 2+ episodes weekly → answer = "inhaled fluticasone + albuterol." Students think this is weird because they jump straight to dual management + are even audacious enough to put the ICS before albuterol in the wording of the answer, but it's what they want. Any patient with 2+ episodes per week needs ICS in addition to the albuterol and can be commenced right away on dual therapy.
Bronchiectasis	- Dilation of the airways (ectasia means dilation) that occurs due to "loss of musculature of the airways." - Transverse CT scan will show cystic dilation of the airways.  - Most common cause worldwide is TB. Most common cause in western countries is CF. But on USMLE, it will usually be a smoker. Asthma does not cause bronchiectasis. - Presents almost always as "cups and cups of foul-smelling sputum." - "Foul-smelling" means anaerobes, such as *Bacteroides*. - One question on 2CK Peds form gives bronchiectasis as the answer in a child who has right middle lobe syndrome, where they say there is scant

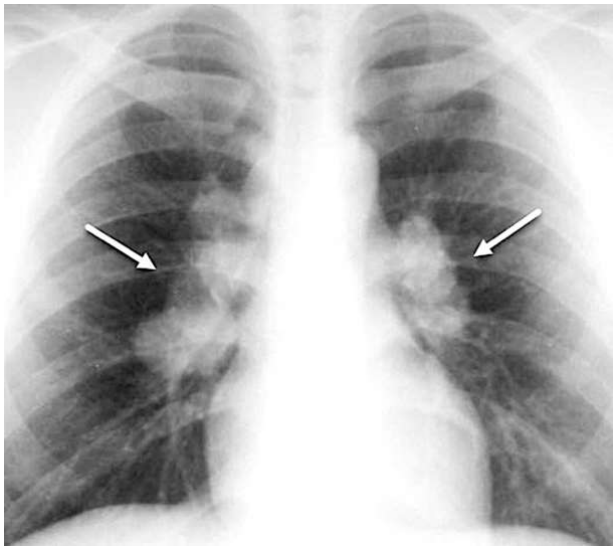
	white sputum (not cups and cups) and a thin, linear opacity visualized in the right middle lobe. If you think it's weird, complain to NBME not me. - **Clubbing** tends to be seen in these patients. It's not mandatory, but it's to my observation USMLE likes it for bronchiectasis.
Atelectasis	- Confusing word that refers to "lung collapse," or "collapse of alveoli." - Highest yield point for USMLE is that it is the most common cause of fever within 24 hours of post-surgery. If this is the first time you're reading this, that might sound weird, but this is pass-level and extremely important for 2CK. - There is one 2CK Q I've seen where they say a woman had a C-section two days ago and the answer was still atelectasis, so even though it's most common <24 hours, just be aware one Q exists where, oh em gee, it's 2 days later. - The mechanism is related to combo of pain meds + sedentation, where breathing becomes slower + shallower in hospital bed, leading to mild collapsing of some alveoli. This is why breathing exercises can be important post-surgery. - Will often present as **bibasilar shadows or opacities.** In other words, patient had surgery yesterday + now has fever + CXR shows mild opacity at the lung bases → answer = atelectasis. - NBME assesses obstructive (aka resorptive) atelectasis. This is when an area of lung distal to an obstruction from, e.g., a tumor, can cause alveoli to collapse. This then increases the chance for pneumonia distal to the obstruction. - 2CK IM form has "endobronchial obstruction" as answer for distal area of lung collapse (i.e., atelectasis) in patient with lung cancer; "vascular occlusion by tumor" is wrong answer (makes sense, as the tumor obstructs the respiratory tree, not blood vessel, in this case, but I've seen students accidentally choose the latter).
Obstructive sleep apnea (OSA)	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). What USMLE wants you to know: - These patients develop chronic respiratory acidosis – i.e., ↑ CO2, ↑ bicarb, pH ↔/↓. - Cor pulmonale can occur as a result of pulmonary hypertension from hypoxic vasoconstriction. JVD or peripheral edema will be seen with cor pulmonale. Descriptors such as RBBB, right-axis deviation on ECG, and wide splitting of S2 all mean RVH. If the patient merely has pulmonary hypertension but not yet cor pulmonale, the vignette can say loud P2 or tricuspid regurg (holosystolic murmur that ↑ with inspiration). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose.
Anaphylaxis	- Acute dyspnea and bilateral wheezes in patient who was gardening (stung by a bee), who ate a particular food (e.g., peanuts), or who was commenced on a recent drug (e.g., TMP/SMX). - Vignette often gives tachycardia and low BP. - Mechanism is IgE crosslinking on surface of mast cells and basophils that leads to degranulation and histamine + prostaglandin release. Eosinophils can be recruited in response. - USMLE wants: ↓ vascular resistance, ↑ CO, ↓/↔ PCWP. - Tx = **intramuscular epinephrine** → the strong β2-agonistic effect opens the airways; the strong α1-agonistic effect constricts the arterioles and restores BP. - NBME 9 for 2CK asks, "In addition to self-injectable epinephrine therapy, what is most appropriate therapy to ↓ recurrences" → answer =

	venom immunotherapy (VIT). What this does is desensitizes the patient to the antigen by allowing him/her to develop neutralizing IgG antibodies against it.
Scombroid	- Often misdiagnosed as seafood allergy. - Bacteria in decaying meaty fish (e.g., mahimahi) can break down histidine in the fish into histamine, which causes an allergic-/asthma-like presentation. - Patients with history of asthma have greater risk of mortality.
Shellfish allergy	- Can cause allergic-/asthma-like presentation. - Occurs literally after eating shellfish (i.e., clams, mussels). - Don't confuse with scombroid. Students get trigger-happy when they learn weird conditions. If USMLE gives shellfish, the answer is shellfish allergy, not scombroid. The latter is from meaty fish and not an allergy.

HY Congenital lung diseases for Surg	
Cystic fibrosis	- Autosomal recessive; chromosome 7; mutations in *CFTR* gene. - Dried up (inspissated) pancreatic secretions lead to "exocrine pancreatic insufficiency" (HY phrase for Surg/IM) and fat-soluble vitamin malabsorption. It is to my observation that vitamin E deficiency is the highest yield vitamin deficiency due to CF across NBME questions, which will present as neuropathy in a child who has CF. - Meconium ileus HY; refers to failure to pass stool in the first 24 hours after birth. - CF can sometimes cause nasal polyps. - *Pseudomonas* pneumonia is HY. - Sweat glands reabsorb Cl^- rather than secrete it. This is impaired in CF, so we have ↑ Cl^- in the sweat (i.e., >60 mEq/L). The sweat chloride test is more accurate for diagnosis than genotyping due to allelic heterogeneity of *CFTR* gene, where many different mutations cause the disease and can be difficult to detect with routine screening panels. - A transepithelial nasal voltage test can also be done, where there is ↑ nasal potential difference. - CF medications not HY for Step, but you could be aware that -caftor agents (e.g., ivacaftor, lumacaftor) facilitate with proper CFTR channel localization to the cell membrane + structural folding. - Dornase-alfa (correct, not alpha) is a nucleotidase that softens mucous. - Guaifenesin is also a mucous-softening agent.
Primary ciliary dyskinesia (Kartagener syndrome)	- Abnormality of cilia function due to defective dynein arm. - Answer on USMLE for a question that sounds like CF but patient has situs inversus or dextrocardia – i.e., patient will have recurrent pneumonias and organs (or just heart) on opposite side of body. - Women can get ectopic pregnancy due to abnormal Fallopian tube cilia. - This diagnosis sounds straightforward enough, but I've seen plenty of students miss these questions. For the situs inversus, they will describe this as cardiac sounds loudest on the right, or there being a large mass palpable beneath the left costal margin (liver on the opposite side).
Primary pulmonary hypertension (PPH)	- Answer on USMLE for a woman 20s-30s, non-smoker, who has increased pulmonary vascularity/markings and either a loud P2 or triscupid regurg (as I talked about in cardio section, these are HY findings for pulmonary hypertension). - "Primary" means the pulmonary hypertension inherently *starts with* the lungs and is not due to a secondary (i.e., external) cause such as smoking, CF, systemic sclerosis, etc. - Due to mutations in BMPR2 gene.

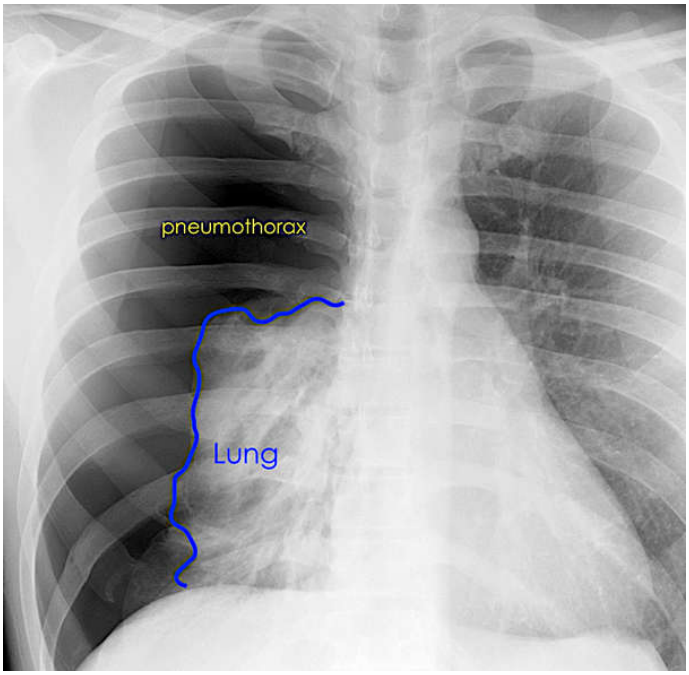
	- Do not confuse this with cor pulmonale. Recall that the latter is right heart failure findings or RV structural changes due to a pulmonary cause. If the question gives evidence of RVH (i.e., RBBB, right-axis deviation on ECG, boot-shaped heart on CXR), or has overt right heart failure findings (i.e., JVD or peripheral edema), then we can say the patient now has cor pulmonale due to primary pulmonary hypertension. A loud P2 and tricuspid regurg, however, are HY findings for pulmonary hypertension that don't necessarily mean cor pulmonale. - One of the highest yield points is that endothelin-1 is ↑ in these patients. This is a vasoconstrictor that is ↑ in pulmonary hypertension from any cause, but notably USMLE likes it for PPH. - Bosentan blocks endothelin-1 receptors and is HY Tx.

HY Autoimmune-related pulmonary DDx for Surg	
Systemic Sclerosis (aka scleroderma)	- Idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Divided into limited and diffuse subtypes. - **Limited scleroderma** = **CREST** syndrome (**C**alcinosis, **R**aynaud's, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias). - USMLE can describe Raynaud as color change of the fingers with cold weather. They describe sclerodactyly as tightening of the skin of the fingers. Esophageal dysmotility presents as GERD. I haven't seen NBMEs give a fuck about calcinosis, which is abnormal Ca^{2+} deposition in tissues. - **Diffuse scleroderma** = CREST syndrome + renal involvement (presenting as ultra-HY BP, e.g., 220/120, due to renal involvement causing a surge in RAAS. If patient doesn't have high BP, it's not diffuse type on USMLE. - Apart from just being able to diagnose these conditions, the highest yield point on USMLE is that both types cause **pulmonary fibrosis,** leading to **pulmonary hypertension.** - Pericardial fibrosis can also occur. - 2CK NBME Q wants "dress warmly in cold weather" as an answer for how to ↓ recurrence of Raynaud. Dihydropyridine CCBs can also be used (e.g., nifedipine), which ↑ arteriolar + capillary diameter. - USMLE wants to know which drugs you avoid in patients with Raynaud → answer = α1 agonists (e.g., phenylephrine, oxymetazoline), since these constrict arterioles/capillaries.
Sarcoidosis	- Idiopathic autoimmune disorder that is one of the highest yield conditions on USMLE. You must know this condition extremely well. - Characterized by non-caseating granulomas within lung tissue. These consist of activated macrophages called epithelioid macrophages, aka histiocytes. - These histiocytes in the lung secrete 1α-hydroxylase (normally produced in the PCT of the kidney in response to PTH), which will convert inactive 25-OH-D3 into active 1,25-(OH)2-D3 (i.e., causing high vitamin D; aka hypervitaminosis D), which then goes to the small bowel and ↑ absorption of calcium, causing **hypercalcemia.** - Even though 1,25-(OH)2-D3 also ↑ small bowel absorption of phosphate along with calcium, it's to my observation that phosphate can be *normal* in sarcoidosis NBME Qs, so do not be confused if you see phosphate in the normal range and only see hypercalcemia. - PTH is *suppressed* in sarcoidosis due to the high Ca^{2+}. - Archetypal presentation is African-American woman 20s-30s with 6+ months of dry cough and red shins (erythema nodosum). Other findings like low-grade fever with flares can occasionally be seen. - **Bihilar lymphadenopathy seen on CXR or CT.** They can also describe this as CXR or CT "shows hilar nodularity."

	 - There is a 2CK Q that gives early-30s African-American woman with dry cough + they say CXR shows no abnormalities → answer = "activation of mast cells" (i.e., asthma is the answer); wrong answer is "non-caseating granulomatous inflammation." So your HY point here is: if the vignette sounds like it could be sarcoidosis but they say CXR is normal or shows "mild hyperinflation" (also buzzy for asthma), choose asthma over sarcoidosis. - Another 2CK Q gives a sarcoidosis stem where they mention bihilar lymphadenopathy on a CT scan + high hepatic AST + ALP + weight loss; they don't mention hypercalcemia. Unusual combo of findings, but the answer is inferable based on the CT. - Lupus pernio is an enlarged nose due to sarcoidosis (not SLE, despite the name). - USMLE wants steroids (i.e., oral prednisone) as treatment.
Rheumatoid lung	- Rheumatoid arthritis can cause a restrictive lung disease known as rheumatoid lung. - In addition, methotrexate, the first-line DMARD for RA, can cause pulmonary fibrosis. So patients with advanced RA can have pulmonary fibrosis from a combo of the two.

HY Pulmonary embolic conditions for Surg	
Pulmonary embolism	- DVT that has embolized up the venous circulation to the pulmonary arteries. - Presents as acute-onset shortness of breath and tachycardia in patient with one of the following risk factors: post-surgery (including C-section); sedentation, e.g., due to trauma such as hip injury; OCP use; and thrombotic disorders (i.e., FVL, prothrombin mutation, MTHFR). - S1Q3T3 is most specific finding on ECG for PE that is nonexistent on USMLE and will get you questions wrong. What you want to remember is: **sinus tachycardia** (i.e., normal ECG but HR is simply high) is the most common ECG finding for PE. - First step in management (2CK) is heparin. - After heparin, do a spiral CT of the chest to diagnose. - If the patient is already on warfarin and gets a PE, the first answer they want is CT to diagnose, followed by IVC filter. Do not choose IVC filter prior to confirming the PE with a CT. - One 2CK NBME Q has tPA as the answer for PE in patient who has obstructive shock (i.e., low BP due to PE). - V/Q scan is done instead of CT first-line in pregnancy. - One 2CK Q gives V/Q that is performed and shows "multiple sub-segmental perfusion defects" → answer = recurrent pulmonary emboli. - Most common cause of death from PE is ventricular fibrillation → acute right heart strain from a saddle embolus precipitates the Vfib.

	- The acid-base arrows USMLE wants for PE are: ↓ CO2, ↑ pH, ↔ bicarb. In other words, we have an acute respiratory alkalosis from high RR. Bicarb is unchanged because the kidney requires minimum 12 hours to accomplish this. These arrows are the same as acute asthma attack. - One 2CK Q gives lab values in PE of ↓ CO2 and ↓ pH. What this means is: patient has lactic acidosis (hence ↓ bicarb) due to ischemia from poor perfusion. It's HY for USMLE that you know any cause of shock, whether that be septic, hypovolemic, cardiogenic (or even obstructive from PE), can cause lactic acidosis and low bicarb. Don't be confused by this.
Amniotic fluid embolism	- The answer on USMLE if they say 30 seconds to 2 minutes after delivery of the placenta, the patient gets shortness of breath and tachycardia → sounds like PE, but answer is amniotic fluid embolism to the lungs. In contrast, if they say a woman had a C-section 2 days ago + now gets up in hospital to use bathroom + now has acute-onset SoB and tachycardia → answer = PE, not amniotic fluid embolism. - Can cause disseminated intravascular coagulation (DIC). It's one of the weird causes for USMLE. They can say the patient has bleeding from IV sites / catheter lines following delivery of the placenta.
Fat embolism	- The answer on USMLE if they say long-bone fracture (i.e., usually femur) in patient who now develops shortness of breath + petechiae on the shoulders/chest. - The patient will also have thrombocytopenia, but only about half of vignettes mention this. - The petechiae on the shoulders/chest are not due to the ↓ platelets, since conditions like ITP, for instance, don't present with this finding. Literature suggests it might be related to microvascular plugging by fat in the dermis. - Question can mention pulmonary biopsy stained with Oil Red O or Sudan black (stains for fat).
Air embolism	- The answer on USMLE if they say patient has sudden death following insertion of a central venous line → air goes to the lungs + kills patient.
Cholesterol embolism	- Will present as two ways on USMLE: 1) violaceous lesions on the feet in someone who just had a AAA repair → cholesterol plaque launches off to feet + they show you image like this one below:  The second way (2CK NBME 10) has cholesterol emboli Q where they don't show an image but describe the violaceous foot lesions in someone with mere Hx of AAA (rather than undergoing repair). They say "Maltese crosses are seen." Obscure, but apparently refers to birefringence of cholesterol esters that can be visualized under polarized light.

HY Trauma and pleural space conditions	
Pulmonary contusion	- Difficult diagnosis on USMLE Surg forms. - Diagnosis of exclusion, meaning all of these 2CK Qs require eliminating the other answer choices to get there. - The Q will say patient was in an accident + now has some form of difficulty breathing and patchy infiltrates in one or more areas of lung. They may or may not mention rib fractures just above the areas of infiltrates. Sounds vague as I said, but that's what USMLE will say. - Textbook descriptors such as "white-out of the lung" are nonsense. I don't think I've ever seen this on NBME material. - Contused (bruised) lung is known to be fluid-sensitive, where the Q can say patient was given IV saline and now has worsening of O2 sats. This is classic, but I've also seen this in myocardial contusion questions, making the Dx difficult as I said. But you'll be able to eliminate myocardial contusion if they don't mention severe bruising or pain over the sternum or post-MVA arrhythmia (e.g., premature atrial contractions).
Flail chest	- The answer on USMLE Surg forms if they say a patient has paradoxical breathing (i.e., the chest wall moves outward with expiration and inward with inspiration).
Ruptured bronchus	- The answer on USMLE Surg forms if they say there's a "persistent air leak despite placement of a chest tube." This will be in a trauma patient.
Diaphragmatic rupture	- The answer on USMLE Surg forms for an MVA followed by CXR showing "obscured left hemidiaphragm and the NG tube present in the lower left side of the chest." - Don't confuse with ruptured bronchus above, or with esophageal rupture, which will tell you water-soluble contrast swallow is visualized in the mediastinum.
Pneumothorax	- Air in the pleural space. The terms almost always refers to a "spontaneous pneumothorax," which will be acute-onset sharp chest pain in a tall, lanky patient in teens or 20s due to "ruptured subapical bleb."  - Pneumothorax + low BP = tension pneumothorax. - Mechanism for tension pneumothorax is compression of the vena cavas leading to ↓ venous return → **can cause JVD.** - Simple/spontaneous pneumothoraces cause ipsilateral tracheal shift. - Tension pneumothoraces cause contralateral tracheal shift.

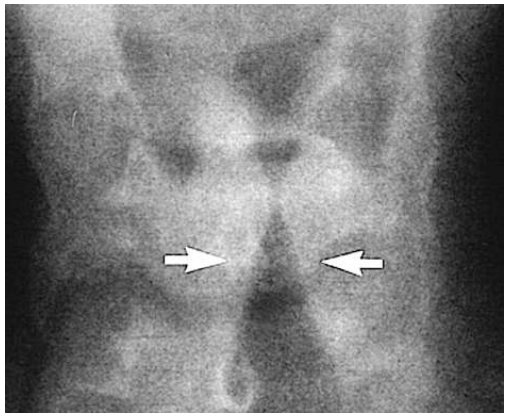
	- Tension pneumothorax need not be due to overt trauma. It will literally just be pneumothorax + low BP 9/10 times on USMLE. I've seen only one Q on a 2CK Surg form where BP is lower end of normal, but the patient is tachy around 120 bpm (implying barely holding BP in normal range). - Patient will have hyper-resonance to percussion, ↓ breath sounds, and ↓ tactile fremitus. Breath sounds are ↓ because the air in the pleural space masks the sounds of underlying air flow in the alveoli. Tactile fremitus is ↓ because the vibration of air in the alveoli is masked by the overlying air. - Treatment for pneumothorax is "needle decompression followed by chest tube." That phrase is HY. - One 2CK Q just has "tube thoracostomy" as the answer without needle decompression being listed. So if they don't list the needle, don't be confused, and just choose the chest tube. Thoracostomy means making a hole in the thorax. - Pleurodesis I've never seen as correct answer on NBME material but can be listed as distractor → means putting talc into pleural space to obliterate it in patients who have recurrent pneumothoraces. - Should be noted that very small pneumothoraces in stable patients with minimal symptoms can be observed. There is one Q on a 2CK NBME form where the answer is observe. But the vignette goes out of its way to emphasize how unremarkable the patient's presentation is. - There is a 2CK Q where pneumothorax is caused by barotrauma from ascending too quickly from underwater. This is different from Caisson disease ("the bends"), where nitrogen bubbles form in the blood. Pulmonary barotrauma from quick ascent, resulting in pneumothorax, can occur if some of the alveoli expand too quickly.
Pleural effusion	- Fluid in the pleural space; often refers to "hydrothorax," which means transudate or exudate of plasma-like fluid.  - USMLE wants: dullness to percussion, ↓ breath sounds, and ↓ tactile fremitus. Similar to pneumothorax, the latter two are ↓ because the fluid in the overlying pleural space masks air movement in the alveoli. - Many causes of pleural effusion on USMLE. I've observed left heart failure as a notable cause (i.e., patient will have pulmonary edema +/- pleural effusion). NBME can also give you pleural effusion with pneumonias, tuberculosis, and aortic dissection. - Meigs syndrome = triad of ovarian fibroma, ascites, right-sided pleural effusion.

	- USMLE wants you to know transudative versus exudative pleural effusions → transudates are more water-like, with fewer solutes; exudates contain more solutes. As per Light's criteria (HY for both Steps 1 and 2CK), an exudate will have: - Pleural fluid protein : serum protein ratio >0.5. - Pleural fluid LDH : serum LDH ratio > 0.6. - Pleural fluid LDH >2/3 upper limit of normal of serum LDH. - In other words, transudate contains less protein and LDH as the two main distinctions. - Transudates contain fewer WBCs than exudates, but I routinely see 500 WBCs/μL in transudate Q on NBME. - Highest yield cause of transudate on USMLE is left heart failure, as discussed in the Cardio PDF. - ARDS, pulmonary emboli, and infections can cause exudative pleural effusions (due to inflammation). - Progression to empyema HY (see below).
Empyema	- For 2CK Surg, it is exceedingly HY you know about empyema. There's two ways it presents on NBME: - 4 out of 5 questions, it will present as a sequence of pneumonia → parapneumonic effusions (i.e., exudative pleural effusions that occur due to pneumonia) → empyema. In other words, if a patient has a pneumonia that then leads to an exudative pleural effusion, we simply call that a parapneumonic effusion. This effusion can become progressively worse and purulent, to the point that the pH of the pleural fluid falls below ~7.1ish. We now call it empyema, which is frank pus in the pleural space. **Low pH of the pleural fluid to < ~7.1** is very HY on Surg forms. I've seen Qs where they say 7.12, 7.11, and 6.99, and they want empyema. - 1 out of 5 Qs will not mention anything about Hx of pneumonia. They will give patient who has **persistent fever + recurrent pleural effusions** in someone who has cirrhosis (i.e., ↓ oncotic pressure → transudative pleural effusions). Then the answer is just empyema. - NBME wants tube thoracostomy to drain the purulence.
Chylothorax	- Chyle (lymph fluid) in the pleural space. - The answer when they tell you a patient has milky or white-cloudy fluid in the pleural space **where the pH is significantly above 7.1.** - For instance, they might give you 15-line paragraph where they say milky fluid + tons of lab values, but you notice the pH of the pleural fluid is 7.40. This means it's chylothorax and empyema is wrong. - Can be caused **iatrogenically by insertion of central line via the left internal jugular vein,** or by cancers, e.g., lymphoma.
Hemothorax	- Blood in the pleural space. - Apart from massive trauma, the way this shows up on USMLE is as **malignant pleural effusion.** - They'll give you massive paragraph with tons of info + tell you patient has history of breast or lung cancer + has dullness to percussion → answer = malignant pleural effusion. - Patient can have hypotension and muffled heart sounds, similar to cardiac tamponade, but rather than JVD (to complete Beck triad), can have **flattened** neck veins (due to loss blood from the circulation).

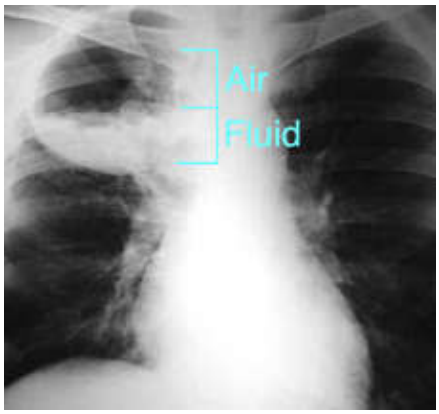
HY Alveolar fluid conditions	
Pulmonary edema	- As discussed in the Cardio PDF, this will be due to left heart failure on USMLE. - There is such thing as "non-cardiac pulmonary edema" (i.e., ARDS, TRALI [discussed below in this table]), but for USMLE purposes, if you see "pulmonary edema" as an isolated phrase on USMLE, it refers to transudation of fluid into the alveolar spaces due to ↑ pulmonary capillary hydrostatic pressure from left heart pathology. - PCWP and LAP are both elevated. - USMLE can give you vignette of, e.g., MI with dyspnea, and they ask for the mechanism of the dyspnea in the patient → answer = "increased alveolar-arteriolar oxygen gradient" – i.e., the patient can breathe just fine so alveolar O2 is normal, but fluid impairs the gas exchange, so we have low arterial/arteriolar oxygen. - "Cephalization of pulmonary vessels" is buzzy and synonymous with pulmonary edema on USMLE. Shows up in some NBME vignettes.
Acute respiratory distress syndrome (ARDS)	- The answer on USMLE for **bilateral** exudative chest infiltrates and ↓ O2 sats in patient following: pancreatitis; **aspiration of vomitus;** near-drowning episodes (aspiration of fresh/sea water); improper insertion of NG tube into the lungs with feeding initiated; toxic shock syndrome; or general trauma / sepsis. - 2CK NBME Q gives vignette of toxic shock syndrome and then asks most likely cause of death in this patient → answer = ARDS. - Pulmonary decompensation associated with pancreatitis is very buzzy. - Another 2CK Q gives patient who is brought to hospital following near-drowning episode + they ask what the patient needs to be monitored for → answer = ARDS. - ARDS is technically defined as a PaO2/FiO2 <300, but USMLE doesn't give a fuck. I've never once seen them apply this ratio or care. - I emphasize up above the bilateral nature of ARDS because many Qs give ARDS as DDx for unilateral conditions and it's wrong. - In theory, patient can be ventilated as follows: prone positioning (patient on stomach) + low-tidal volume setting + permissive hypercapnia. IIRC this is asked once on a 2CK form.
Transfusion-associated lung injury (TRALI)	- The answer on USMLE if a patient has an ARDS-like presentation with bilateral crackles and low O2 sats **<6 hours** following a transfusion. - Mechanism is abnormal priming of neutrophils in the lung that react to cytokines within transfused blood products (this is straight from a CMS Surg form 8 explanation). - This is technically a type of non-cardiogenic pulmonary edema.
Transfusion-associated circulatory overload (TACO)	- Aka transfusion-induced hypervolemia. - Differs from TRALI in that this is a type of cardiogenic pulmonary edema (i.e., the left heart can't handle the ↑ hydrostatic pressure from ↑ volume, so transudation into the alveoli occurs). - Annoying diagnosis but presents two ways: 1) If a patient *with Hx of heart failure or MI* develops respiratory distress following transfusion of repeated blood products. - A 2CK NBME Q gives a 72-yr-old with Hx of MI ten years ago who gets shortness of breath and bilateral crackles 30 minutes after transfusion with crystalloid solution and 4 packs of RBCs (they don't specify the volume of crystalloid in the Q). 2) They don't mention Hx of cardiovascular disease + the vignette will sound exactly like TRALI **> 6 hours** following a transfusion. - Surgery form 7 for 2CK gives an elderly dude who received only 3 packs of RBCs + he develops bilateral crackles and low O2 sats, but they say this occurs 12 hours after admission (not <6 hours as with

	TRALI), where the answer is "X-ray of the chest" as the next best step in diagnosis. The Q doesn't rely on you discerning TACO vs TRALI to get it right, but the explanation says it's TACO. In summary, if ARDS-like picture after transfusion: If <6 hours → TRALI. If >6 hours → TACO. If heart disease → TACO regardless of time frame.

HY Infection-related stuff	
Pneumonia	- Lobar pneumonia = *Strep pneumoniae* on USMLE (right lower lobe consolidation with dullness to percussion). - Bilateral interstitial pneumonia (aka atypical pneumonia) in immunocompetent patients = *Mycoplasma* on USMLE. - Lobar pneumonia where they say "interstitial markings" and *Strep pneumo* isn't listed → answer = *Mycoplasma* (the word "interstitial" wins over location). - Bilateral interstitial / "ground-glass" pneumonia in AIDS patient → *Pneumocystis jirovecii* pneumonia (PJP). - Lobar pneumonia in AIDS patient → *Strep pneumo*, not PJP. - Bilateral pneumonia + low Hb or (+) Coombs test → *Mycoplasma* → can cause cold agglutinins, which means IgM against RBCs → hemolysis). - Pneumonia + hyponatremia and/or diarrhea → *Legionella*. - Pneumonia + business conference or residential home (implies air conditioners) → *Legionella.* - Pneumonia in patient with central venous catheter + right upper lobe lesion → answer = *Staph epidermidis* (on NBME). Cather = biofilms. - USMLE wants for pneumonia: adventitious/bronchial (i.e., abnormal) breath sounds + ↑ **tactile fremitus** (air vibrates due to movement through infective consolidation within alveoli). - Community-acquired pneumonia (CAP) empiric Tx = azithromycin on 2CK (on NBME). This covers the atypicals (*Mycoplasma, Legionella, Chlamydia)* as well as *S. pneumo*. - If patient has been on antibiotics in the past 3 months or has severe lung disease, levofloxacin (respiratory fluoroquinolone) can be given first-line. - CAP that results in sepsis or septic shock → give ceftriaxone (if listed, choose cefotaxime for peds). - Nosocomial pneumonia (i.e., hospital- or ventilator-acquired) requires coverage for MRSA and *Pseudomonas*. USMLE wants vancomycin PLUS either ceftazidime (a 3rd-gen cephalosporin) or cefepime (a 4th-gen). - For fungal pneumonia, Tx = fluconazole. - For fungal pneumonia + fungemia (high fever, chills) → Amphotericin B. - NBME 9 for 2CK wants you to know **sputum culture, followed by blood cultures** are done in all patients with pneumonia who are septic. What they do in this patient is give you patient who has sputum culture performed, then they ask what should be done next for diagnosis? → answer = blood culture. - If you get *Pneumocystis* pneumonia, jump straight to **bronchoalveolar lavage** as the answer. - If you get a patient who has CXR or CT showing cavitary lesions in the lungs filled with a mass (likely *Aspergillus* fungus ball), they want "open lung biopsy" as the most confirmatory test.

	- When a pneumonia fully resolves and USMLE asks why the CXR is normal after the fact, the answer is "maintenance of integrity of basement membranes," as mentioned earlier. - 2CK wants you to know that pneumonias can occur in patients distal to areas of lung obstructed by tumors; the answer will be "endobronchial obstruction" as the reason the patient has distal pneumonia (same answer for why there can be distal atelectasis).
Bronchiolitis	- Answer on USMLE for a kid <18 months old who has low-grade fever and bilateral wheezes. - Caused by respiratory syncytial virus (RSV). - Tx is supportive care on USMLE. Don't choose answers like ribavirin or palivizumab.
Laryngotracheal bronchitis (aka croup)	- Caused by paramyxovirus (aka parainfluenza virus). - Presents as hoarse, barking, or seal-like cough in school-age kid. The Q can say the cough gets better when his dad brings him out into the cold air. - Neck x-ray shows "steeple sign," which is sub-glottic narrowing.  - Sometimes the Q can give you easy vignette of croup, but then the answer is just "larynx" (literally inflammation of the larynx, trachea, and the bronchi). "Sub-glottic" means below the area of the vocal cords. The larynx is the area encompassing the vocal cords. - Tx is supportive. If they force you to choose an actual Tx, the answer is steroids if listed, followed by nebulized racemic epinephrine.
Epiglottitis	- Caused by *Haemophilus influenzae* ***type B***. - Seen in unvaccinated and immigrants (can be unvaccinated), as well as patients with asplenia or sickle cell (auto-splenectomy). - Can also cause meningitis. - X-ray of neck shows "thumbprint sign." 

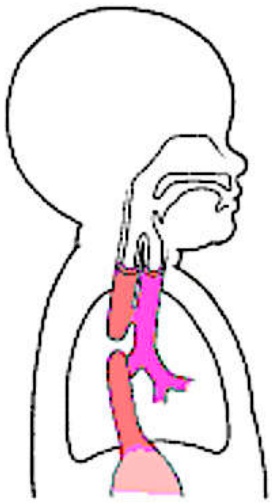
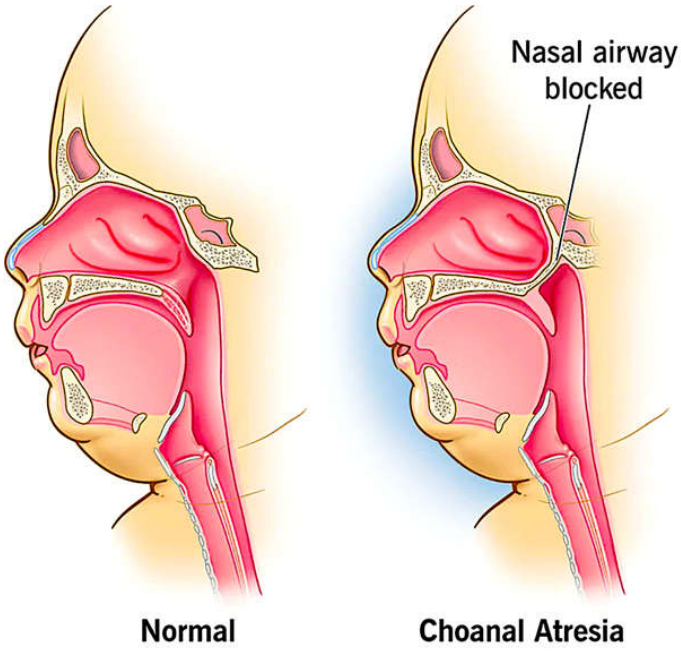
	- Presents as child who has fever + difficulty breathing. They can say the kid is drooling and/or in tripod positioning (facilitates use of accessory muscles). - USMLE wants intubation as answer. Epiglottitis is a medical emergency that can lead to sudden occlusion of the airway. - Tx = 3rd-gen cephalosporin (cefotaxime in peds, or ceftriaxone); give rifampin to close contacts.
Bacterial tracheitis	- Diagnosis of exclusion on USMLE (i.e., you eliminate to get there). - Classically caused by *S. aureus* following a viral URTI. - Shows up on 2CK form as patient with recent viral URTI who now has stridor (suggesting upper airway narrowing) + no drooling (suggests against epiglottitis) + **can fully open the mouth without difficulty** + does not improve with racemic epinephrine (suggests against croup). - Being able to open the mouth without difficulty is important because this contrasts with peritonsillar abscess, where the patient has difficulty fully opening the mouth.
Pertussis	- Classic whooping cough presents as succession of many coughs followed by an inspiratory stridor. - What you need to know for USMLE is that this can absolutely present in an adult and that they can be vague about it, just describing it as a regular cough. The way you'll know it's pertussis, however, is they will say there's either **hypoglycemia** or **post-tussive emesis,** which means vomiting after coughing episodes. - Pertussis can cause super-high WBC counts in the 30-50,000-range, where there are >80% lymphocytes. *This makes it resemble ALL.* So you should know for Peds that ALL-like laboratory findings + cough = pertussis. - Q will ask number-one way to prevent → answer = vaccination (not hard, but they ask it). Pertussis is part of TDaP. The pertussis component is killed-acellular; the tetanus and diphtheria are toxoid. - Erythromycin can be given to patients with active cough; USMLE doesn't give a fuck about pertussis stages. - Close contacts should also receive erythromycin.
Pleurodynia	- As mentioned in the Cardio section, this is an MSK condition I've seen asked twice on 2CK material that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Tuberculosis	- *Mycobacterium tuberculosis* has unique cell wall composed of mycolic acid that is difficult to gram stain. Requires acid-fast stain. - Can present similar to lung cancer, where patient can have B symptoms (i.e., fever, night sweats, weight loss) and hemoptysis. - Living in a homeless shelter or immigrant status from endemic area is buzzy. I've seen rural India and Albania as two locations in NBME Qs. - Can cause cavitations and calcification in the lung grossly; on histo, causes caseating granulomatous inflammation. - Ghon foci/complexes are textbook descriptors for TB lesions but not assessed eponymously on USMLE. - Can cause constrictive pericarditis (can also calcify). - Disseminated TB (miliary TB) can affect multiple organ systems, leading to **psoas abscess, Pott disease** (TB infection of the vertebrae), adrenal insufficiency, meningitis, osteomyelitis, and arthritis. - First step in diagnosis is PPD test (type IV hypersensitivity). - If PPD test is (+), the next best step is CXR. - If PPD is (+) but CXR (-), next best step is "treat for latent TB, " or "give TB prophylaxis." This is isoniazid (INH) for 9 months + vitamin B6 (since INH can

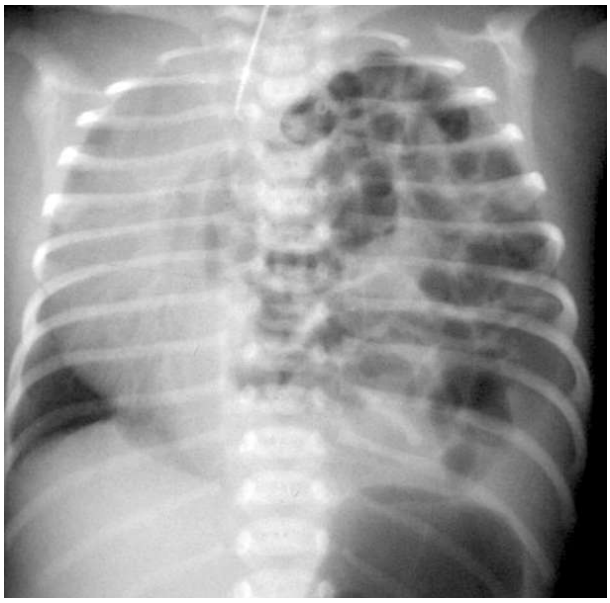
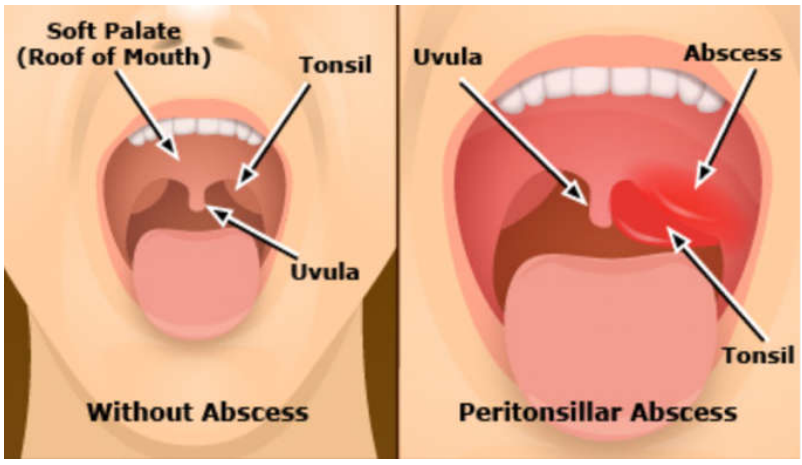
	cause B6 deficiency). It is exceedingly HY you know that neuropathy in a patient being treated for TB has B6 deficiency. - If PPD and CXR are both (+), the next best step is "treat for active TB," which is RIPE for 2 months + RI for 4 more months (6 months total). RIPE = rifampin, isoniazid, pyrazinamide, ethambutol. - BCG vaccine is live-attenuated. USMLE wants you to know Hx of BCG vaccine does not change management based on PPD guidelines. - If USMLE asks you how long after TB exposure will someone's sputum cultures be positive, the answer is 2-5 weeks. - Interferon-gamma release assay (Quantiferon Gold) can be used in patients who have Hx of BCG to ↓ false (+)s, but USMLE doesn't assess it. The reason I mention it is because they want you to know *interferon-*γ is required for stimulation of alveolar macrophages to control TB. - TNF-α is also required to suppress TB. Therefore drugs such as infliximab, adalimumab, and etanercept ↑ risk of TB, which is why they should be avoided in silicosis patients, as mentioned earlier. - What is considered a positive PPD test differs depending on risk factors: **- >5mm (+):** Hx of close contact to someone with active TB; immunocompromised patient (AIDS, organ transplant recipient receiving immunosuppressants, chronic corticosteroid user); calcification on CXR. **- >10mm (+):** Health care worker or prisoner/prison worker; immigrant from endemic area; TB laboratory personnel, children <4. **- >15mm (+):** everyone else. - If a PPD test is (+), never repeat it. If it is negative, it must be repeated in 1-2 weeks (i.e., sometimes false-negatives).
Hot tub lung	- Answer on a 2CK form for a guy who moves into a new apartment building with a hot tub and gets bilateral chest infection → inhalation of fumes with *Mycobacterium avium complex* (MAC).
Pulmonary abscess	- Exceedingly HY on USMLE. - USMLE wants "aspiration of oropharyngeal normal flora," or "aspiration of oropharyngeal anaerobes" as the cause. - Q will give aspiration risk factor, such as alcoholism, dementia (can cause loss of gag reflex), Hx of stroke (leading to dysphagia), or epilepsy. - Q can also mention broken or missing teeth (hypodontia) as risk factor. - Often described on NBME as pulmonary lesion with an **air-fluid level.** This is buzzy, but not a mandatory descriptor. This refers to the top half of the circle being air, and the bottom being pus, the latter settling due to gravity. 

	- The stem can say the patient has "foul-smelling sputum." This descriptor is exceedingly HY and is synonymous with anaerobes on USMLE. - Oropharyngeal normal flora = *Bacteroides* (strictly anaerobic gram-negative rods); as well as *Peptostreptococcus* and *Mobiluncus*. The latter two are not HY, but *Bacteroides* is. I mention all three, however, because the Q can say sputum sample shows "gram-negative rods, gram-positive cocci, and gram-positive rods," which refers to all three. But the bigger picture concept is, this = mixed normal flora. - **Tx = clindamycin**. USMLE loves this. - If Q tells you patient was treated for pulmonary abscess + a year later there's still a lesion seen on CXR → answer = "failure of maintenance of basement membranes."
Acute bronchopulmonary aspergillosis (ABPA)	- Presents as asthma-like presentation in patient with ↑ sensitivity to aspergillus skin antigen.
Hypersensitivity pneumonitis	- Answer on USMLE for bilateral lung condition + fever in farmer who has exposure to hay (on new NBME). - They will tell you the fever self-resolves after 2 days and he now is afebrile. - Byssinosis (pneumoconiosis from hemp) is wrong answer, since this won't present with fever + classically presents in textile workers.
Cryptogenic organizing pneumonia (COP)	- Idiopathic restrictive lung disease where patient has pneumonia-like presentation that fails to improve with antibiotics. Not actual pneumonia. - Formerly known as bronchiolitis obliterans organizing pneumonia (BOOP). - Nonexistent yieldness on USMLE, but I mention it because you will sometimes see this as a *wrong* answer choice, particularly on hard 2CK Qs, and I've seen enough students erroneously pick it.
Otitis media (OM)	- Infection of portion of ear just deep to tympanic membrane. - Most commonly *Strep pnuemo*. - Will present as red, **immobile** tympanic membrane. Immobility of the tympanic membrane is highly sensitive for OM, meaning that if the Q says mobility is normal, we can rule out. - "Ear tugging" can be a sign in children of either otitis media or externa. - Tx is amoxicillin or penicillin. - Augmentin (amoxicillin/clavulanate) is classically given for *recurrent* OM. So if you are forced to choose between amoxicillin/penicillin alone or Augmentin, go with the former. - Tympanostomy tube (aka grommet) is used if the kid has >3 OM occurrences in 6 months, or >4 in a year.
Serous otitis media	- Aka otitis media with effusion. - Presents as fluid behind the tympanic membrane in a kid weeks after resolution of 1 or 2 otitis media infections. - Almost always benign and self-resolves in 4-8 weeks. Answer is observation.
Tympanic membrane perforation	- "Tympanic membrane perforation" is the answer on new 2CK NBME for 2-year-old who had 3-day Hx of viral infection followed by awakening with severe ear pain + has dried blood on ear lobe and pillow + otoscopy cannot visualize tympanic membrane because of seropurulent fluid draining from the ear canal. - Can occur due to otitis media, although vignette on NBME doesn't sound like classic OM and is as described above.
Mastoiditis	- Inflammation of mastoid bone caused by untreated otitis media. - The mastoid process is the posterior part of the temporal bone that is felt just behind the ear. - Can present as a painful ear pinna that is displaced (e.g., upward and outward). - Diagnosis is made by **CT or MRI.** X-ray is wrong answer.

	- 2CK IM Q gives a 2-year-old with mastoiditis where the answer is "CT of the temporal bone." Sounds wrong, since this is ↑ radiation for a kid, but it's what they want.
Myringitis	- Isolated inflammation of the tympanic membrane. - Can be bullous (i.e., bullous myringitis). - Caused by *Strep pneumo* or *Mycoplasma*.
Otitis externa (OE)	- Infection of ear superficial to tympanic membrane. - Classically caused by *Pseudomonas*. - Increased risk in swimmers and diabetics. - An NBME form has "necrotizing otitis externa" as answer for black skin within the ear canal in a patient. This is aka "malignant otitis externa." - USMLE wants "acetic acid-alcohol drops" as prophylaxis in college student who does crew + continues to have water exposure. - Tx (not prophylaxis) = "topical ciprofloxacin-hydrocortisone" drops.
Sinusitis	- The answer if they tell you a school-age kid has a lingering fever after an upper respiratory tract infection (URTI) for 10-14+ days. - Whenever a URTI lingers for more than ~10ish days, you want to think about sinusitis as a differential. - A 2CK vignette gives nocturnal cough (reflects aspiration; in this case, from the sinuses) and grey membranes in the oropharynx. - The grey oropharyngeal membranes detail sounds weird, since that is normally buzzy for *Diphtheria*, but it shows up on an NBME Q where the answer is sinusitis and *Diphtheria* isn't listed. - IgA deficiency Qs, which presents as recurrent sinopulmonary infections, can say patient has Hx of pneumonias + presents today with sore left cheek → reflects sinusitis. - For 2CK, **CT scan** is done if chronic sinusitis >12 weeks. After CT is performed for chronic sinusitis, nasal endoscopy can be performed. - Tx is amoxicillin/clavulanate (Augmentin). This is in contrast to OM and *Strep* pharyngitis, which are treated with just amoxicillin or penicillin alone, without the clavulanate (unless recurrent).

Vaccine stuff	
S. pneumo	- PCV15 at 2, 4, 6 months, then again 12-15 months. - Age ≥65: PCV20 once; OR, PCV15 then PPSV23 a year later. - For **asplenia/surgical splenectomy**, sickle cell, cochlear implant, CSF leak, chronic renal failure/nephrotic syndrome, HIV, and immunosuppressed patients: extra dose of PCV15, followed by PPSV23 8+ weeks later.
Influenza	- IM killed vaccine: start age 6 months, then give yearly through life; safe to give during pregnancy. - Intranasal live-attenuated: ages 2-45; immunocompetent, non-pregnant persons.
Coronavirus	- IM vaccine starting at 6 months; 2-3-doses.

HY Peds-Surg respiratory DDx	
Tracheoesophageal fistula (TEF)	- HY pediatric condition in which the proximal esophagus ends in a blind pouch + the distal esophagus connects to the trachea.  - Will present as a neonate who coughs up milk with initial feeding. - Highest yield point about TEF is that diagnosis is made via **insertion of nasogastric tube** (which cannot be inserted fully because it hits the blind pouch of the esophagus).
Choanal atresia	- Weird condition in which the nasal passages don't develop patency, so the neonate is an obligate mouth-breather.  - Will present as a child who becomes blue/hypoxic while breast feeding (because he can't breathe through the nose), then cries/becomes pink once detaching from the breast. - Similar to TEF, diagnosis is made via insertion of nasogastric tube. - Part of **CHARGE** syndrome → **C**oloboma of the eye (hole in the eye), **H**eart defects, **A**tresia of the choanae, **R**enal defects, **G**enitourinary anomalies, **E**ar anomalies.
Congenital diaphragmatic hernia	- One of the highest yield Peds conditions on USMLE. - Caused by **failure of formation of pleuroperitoneal membranes.**

	- Always occurs on the left, where bowel from the abdomen can herniate up into the left-chest. - Can present as ↓ bowel sounds in the abdomen + ↑ bowel sounds in the left hemithorax.  - Vignette might say there are cystic-appearing areas in the left hemithorax seen on CXR (loops of bowel).
Foreign body aspiration	- Answer on 2CK Peds for kid under 2 who was crawling around on the floor who now has acute-onset respiratory distress. The vignette will then add one of two features: 1) Unilateral hyperresonance in one lung (but not pneumothorax); 2) Unilateral atelectasis + ipsilateral tracheal shift. - Exam will ask for "bronchoscopy," or "fiberoptic examination of the airways" as the answer.
Adenoid hypertrophy	- Just know that enlarged tonsils (adenoids) are a known cause of stridor in pediatrics. Answer = tonsillectomy.
Peritonsillar abscess (Quinsy)	- Pus collection causing pain and swelling at posterior oropharynx. - Can be seen as complication of tonsillitis. - Patient can have **difficulty opening his/her mouth;** this contrasts with bacterial tracheitis, where the patient can fully open his/her mouth (on 2CK NBME, where this detail is important in an otherwise vague vignette). 

Laryngomalacia	- Most common cause of stridor in infants. - Softening of airway cartilage. - Vignette will give kid under the age of 1 who has noisy breathing that is mitigated when placed prone (on stomach) or upright. - Self-resolves almost always (i.e., don't treat).
Vascular ring	- Weird cause of stridor that USMLE likes to contrast with laryngomalacia. - Vascular embryology variant can lead to airway compression. - In contrast to laryngomalacia, mitigated when neck is extended. - Tx = surgery.
Velopharyngeal insufficiency	- High-pitched, hyper-nasal, or whistling speech in patients with Hx of cleft palate due to shortened palate, with or without repair. - It's the answer on a 2CK NBME, so know that it exists. They will simply just give you vignette of cleft palate + ask what the kid is at risk for.
Cholesteatoma	- Idiopathic squamous proliferation in middle ear behind the tympanic membrane that presents as an enlarging mass. - Can cause obstructive hearing loss + grow into inner ear. - Requires surgery. - Sounds like an obscure diagnosis, but it's not. Be aware it exists for Peds.
Tracheomalacia	- Not pediatric, but I'm putting it here because I see this confused with laryngomalacia. - This is a cause of stridor in (usually) adults who have stridor developing months post-surgery. - Caused by Hx of intubation leading to softening of airway cartilage. - Just know this diagnosis / it exists + don't confuse with laryngomalacia.

Ventilator settings nonsense for USMLE

- Students will sometimes nag about pedantic ventilator settings info. The vast majority of it is garbage for USMLE. What I can basically say is that **increasing PEEP** (positive-end expiratory pressure) is probably the answer 4 out of 5 times for any ventilator settings-type of question. The Q might give you a big 15-line rambling paragraph where you're not sure what's going on, and then the answer is just "increase PEEP."
- PEEP keeps the alveoli open longer, thereby facilitating gas exchange.
- When the patient is on a ventilator, the general aims are:

    - Keep FiO2 as low / as close to room air as possible (i.e., high O2 can cause free radical damage);
    - Keep tidal volume as low/normal as possible (i.e., 400-500-ish mL); high tidal volumes can cause ventilator-associated barotrauma.
    - Increase PEEP if it means lower FiO2 and tidal volumes can be achieved.

- But the above needs to be taken with a caveat. For example, the Q might say the patient is on a ventilator and has arterial pO2 of 40 mm Hg (normal is 80-100) + FiO2 is 100% + tidal volume is 1000 mL, and the answer will be "increase PEEP." The student says, "Wait but I thought you said we want to reduce FiO2 and tidal volume." → Sure, but we can't do that here because pO2 is super-low, so that would only exacerbate it further.
- As I talked about for ARDS earlier, you could be aware the triad of 1) prone positioning, 2) low-tidal volume mechanical ventilation, and 3) permissive hypercapnia can be implemented.
- A very HY point is that ↓ CO2 causes ↓ cerebral perfusion (this is why patients faint in panic attacks, which is asked on NBME). The first step in managing ↑ intracranial pressure is "intubation + hyperventilation."
- They also ask the inverse of this on NBME – i.e., they say an anesthesiologist wants to ↑ a patient's cerebral perfusion, and the answer is "decrease respiratory rate" → causes ↑ CO2 → ↑ cerebral perfusion.
- Ventilator-acquired pneumonia (VAP) should be treated with vancomycin PLUS either ceftazidime or cefepime. Vancomycin not only covers MRSA but also is effective against some high-resistance *S. pneumo* strains. Ceftazidime (3rd generation ceph) and cefepime (4th gen ceph) are very effective against *Pseudomonas*, which is a HY nosocomial organism for VAP.

- Patients should ideally be weaned from ventilators as quickly as possible. USMLE also likes patients who are sick on ventilators as sometimes having euthyroid sick syndrome – i.e., they will say patient cannot be weaned from ventilator + has ↓ T3 and normal TSH → answer = euthyroid sick syndrome. The full array of arrows are: ↓ T3, ↑ rT3, ↔ T4, ↔ TSH. If you have no idea what I'm talking about, then I recommend going through my HY Arrows PDF.
- If a patient ever has a high CO2 (NR 33-44 mmHg) → answer can = "ventilatory insufficiency." This applies to both patients on and off ventilators. Don't be confused if they give you ↑ respiratory rate. New Surg form gives RR of 40 with ↑ CO2 as an example. Sometimes patients have shallow breathing despite ↑ RR.

HY Pulmonary drug points	
Bosentan	- Endothelin-1 receptor antagonist used for pulmonary hypertension. - Endothelin is a vasoconstrictor ↑ in pulmonary hypertension.
Methotrexate	- Dihydrofolate reductase inhibitor. - 1st-line DMARD for RA + 1st-line oral agent for psoriasis that fails topicals. - Causes pulmonary fibrosis (and hepatotoxicity and neutropenia). - Toxicity can be mitigated with "leucovorin rescue," aka folinic acid (not folic acid).
Albuterol	- Short-acting β2-agonist used for asthma. - Can cause tremor (asked on NBME).
Fluticasone, Beclomethasone	- Inhaled corticosteroids (ICS) used for asthma and COPD. - Used for prevention / to decrease recurrence of episodes. They have no role in acute exacerbations. For the latter, IV methylprednisolone is used.
Prednisone	- Most effective at ↓ recurrent asthma attacks, but used last resort because oral steroids can cause Cushing syndrome and growth stunting.
Montelukast, Zafirlukast	- Leukotriene receptor antagonists (LTC, D, and E4). - Leukotrienes normally cause bronchoconstriction. So these agents ↓ airway constriction. - Can be used for asthma in patients with aspirin-induced asthma (blockage of COX shunts arachidonic acid down the lipoxygenase pathway toward leukotrienes), or for asthma in general in between a LABA and prednisone.
Zileuton	- Lipoxygenase inhibitor. Blocks synthesis of leukotrienes. - Use-case same as the -lukasts.
Nedocromil, Cromolyn sodium	- Mast cell stabilizers used for asthma. Prevent release of histamine.
Tiotropium	- Long-acting muscarinic receptor antagonist (LAMA) used first-line for COPD.
Olodaterol	- Long-acting β2-agonist (LABA) used first-line for COPD. - As discussed earlier, either a LAMA or LABA can be used first-line for COPD.
Ipratropium	- Short-acting muscarinic receptor antagonist (SAMA) used for COPD. - Not first-line anymore, but still shows up on USMLE as an agent that's used. Some old-school practitioners still use it, so don't disregard it.
Epoprostenol	- PGI2 prostacyclin that can be used for pulmonary hypertension.
Sildenafil	- Viagra; PDE-5 inhibitor that can also be used for pulmonary hypertension.
Nifedipine	- Dihydropyridine CCB that can be used for pulmonary HTN and Raynaud. - As discussed in the Cardio section, can cause peripheral edema/fluid retention.
Omalizumab	- Monoclonal antibody against IgE. - Can be used in theory in severe asthmatics who have high IgE levels.
Dornase alfa	- Nucleotidase used to break down airway mucous in cystic fibrosis patients.
Guaifenesin	- Mucolytic agent that softens mucous in cystic fibrosis.
Pirfenidone	- Used for idiopathic pulmonary fibrosis. - Anti-fibrotic agent that inhibits TGF-β-mediated collagen synthesis.

Surg Renal

Basic HY Renal lab values for Surg	
Blood urea nitrogen (BUN)	- Should be <20 mg/dL. - High urea nitrogen in the blood is called azotemia. - "Uremia" technically means "urea in the blood," but is used to refer to patients who have symptomatic renal failure with poor lab values. This has absolutely zero to do with hyperuricemia (gout) and is sometimes confused by students.
Creatinine (Cr)	- 0.7-1.2 mg/dL - Once you've hit a creatinine of 2, you've lost ~90% of your renal function.
Potassium (K^+)	- 3.5-5.0 mEq/L. - Will always be elevated in renal failure on USMLE. - This is because the kidney normally secretes (i.e., excretes through tubular walls into the urine) K^+ in the cortical collecting duct under the action of aldosterone. So if the kidney is fucked up, we can't do that, so serum K^+ rises.
Sodium (Na^+)	- 135-145 mEq/L. - Can be variable in renal failure as per my observation on NBMEs. - Aldosterone normally reabsorbs sodium in the cortical collecting duct in exchange for potassium being secreted, so one might think it should automatically be low if the kidney can't reabsorb it. But the kidney has a multitude of sodium regulation mechanisms (i.e., vasopressin), so Na^+ can be normal.
Calcium (Ca^{2+})	- Always ↓ in renal failure. This is due to 2 reasons: 1) Ca^{2+} is normally reabsorbed in the late-distal convoluted tubule (DCT), so if we can't do that, serum Ca^{2+} falls. 2) Proximal convoluted tubule (PCT) of the kidney is normally where 1α-hydroxylase, under the action of PTH, activates inactive 25-OH-D3 into active 1,25(OH)2-D3. The latter then goes to the small bowel, where it ↑ absorption of Ca^{2+}. So if we can't do that, serum Ca^{2+} falls. - Low serum Ca^{2+} means ↓ negative feedback at the Ca^{2+}-sensing receptors at the parathyroid glands → PTH goes up. We call this secondary hyperparathyroidism, which means PTH is high due to a cause external to the parathyroid glands themselves. You need to know secondary hyperthyroidism is due to renal failure.
Phosphate ($PO4^{3-}$)	- Always ↑ in renal failure. - This is for 2 reasons: 1) The kidney cannot filter it out sufficiently. Lots of phosphate is normally filtered by healthy kidneys. 2) PTH normally acts to ↓ phosphate reabsorption in the PCT via downregulation of 3 different types of phosphate pumps. Since the function of PTH is impaired, the kidney *reabsorbs too much* phosphate, so serum levels rise. - Combo of ↓ Ca^{2+} and ↑ $PO4^{3-}$ in renal failure = secondary hyperparathyroidism.
Bicarbonate ($HCO3^-$)	- Always ↓ in renal failure. - This is for 2 reasons: 1) The kidney cannot reabsorb bicarb in the PCT as well, so ↑ bicarb in the urine → ↓ bicarb in serum → metabolic acidosis. 2) The kidney cannot secrete protons (H^+) in the cortical collecting duct as readily → retention of H^+ mops up $HCO3^-$ → ↓ serum bicarb. - Renal failure = Uremia = the U in MUDPILES for high anion-gap metabolic acidoses.

Quick afferent vs efferent review for Surg	
Afferent arterioles	- Go to the kidney. - Patency maintained by **vasodilating prostaglandins.** - NSAIDs inhibit prostaglandin synthesis → ↓ dilation of afferent arterioles → ↓ renal blood flow. - NSAIDs don't actively constrict the afferent arterioles. This is incorrect wording. They just merely ↓ vasodilation / ↓ diameter of the afferent arterioles.
Efferent arterioles	- Leave the kidney. - Angiotensin II normally constricts these. - ACEi/ARBs therefore ↑ diameter of the efferent arterioles. They are not dilating them; they are merely ↓ constriction of them. As mentioned above, prostaglandins will actively dilate the afferent arterioles. - Some students might say, "Well doesn't AT II act on the afferent arterioles as well?" In theory, a very minor amount, but on USMLE, this will get you questions wrong. You want to think: AT II = constricts efferent; ACEi/ARBs = ↓ constriction of efferent (↑ diameter). Prostaglandins = dilate afferent; NSAIDs = ↓ dilation of afferent (↓ diameter).

Pre-, intra-, and post-renal azotemias	
BUN/Cr, FENa	- BUN/Cr = Blood urea nitrogen to creatinine ratio. - FENa = fractional excretion of sodium (i.e., amount of sodium in the urine). - There's no "normal" range you should be looking for here for either of these. - What matters is that when there is an actual kidney problem, we look at the values to help us figure out what the diagnosis is. But we never actually say, "Oh the BUN/Cr or FENa is out of the normal range." - BUN/Cr >20 and FENa <1% mean pre-renal azotemia. - BUN/Cr <20 and FENa >1% just simply mean "not pre-renal." - The notion of intra- vs post-renal having slightly different ranges is nonsense for USMLE. - Also, what I've come to learn from NBME Qs is that only ~9/10 times the BUN/Cr will be what we expect. There are a couple 2CK NBME Qs that give BUN/Cr <20 for pre-renal azotemia and >20 for acute tubular necrosis. If the NBME ever "breaks the rules" this way, the vignette will be overwhelmingly obvious what the diagnosis is anyway.
Pre-renal	- Pre-renal azotemia means there is renal pathology because of reduced renal blood flow over the sub-acute to chronic time frame. The kidney will do everything it can to ↑ retention of fluid (because it thinks blood volume is low). The way it accomplishes this is by ↑ reabsorption of urea and sodium, since water will follow. This is why blood *urea* nitrogen is high (i.e., BUN/Cr >20) and sodium in the urine is low (FENa <1%). - Notice above I stress *sub-acute to chronic time frame* (i.e., days-months), since one of the highest yield points you need to know for USMLE is that *acute* ↓ perfusion (i.e., seconds to minutes) to the kidney from blood loss, acute heart failure exacerbation, or arrhythmia (i.e., VFib for 30 seconds before resusc) **causes acute tubular necrosis, not pre-renal.** - This is all over the NBME exams. For example, if they say a guy loses lots of blood during surgery and receives 20 packs of RBCs, then two days later while recovering in hospital he gets oliguria + deterioration of renal function, the answer is acute tubular necrosis, not pre-renal. Student says, "Wait, but there was ↓ renal perfusion though, so how does that make any sense. Isn't pre-renal caused by ↓ perfusion?" → *The PCT of the kidney is the most susceptible to anoxic/hypoxic injury due to the ↑concentration of ATPase transporters.* So acute ↓ drop in blood flow → acute hypoxia → PCT sheds.

	- There is a 2CK NBME Q where they mention a guy post-op who had *not* lost any blood during surgery + had *not* experienced any episodes of low BP + was started on ketorolac (an NSAID) for pain post-op + now has oliguria and ↑ creatinine → answer = "hypoperfusion" as cause; this is example of where NSAIDs, not ATN, can cause post-op oliguria from pre-renal. - HY causes of pre-renal for USMLE are NSAID or diuretic use, chronic left heart failure, or dehydration from days of vomiting/diarrhea. - USMLE will give patient who's been on an NSAID (e.g., naproxen) for several weeks, or who was commenced on furosemide (loop diuretic) a few days ago. - Vignette can give patient who is on an NSAID + now has peripheral edema + they ask why there's edema → answer = "↓ renal excretion of sodium." This is because the PCT ↑ reabsorption of sodium in pre-renal (FENa <1%). They will also sometimes give NSAID + edema + ask what you do, and the answer is just "discontinuation of ibuprofen."
Intra-renal	- Essentially synonymous with acute tubular necrosis on USMLE. - The kidney can't reabsorb urea and sodium as easily, so BUN is lower (i.e., BUN/Cr is <20) and sodium is higher in the urine (i.e., FENa is >1%). - I discuss acute tubular necrosis in more detail in its own table below, but as I mentioned above in red font, the most important point for USMLE is that you remember *acute* drop in perfusion to the kidney causes acute tubular necrosis, *not* pre-renal. I need to be an asshole and inculcate that.
Post-renal	- Almost always due to BPH on USMLE, but can also be due to ovarian or cervical cancer impingement on the ureter(s). - What you need to know is: old dude + high creatinine = BPH till proven otherwise. - The answer on USMLE is often just "increased Bowman capsule hydrostatic pressure." - Next best step is "insertion of catheter" to relieve the urinary retention, even if the patient has bacteriuria (i.e., choose catheter insertion over antibiotics). - Another 2CK Q has "check post-void volume" as answer in elderly male with high Cr. USMLE won't force you to choose between catheter insertion of checking post-void volume. - Normal post-void volume is <50ish mL. If the USMLE wants overflow incontinence, they'll give post-void volume ~300-400+ mL. - Diabetic neurogenic bladder (i.e., hypotonic bladder from neuropathy to the detrusor muscle) can also lead to overflow incontinence and post-renal azotemia, but as I said, most Qs on USMLE will focus on the old dude with BPH. - Hydronephrosis can be seen as large, dilated kidneys in patients with obstruction, but this is rare diagnosis.

Nephritic vs Nephrotic syndromes
- Nephritic conditions have blood in the urine; nephrotic syndromes don't. - You need to memorize for USMLE which HY conditions are nephritic vs nephrotic. Don't worry, I keep things real clean and concise in the tables below without the bullshit nonsense of other resources. - There are four main points you should memorize as being part of each syndrome type. **Nephritic syndromes:** 1) **Hematuria** (blood in the urine). 2) **Oliguria** (↓ urinary output; the definition is <400 mL/day, but USMLE won't assess that number). 3) **Azotemia** (↑ blood urea nitrogen; this is because urinary output is ↓). 4) **Hypertension** (↑ RAAS due to inflammation of renal microvasculature). **Nephrotic syndromes:** 1) **Proteinuria** (nephrotic level by definition is >3.5g/day, but USMLE won't assess that number). 2) **Hypoalbuminemia** (due to the proteinuria; will be <3.5 g/dL, but USMLE doesn't care about the #). 3) **Peripheral edema** (due to the hypoalbuminemia → ↓ serum oncotic pressure → transudation of fluid into interstitial spaces; severe can present with ascites; if the stem says "pre-sacral edema," this is nephrotic syndrome till proven otherwise on USMLE).

4) **Hyperlipidemia** (liver pumps out apolipoproteins in an attempt to preserve serum osmolality / oncotic pressure).

- Should be noted that nephritic syndromes often have proteinuria, just usually not at nephrotic levels. Students sometimes erroneously think nephritic syndromes don't have proteinuria because it's not part of the 4-point categorization above.
- You don't have to worry about the notion of which conditions are "both nephritic and nephrotic." What is most important is: simply know which conditions have hematuria and which ones don't (via tables below).
- Nephrotic syndromes can ↑ risk of DVT, renal vein thrombosis, and varicocele due to loss of antithrombin III in the urine → hypercoagulable state. This is because nephrotic syndrome usually entails non-specific massive protein loss due to loss of size and charge barrier, and antithrombin is a protein. The NBME wants you to know left renal vein thrombus accretion from hypercoagulable state can cause varicocele.

HY Nephritic syndromes for Surg	
Post-streptococcal glomerulonephritis (PSGN)	- Aka "proliferative glomerulonephritis" (asked on NBME, where they give vignette of PSGN, and answer is just "proliferative glomerulonephritis"). - Type III hypersensitivity. - Answer can sometimes be written by NBME as just "acute glomerulonephritis." - The answer on USMLE for red urine 1-3 **weeks** following a sore throat caused by Group A Strep (*Strep pyogenes*). This is in contrast to IgA nephropathy (which I discuss in detail below), which is red urine **1-3 days** following a sore throat. - PSGN can be caused by skin infections. This can be impetigo (school sores), cellulitis, or erysipelas (I discuss these in detail in my HY Derm PDF). For example, the Q might say a 10-year-old has yellow crusties on his arm for the past 7 days (impetigo) + now has red urine → answer = "acute glomerulonephritis." - Serum complement protein C3 can be ↓. - Streptolysin O or A titers will be ↑.
IgA nephropathy	- Aka Berger disease; post-viral IgA deposition in the mesangium. - Red urine 1-3 days after a sore throat. This is in contrast to PSGN, which is red urine 1-3 weeks after a sore throat. I just mentioned it above obviously. But students fuck this up despite the inculcation so I'm reiterating it like an asshole. - Caused by **viral infection,** not Group A Strep. - Can sometimes be caused by GI infections, but USMLE usually avoids this etiology unless including it in the Henoch-Schönlein purpura constellation. - Henoch-Schönlein purpura is a tetrad: 1) Palpable purpura (usually on buttocks/thighs). 2) IgA nephropathy (red urine). 3) Arthralgias. 4) Abdominal pain. - All 4 need not be present for HSP, but the abdo pain component here is presumably viral gastroenteritis leading to IgA nephropathy.
Alport syndrome	- X-linked disease; *mutation in* collagen IV gene. - Do not confuse with Goodpasture syndrome, which is antibodies against type IV collagen. - The answer on USMLE for a male who has red urine + an eye or ear problem (collagen IV is present in basement membranes in the kidney, ear, and eyes).
Membranoproliferative glomerulonephritis (MPGN)	- The answer on USMLE for red urine in a patient with hepatitis C, heroin use, or malignancy. - Answer is "renal biopsy" as the next best step. This guides our management. - A 2CK form has "heroin" as the answer (heroin-induced nephropathy) for patient with protein and blood in the urine. In contrast, FSGS (discussed below) from heroin has no blood in the urine.

Diffuse proliferative glomerulonephritis (DPGN)	- The answer on USMLE for red urine in a patient with SLE. - Same as with MPGN, USMLE wants "renal biopsy" as the next best step, since this guides our management.

HY Nephrotic syndromes for Surg	
Minimal change disease	- The answer on USMLE for otherwise unexplained edema in a kid who doesn't have blood in the urine. - Presents as peripheral edema, periorbital edema, and/or ascites. - Almost always this is pediatric. Very rarely it can be due to Hodgkin in adult. - Classically post-viral (i.e., URTI), but ~50% of vignettes won't mention that. - In other words, textbook vignette is an 8-year-old who has the sniffles for 4 days, followed by peripheral and periorbital edema a week later, without blood in the urine. Once again though, the stem need not mention the viral infection. - Called minimal change disease because light microscopy (LM) shows no abnormalities. EM, however, shows effacement of the podocytes. - Mechanism for nephrotic syndrome is "loss of size and charge barrier." - Corticosteroids are the treatment and are highly effective. - A 2CK Q gives minimal change disease as etiology for spontaneous bacterial peritonitis (i.e., any cause of ascites can cause SBP; I discuss this stuff in the Gastro section in extensive detail).
Membranous glomerulonephritis / nephropathy	- The answer on USMLE for nephrotic syndrome in patient who has: - Exposure to drugs: dapsone, gold salts, sulfonamides. - Infection: hepatitis B (Hep C can cause it too, but rare). - Visceral cancers, e.g., breast, pancreatic. - Autoimmune (primary): antibodies against phospholipase A2 receptor.
Renal amyloidosis	- The answer on USMLE for nephrotic syndrome in multiple myeloma. - Amyloidosis means protein depositing where it shouldn't be depositing. - Multiple myeloma = cancer of plasma cells that produce too much immunoglobulins → high serum IgG kappa/lambda light chains. Immunoglobulins are proteins → these fly through the kidney and cause Bence-Jones proteinuria; they also deposit in the renal parenchyma, causing renal amyloidosis. - Biopsy shows classic apple-green birefringence with Congo red stain. - An NBME Q has "renal parenchymal disease" as answer for kidney issue due to multiple myeloma.
Diabetic glomerulosclerosis	- Diabetes is the most common cause of chronic renal failure. - Kimmelstiel-Wilson nodules are composed of hyaline and look like pink circles on light microscopy. This image is very HY. - Hyaline arteriolosclerosis is deposition of hyaline (protein) in the walls of the renal vasculature. Although this can be due to many things, on USMLE, they want you to know this is caused by diabetes. - ACEi (e.g., lisinopril) or ARBs (e.g., valsartan) are the first drugs used in diabetes if there is HTN (>130/80 in diabetes, not 140/90), proteinuria, or ↑ creatinine or renin. - AT-II normally constricts efferent arterioles leaving the kidney, causing a backup of pressure at the glomerulus. This serves the purpose of maintaining GFR (i.e., filtration through Bowman capsule) when renal blood flow has ↓. - In other words, when renal perfusion ↓, RAAS goes ↑, AT-II goes ↑, and we get constriction of efferent flow leaving the kidney → ↑ filtration fraction (GFR/renal plasma flow), where we have GFR staying the same despite ↓ perfusion. - ACEi/ARBs ↓ constriction of efferent arterioles (they don't "dilate"; this refers to the action of prostaglandin on the afferent arterioles going to the kidney), thereby ↓ filtration fraction and reducing GFR. This might sound like a bad thing – i.e., "Why would we want to ↓ GFR in diabetes if it's the most common cause of chronic renal failure?" → The answer is because the slightly reduced GFR and filtration fraction early on ↓ the rate at which excess glucose is filtered across the glomerulus. Over

	time, it is inevitable that the glomerular basement membrane will non-enzymatically glycosylate and thicken, so ACEi/ARBs slow this process (i.e., slightly reduce GFR early in order to prevent a massive decrease later).
Focal segmental glomerulosclerosis	- The answer on USMLE for nephrotic syndrome in sickle cell, heroin use, and HIV. - Not as responsive to steroids as MCD.

Acute tubular necrosis	
- Sloughing of PCT tubular epithelial cells due to chemical or ischemic insult. - Only ~50% of Qs will mention "muddy brown granular casts," or "dirty brown granular casts," or "brown, pigmented casts." - The other ~50% of Qs will just mention oliguria or acutely worsening renal function in patient with classic ATN etiologies. Next best step in diagnosis (answer on Surg form 8) is "measurement of urine sodium and creatinine concentrations." We expect FENa >1% and BUN/Cr <20 almost always for ATN.	
Cause	**HY points**
Drugs	- Can be caused by drugs: **aminoglycosides** (gentamicin, tobramycin, amikacin); cisplatin (a chemo agent), or IV contrast. - Any of the above drugs + new-onset oliguria or ↑ creatinine (there need not be mention of casts) = ATN. - For instance, new NBME simply tells you patient was given many drugs and now has creatinine of 2 (NR 0.1-1.2) + no mention of casts + they ask for what was given → answer = gentamicin. Not hard. - Patient who's recently been treated for endocarditis + now has new-onset oliguria = ATN due to gentamicin (empiric Tx for endocarditis = gentamicin + vancomycin [new literature suggests vancomycin is not reliably nephrotoxic]). - Contrast nephropathy is exceedingly HY on 2CK, where they want you to know that **adequate hydration** prior to IV contrast is the number-one way to prevent it. For instance, they'll say patient got a CT + now has ↑ creatinine + they'll ask how it could have been prevented → answer = "0.9% saline," or just "IV hydration therapy."
Rhabdomyolysis	- Myoglobin is nephrotoxic. Big risk factors are **alcoholism, falls,** intense training (e.g., marathon), McCardle syndrome, statins/fibrates. - **Rhabdo causes a false (+) blood on urine dipstick.** They will say urinalysis shows 2+ blood but only 1-2 RBCs/hpf. This is because the dipstick can't differentiate between free myoglobin and hemoglobin on RBCs. - One Surgery Q gives 3-4 RBCs/hpf as negative. I've seen this fool some students who were already aware of the false (+) blood point about rhabdo. If the Q wants (+) RBCs, they'll say like 10-20+/hpf. - The other half of rhabdo vignettes won't mention the false (+) urinalysis and will instead just say oliguria. You need to know that refers to ATN. - For example, an NBME-favorite vignette is patient being found at bottom of stairs in his/her house + has false (+) blood on urine dipstick and/or oliguria → answer = rhabdomyolysis. - They also like ↑ K^+ as part of rhabdo vignettes (makes sense if we have renal failure from ATN) and ↑ serum creatine kinase (CK). - Q can say 44-year-old guy found on bench in park has ↑ K^+, ↑ CK, and oliguria → diagnosis = rhabdo. Or they'll say just ↑ CK + oliguria, and they ask what is most likely to be seen in this guy → answer = ↑ K^+.
Acute ischemia	- It's to my observation **acute ischemia** is the highest yield cause of ATN on NBME exams. - Even though the medulla of the kidney technically receives less blood flow compared to the cortex, as I mentioned earlier, the USMLE is obsessed with the PCT of the kidney as most susceptible to anoxic / hypoxic injury. This is due to the

	↑ concentration of PCT ATPase transporters, so acute ↓ drop in blood flow → acute hypoxia → PCT sheds. - NBME Qs love *cellular swelling* as part of this mechanism – i.e., acute ischemia → ↓ activity of PCT ATPases → buildup of intracellular Na^+ → water stays with sodium → tubular cell swelling. Answer on NBME for why there's swelling simply = ↓ ATPase activity (this is similar mechanism for why we get swelling + hemolysis in pyruvate kinase deficiency). - Acute ischemia to the kidney can be due to loss of blood from trauma/surgery, acute arrhythmia (e.g., 30 seconds of Vfib), MI, or **acute** exacerbation of heart failure (on 2CK Free 120). **These conditions cause ATN, not pre-renal.** Once again, ↓ flow to the kidney causing pre-renal azotemia will be more subacute/chronic, such as due to NSAID use, recent initiation of diuretic, or dehydration. There are rare Qs that are exceptions, but that is the general principle. - As mentioned earlier, USMLE will give a one-liner where they say, "Dude had surgery where he lost a lot of blood + received many packs of RBCs. 2 days later, he's now recovering in hospital + gets oliguria" (they don't say anything about brown casts) → answer = ATN, not pre-renal. - Likewise, they'll say woman had surgery + had 30-second episode intra-operatively where BP fell to 80/40 → answer = ATN, not pre-renal. - Guy has an MI (cardiogenic shock) + new-onset oliguria → ATN, not pre-renal. - Guy has burns covering 50% of his body + develops oliguria → ATN due to excessive fluid loss → acute ↓ perfusion to kidney. - Acute exacerbation of heart failure causing ATN is a difficult one since heart failure is classically associated with pre-renal from chronic ↓ blood flow to the kidney. But in acute exacerbation of heart failure, we have acute ↓↓ blood flow, leading to ATN (as I said, it's on Free 120).

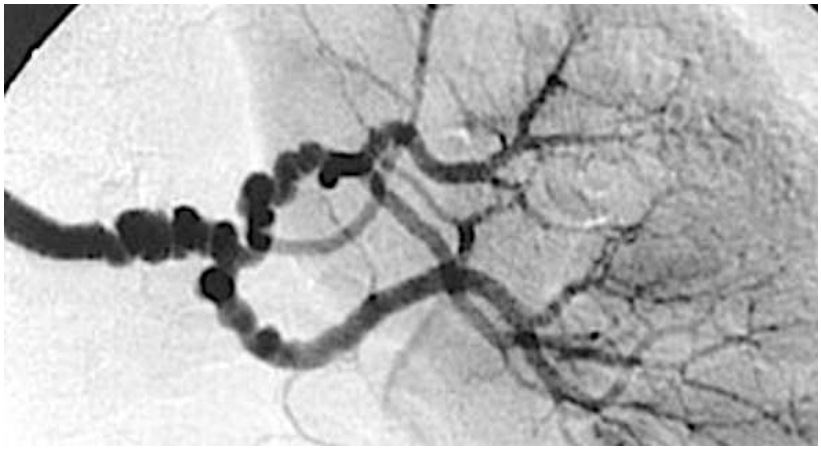
Interstitial nephritis
- Exceedingly HY renal condition often confused with ATN. - Aka interstitial nephropathy, or tubulointerstitial nephritis/-nephropathy. - Think of this as "an allergic reaction of the kidney." - 4/5 Qs will be an **NSAID, β-lactam, or cephalosporin,** followed by getting a maculopapular rash and WBCs (eosinophils) in the urine. This presentation is textbook/pass-level. - Only ~50% of vignettes will mention the maculopapular rash. - The stem need not say "eosinophils" either. They can just say patient is on β-lactam + now has WBCs in the urine + no mention of rash → answer = interstitial nephritis. If they say the rash, it's even easier. - 1/5 Qs will just say a patient was on an NSAID, β-lactam, or cephalosporin and now has mild proteinuria and hematuria. They don't mention a rash or WBCs in the urine. - There is also a singular NBME Q where they just say a patient has simple peripheral edema due to an NSAID, with no mention of anything else, and the answer is interstitial nephropathy. This is more unusual, since "NSAIDs + edema" classically = pre-renal, but I've seen the NBME assess interstitial nephropathy for this as well. - NBME also can give you simple vignette of interstitial nephritis, and then ask for the location in the kidney that's affected (e.g., efferent arterioles, etc.) → answer = "renal tubule." - For example, 40M + treated with nafcillin for 6 weeks for MSSA endocarditis + now has maculopapular rash and eosinophils in the urine; diagnosis? → interstitial nephritis. - 60F + using naproxen (an NSAID) for 6 weeks for her osteoarthritis + has peripheral edema → answer = interstitial nephropathy. - 35F + just recently finished 10-day course of cephalexin + has mild proteinuria and hematuria → answer = interstitial nephropathy. - 35F + taking ibuprofen + has maculopapular rash and eosinophils in the urine; Q asks where in the kidney is fucked up → answer = "renal tubule." Student says, "But don't NSAIDs affect the afferent arteriole?" → Yes, but this particular presentation is clearly tubulointerstitial nephropathy. I haven't seen them be

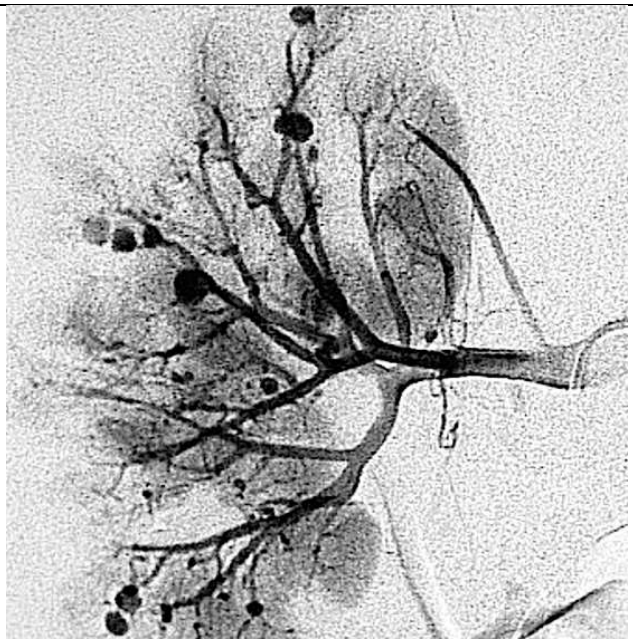
ambiguous here with "NSAID + peripheral edema alone," where you have to debate whether it's interstitial nephropathy or pre-renal. They'll either ask one or the other.

Renal tubular acidosis
Quick notes before discussing RTA: **MUDPILES** = mnemonic for high anion-gap metabolic acidoses = **M**ethanol, **U**remia (renal failure), **D**KA, **P**henformin (a drug you don't have to worry about), **I**ron/**I**soniazid, **L**actic acidosis, **E**thylene glycol, **S**alicylates (aspirin). Anion-gap is calculated as Na^+ - (Cl^- + HCO_3^-). Normal range is 8-12. High anion-gap = 13 or greater.
- USMLE really doesn't give a fuck that you know the specifics of types I, II, and IV. - The way they assess this is by you knowing this is a type of **normal** anion-gap metabolic acidosis (i.e., it is not part of MUDPILES). - For example, you'll get a 15-line massive paragraph + tons of lab values + have no idea what's going on, but then you calculate the anion gap as 12 (NR 8-12), so you can eliminate all of the MUDPILES answer choices, such as lactic acidosis, DKA, and ethylene glycol poisoning, and you're left with, e.g., Crohn disease or renal tubular acidosis. Then you just say, "Well this clearly ain't Crohn, so it must be RTA." That is how I would say 4/5 RTA Qs show up. - RTA type I = ↓ ability to secrete H^+ in the cortical collecting duct. - RTA type II = ↓ ability to reabsorb HCO_3^- in the PCT. - RTA type IV = renal resistance to aldosterone, or hyporeninemic hypoaldosteronism. - Type IV presents with hyperkalemia. The others do not. Type III apparently is hyper-rare. - Type IV will be a patient who has Addisonian-like picture (i.e., ↓ Na+, ↑ K+, ↓ $HCO3^-$), but the vignette will present in a patient who has chronic renal failure. - Type II can be caused by Fanconi syndrome (discussed more in table below). - Type I can apparently have renal parenchymal stones (i.e., stones in the actual tissue of the kidney, rather than within the tubular lumina as with traditional nephrolithiasis). - As I said, the key point is you just know RTA is normal anion-gap / not part of MUDPILES. That'll cover you like 4/5 times.

Other random renal conditions	
Tumor lysis syndrome	- Lysis of tumor cells (especially leukemic) by chemoradiotherapy leads to release of uric acid and phosphate that can overwhelm the kidneys and cause crystals, namely from uric acid. - USMLE loves asking what can be given to prevent renal failure in patient undergoing chemo → answer = "allopurinol."
Hepatorenal syndrome	- Nonsense diagnosis on USMLE that will sometimes show up as an incorrect / distractor answer choice. - Apparently liver failure can sometimes cause renal failure via obscure mechanisms. - Renal biopsy will be normal almost always.

Renovascular hypertension	
Renal artery stenosis (RAS)	- Narrowing of one or both renal arteries due to atherosclerosis that causes ↑ renin-angiotensin-aldosterone system (RAAS) and ↑ BP. - Q will be patient over the age of 50 with cardiovascular disease risk factors, such as diabetes, HTN, and/or smoking.

	- Patients who have pre-existing HTN causing atherosclerosis leading to RAS will often have 10-20 years of background HTN that then *becomes accelerated* over a few-month to 2-year period. What this means is: the slowly developing atherosclerosis in the renal arteries finally reaches a point at which the kidney is unable to maintain autoregulation, and the RAS is now clinical (i.e., accelerated HTN of ↑↑ BP within, e.g., 3 months). - Another way USMLE will give RAS is by giving ↑ BP in patient with significant evidence of atherosclerotic disease (i.e., Hx of coronary artery bypass grafting, intermittent claudication), and then ask for the most likely cause → RAS. You have to say, "Well he clearly has atherosclerosis in his coronaries and aortoiliac vessels, so that means he'll have it in the renal arteries too." - Q can say older patient with carotid bruit has recent ↑↑ in BP and then ask for diagnosis → answer = RAS. Similar to above, if the patient has atherosclerosis in one location (i.e., the carotids), then he/she will have it elsewhere too. - HY factoid about RAS is that **ACEi or ARBs will cause renin and/or creatinine to go up.** This is a HY point that is often overlooked and is asked on NBMEs. I have not seen NBME care whether it's uni- or bilateral in this case. → Kidney can autoregulate across flux in perfusion. Patients with already-compromised renal blood flow are more sensitive to the subtle ↓ in filtration fraction that occurs secondary to ACEi/ARB use, so renin/creatinine ↑. - If USMLE gives you unilateral RAS, renin is only ↑ from that kidney. The other kidney will not produce ↑ renin. - After renin and aldosterone levels are obtained, **MR angiography** of the renal vessels is what USMLE wants for the next best step in diagnosis.
Fibromuscular dysplasia (FMD)	- The answer on USMLE for narrowing of the renal arteries in a woman 20s-40s. - **Not** the same as renal artery stenosis, and **not** caused by atherosclerosis. - If you broadly say "renal artery stenosis," that specifically refers to atherosclerosis of the renal arteries in patient >50 with CVD. - FMD is tunica media hyperplasia (not dysplasia, despite the name) that results in a "string of beads" appearance on renal angiogram.  - MR angiography is answer on NBME for diagnostic modality. - Can affect the carotid vessels. A 2CK Surg Q gives FMD vignette and also says there is 25% occlusion of one of the carotids.
Polyarteritis nodosa (PAN)	- As discussed in the cardio PDF, this is a medium-vessel vasculitis that causes a "string of pearls" appearance of the renal vessels. This can be confused with FMD, but note the difference in the imaging. FMD shows more beading along the larger/proximal arterial sections, whereas PAN shows pearls more at the vascular termini, with the more proximal parts more likely to be spared.

	 Angiogram of PAN shows pearls that are more distal in the renal vasculature; FMD, in contrast, shows more proximal beading, with the termini not as conspicuously involved. - There is an NBME Q where they list both PAN and FMD as answer choices and it relies on you knowing the angiogram to get it right. They don't mention hepatitis B, but answer is PAN. That's why I'm harping on this here. - Causes fibrinoid necrosis. - **Can be caused by hepatitis B.** - Spares the pulmonary vasculature.
Renal artery thrombosis	- The answer on USMLE for hypertension in a neonate following umbilical artery catheterization. USMLE simply wants "↑ renin, ↑ aldosterone" as the answer. - Sounds weird, but you need to know umbilical artery catheterization is a major risk factor for renal artery thrombosis in neonates. - 2CK NBME Q gives brief umbilical artery catheter insertion in kid born 26 weeks' gestation in order to monitor blood pressure for a pneumonia. 3 weeks later, he has ↑ BP (128/86) → answer = "↑ renin, ↑ aldosterone" as answer. - BP in term neonates should be ~60/40. In a preemie 29 weeks' gestation, it should be ~50/30 according to Google.

HY Uremia points	
- As mentioned earlier, "uremia" means "urea in the blood," but is used to refer to patients who have symptomatic renal failure with poor lab values. - On USMLE, uremic patients will have: ↑ K^+, ↓ HCO_3^-, ↓ Ca^{2+}, ↑ PO_4^{3-}. Na^+ is variable.	
Uremic pericarditis	- Friction rub in patient with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic encephalopathy	- Mental status change in patient with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic platelet dysfunction	- Nosebleeds and/or petechiae in patient with renal failure. - **Qualitative, not quantitative,** platelet problem. In other words, bleeding time is ↑, but platelet count is normal. - Mechanism is ↑ BUN causing impairment of platelet function. - Answer on USMLE can be written as "acquired platelet dysfunction." - Treatment = hemodialysis.

	- On 2CK NBME 10, Surg Q wants "initiation of dialysis" as answer in uremic patient pre-op. Improving RFTs will ↓ peri-op morbidity/mortality + ↓ risk of intra-op bleeding due to uremic platelet dysfunction.
Uremic frost	- Yellow skin sometimes seen in severe renal failure due to ↓ ability to excrete urea. Excess is excreted in sweat, precipitating out as white crystalline deposits (frost). - Can be associated with itchy skin (uremic pruritis).

Urolithiasis	
- Urolithiasis is broad, umbrella term that refers to both nephrolithiasis and ureterolithiasis. - Diagnosed with non-contrast CT of abdomen and pelvis. - KUB (X-ray of kidney, ureters, bladder) can be done but isn't as sensitive (although below I show an X-ray).	
Stone type	**HY points**
Calcium	- Most common type of stone. Can be calcium oxalate or calcium phosphate, although I've never seen USMLE once assess or give a fuck about phosphate stones. - Most young adults with idiopathic kidney stones will have "normocalcemia and hypercalciuria" – i.e., normal serum calcium but elevated urinary calcium. - USMLE will give you healthy male in his 20s with sharp pain in the flank or groin + RBCs in the urine → answer = urolithiasis. - **Crohn disease** and disorders causing fat malabsorption increase the risk for calcium oxalate stones (↑ fat retained in GI tract → ↑ chelation with calcium in GI tract → ↓ calcium available to bind oxalate → ↑ oxalate absorption by GI tract → ↑ urinary oxalate). - Primary hyperparathyroidism and malignancy causing hypercalcemia are HY etiologies. - Milk-alkali syndrome: ↑ calcium + ↑ bicarb + calcium stones. Textbook scenario is a patient taking too many antacids, but vignette will usually not mention this and it will simply be a diagnosis of exclusion (i.e., you eliminate to get there). - Ethylene glycol (Anti-Freeze) and hypervitaminosis C can cause oxalate stones. - First step in prevention and treatment of stones is "adequate hydration." - **Thiazides** can be used to prevent recurrent stones by ↓ urinary calcium. NBME has a Q that asks why → answer = ↑ reabsorption of calcium. - In theory, alkalinization of the urine can help treat oxalate stones, and acidification of the urine can help treat phosphate stones.
Struvite	- Composed of ammonium magnesium phosphate. - Formed in ↑ pH in the presence of urease (+) bacteria. - *Klebsiella, Serratia,* and *Proteus* are HY causes. - These stones are large, ram-horn like. The image is HY.  - Apart from adequate hydration, Tx is acidification of urine with NH_4Cl + surgery.

Uric acid	- Seen in patients who have gout. Not complicated. - First-line Tx for *chronic* gout is xanthine oxidase inhibitors (i.e., allopurinol or febuxostat). They are preferred over uricosurics (meaning ↑ urinary excretion of uric acid) like probenecid and sulfinpyrazone because the latter agents ↑ risk of uric acid stones. - Probenecid and sulfinpyrazone inhibit organic anion transporter (OAT), which normally functions to ↑ absorption of uric acid. - Weird factoid about probenecid is that it can help maintain serum β-lactam levels, since the latter are excreted by OAT. There's an NBME Q around somewhere that asks about this. Sometimes old-school docs will put a patient on a β-lactam and probenecid. Don't be confused by this. The patient doesn't have gout. It's just a way of ↓ renal excretion of the β-lactam.
Cystine	- Cystine stones (hexagonal) can occur in young adult due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystinuria can be diagnosed with cyanide-nitroprusside test.
Miscellaneous	- Acyclovir (for HSV and shingles) and indinavir (for HIV) can cause crystal nephropathy. Not particularly HY, but worthy of 1-2 lines here.

Cystic kidneys	
ADPKD	- Autosomal dominant polycystic kidney disease. - The answer on USMLE if disease starts as an adult (i.e., 30s-40s). - Cysts are technically present early in life, but only become clinical as adult (i.e., ↑ BP and ↑ RFTs). - These patients have ↑ BP due to ↑ RAAS (compression of microvasculature of kidney due to enlarging cysts). - Can cause saccular (berry) aneurysms of the circle of Willis → risk for subarachnoid hemorrhage. - Highest yield point is that **serial blood pressure checks** are correct over circle of Willis MR angiogram screening. Latter is wrong answer on USMLE. MR angiogram screening of circle of Willis is only done when there is (+) family Hx of SAH or saccular aneurysms. - Most common extra-renal location for cysts is the liver (85% by age 30).
ARPKD	- Autosomal recessive polycystic kidney disease; chromosome 6. - The answer on USMLE for cystic kidneys in pediatrics. - Can be associated with **hepatic fibrosis.**
Miscellaneous	- Just be aware renal cysts can occasionally develop in other conditions, such as chronic dialysis patients. - Conditions such as medullary sponge kidney are garbage for USMLE. Waste of time.

HY anatomic abnormalities	
Posterior urethral valves (PUV)	- Most common genitourinary abnormality in neonatal males. - Urethra has abnormal presence of valves at the posterior (prostatic) urethra, preventing the outflow of urine. - USMLE will give Qs of varying severity. - Most severe is *in utero* oligohydramnios. - Can present as 12-hour-old neonate who hasn't yet urinated + has suprapubic mass (i.e., full bladder). - Can also present as 6-week-old boy who has full bladder (i.e., possibly slower accumulation due to only partially obstructed outflow). - Obstructed outflow ↑ risk for UTIs, cystitis, and pyelonephritis. - Apart from knowing this diagnosis, highest yield point is that we do **ultrasound followed by voiding cystourethrogram** to diagnose. - NBME Qs might already mention negative ultrasound in the vignette, or might omit it altogether, where you just select "voiding cystourethrogram."

Vesicoureteral reflux	- Congenital abnormality in which urine from the bladder can go back up into the ureter toward the kidney. - This ↑↑ risk for recurrent acute pyelonephritis → chronic pyelonephritis (as I discuss below). - Can also occur in pregnancy due to 2 reasons: 1) larger uterus in 3rd trimester can compress the ureters; 2) progesterone ↓ ureteral peristalsis. - This is why pregnant women can get pyelonephritis, and also why we always treat asymptomatic bacteriuria in pregnancy, whereas we don't treat it if woman is not pregnant.

Genitourinary infection-related stuff	
Acute pyelonephritis	- Infection of the kidney. "Pyelo" means kidney. - 9/10 questions will mention fever + **costovertebral angle (CVA) tenderness,** which is pain with percussion of the flank. - "Granular casts" can be seen. Do not confuse this with acute tubular necrosis. "General granular casts" can sometimes be seen in pyelonephritis and dehydration. It's on one of the 2CK IM forms, where students say, "Wait, but they say granular casts; how is this not ATN?" - Major risk factors are vesicoureteral reflux (especially in pregnancy) and posterior urethral valves. - Most common organism is *E. coli*. Other organisms, i.e., *Klebsiella, Serratia,* and *Proteus* cause struvite stones. - It's to my observation **bacteria can be few in the urine** in acute pyelo. This confuses some students. But it's typically what I see on NBME forms. If the infection is further down, i.e., UTI in the urethra, then bacteria are more copious. - Treatment for pyelo is **ciprofloxacin** or **ceftriaxone.** USMLE is known to ask these. - Persistent fever despite Abx Tx → do CT to look for **perinephric abscess.** - For example, old dude + high Cr (caused by post-renal from BPH) + treated for pyelo → now gets sore ankle → was treated with cipro (causes tendonopathy). - 2CK form has ceftriaxone as an answer in pyelo Q where cipro isn't listed. But cipro is classic Tx. It's to my observation that ceftriaxone is HY drug on 2CK for community-acquired "general sepsis" or "general complicated/severe infections" – i.e., it is hard-hitting and covers wide array of community organisms. - There is one 2CK Q where they mention pyelo is treated with amp + gent in the stem, but I've never seen this assessed as an answer you need to choose. I've only ever seen ciprofloxacin or ceftriaxone as actual NBME answers for pyelo.
Chronic pyelonephritis	- Due to recurrent acute pyelonephritis. - Almost always pediatric, where they give a 4-year-old who has a small, shrunken kidney with tubular atrophy and blunting and scarring of the renal calyces. - Q can give child with recurrent UTIs, and then the answer is just "tubular atrophy" as what is seen, or "congenital ureteral obstruction" as the cause. - Chronic pyelo doesn't present with cellular infiltration the way acute pyelo does. It refers to the kidney being destroyed/shrunken from repeated prior infections.
Cystitis	- Suprapubic tenderness in female. Patient need not have fever. - *E. coli* most common cause. - Will have bacteria and WBCs in the urine (i.e., bacteriuria + pyuria). - Urinary nitrites and leukocyte esterase can be positive. These just reflect bacterial infections. Some students get pedantic about which organisms result in which combo of (+) or (-) findings here, but that's low-yield for USMLE. - Can be caused by **suprapubic catheters** (on 2CK Surg). - **Nitrofurantoin** is HY drug to treat cystitis on USMLE.
Chronic interstitial cystitis	- Not an infection.

	- This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care" and "physiotherapy."
General UTI	- Classically *E. coli* infection of the urethra. - Common in women due to shorter urethras. - Inoculation of the urethra following sexual activity is most common mechanism. - Can be caused by catheters. You need to know **dysuria + Hx of catheter = UTI.** - Tangentially, this is also HY for general sepsis, where if they mention Hx of IV line/catheter, USMLE wants you to be able to say, "Got it. That's the cause." - Trimethoprim, trimethoprim/sulfamethoxazole (TMP/SMX), or nitrofurantoin are standard treatments for UTI. - For prevention, post-coital voiding confers ↓ risk of recurrence. - If post-coital voiding doesn't work, "post-coital nitrofurantoin therapy" is answer on 2CK NBME form. - If post-coital nitrofurantoin therapy doesn't work, "daily TMP/SMX prophylaxis" is the answer. Sounds absurdly wrong / like a bad idea, but it's an answer a couple times on Obgyn assessments. - There's also a Q where they say a girl was treated successfully in the past with TMP/SMX for a UTI + ask how to prevent recurrent UTIs now, and they just jump straight to "daily TMP/SMX prophylaxis." - If patient is treated for a UTI and dysuria persists, the next best step is testing for *Chlamydia* and *Gonorrhea*.

HY Bladder incontinences	
Stress	- The answer for loss of urine with ↑ intra-abdominal pressure from laughing, sneezing, coughing. - Stereotypical risk factor is grand multiparity (i.e., Hx of many childbirths) leading to weakened pelvic floor muscles. - I'd say only ~50% of Qs will mention Hx of pregnancy. The other ~50% are idiopathic. - Buzzy vignette descriptors and answer choices are: "downward mobility of the vesicourethral junction," "urethral hypermobility," and "urethral atrophy with loss of urethrovesical angle." - Treatment is pelvic floor (Kegel) exercises. These notably strengthen **levator ani, pubococcygeus,** and the **external urethral sphincter.** - USMLE is known to ask which muscle is *not* strengthened by Kegel exercises, which sounds obscure, since any muscle could theoretically be the answer ("Well the deltoid isn't strengthened."). But a favorite answer here is **internal urethral sphincter.** The way you know this is the answer is because internal sphincters are under sympathetic (i.e., involuntary; autonomic) control, which means it's impossible to strengthen it via a voluntary (i.e., somatic) exercise. USMLE doesn't expect you to be an obstetrician. The bigger picture concept is simply knowing internal sphincter control is involuntary. It is external sphincter control that is voluntary (somatic). - Do **not** give medications for stress incontinence on USMLE. - If Kegel exercises fail, patients can get a mid-urethral sling (LY; asked once).
Urge	- The answer on USMLE for patient who has an "urge" (NBME will literally say that word and I've seen students get the Q wrong) to void 6-12+ times daily **unrelated** to sneezing, coughing, laughing, etc. (otherwise stress incontinence). - Ultra-HY for **multiple sclerosis.** I've had students ask whether MS is urge or overflow. It shows up repeatedly on the NBMEs as urge; I've never seen it associated with overflow. I'd say ~1/3 of urge incontinence vignettes on NBME forms are MS. - Other vignettes will be peri-menopausal women, or idiopathic in old women.

	- Mechanism is "detrusor hyperactivity," or "detrusor instability." - Vignette can mention woman has urge to void when stepping out of her car, or when sticking her key in the car/front door of her house. Sounds weird, but these are important Qs to ask when attempting to diagnose urge incontinence. - UTIs can present similarly to urge incontinence. Some students have asked, "Well isn't that because UTIs are a cause of urge incontinence?" Not really. It just happens to be that UTIs can sometimes cause transient urinary urgency. For example, if they give a Q where they say patient had Hx of urinary catheter + now has dysuria and urinary urgency, answer = "urinary tract infection" on NBME; "detrusor hyperactivity" is wrong answer. - Treatment is **oxybutynin** (muscarinic receptor antagonist); this ↓ activity of the detrusor muscle of the bladder. - Some students get hysterical about mirabegron (β3-agonist), but I've never seen NBME forms assess this.
Overflow	- Will be due to either BPH or diabetes on USMLE. - Will have ↑ **post-void volume.** Normal is < ~50 mL. On USMLE for overflow, they'll give you 300-400 mL as post-void volume. - As I talked about earlier for BPH, they will give old dude + high creatinine (post-renal azotemia). Next best step is "insertion of catheter" to relieve the obstruction. If they don't have this listed, "measurement of post-void volume" can be an answer. We then treat the BPH with finasteride (5α-reductase inhibitor) or an α1-blocker (tamsulosin, terazosin). - For diabetes, the mechanism is neuropathy to the bladder causing "neurogenic bladder," or "hypotonic bladder," or "hypocontractile bladder/detrusor muscle." - For neurogenic bladder causing overflow incontinence +↑ post-void volume, remember that USMLE is first obsessed with "measure post-void volume" and "insertion of catheter" if they are listed. They will not force you to choose between the two. But they like these answers prior to giving medications. - Give **bethanechol** (muscarinic receptor agonist); this stimulates the detrusor muscle. - Making sure you don't confuse oxybutynin and bethanechol is pass-level for USMLE.
Normal pressure hydrocephalus (NPH)	- Patient who has "wet, wobbly, wacky" presentation (i.e., urinary incontinence, ataxia, cognitive changes) +/- Parkinsonism (e.g., short-steppage gait). - Caused by failure of reabsorption of CSF by the arachnoid granulations, resulting in impingement on the zona radiata and **"failure to inhibit the voiding reflex"** (answer on NBME). - Enlargement of the lateral ventricles on head CT.
AIDS complex dementia	- Presents similar to NPH – i.e., "wet, wobbly, wacky" in AIDS patient.

Urinary retention diagnoses	
Bladder neck dyssynergia	- Lack of coordination between bladder sphincters and detrusor muscle. - HY cause is spinal cord injury, which can cause either inability to void or urinary incontinence. - Multiple sclerosis is another theoretical cause, which is where "overflow incontinence" as a result of MS has been asked by students, but once again, I've only ever seen urge incontinence for MS on NBME exams. Qbank ≠ NBMEs.
Bladder outflow obstruction	- Acute urethral obstruction due to **catheter removal** is HY on Surg forms. - They'll say patient has *anuria* (i.e., no urinary output at all) following removal of a catheter. - "Acute urethral obstruction" or "bladder outflow obstruction" is the answer. - Treatment on NBME is "re-insertion of urinary catheter."
Cauda equina syndrome	- Will present as leg pain + urinary retention on USMLE.

	- Can be idiopathic, from disc herniation, or other trauma. - Metastases is HY cause (i.e., random guy over 50; or female with Hx of mastectomy 5 years ago). - "Saddle anesthesia" is an overrated detail and doesn't show up in the vast majority of NBME Qs, as per my observation. - Cauda equina is the collection of nerves at the end of the spinal cord. Impingement on these nerves is what causes the syndrome. I cover this stuff in more detail in my HY Neuroanatomy PDF.

HY Genitourinary injuries for Surg (high-yield)	
Urethral injury	- Always presents with blood at the urethral meatus. - 3 ways this presents on USMLE: 1) Blood at urethral meatus + saddle injury (dude falls on balance beam). 2) Blood at urethral meatus + pelvic fracture/instability. 3) Blood at urethral meatus + "prostate not palpable." - Next best step is **retrograde urethrogram.**
Bladder injury	- Answer on USMLE for blunt force trauma to abdomen + exquisite suprapubic tenderness + catheter insertion yields 10-30 mL of dark bloody fluid. - Next best step is **retrograde cystourethrogram.** Do not confuse with voiding cystourethrogram, which is for posterior urethral valves.
Ureteral injury	- Just be aware this is USMLE-favorite injury in Obgyn for any surgery done in the gynecologic area. - Diagnosis is with retrograde pyelogram or CT urogram, although I haven't seen the USMLE give a fuck. Any ureter injury Q I've seen just has "ureter" as answer.
Renal injury	- Exceedingly HY on both IM and Surg forms on 2CK. - Will present as bruising or pain over a flank following accident. - Blood in the urine is highly sensitive, meaning if it's not there, we can rule-out. - First step in diagnosis: "Is there gross blood in the urine?" If yes → do **CT of the abdomen with contrast** to diagnose (sounds wrong, since we're worried about kidney injury, so student says, "Why would we use contrast?" But it's what USMLE wants. **Ultrasound is wrong answer**). - If patient does not have gross blood in the urine, next best step is **urinalysis looking for microscopic blood.** If present → answer = CT of the abdomen with contrast. If negative → "no further diagnostic studies indicated."

Urothelial malignancies	
Renal cell carcinoma (RCC)	- Aka "clear cell carcinoma"; this is most common variant of RCC. - Classic Q is a smoker over 50 with red urine and a painful flank mass. - Q need not mention smoking, but it is biggest risk factor. - Can cause **polycythemia** due to ↑ EPO and **hypercalcemia** due to PTHrp secretion (correct, same as squamous cell carcinoma of the lung). - For example, they say 55-year-old male + red urine + polycythemia + hypercalcemia + show you above image; they ask for the source of the malignancy → answer "kidney," not lung. Squamous cell carcinoma of the lung won't cause hematuria, nor does it metastasize to the kidney.
Wilms tumor (Nephroblastoma)	- The answer for painless flank mass in a kid age 2-4 years. - Does **not** present in newborns. If they give you a 12-hour-old male, for instance, with a midline mass, that is posterior urethral valves causing a full bladder (let alone the fact they say "midline," not flank). I've seen tons of students select Wilms for this, and I'm like, is the kidney in your midline? - If they give you a kid 2-4 years with a midline mass, that is neuroblastoma.

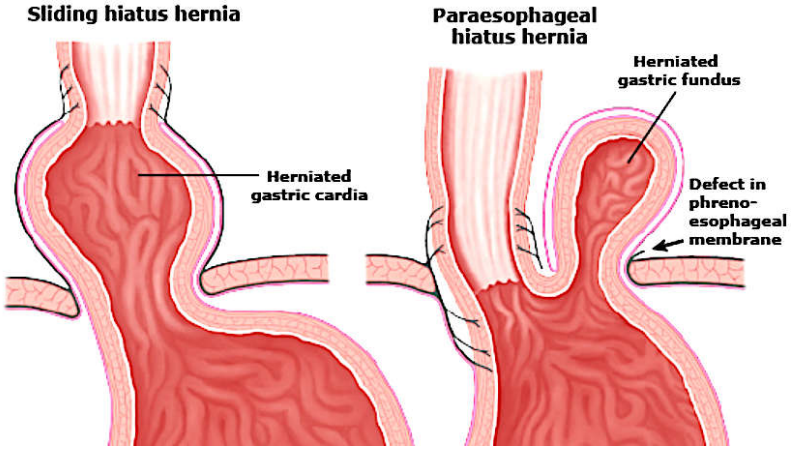
	- For whatever magical reason, you need to know Wilms tumor is sometimes caused by mutations on **chromosome 11.** They ask this factoid on 2CK as well. There are various Wilms tumor syndromes you need to be aware of: - **Beckwith-Wiedemann syndrome** - Wilms tumor + macrosomia (big baby >4,000g) + hemi-hypertrophy (half of the body is bigger than the other) + macroglossia + omphalocele + hypoglycemia. - Hard 2CK Peds Q gives newborn with macrosomia + hemi-hypertrophy + macroglossia + omphalocele; they don't mention a Wilms tumor (makes sense, since we said we won't see it in newborns); they ask for what else could be seen in this patient → answer = "hypoglycemia." I say hard Q because Wilms plays no role in the vignette. **- WAGR syndrome** - **W**ilms tumor, **A**niridia (iris abnormalities), Genitourinary anomalies, Retardation. - Usually an easy Q, where they mention "aniridia" straight up in the vignette and then the answer is just "Wilms tumor" for what the kid can go on to develop. **- Denys-Drash syndrome** - Ambiguous genitalia + Wilms tumor.
Collecting duct carcinoma	- Transitional cell in origin. - "Naphthylamine" (moth balls) is a causative agent asked on NBME.
Transitional cell carcinoma of the bladder	- Most common bladder cancer; can have papillary structure on histo. - Classic vignette is hematuria in smoker *without* a painful flank mass or polycythemia/hypercalcemia (otherwise RCC). - USMLE wants you to know smoking is most common risk factor, but aniline dyes (industrial clothing dyes) and naphthylamine are important causes. - 2CK Surg Q gives Stage 0 bladder cancer vignette; answer is just "endoscopic resection." - I've never seen BCG vaccine as correct answer on NBME. I've only seen it as a distractor. For whatever reason, some studies in the past have suggested intra-cystic instillation of the TB vaccine can help with bladder cancer. But this is wrong on USMLE.
Squamous cell carcinoma of the bladder	- Can be caused by *Schistosoma hematobium* (a trematode or fluke, which is a type of helminth). - *S. hematobium* lays its eggs in the bladder wall and cystic veins draining the bladder. - Vignette will be young guy who has red urine months after returning from Africa where he went swimming in a lake → answer is just *Schistosomiasis*. Or they ask what he's at ↑ risk of → answer = squamous cell carcinoma.

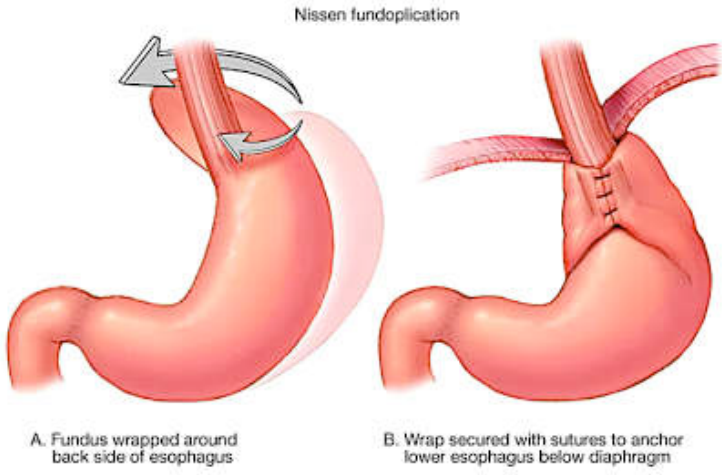
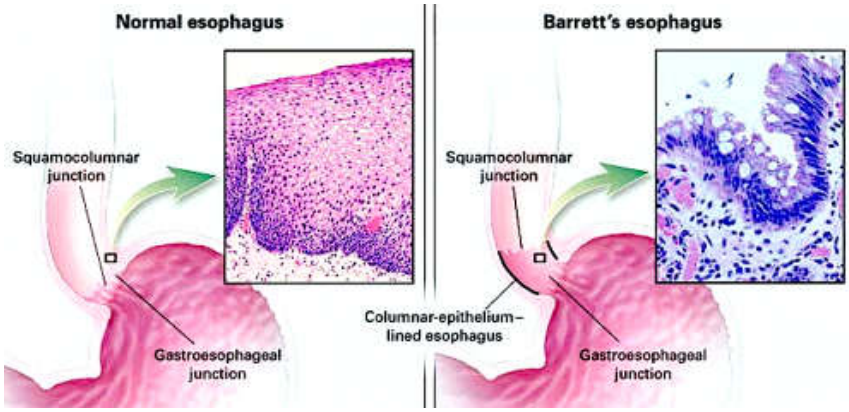
Hyper-terse summary of drugs causing HY renal conditions	
- Discussed all this stuff earlier. But students confuse this stuff a lot. So the summary is warranted.	
Acute tubular necrosis	- Aminoglycosides (gentamicin, tobramycin, amikacin). - Cisplatin.
Interstitial nephritis	- NSAIDs, β-lactams, cephalosporins.
Membranous glomerulonephropathy	- Dapsone, gold salts, sulfonamides.

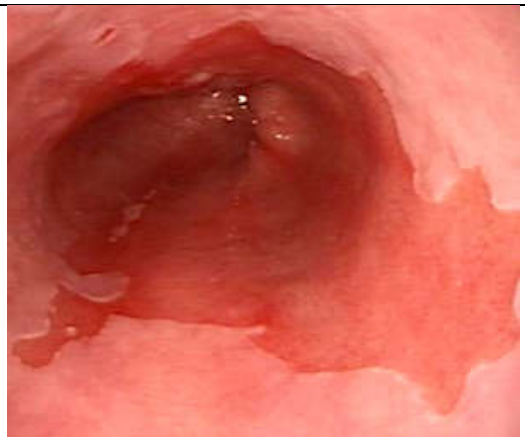
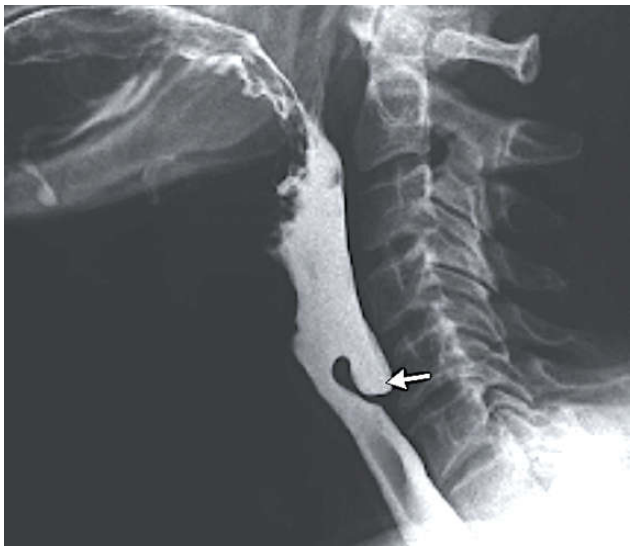
Diuretics/Renal drugs for Surg	
Mannitol	- Osmotic diuretic that ↑ osmolality of urine, thereby retaining free water within the PCT and descending limb of loop of Henle. - Can be used as Tx for ↑ intracranial pressure after patient has been intubated + hyperventilated (↓ CO2 → ↓ cerebral perfusion). - Avoid in heart failure (transient ↑ in serum osmolality prior to renal excretion → ↑ retention of free water within vascular space → transient ↑ preload on heart).
Acetazolamide	- Carbonic anhydrase inhibitor (enzyme that enables $H_2CO_3 \leftrightarrow HCO_3^- + H^+$) that acts within the tubular cells of the PCT. - Classic use is altitude sickness → causes ↓ reabsorption of HCO_3^- in the PCT → results in ↑ urinary pH and volume + ↓ serum pH and volume. - In altitude sickness, patients hyperventilate (↓ CO2 → ↑ pH) to take in more O2, but it takes a minimum of 12-24 hours for the kidney to ↓ HCO_3^- reabsorption, so acetazolamide forces the ↓ reabsorption by causing metabolic acidosis.
Furosemide	- Classic loop diuretic that inhibits the apical $Na^+/K^+/2Cl^-$ ATPase symporter on the thick ascending limb. Apical means side of urine. Basolateral means side of blood. - Most efficacious diuretic at enabling fluid unloading for pulmonary and peripheral edema. - Answer on USMLE for patient with heart failure who has ↓ O2 sats who "refuses to lie supine/back on the gurney" because he/she can't breathe. Q might say CXR or exam shows fluid 2/3 up the lung fields. - Also answer for first drug to ↓ peripheral edema. - Application of loops can be confused by students with ACEi/ARB. For instance, the latter are used first-line to ↑ EF in heart failure even if O2 sats are low (↓ AT-II constrictive effect on arterioles → ↓ afterload → heart pumps easier), but loops are first-line if the vignette *specifically emphasizes* the dyspnea and low O2 sats of the patient as the focus. - Loops do not decrease mortality in cardiac patients. They are simply used for fluid unloading for pulmonary and peripheral edema. - Can cause ototoxicity (tinnitus, vertigo). - HY point is that they can cause **hypokalemia.** What USMLE loves to do is tell you a patient is initiated on furosemide but it is insufficient + now needs a second diuretic → answer = anything that is potassium-sparing (i.e., ENaC inhibitor such as amiloride or triamterene; or aldosterone receptor antagonist such as spironolactone or eplerenone). - Loops ↑ urinary calcium. This is in contrast to thiazides, which ↓ it. - Ethacrynic acid is another loop you can be aware of. It is used in patients with sulfa allergies, since furosemide is a sulfa drug. It is also ototoxic.
Thiazides	- Hydrochlorothiazide (HCTZ) is HY example; chlorthalidone is "thiazide-like." - Inhibit the Na^+/Cl^- symporter on the apical membrane of the early-DCT. - Thiazides and dihydropyridine calcium channel blockers are used first-line for HTN in patients without any renal or cardiovascular issues. If patient has proteinuria, ↑ in creatinine or renin, or pre-diabetes or diabetes, ACEi or ARBs are used first. - Used to ↓ recurrence of calcium stones by promoting ↑ reabsorption of urinary calcium. In turn, they can sometimes cause hypercalcemia. - Can be used in heart failure in patients with diabetic nephropathy who are already on ACEi and β-blocker. Sounds specific, but there's a new 2CK Q that has thiazide as correct over spironolactone in a diabetic. Spironolactone can cause hyperkalemia in patients who have worrisome kidney function. - Thiazides can cause gout (i.e., contraindicated in gout).
Amiloride, Triamterene	- ENaC inhibitors → block apical sodium channel in cortical collecting duct → ↓ Na^+ reabsorption → ↓ water reabsorption. - These are potassium-sparing, which means they do not ↓ serum K^+. This is because by inhibiting the apical ENaC channel, they indirectly inhibit the basolateral Na^+/K^+ ATPase → ↓ Na^+ reabsorption + ↓ K^+ secretion.

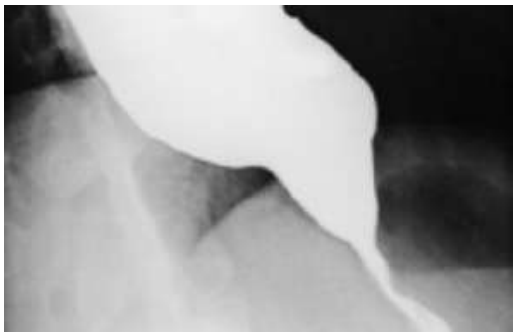
	- The answer on USMLE for a second diuretic given in a patient already on furosemide who needs additional fluid unloading. Be careful however. I've seen one Q where the patient is on furosemide, but the point of the Q is he/she needs HTN control, and the answer is a thiazide, not the ENaC inhibitor. You want to select an ENaC inhibitor specifically if the Q says, "We have a patient who has peripheral/pulmonary edema due to X cause + is already on furosemide; what do we do now?" → answer = amiloride or triamterene.
Spironolactone, Eplerenone	- Aldosterone receptor antagonists. - By blocking aldosterone receptor, it ↓ activity of the basolateral Na^+/K^+ ATPase → leads to indirect ↓ activity of apical ENaC → ↓ sodium reabsorption → ↓ water reabsorption. - Potassium-sparing diuretic. Same as with ENaC inhibitors, it is used for fluid unloading in patient who is already on furosemide in whom we worry about dropping their K^+ too much. - Used in heart failure up the hierarchy of meds – i.e., ACEi/ARB first-line, followed by adding β-blocker, followed by spironolactone (but if patient is diabetic with poor renal function, don't add spironolactone here; give thiazide instead as per new 2CK NBME). - Can be for aldosteronoma (Conn syndrome) prior to surgery. - Can cause **gynecomastia** (anti-androgenic effect by blocking androgen receptors). - Eplerenone has ↓ risk of gynecomastia compared to spironolactone, but USMLE doesn't give a fuck. Resource masturbation garbage that's been regurgitated at times over the years.
Tolvaptan, Conivaptan	- ADH receptor antagonists at V2 receptor in the medullary collecting duct. - Can be used for SIADH. - However, on 2CK NBME, demeclocycline is also answer for SIADH.

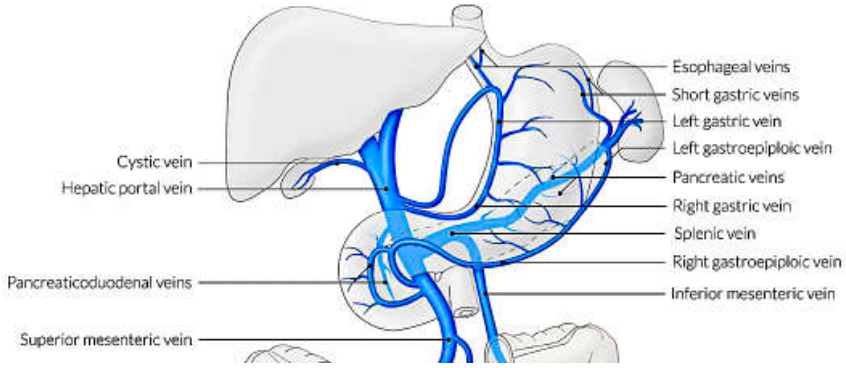
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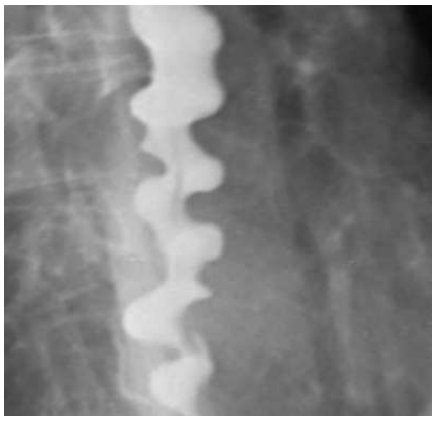
HY Esophageal conditions for USMLE	
Hiatal hernia	- Herniation of the stomach upward through the diaphragm. - 95% of the time, hiatal hernia is known as "sliding," where the cardia of the stomach rises upward through the esophageal hiatus (opening in diaphragm where esophagus passes). - "Abnormal relation of the cardia to the lower end of the diaphragm" is answer on NBME. - 5% of the time, it is paraesophageal, where the fundus of the stomach herniates upward through a hole in the diaphragm adjacent the esophagus. - "Protrusion of the fundus into the chest above the level of T10" = answer for paraesophageal hernia on NBME.  - Often associated with gastroesophageal reflux disease (GERD). - Diagnose with upper endoscopy (asked on Surg form). - Operative repair is the Tx (also asked on Surg form).
Gastroesophageal reflux disease (GERD)	- Irritation of esophageal mucosa by gastric acid; usually caused by ↓ **lower esophageal sphincter (LES) tone.** - USMLE wants you to know **obesity** is a risk factor for GERD, likely due to ↑ stretching and pressure applied to the LES. - As mentioned above, hiatal hernia is also a risk factor. - Classic presentation is burning in the throat/chest after eating a meal. - Can present with **nocturnal cough** or **recurrent pneumonitis.** These are HY findings in patients who don't have classic esophageal irritation symptoms. - Can lead to Barrett esophagus (discussed below), which leads to esophageal adenocarcinoma. - First step in diagnosis is 2-week trial of proton pump inhibitor (PPI), such as omeprazole, which will ↓ symptoms. - PPIs are more efficacious than H2-blockers (e.g., cimetidine). If you see both a PPI and H2-blocker as answers for a GERD Q, choose the PPI. - One 2CK Surg form has "2-week trial of H2-blocker" as an answer, but PPI isn't listed, so just know the Q is out there. - If trial of PPI doesn't work, 24-hour esophageal pH monitoring is the answer. - Nissen fundoplication is used last resort, but is asked on NBMEs.

	 - Answer can be written as "fundoplication" alone in patient with severe GERD. This also shows up on Peds, where they say infant with severe GERD and fundoplication is answer. - Pediatric GERD will present as coughing up milk 2-3x daily. It will not be described as high-energy or forceful (i.e., pyloric stenosis); it will not be bilious (i.e., duodenal atresia; annular pancreas); it will not occur as choking or spitting up during the first feed (i.e., tracheoesophageal fistula). - Answer to pediatric GERD Q is "immature lower esophageal sphincter". It's not complicated, but I've seen students say, "Wait, you can get GERD in kids?" - Tx for Peds GERD is thickened feeds (addition of dry rice to formula) and positional change prior to any use of PPIs.
Barrett esophagus	- Metaplasia of distal esophagus characterized by conversion of non-keratinized stratified squamous epithelium into columnar epithelium with goblet cells (which secrete mucous). - You'll sometimes here the latter described as "intestinal columnar epithelium." This is because the stomach doesn't have goblet cells; it has mucous neck (foveolar) cells. - Major risk factor for esophageal adenocarcinoma. 

	 - Can appear grossly red on upper endoscopy. Once visualized, the next best step is biopsy to confirm the presence of Barrett metaplasia.
Adenocarcinoma	- Affects lower 1/3 of esophagus. - HY pathogenesis is: GERD → Barrett esophagus → adenocarcinoma. - Will present in patient over 50 who has Hx of GERD with either 1) new-onset dysphagia to solids, or 2) dysphagia to solids *that progresses* to solids and liquids. - The "new-onset dysphagia" can refer to 3-6-month Hx in patient with, e.g., 10-20-year Hx of GERD. - 2CK wants immediate endoscopy in either of the above situations (i.e., don't choose barium first). - Sometimes rather than making you choose endoscopy straight up, they'll tell you in the last line an endoscopy was performed and shows a stricture, then they'll ask for next best step → answer = biopsy of the stricture.
Squamous cell carcinoma	- Affects upper 2/3 of esophagus. - Biggest risk factors for USMLE are **heavy smoking/alcohol use.** - Other risk factors like burns, chemicals, achalasia, etc., are nonsense. - Will present in patient over 50 who is heavy smoker/drinker who has 1) new-onset dysphagia to solids, or 2) dysphagia to solids that *progresses* to solids and liquids. - Same as with adenocarcinoma, USMLE wants immediate endoscopy as the answer, followed by biopsy of a stricture or lesion if present.
Zenker diverticulum	- Outpouching of esophagus above the cricopharyngeus muscle (just above the upper esophageal sphincter). 

	- Mechanisms on USMLE are numerous and include, “cricopharyngeal muscle spasm” (Surg form 5), “uncoordinated swallowing,” “dysphagia,” and “increased oropharyngeal pressure.” - False diverticulum (contains only mucosa + submucosa; true diverticula, in contrast, such as Meckel, include the muscular layer as well). - Often presents as overweight male over 40-50 with gurgling sounds when swallowing, or regurgitation of undigested food. - Halitosis (bad breath) is buzzy but only mentioned in maybe 1/3 of Qs. - Diagnosis is made with barium swallow. - Treatment on 2CK NBME is “diverticulectomy.”
Achalasia	- Tightening (i.e., ↑ tone) of the LES due to loss of NO-secreting neurons in the Auerbach (myenteric) plexus. - New NBME wants **“enteric ganglia”** as the answer for site of problem. - Can be described on NBME as “dilated esophagus with constriction of the gastroesophageal junction.” - Presents usually as dysphagia **to both solids and liquids from the start.** This is in contrast to cancer, which will be new-onset dysphagia to just solids, or *progression* from solids only to solids + liquids. The HY point is that neurogenic causes of dysphagia (i.e., achalasia) lead to solids + liquids dysphagia. - Almost always idiopathic; can rarely be caused by Chagas disease. - New 2CK NBME has “South American trypanosomiasis” as the answer for achalasia, which is aka Chagas disease. - First step in diagnosis is barium swallow, which shows a classic “bird’s beak” appearance.  - After the barium is performed, esophageal manometry (a pressure study) is confirmatory. Biopsy is not traditionally performed. - Treatment is pneumatic (balloon) dilation, followed by myotomy (cutting of muscle fibers of the LES).
Systemic Sclerosis (aka scleroderma)	- As discussed in detail in the Pulm section, this is an idiopathic autoimmune disease characterized by multi-organ system fibrosis and hardening of tissues (i.e., sclerosis). - Can lead to GERD. An offline NBME wants ↓ LES sphincter tone and ↓ esophageal peristalsis as an answer. This refers to the esophageal dysmotility component of **CREST** (**C**alcinosis, **R**aynaud’s, **E**sophageal dysmotility, **S**clerodactyly, **T**elangiectasias).
Esophageal perforation	- Can either be iatrogenic (i.e., from endoscopy) or Boerhaave syndrome (from increased pressure due to retching, vomiting, or straining). - USMLE likes to say water-soluble contrast swallow shows accumulation in the mediastinum. If they don’t mention this, they’ll give either recent endoscopy or straining + acutely low BP and high HR, where you have to eliminate the other answers out of reasoning. - Water-soluble contrast (Gastrografin) is used in suspected perforation because the contrast will leak into the mediastinum and barium can cause mediastinitis.

	- In contrast (no pun intended), if patient has aspiration risk, we use barium and not Gastrografin because the latter can cause pneumonitis. - "Esophageal repair" is the Tx on NBME. Pretty straightforward.
Mallory-Weiss tear	- Classically retching/vomiting in an alcoholic that causes stretching and tearing of the gastroesophageal junction leading to **small amounts** of blood in vomitus. - Can also occur in patients with eating disorders or hyperemesis gravidarum. - It's to my observation on NBME material that the highest yield point about MW tear is that it does **not** cause high-volume hematemesis. If the vignette mentions high-volume blood, that is ruptured varix instead (discussed below).
Esophageal varices	- Enlarged and tortuous superficial esophageal veins due to portal hypertension in patient with cirrhosis or splenic vein thrombosis.   - The mechanism in the setting of cirrhosis: the **left gastric vein** connects the esophageal veins to the portal vein. In the setting of ↑ portal pressure, this pressure backs up to the left gastric vein, which backs up to the esophageal veins. Many USMLE Qs want "left gastric vein" as the answer for the vessel responsible for the varices. - In splenic vein thrombosis, the ↑ splenic venous pressure causes formation of collaterals to circumvent the thrombosis. This means nearby veins will form small tributaries/branches from the splenic vein. The left gastric vein is one of them → ↑ esophageal venous pressure. - It is to my observation on NBME Qs that 4/5 varices Qs will give a patient with **high-volume hematemesis.** In contrast, MW tear is 'just a little blood." - Ruptured varices are lethal ~50% of the time. The patient will vomit high volumes of blood. - 1/5 Qs might say "just a little blood" if the varix is friable but not overtly ruptured, but the vignette will be obvious (i.e., cirrhosis) + they might just ask for "left gastric vein" as the answer. - Propranolol is prophylaxis, not for acute treatment. - Treatment is endoscopy + banding. - Octreotide can also be used for acute bleeding, but endoscopy + banding is best answer on USMLE if you are forced to choose.

	- There is a 2CK Q where patient has ↓↓ BP due to ruptured varix and the answer is "IV fluids," where endoscopy is the wrong answer. I've had a student say, "Wait but I thought you said endoscopy and banding is what we do first." And my response is, yeah, but you always have to address ABCs first on 2CK. They could by all means say patient has O2 sats of 50% and the answer would be give oxygen before fluids.
Diffuse esophageal spasm	- Idiopathic spasm of the esophagus. - All you need to know is that this that causes pain that can mimic angina pectoris, but patient will be negative for cardiac findings/disease.  - Barium shows a corkscrew appearance. - "Nutcracker esophagus" is a similar condition that is due to idiopathic hyperperistalsis and causes pain similar to DES.
Eosinophilic esophagitis	- Idiopathic allergic condition of the esophagus that presents with difficulty swallowing. - Biopsy shows eosinophils in epithelium (it's like no shit).  - Endoscopy shows "trachealization" of the esophagus with concentric rings. - Patient might have history of asthma or atopy.
Schatzki ring	- Garbage diagnosis for USMLE. Doesn't exist. Mentioning it here as a negative for USMLE giving a fuck cuz otherwise someone will be like, "Mike u didn't mention Schatzki ring." - Technically a narrowing of the distal esophagus that could be considered as a differential alongside achalasia. Once again, never seen this assessed as a correct answer on any NBME material. Maybe just once as a distractor. Garbage diagnosis.

Peptic ulcers HY causes	
- "Peptic ulcer" is an umbrella term that refers to both duodenal and gastric ulcers. - Gastric ulcers cause pain immediately with meals (due to ↑ acid secretion). Patients can sometimes lose weight due to aversion to pain from eating. - Duodenal ulcers cause pain 1-2 hours after meals (due to ↑ acid entering duodenum). With meals, the pylorus tightens, thereby relieving any residual discomfort. Patients may gain weight since eating ↓ pain.	
Helicobacter pylori	- Responsible for almost all duodenal ulcers (>95%), whereas it causes a lower % (only >60%) of gastric ulcers. This is merely because the latter are caused by many other things as well, so we simply have ↓ fraction caused by *H. pylori.* In other words, there's no special tropism of *H. pylori* toward duodenal over gastric mucosa. - Mechanism for ulcers that shows up on USMLE is: "secretes proteinaceous substrates that damage mucosal lining." This is correct over "↑ gastric acid secretion" if both are listed side-by-side, even though *H. pylori* does ↑ gastrin levels, which ↑ acid secretion. - Produces urease, which causes ↑ ammonia production around the organism, allowing it to survive in the ↓ pH of the stomach. - Antral/pyloric ulcers can lead to gastric outlet obstruction. They will mention this in a Surg Q as a "succussion splash." - ↑ risk of MALT lymphoma, a type of B-cell lymphoma. - Diagnose *H. pylori* with urease breath test or stool antigen. - Treat *H. pylori* with CAP → clarithromycin, amoxicillin, PPI (e.g., omeprazole). - USMLE really doesn't give a fuck about the treatment, but CAP is safe to know. Metronidazole, tetracycline, bismuth, and PPI tetrad is used if CAP fails (students ask about those other drugs). - Perforated duodenal ulcer will present as sudden-onset rigid abdomen (involuntary guarding). Patient will often have SIRS, with abnormal vitals due to sympathetic activation. USMLE wants "X-rays of the chest and abdomen" to look for air under the diaphragm (HY finding that indicates ruptured viscus).
Gastrinoma	- Zollinger-Ellison syndrome causes recurrent duodenal ulcers and sometimes jejunal or ileal ulcers. - Can be part of MEN1 or idiopathic. - *H. pylori* is more common than gastrinoma. As mentioned above, >95% of duodenal ulcers are due to *H. pylori*. There is an NBME Q where they give older male with a duodenal ulcer + no other information + they ask for most likely cause → answer = testing for *H. pylori*; gastrinoma is wrong. - Vignettes can be tricky with gastrinoma and tell you the patient has 8-10 watery stools daily, where you say, "That sounds like VIPoma." But they'll also tell you the patient has history of abdo pain after meals. Q on IM form 8 does this as example.
NSAIDs	- Cause gastric ulcers. I haven't seen these cause duodenal ulcers on USMLE. - Prostaglandins are necessary for stimulation of gastric alkaline mucous production and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both ulcers and irritation (gastritis). - USMLE wants you to know that PPIs are first-line for ulcer treatment in general, but that **misoprostol** is a PGE1 analogue that is used in NSAID-induced ulcers (i.e., we're replenishing the ↓ prostaglandin from NSAIDs).
Burns	- Rare, but just know they exist. - Called Curling ulcers (think: curling irons are hot). - Loss of fluid post-burns from evaporation → ↓ blood flow to stomach → ischemic ulcers.
Head trauma	- Rare, but just know they exist. - Head trauma or brain tumor → ↑ intracranial pressure → ↑ parasympathetic outflow to stomach → ↑ ACh binding to muscarinic receptors at parietal cells → ↑ acid production. - Don't forget that the 3 synergistic mechanism for acid production are 1) gastrin binding directly to gastrin receptors on parietal cells; 2) gastrin stimulating enterochromaffin-like cells to secrete histamine, which then binds to H2 receptors on

	parietal cells; and 3) direct parasympathetic activity, where ACh binds to muscarinic receptors on parietal cells.
Smoking/Alcohol	- Not tested as overt causes of ulcers on USMLE. But you should know that smoking and alcohol are believed to *decrease healing* of pre-existing ulcers.

Gastritis causes + HY points	
- Acute gastritis will present as irritation leading to bleeding of gastric mucosa (e.g., from NSAIDs). - Chronic gastritis will be an atrophy or autoimmune destruction of the mucosa associated with ↓ mucosal thickness, ↓ HCl production (achlorhydria), ↑ gastrin production, and enterchromaffin-like cell hyperplasia (↑ histamine production to compensate for ↓ acid). NBME will give you this constellation of findings and then just simply have "chronic gastritis" as the answer, with acute gastritis not even listed. It's not hard, but I've seen students miss this a lot.	
H. pylori	- Causes what is referred to as "Type B gastritis," which is inflammation tending to affect the antrum of the stomach. - Can lead to pyloric channel ulcers + gastric outlet obstruction. This will present on 2CK Surg forms as a "succussion splash."
Pernicious anemia	- Causes what's referred to as type A gastritis," which affects mostly the fundus/body of the stomach. - Autoimmune antibody-mediated destruction of parietal cells. - Sometimes antibodies can be against intrinsic factor. - Ultra-HY cause of B12 deficiency (↑ MCV + hypersegmented neutrophils +/- neuropathy) on USMLE. - Associated with other autoimmune diseases, e.g., vitiligo. So if they give you ↑ MCV in patient with, e.g., type I diabetes, you should think, "Autoimmune diseases go together, so if the patient has one AA disease, he/she has ↑ propensity for others." Don't worry about strict HLA associations here.
NSAIDs	- Answer on USMLE for GI bleeding in someone taking, e.g., indomethacin or naproxen. - Causes type A gastritis, affecting the fundus/body. USMLE doesn't specifically give a fuck, but you should basically be like, "*H. pylori* causes antral gastritis, whereas other causes like NSAIDs and pernicious anemia are fundus/body of stomach." - Prostaglandins are necessary for stimulation of gastric alkaline mucous production and maintenance of the gastric lining. NSAIDs → ↓ prostaglandin production → disruption of gastric lining. This can lead to both gastritis (inflammation) and ulcers. - As mentioned above for ulcers, **misoprostol** can be used in these patients following PPI.

Gastric cancer points
- Perpetuated over the years as "intestinal vs diffuse" types. USMLE doesn't care. What you need to know is that histo will show **signet-ring cells that contain mucin.** And you don't need to know the histo for this either. - Grossly, can cause a linitis plastica, which is a leather bottle appearance to the stomach. - Gastric metastases can spread hematogenously to the ovaries. These will show up as bilateral ovarian lesions that show, you guessed it, **signet-ring cells that contain mucin.** - ↑ incidence of gastric cancer in Japan due to ↑ nitrosamines/smoked foods. - MALT lymphoma is a B cell lymphoma that can be caused by *H. pylori*. This does not have signet ring cells. - Acanthosis nigricans can be associated with gastric cancer (even though most of the time, it just means insulin resistance). - Virchow node (Troisier sign) → palpable supraclavicular lymph node that can reflect visceral malignancy, especially gastric cancer.

Hyperbilirubinemia	
Unconjugated (indirect)	- Bilirubin is produced from breakdown of heme from RBCs at the spleen. It will initially be "unconjugated" in this form, where it merely leaves the spleen non-covalently bound to albumin and is not water-soluble – i.e., it won't show up in the urine. Unconjugated bilirubin is aka indirect bilirubin. - In the setting of hemolysis or ↑ RBC turnover (i.e., sickle cell, **hereditary spherocytosis, blood given during surgery**), where we have ↑ RBC breakdown, we get ↑ indirect bilirubin. - Once it arrives at the liver, it is taken up by the liver and conjugated to glucuronide, making it water-soluble. Conjugated bilirubin is aka direct bilirubin. - If the there's a problem with uptake at the liver (i.e., acute hepatitis), or there is deficient conjugation enzyme (Gilbert syndrome, Crigler-Najjar), indirect bilirubin also goes ↑. - Hemolysis or ↑ RBC turnover can cause direct bilirubin to go ↑ sometimes as well, but the shift will be more toward indirect being ↑. I point this out because students often get confused by this, but if we have ↑ indirect bilirubin going into the liver, then this means ↑ direct bilirubin going out. A 2CK NBME Q gives 9 packs of RBCs given during surgery → days later, total bilirubin is 5.0 and direct bilirubin 2.3 → answer = "overproduction of bilirubin." - In regard to neonatal labs (i.e., first month of life), I should note that hematocrit % can be in the 50s (NR 45-61; in adults, NR is 40-50). 2CK, for instance, will often show Hct as 56% in a newborn and the student is like "Wow that's high!" where they think there's neonatal polycythemia or some other pathology, but it's actually normal. Unconjugated jaundice can occur in neonates because of ↑ RBC turnover where HbF is replaced with HbA.
Conjugated (direct)	- Direct bilirubin, since it is water-soluble, shows up in the urine. It will also be secreted into bile. Therefore, if we have a bile duct obstruction, we get ↑ direct bilirubin. ALP also goes ↑ in bile duct obstruction because it is secreted by bile duct epithelium. This means "↑ ALP + ↑ direct bilirubin" is very buzzy for bile duct obstruction. If ALP is high but direct bilirubin not ↑, this can be due to things like bone fractures or Paget disease. - GGT will also go up with bile duct obstruction, since it is also secreted by bile duct epithelium, but USMLE actually rarely mentions this one. What they want you to know is GGT spikes with acute alcohol consumption / binge drinking. - In the event of bile duct obstruction, not only will ALP and direct bilirubin go ↑, but the urine becomes darker from ↑ direct bilirubin in it (and urobilin, which comes from direct bilirubin, but USMLE doesn't assess this). In addition, stools become lighter/pale (aka "acholic stools"), since there is ↓ direct bilirubin making it to the intestines, which means ↓ stercobilin production (pigmentation in stool). In other words, "dark urine + pale stools" is buzzy for bile duct obstruction the same way "↑ ALP and ↑ direct bilirubin" is. - Acute hepatitis can also cause ↑ direct bilirubin due to ↓ secretion into bile, in addition to ↑ indirect. But this makes sense, since it is literally an intra-hepatic pathology. - Highest yield cause of ↑↑ direct bilirubin on USMLE is biliary atresia in neonates (discussed later). This is all over the place, whereas Crigler-Najjar (↑↑ indirect bilirubin in neonates) is nonexistent. - Dubin-Johnson and Rotor syndromes are virtually nonexistent on USMLE but cause ↑ direct bilirubin in adults due to ↓ bile excretory pumps at the liver. Students get hysterical about these because they sound weird, but they're LY. Dubin-Johnson is asked once on a new 2CK NBME form. But apart from that, LY. - Cholangitis (inflammation of bile ducts), choledocholithiasis (stone in biliary tree), choledochal cyst, head of pancreas cancer (impingement on common bile duct), and cholangiocarcinoma (bile duct cancer) are all HY causes of bile duct obstruction. I discuss all of these conditions in detail below.

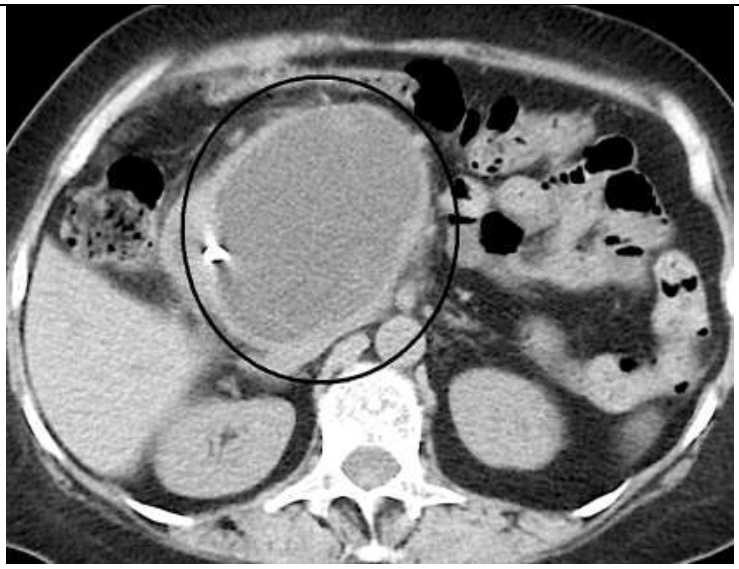
Bilirubin uptake/secretion diseases	
Gilbert	- Pronounced "Jeel-BEAR, not "GILL-burt." Gastroenterologist I met once tripped out over med students pronouncing this wrong. - Partial deficiency of bilirubin uptake enzyme at the liver (UDP glucuronosyltransferase). - Presents as isolated ↑ indirect bilirubin with yellow eyes in young adult with stress factor, such as studying for exams, or recent surgery/trauma. The patient will otherwise be completely healthy. - No treatment necessary.
Crigler-Najjar	- Near-absence of UDP-glucuronosyltransferase causing ↑↑ indirect bilirubin in neonate. - Nonexistent yieldness but students ask about this.
Dubin-Johnson	- ↓ excretion of bilirubin into bile at the liver due to ↓ secretory pumps. - Causes ↑ direct bilirubin in otherwise healthy adult. - Can cause black liver in theory. - Asked once on new 2CK NBME. Apart from that, nonexistent on USMLE material.
Rotor	- Same as Dubin-Johnson but no black liver. - Nonexistent diagnosis on USMLE. Students get maniacal over this supposedly not having a black liver, whereas Dubin-Johnson has a black liver – "Rotor! That's the one that has no black liver but Dubin-Johnson does!" Relax. Take two steps back, chill the fuck out for two seconds. USMLE doesn't give a fuck.
Biliary atresia	- One of the highest yield Peds diagnoses. Answer on USMLE for ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - Ultrasound will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis. - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

Autoimmune liver conditions	
Primary biliary cirrhosis	- Inflammation of bile ducts within the liver, leading to their destruction. - Answer in a woman 20s-50s who has generalized pruritis, ↑ serum cholesterol, ↑ ALP, ↑ direct bilirubin. - USMLE loves to mention **Hx of autoimmune disease in the patient or family member** (because autoimmune diseases go together). So they'll say she has type I diabetes mellitus, SLE, or vitiligo; or her brother has RA, etc. - They can mention a stone is present in the gallbladder on ultrasound, which gets some students real emotional / confused, but it makes sense since patient has ↑ cholesterol. It's just a distractor point and doesn't relate to the actual diagnosis at hand. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol). - New NBME asks a couple Qs on fat-soluble vitamin malabsorption for PBC (deficiency of A and D on NBME for each Q, respectively, with no info supporting those presentations; they just ask for hypothetical vitamin deficiency). In theory, would be due to biliary obstruction, where ↓ bile entering small bowel merely means ↓ fat absorption. This is nothing special to PBC, but I mention it because it's asked twice.
Autoimmune hepatitis	- Young adult with ↑ LFTs who has (+) anti-smooth muscle antibodies.

HY Hepatobiliary conditions	
Pancreatic cancer	- Head of pancreas cancer impinges on common bile duct, resulting in obstructive jaundice (↑ ALP + ↑ direct bilirubin) in smoker with weight loss, or in patient who had gallbladder taken out years ago (so it clearly can't be due to a stone in common bile duct). - Patient will not be febrile; can have dull abdominal pain. - Courvoisier sign is a painless, palpable gallbladder in an afebrile patient who's jaundiced. This is pancreatic cancer until proven otherwise and is pass-level. - **Pancreatic enzymes are normal in pancreatic cancer.** - USMLE wants **CT of the abdomen** to diagnose. - Whipple procedure is done to remove head of pancreas. If the cancer is isolated to the tail, distal pancreatectomy is the answer. - Patients with pancreatectomy need pancreatic enzyme supplementation. The exam can write this as "pancrelipase."
Cholangiocarcinoma	- Bile duct cancer. Answer on USMLE if the vignette sounds like pancreatic cancer but they tell you in the last line CT is negative. - Answer is ERCP as next best step. - Smoking is risk factor, same as pancreatic cancer. - Can be caused by *Clonorchis sinensis* (trematode).
Cholelithiasis	- Stones in the gallbladder. - Presents with biliary colic, which is acute-onset waxing/waning spasm-like pain in the epigastrium or RUQ. - Pain is due to **cholecystokinin** causing gall bladder contractions, where a stone within the gallbladder transiently obstructs flow of bile into cystic duct. They ask this on 2CK as well, where patient will biliary colic and answer is "obstruction of cystic duct opening by stone." Not dramatic. - Classic demographic is 4Fs = Fat, Forties, Female, Fertile for cholesterol stones. This is because estrogen upregulates HMG-CoA reductase, causing ↑ cholesterol synthesis and secretion into bile. NBME will give you standard vignette of 4Fs, and then the answer will just be "increased secretion of cholesterol into bile." - Another NBME Q gives vignette of cholelithiasis, and then rather than asking for the diagnosis, they ask what the patient most likely has → answer = "lithogenic bile." Slightly awkward, but means promoting the formation of stones. - ↑ cholesterol stones in pregnancy not just due to estrogen effect but also because progesterone slows biliary peristalsis (biliary sludging). Tangentially, progesterone also slows ureteral peristalsis, which is why there's ↑ risk of pyelonephritis (as discussed in the HY Renal PDF). - Cholesterol stones most common, but pigment (calcium bilirubinate) stones are exceedingly HY for hereditary spherocytosis and sickle cell, due to ↑ RBC turnover. - USMLE wants you to know that splenectomy is Tx for hereditary spherocytosis to ↓ incidence of cholelithiasis. Sometimes the vignette wants "cholecystectomy + splenectomy" as combined Tx. Or if they say this was in a parent's Hx in patient with low Hb, you know the Dx is hereditary spherocytosis (autosomal dominant). - Infections can also sometimes cause pigment stones, but LY. - Diagnose with abdominal ultrasound. - Ursodeoxycholic acid (ursodiol) ↓ **secretion of cholesterol into bile.** USMLE just wants you to know this MOA + that it can be used in patients with asymptomatic cholelithiasis, those declining cholecystectomy, and in pregnancy.
Cholecystitis	- Cholelithiasis + fever. - Vignette will sound exactly like cholelithiasis, but if we add a fever on top of it, we now just call it cholecystitis. - Virtually always due to obstruction by a stone + infection as a result. - Diagnose with abdominal ultrasound, showing the stones. - If ultrasound is negative, do a HIDA scan. - Tx is cholecystectomy.

	- Emphysematous cholecystitis is air visualized within the biliary wall. USMLE wants *Clostridium perfringens* as the organism (can cause gas gangrene anywhere) in this case. - Chronic cholecystitis is calcification of the gallbladder (aka chronic calculous cholecystitis, or "porcelain gallbladder"). It is due to repeated bouts of acute cholecystitis. - ALP and bilirubin will not be increased 19/20 questions. This is because inflammation of the gallbladder doesn't mean we have any form of common bile duct obstruction. However there is one nonsense Q on a 2CK NBME where 4 of the answers are wildly wrong, with correct answer being cholecystitis in setting of high ALP + direct bilirubin. Since cholecystitis is caused by stones virtually always, the implication is patients can occasionally have concurrent choledocholithiasis.
Choledocholithiasis	- Stone anywhere within the biliary tree. Don't confuse with cholelithiasis. - Obstructive jaundice (↑ ALP + ↑ direct bilirubin) in patient who has Hx of cholelithiasis or Hx of cholecystectomy performed within the past week. - Regarding the latter, the Q can say patient had cholecystectomy performed a week ago + "intra-operative cholangiography was not performed." The implication is bile duct patency is supposed to be visualized during cholecystectomy to ensure there isn't a stone retained within the biliary tree. However, the Q need not say "intra-operative cholangiography was not performed." This makes the Q pass-level. - **USMLE wants abdominal ultrasound followed by ERCP** as the answer. ERCP tends to show up as what they want, but there is a Q floating around somewhere that asks for ultrasound first, so know that sequence. - As I talked about earlier, I've never seen MRCP as a correct answer. - Being able to discern choledocholithiasis from pancreatic cancer is vital for USMLE. If they give you obstructive jaundice but say the patient is a heavy smoker with weight loss, or hasn't had a gallbladder for 25 years, you know it's head of pancreas cancer, not choledocholithiasis. And then, once again, if it sounds like pancreatic cancer but the CT is negative, that's cholangiocarcinoma and we do ERCP.
Gallstone pancreatitis	- **Gallstone pancreatitis** is a type of choledocholithiasis in which the stone has descended distally enough in the common bile duct that it now obstructs the **hepatopancreatic ampulla.** This results in a backflow of pancreatic enzymes to the pancreas causing damage → acute pancreatitis. - USMLE wants "hepatopancreatic ampulla" as correct over "common bile duct" if they ask you to choose location of the obstruction. Slightly odd, since the stone is still in the common bile duct, but hepatopancreatic ampulla blockage is *why* we have the pancreatitis. - Vignette will give obstructive jaundice *but also high pancreatic enzymes*. - In other words, the Q will give ↑ ALP, ↑ direct bilirubin, and ↑ amylase / lipase. - I repeat that pancreatic cancer will never have elevated enzymes on USMLE, so this is a pass-level means to distinguish. - New 2CK NBME has **CT of the abdomen** as done first, prior to ERCP. This is because even though we have a stone requiring removal from the biliary tree, CT of the abdomen first looks for pancreatic fluid collection.
Cholangitis	- Inflammation of the bile ducts. You must know **Charcot triad for cholangitis: 1) fever, 2) jaundice, 3) abdominal pain.** We classically learn this as "RUQ/epigastric pain," but I can tell you USMLE doesn't give a fuck and will just say "abdominal pain." I've had students get cholangitis Qs wrong because they're like, "I thought it had to be RUQ pain though." And I'm like, yeah, I agree, but USMLE doesn't give a fuck. - Will present 3 ways on USMLE: - 2/5 Qs will be ascending cholangitis, which is infection from ascension of bowel flora (*E. coli*, *Bacteroides*) up the common bile duct.

	- 2/5 Qs will be primary sclerosing cholangitis, which is inflammation of the bile ducts in the setting of IBD (always due to ulcerative colitis on USMLE, but can rarely be due to Crohn in real life). p-ANCA can be positive, and there can be a beaded appearance of the common bile duct on cholangiogram. On 2CK NBME 10, there is a Q on primary sclerosing cholangitis where the patient is afebrile (rare; i.e., we don't have full Charcot triad), but they say patient has UC + obstructive jaundice + ERCP shows "narrowing of the bile ducts," so the answer is evidently PSC. - 1/5 Qs will just give you a patient with history of cholelithiasis, where the implication is a stone has descended causing choledocholithiasis, which in turn led to obstruction + inflammation of the common bile duct. - USMLE wants antibiotics + **ERCP to diagnose and treat.** Biliary drainage by ERCP ↓ morbidity and mortality.
Acute pancreatitis	- ↑ amylase and/or lipase in patient who has abdominal pain. - Pain need not radiate to the back. - Alcoholism and stones are two biggest risk factors. - Hyperlipidemia and hypercalcemia can also occasionally be causes. - Can be caused by drugs (didanosine is HIV NRTI; exenatide is a GLP-1 analogue for diabetes). - Scorpion sting and mumps as causes are nonexistent on USMLE. - Can be caused by choledocholithiasis (gallstone pancreatitis; ↑ pancreatic enzymes in setting of ↑ ALP and ↑ direct bilirubin). - Results in enzymatic fat necrosis. - You don't need to memorize Ranson criteria (elaborate criteria for Surg to determine prognosis in pancreatitis). What you need to know are two main points: 1) ↓ Serum calcium and ↑ glucose are bad prognostic indicators. In fact, I'd say low calcium is quite possibly the highest yield variable on USMLE for pancreatitis. There are many long/vague pancreatitis Qs out there, where they'll mention ↓ calcium in the stem, and you're like "Boom pancreatitis." 2) The degree of lipase and/or amylase elevation doesn't correlate with prognosis. In other words, if amylase is 1200 versus 10,000, that's not important for patient outcome. 3) The rest of the Ranson criteria is more Surg rotation masturbation info and otherwise nonsense for USMLE. - It's to my observation NBME Qs will sometimes list high amylase without mentioning lipase for whatever reason. Lipase is more specific for pancreatitis, but it's what NBME sometimes does, so don't be confused. - HY initial management for 2CK is a triad: 1) NPO (nil per os; or nothing by mouth; this can be written as simply **bowel rest** as NBME answer); 2) NG tube decompression; 3) IV fluids with normal saline. This triad is not specific for pancreatitis, and can be done for a number of GI-related conditions, including cholangitis and stones. But it just tends to get tested a lot for pancreatitis. - After this initial triad, "CT scan of the abdomen" is answer on NBME as next best step to look for any degree of fluid collection. If a pancreatic pseudocyst is present, NBME answer is to drain this by ERCP. - It's to my observation USMLE doesn't assess antibiotics as part of pancreatitis Tx, but apparently the carbapenems (like imipenem) have fantastic penetration of pancreatic tissue.

	 - A 2CK NBME Q gives pancreatitis + similar image as above, and the answer is just ERCP (for drainage of the pseudocyst within black circle). - A necrosectomy is removal of pancreatic tissue in the setting of high fraction of necrosis. Patients with pancreatectomy require enzyme replacement (pancrelipase). - Acute pancreatitis is HY cause of ARDS.
Chronic pancreatitis	- **Steatorrhea in alcoholic** or patient with recurrent acute pancreatitis episodes → results in pancreatic burnout with ↓ production of pancreatic enzymes → ↓ proteases/lipases → malabsorption. - In other words, we have **normal** pancreatic enzymes – i.e., they are **not** elevated. - Question can say alcoholic has ↑ fat and protein seen in stool. - D-xylose test is normal. This is a monosaccharide that is readily absorbed by the small bowel insofar as the intestinal lining is intact. So if we get steatorrhea + normal D-xylose, this can be due to pancreatic insufficiency or lactose intolerance. If D-xylose test is abnormal, we know the malabsorption is due to an intestinal lining issue, such as Celiac or Crohn. - If the USMLE gives you a CT for chronic pancreatitis, they will show you calcification within the pancreas.  - Tx = pancreatic enzyme supplementation (pancrelipase).

Bile duct leak	- The answer on 2CK Surg if they give you patient with cholecystectomy within the past week who now has fever, abdominal pain, and ↑ direct bilirubin. - During the surgical anastomoses created during cholecystectomy, sometimes there can be a post-op bile leak, literally. Weird diagnosis but it shows up on occasion.
Sphincter of Oddi dysfunction	- The answer on USMLE if the Q sounds like gallstone pancreatitis (i.e., ↑ ALP, ↑ direct bilirubin, ↑ pancreatic enzymes) but the patient hasn't had a gallbladder for many years (so it can't be a stone causing the obstruction).
Gallstone ileus	- Gallstone enters small bowel and causes an intestinal obstruction. - The answer on USMLE if they say "air in the biliary tree" (pneumobilia). - Can be caused by cholecystoduodenal fistula, which is when a patient with cholelithiasis can develop a conduit between the gallbladder and small bowel. - What the USMLE will do is give you a long, rambling paragraph where they say "air in the biliary tree" in the last line. The answer will then just be "gallstone ileus" straight up, or it will be "cholecystoduodenal fistula."
Choledochal cyst	- Asked on 2CK Surg. What they're going to do is give you a long, nonsense paragraph with ↑ ALP and ↑ direct bilirubin + they say "CT shows a cystic structure within the biliary tree." Answer = "simple excision of the cyst." Students are huh, what's going on here? → choledochal cyst. You have to just excise it. Bullshit/dumb diagnosis. Take it up with NBME not me.
Primary biliary cirrhosis	- As mentioned earlier, this will be a woman 20s-50s who has generalized pruritis, ↑ cholesterol, ↑ ALP, ↑ direct bilirubin, and Hx of autoimmune disease in her or a relative. - Diagnose with anti-mitochondrial antibodies as first step, followed by liver biopsy to confirm. - Initial Tx = ursodeoxycholic acid (ursodiol).
Spontaneous bacterial peritonitis (SBP)	- One of the highest yield diagnoses on 2CK. - Bacterial infection of peritoneal fluid by mixed enteric flora. - The answer on USMLE when they give you diffuse abdominal pain and fever in one of the three following scenarios: 1) cirrhosis; 2) recent peritoneal dialysis; 3) nephrotic syndrome. - Shifting dullness" or a "fluid wave" are buzzy for ascites, but questions will often omit these descriptors from stems. What they care about is you identifying when a patient either has a major risk factor for ascites, or are aware of a peritoneal intervention as the cause for the presentation. - In other words, they can tell you a patient has cirrhosis + abdo pain + fever, where they don't have to mention a fluid wave, but you just have to infer, "Well he/she clearly has major risk factor for ascites, so this sounds like SBP." - There is a 2CK Peds Q where they say kid with minimal change disease (nephrotic syndrome) has abdo pain + fever → answer = spontaneous bacterial peritonitis. - Next best step in diagnosis is abdominal **paracentesis**. This refers to aspiration of fluid from the peritoneal cavity. Do not confuse this with pericardiocentesis. "Paracentesis" as an answer shows up everywhere, and I've seen students avoid it because they're like, "What? I thought that meant pericardiocentesis." - After the paracentesis is done, USMLE wants a very specific order for what to do next. **Choose "white cell count and differential" first if it's listed, followed by "gram stain and culture of the fluid."** - This order is important because "gram stain and culture of the fluid" is a correct answer in one Q but wrong answer in another Q, where "white cell count and differential" is correct to do first. - The reason white cell count and differential is done first is because SBP is diagnosed when paracentesis shows >250 WBCs/μL. - There is a new 2CK Q where they give ascites in patient with cirrhosis, but do not mention fever or abdominal pain. However, they say paracentesis shows 900

	WBC//μL → answer = "antibiotic therapy." It should be made clear though that ~6/7 SBP Qs will give abdo pain and fever. - Treatment is ceftriaxone.
Focal nodular hyperplasia	- Obscure diagnosis where they tell you adult has an abdominal CT for unrelated reason and has a 1-2-cm hepatic lesion **with a central scar.** There will be zero mention of trauma or infection. Every student says WTF. - This is a lesion of hepatocellular hyperplasia **that does not require treatment.** This is the answer on NBME, where they tell you CT shows hepatic lesion with central scar. Answer = "no further diagnostic studies indicated."

HY cirrhosis points
- Cirrhosis is a small, shrunken, burnt out liver due to chronic disease. - HY causes are alcoholism, HepB/C, Wilson disease, hemochromatosis, NASH, α1-antitrypsin deficiency etc. - "Burned out" means LFTs are normal or low – i.e., there is not transaminitis as with acute hepatitis. - **USMLE likes ↑ PT** and ↓ clotting factor synthesis in cirrhotic patients. - Hyperammonemia occurs due to ↓ urea cycle activity (normally occurs in liver). This can cause hepatic encephalopathy (confusion) and asterixis ("hepatic flap" of the hands). - USMLE likes acute exacerbations of hyperammonemia caused by **GI bleeds** → ↑ ammonia absorption. - Spontaneous bacterial peritonitis (SBP) is ↑↑ HY on 2CK, as discussed above. - Esophageal varices from ↑ portal pressure that backs up to **left gastric vein** (discussed earlier). - Caput medusae are visible periumbilical veins (superior epigastric veins).

Systemic inflammatory response syndrome (SIRS)	
- Knowing this short table is vital for understanding many Surg Qs on 2CK. - In the setting of stress (i.e., due to trauma, surgery, autoimmune flare, infection), catecholamines and ↑ sympathetic activity might shift the patient's vitals out of the normal range. - The reason knowing SIRS is important is because the patient can have abnormal vitals **without** having an infection.	
SIRS	- 2 or more of the following: - Temperature <36C or >38 (<96. 8F or >100.4). - HR >90. - RR >20. - WBCs <4,000 or >12,000.
Sepsis	- SIRS + source of infection.
Septic shock	- Sepsis + low BP.
- Sometimes you will see a patient's vitals slightly out of the normal range in the setting of trauma, surgery, or autoimmune flares, and you have to be able to say, "There's no infection. That's just SIRS from sympathetic activation." - Knowing if a patient is septic is important for management of patients on 2CK, where sometimes antibiotic regimens are stepped up. For example, when treating PID, if the patient is septic, intravenous ceftriaxone and azithromycin is correct on one of the 2CK NBMEs; IM ceftriaxone and oral azithromycin is wrong. This is because the latter is for most patients who have PID but aren't septic. - Ceftriaxone is frequently an answer on 2CK for in-hospital patients who are septic from a variety of community-acquired conditions, e.g., pneumonia, pyelonephritis, prostatitis. For instance, community-acquired pneumonia is empirically treated with azithromycin (on 2CK NBME 8), but if patient is septic, we can go straight to ceftriaxone (have seen this more than once on 2CK NBMEs). - For hospital-acquired infections in which patients are septic, NBME goes hard-hitting with vancomycin PLUS ceftazidime or cefepime. This regimen covers MRSA and *Pseudomonas*.	

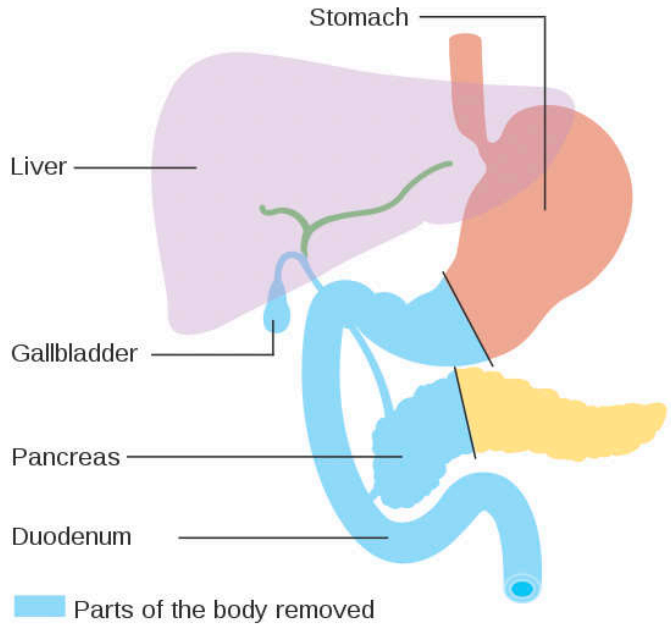
Miscellaneous Surg intestinal diagnoses

Adhesions	- One of the highest yield causes of small bowel obstruction (SBO) on USMLE. - Occur as a result of major abdominal surgery – i.e., hysterectomy, laparotomy, appendectomy. - Prior surgery results in inflammation that yields fibrin deposition within the peritoneal cavity. This causes "fibrous adhesions" over time that the small bowel can get caught in. - Vignette will say, e.g., 50-year-old who had laparotomy 10 years ago for gunshot wound now has obstipation (failure to pass flatus in addition to stool) for 2 days + SIRS. This is pass-level and answer is just "adhesions." - New 2CK NBME has "mechanical obstruction of small intestine" as the answer in patient with Hx of appendectomy. Pass-level, but that's another way they can write the answer. - As mentioned earlier, for many abdo conditions – i.e., SBO, pancreatitis, cholangitis, etc., a general "triad" of NPO, IV fluids, and NG tube is done first. In other words, you might get a simple adhesions Q, and then the answer is just "0.9% saline" as the first step, where bowel rest and NG tube aren't listed. - If the patient is stable, do abdominal X-ray if SBO is suspected to look for gas / obstruction. If the patient is unstable (i.e., low BP), go straight to laparotomy. Never do a CT scan in an unstable patient on USMLE (this includes patients with normal BP *after* IV fluids given to restore low BP – i.e., the fluids are just acting as a temporizing measure and you need to find source of problem asap). - Crohn disease can cause *intraluminal fibrotic strictures*, **not** adhesions. There is a 2CK Q that lists both as answer choices. Strictures due to Crohn are internal / within the lumen; adhesions are external fibrotic bands.
Post-op ileus	- Surgery is a stressor / form of trauma. Especially post-abdominal surgery in which the bowel has been touched/manipulated, there can be acute diminution of peristalsis resulting in pseudo-obstruction (i.e., the patient is obstipated but there's no physical obstruction). - Opioids can ↓ peristalsis / cause constipation. These should be considered as a contributing factor to ileus. An important rule for USMLE is that we **always treat pain fully,** even if patient has Hx of drug abuse or ileus. There is a 2CK Q where a patient has ileus post-major surgery; answer is "maintain dose of opioids + add stool softeners." - If the Q gives you Bristol 7 stool (watery diarrhea; no solid pieces) with laxatives but Bristol 1 (severe constipation; pellet-like stools) when on nothing, answer on NBME is "add fiber," with the aim being Bristol 3-4.
Ogilvie syndrome	- Large bowel pseudo-obstruction. - **Classically follows hip surgery** for whatever reason. - Think of this as ileus but of the large, not small, bowel. "Ileus" sounds like ileum, which is part of small bowel. So ileus isn't used to refer to large bowel pseudo-obstruction. - Abdominal x-ray is what we do first in stable patients when we are looking for suspected obstruction. If patient is unstable, always go straight to laparotomy.
Duodenal hematoma	- Shows up on 2CK NBME. You just need to know this is the answer for someone who has ↓ hemoglobin + bilious vomiting. - The ↓ Hb is because the patient is bleeding internally. - Bilious vomiting implies obstruction (almost always duodenal).
Duodenal perforation	- Ultra-HY on 2CK; caused by Hx of duodenal ulcers. - Highest-yield point is that this is diagnosed with **chest and abdominal x-rays** showing **air under the diaphragm.** Presents two ways: 1) Q will give patient who has Hx of abdo pain after meals (implying Hx of duodenal ulcer) + now has SIRS + acutely worse abdo pain + rigid abdomen → answer = "x-rays of the chest and abdomen." 2) Q will say acute-onset epigastric pain as though the patient was kicked in abdomen + board-like rigidity + x-rays show air under the diaphragm → answer = "immediate surgical exploration of upper abdomen."

	- Not a hard presentation or management, but it shows up all over the place + students get this wrong all the time. Easy points.
Subphrenic abscess	- Complication of splenectomy. Shows up on 2CK Surg. - Patient will have fever + abdo pain + leukocytosis post-splenectomy + they'll ask for diagnosis → answer = subphrenic abscess. Tx = drain.
Dumping syndrome	- Answer on 2CK in patient who has diarrhea + hypoglycemia post-major surgery of the stomach/small bowel. - Post-gastrectomy, for instance, "rapid transit of hyperosmolar chyme" is an answer on NBME for the mechanism. This triggers a ↑↑ spike in insulin due to glucose-dependent insulinotropic peptide.
Blind loop syndrome	- Answer on 2CK in a patient who has diarrhea post-major surgery of the stomach/small bowel **without** hypoglycemia. The vignettes can otherwise sound pretty similar. - This is when a part of the proximal small bowel forms a crevice or nook in which peristalsis bypasses this "blind" segment, thereby creating stasis within it. This can lead to small intestinal bacterial overgrowth (SIBO) and diarrhea.

HY Referred pain	
Spleen	- Splenic laceration → ULQ pain +/- can refer to left shoulder (Kehr sign).
Diaphragm	- Diaphragmatic irritation can cause pain going to left shoulder (asked on NBME); spleen is wrong answer. The key here is they ask "irritation."
Gallbladder	- RUQ or epigastric pain +/- can refer to right shoulder.
Appendix	- Epigastric pain initially (visceral peritoneal inflammation) that migrates to RLQ (parietal peritoneal inflammation).

GI Surgeries	
- There are countless fancy GI surgeries that could be discussed just for the sake of it, but it's a vehement waste of time. I'm going to comment on just stuff I think is HY for USMLE, without the nonsense.	
Gastrectomy	- The Billroth II, or gastrojejunostomy, is a partial gastrectomy that shows up sometimes on Surg forms as causing dumping or blind loop syndromes. - In other words, they'll say a patient has diarrhea following a Billroth II gastrojejunostomy and then ask for diagnosis → answer = blind loop syndrome. - Or they'll give diarrhea + hypoglycemia (dumping syndrome), and the answer is just "rapid transit of hyperosmolar chyme." - Can cause B12 deficiency due to removal of parietal cells. Can also cause B1 deficiency (asked twice on new 2CK NBMEs) → confusion/nystagmus (i.e., Wernicke findings; B12 deficiency doesn't present with these); can have (+) Romberg sign in both B1 and B12 deficiency Qs (for B1, this is part of dry beriberi; for B12, this is part of subacute combined degeneration). - Done for gastric cancer and perforations.
Gastric bypass	- In contrast to gastrectomy, no part of the stomach is removed here. A pouch is made where part of the stomach is merely walled off from food passage. - Answer on USMLE as the most effective way to manage weight in someone who is morbidly obese, albeit last resort. - Sequelae same as for gastrectomy. B12 deficiency can occur due to disruption of parietal cells, even though stomach is not removed. - Another name for gastric bypass is roux-en-Y, which will show up tons on 2CK Surg forms as a wrong/distractor answer. I've only seen "gastric bypass surgery" as correct, never "roux-en-Y" on Surg forms.
Whipple procedure	- Aka pancreaticoduodenectomy. - Used for **head of pancreas** cancer.

	- Because of interconnected blood supply, removal of head of pancreas requires removal of the gall bladder, duodenum, and proximal jejunum.  - As I said, just know that this is done for head of pancreas cancer.
Distal pancreatectomy	- Answer on Surg form for tail of pancreas cancer. Not complicated, but it's asked.
Colectomy	- Even though segmental colectomies (many variants exist) can be used to treat localized cancers of the colon, it's to my observation that basically all colon cancer Qs on NBME have total colectomy as the answer if they force you to choose. - A 2CK NBME Q gives an 18-year-old with recently diagnosed FAP, and the answer is "total proctocolectomy." Serial colonoscopy is wrong answer, since chance of cancer is 100%. USMLE wants total removal once patient is 18. - A difficult newer 2CK NBME Q gives patient with uncontrolled colonic bleeding and severely low Hb despite transfusions + they say a radiouptake scan shows localization to one part of the colon → answer = total colectomy; hemicolectomy is wrong answer.
Laparoscopy	- Keyhole surgery used for appendectomy, cholecystectomy. - Highest yield indication on USMLE is diagnosis/treatment of endometriosis.
Laparotomy	- Used for unstable (low BP) patients after abdominal trauma (e.g., ruptured spleen from MVA); also for ruptured ectopic pregnancies where patient is unstable. - If patient is unstable, **never pick CT** on USMLE. Go straight to laparotomy. - If patient has abdominal trauma + low BP + was given fluids and now has normal BP, do not do CT. Go straight to laparotomy. The fluids are just a temporizing measure (buying time), but the underlying problem still needs to be fixed asap. - Penetrating trauma to the abdomen (usually gunshot wounds) requires immediate laparotomy, even if the patient is stable. Below the level of the nipples is considered "abdomen" in this scenario. - USMLE loves this as major cause of post-surgical adhesions (SBO months to years later). - Celiotomy is another name for laparotomy that shows up on some of the 2CK CMS Surg forms. It's a correct answer on one of the forms, where every student says wtf. Old-school surgeons might remark on differences between laparotomy and celiotomy, but USMLE uses the terms interchangeably.
Splenectomy	- Done in setting of trauma (splenic laceration), hereditary spherocytosis, or ITP. - "Autosplenectomy" refers to loss of spleen due to repeated microinfarcts in sickle cell.

	- "Asplenia" can refer to anyone who's had splenectomy, autosplenectomy, or who didn't develop a spleen from birth (Ivemark syndrome on 2CK NBME). - The white pulp of the spleen ordinarily contains 50% of the immune system's reservoir of macrophages. - Patients are susceptive to encapsulated organisms because the spleen has a high phagocytic capacity, where opsonization with C3b and IgG, followed by phagocytosis in the white pulp, is how they are cleared. - Patients need vaccines against *Strep pneumo, Neisseria meningitidis,* and *Haemophilus influenzae* type B. - Penicillin prophylaxis is given to patients with asplenia until age 5, or until 1 year post-surgical splenectomy. If the USMLE forces you to choose which organism this protects against the most, the answer is *Strep pneumo.* - However, if USMLE vignette tells you a patient with asplenia misses a penicillin prophylaxis dose + now has sepsis, the Tx is a 3rd generation cephalosporin (i.e., cefotaxime or ceftriaxone), not penicillin. The latter is only for prophylaxis. - Howell-Jolly bodies (nuclear remnants) are seen within RBCs on a smear in patients with asplenia. This is because senescent and abnormal RBCs are ordinarily cleared out by the red pulp of the spleen. - USMLE wants you to know the lifespan of a RBC is 4 months (118 days). They ask this on NBME as how long it will take for all carbon monoxide to be cleared out of the blood in the setting of exposure (i.e., not until all RBCs exposed are removed).

HY GI Diagnostic modalities for Surg/IM (2CK mostly)	
- Same as with the GI surgeries table above, we could discuss endless indications for the below modalities, but I'm staying concise here with just the highest yield points.	
Nasogastric tube	- "Insertion of NG tube" is the answer for tracheoesophageal fistula and choanal atresia (discussed in HY Pulm PDF).
Barium swallow	- For suspected Zenker; shows outpouching. - First step in achalasia (before monometry); shows bird's beak. - Do not use for esophageal perforation. Can cause mediastinitis if it leaks out through a hole in the esophagus. - Can be used for patients with aspiration risk, since it doesn't cause pneumonitis.
Gastrografin swallow	- Aka water-soluble contrast swallow. - Used for esophageal perforations because it doesn't cause mediastinitis. - Do not use if patient has aspiration risk; causes pneumonitis. Use barium instead.
Esophageal manometry	- Answer for achalasia after barium swallow shows bird's beak. - A pressure study of the esophagus.
Endoscopy	- Aka esophagogastroduodenoscopy. - Immediate answer for any patient with new-onset dysphagia and Hx of GERD or heavy smoking/alcohol (for esophageal cancer). Then biopsy any lesion/stricture. - Endoscopy + banding for esophageal varices emergent management. - Diagnosis of hiatal hernia (asked on NBME). - Done in patients over 50 who have *H. pylori* positivity (on newer 2CK form).
Capsule endoscopy	- Always wrong fucking answer on USMLE.
Upper GI series	- Congenital midgut volvulus in pediatrics; will show a corkscrew. - This is a contrast swallow followed by X-rays to visualize the upper GI tract.
Abdominal ultrasound	- Cholelithiasis. - First step to diagnose cholecystitis; if negative, do HIDA scan. - First step for choledocholithiasis, then do ERCP. - Intussusception Dx, then do enema (definitively diagnostic and therapeutic). - Pyloric stenosis.
HIDA scan	- Answer for confirmatory diagnosis of cholecystitis (not cholelithiasis alone) if ultrasound is negative.

	- Radiocontrast is injected + secreted into bile. If gallbladder lights up, there is no obstruction of the cystic duct and it is negative; if gallbladder doesn't light up, we know there's an obstruction by a stone and it confirms cholecystitis.
CT of abdomen	- USMLE won't force you to choose, but just assume contrast CT is always used. The only times non-contrast CT will be an answer is for urolithiasis and intracranial bleeds. - Diagnosis of pancreatic cancer (highest yield indication on USMLE). - Diagnosis of liver cancer and focal nodular hyperplasia. - Renal injury (ultra-HY; discussed in HY Renal PDF). - Blunt force trauma to abdomen in patient who is stable. - Diverticulitis.
Abdominal x-ray	- Used to look for bowel gas in suspected obstruction (e.g., sigmoid volvulus showing coffee bean sign). - Duodenal atresia (double-bubble sign). - Congenital diaphragmatic hernia (bowel gas in left hemithorax). - Necrotizing enterocolitis (pneumatosis intestinalis). - Toxic megacolon if patient is stable. - "X-rays of chest and abdomen" used for duodenal ulcer perforation to look for air under the diaphragm.
ERCP	- Endoscopic retrograde cholangiopancreatography; type of EGD that can also enter the biliary tree, remove stones there, and inject contrast if necessary. - Answer on USMLE for choledocholithiasis (including gallstone pancreatitis) after ultrasound is performed. - Answer for diagnosis of cholangitis. - Answer for drainage of pancreatic pseudocyst.
MRCP	- Magnetic resonance cholangiopancreatography. - Never seen this as correct answer on NBME, but I observe that students always pick it when they don't know what's going on, maybe because it sounds weird and specific. - Can visualize biliary tree much more safely than ERCP, but unlike ERCP, it isn't a form of treatment (ERCP is both diagnostic and therapeutic).
Endoscopic ultrasound	- Never seen this as correct answer, only wrong answer. - Can be done to diagnose pancreatic cancer if CT is negative or to drain pancreatic fluid collections in place of ERCP. But once again, never seen this as correct.
Paracentesis	- Used for spontaneous bacterial peritonitis. - As mentioned earlier, choose "white cell count and differential" before "gram stain and culture of the fluid." Do not confuse with pericardiocentesis.
Meckel scan	- Radiocontrast uptake scan that localizes to the diverticulum at terminal ileum.
Colonoscopy	- Done for diagnosis of colorectal cancer, IBD, pseudomembranous colitis. - Do **not** do in diverticulitis (can cause perforation; do CT instead). - Do if patient has anal cancer prior to excision in order to first see if there's greater extent of cancer (might change management). Asked on 2CK form, where they say there is cancer at anal verge, and excision is wrong; answer is colonoscopy. - Commence at age 45-50 (guidelines are evolving) and then do every 10 years. - USMLE will not force you into a position where ohemgee is it 45 or 50. If they force you to choose 45 as a new guideline, it will be obvious the other answer choices are wrong. - If first-degree relative (parent or sibling) has colon cancer, commence at age 40 or 10 years prior to diagnosis in that family member, whichever is earlier. In other words, never later than 40. For example, if dad was diagnosed at 58, commence at age 40. If dad was diagnosed at 44, commence at age 34. Then do every 5 years. - If patient has IBD (UC or Crohn), commence 8 years after the diagnosis was made and then do every 2-3 years. 2CK NBME is real slick about this. They give

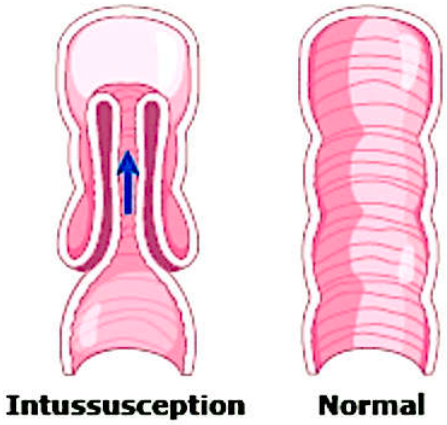
	mid-30s patient with IBD diagnosed in 20s + also has dad diagnosed with colon cancer in 50s; wrong answer = colonoscopy at age 40; correct answer is "colonoscopy now," since patient has IBD and should have had one done 8 years after his/her diagnosis. - Patients who have history of dysplastic polyps need repeat colonoscopies every 2-5 years, depending on size/morphology of polyp(s). - If patient has HNPCC (Lynch syndrome), start at 20-25, then do every 1-2 years. - For FAP, start at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - For Peutz-Jeghers, start at age 8, then every 1-2 years.
Sigmoidoscopy	- "Sigmoidoscopy-guided insertion of rectal tube" is answer on NBME for treatment of sigmoid volvulus. - I've never seen sigmoidoscopy as a diagnostic tool. I make this point because there are alternative theoretic regimens to colon cancer screening – i.e., sigmoidoscopy + barium enemas, etc., but I've never seen these assessed.
Anoscopy	- Shows up on newer 2CK form as answer for diagnosis of hemorrhoids before banding is performed. This will help determine extent of hemorrhoids.

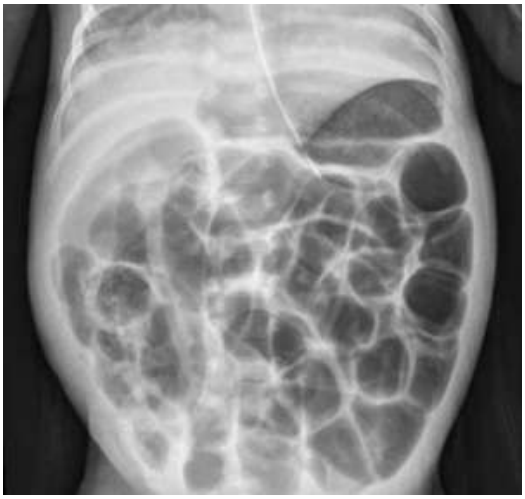
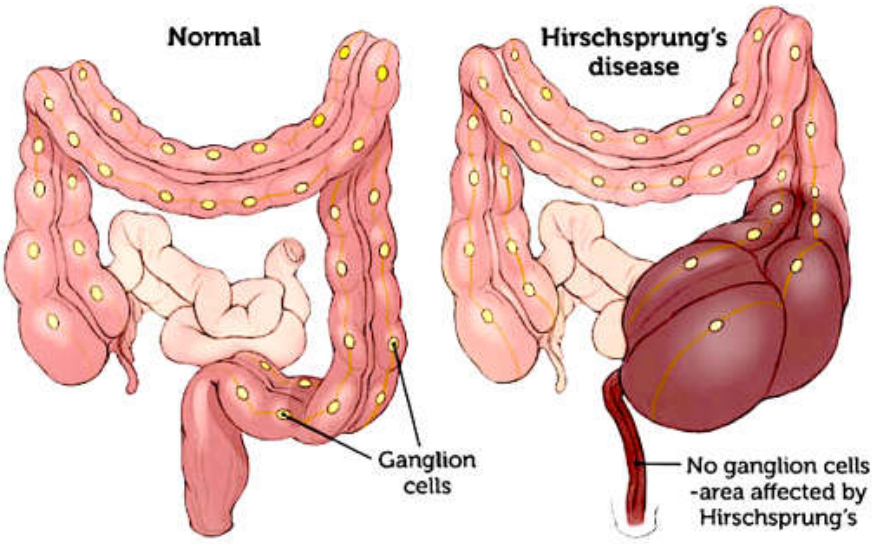
HY Polyp conditions for Surg	
Lynch syndrome	- Hereditary non-polyposis colorectal cancer (HNPCC). - Mismatch repair genes *MSH2/6, MLH1, PMS2.* - Mutations cause "microsatellite instability." - Colonic polyps/cancer; also associated with gynecologic cancer. - Start colonoscopy at age 20-25, then do every 1-2 years.
Familial adenomatous polyposis	- Familial Adenomatous Polyposis (FAP); chromosome 5; AD. - Hundreds to thousands of polyps on colonoscopy; 100% cancer risk. - Start colonoscopy at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - FAP + soft tissue (e.g., lipoma) or bone tumors (e.g., of the skull) = Gardner syndrome. - FAP + CNS tumors = Turcot syndrome.
Peutz-Jeghers	- As talked about at the start of this PDF, this is combo of perioral melanosis and **hamartomatous** colonic polyps. - Start colonoscopy at age 8, then do every 1-2 years.
Juvenile polyposis	- Shows up on a 2CK NBME. - Q will tell you there's a teenager (i.e., juvenile, LOL!) with intermittent bleeding per rectum + colonoscopy shows scattered polyps + biopsy shows **"dilated, cystic, mucus-filled glands** with abundant lamina propria and inflammatory infiltrates" → answer = juvenile polyposis.
Hyperplastic polyps	- All you need to know is these are **not pre-cancerous /** have **no dysplasia.**
Tubulovillous	- Dysplastic polyps that precede full-blown colorectal cancer are classically either **tubular or villous,** AND either **pedunculated or sessile.** - Villous is worse than tubular. Sessile is worse than pedunculated. - This means pedunculated tubular polyps are "best" and sessile, villous polyps are "worst." - Sessile means flat; pedunculated means "sticks out." - Polyps can have mixed chacteristics and hence be tubulovillous. - What you need to know is: USMLE will give you a random 65-year-old with a polyp + show you a pic + ask you what it is → answer = "tubular polyp." Student freaks out and says, "Wait, we need to know polyp pics??" → No. The big picture concept is that older people who get colorectal cancer will have tubular, villous, or tubulovillous polyps as I just said. Wrong answers would be things like hamartomatous, juvenile, and hyperplastic. You can just eliminate to get there without even knowing the image.

	- 2CK wants you to know that patients with history of dysplastic polyps need repeat colonoscopies every 2-5 years, depending on size/morphology of polyp(s).

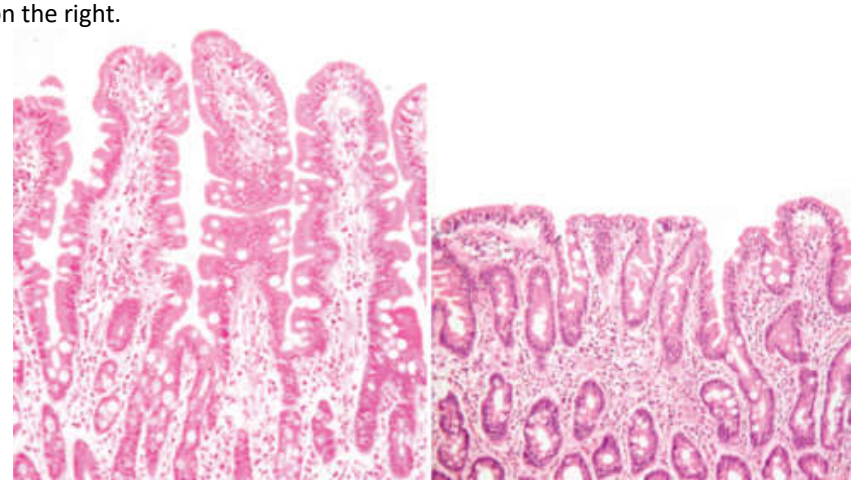
Colon cancer short points
- *KRAS* gene is the answer on USMLE for the first gene mutated in colonic polyps, prior to progression to overt colon cancer. - Colon cancer often develops as a result of *progressive* mutations, rather than one mutation straight-up. In other words, first *KRAS*, then *PTEN*, then *DCC*, then *TP53*. - If they tell you a colon cancer has metastasized and force you to choose a gene that's mutated, go with *TP53* for p53 protein. - If they tell you a polyp is seen and there is no evidence of invasion of the stalk, choose *KRAS*. This reflects earlier changes in the sequence as I just wrote above. - Can cause **colovesicular** fistulas, where a passageway develops between the GI tract and bladder, leading to UTIs and mixed enteric flora in the urine. This is also assessed for diverticulosis on 2CK and seems to be a new diagnosis USMLE likes. This could also be due to Crohn, in theory, but I haven't seen it on NBMEs yet. - Can cause *Strep bovis* endocarditis. Obscure, but rare point assessed on USMLE. - I already discussed colon cancer screening extensively above.

HY Peds GI diagnoses	
Pyloric stenosis	- Forceful/projectile **non**-bilious vomiting in neonate days to weeks old. Obstruction is above level of duodenum, so we won't see bile. - Students fixate on exact age of the kid for pyloric stenosis versus duodenal atresia. USMLE doesn't give a fuck and will give variable ages. - Hypertrophic pylorus; "olive-shape mass" in abdomen is buzzy but rarely seen in Qs. - **Ultrasound** done to diagnose; myotomy to treat. - USMLE wants you to know this is almost always a one-off / sporadic developmental defect, where the neonate will not have a broader syndrome. In contrast, duodenal atresia is usually Down syndrome.
Duodenal atresia	- Bilious vomiting in neonate. Bilious = obstruction at level of duodenum or lower. - Associated with **Down syndrome,** albeit not mandatory. - Will cause **double-bubble sign** on **abdominal x-ray.** 
Annular pancreas	- Another cause of duodenal obstruction with double-bubble sign, but much more rare than duodenal atresia.

Intussusception	- Intussusception is telescoping of the bowel into itself.  - Almost always under age 2. - Idiopathic, but can be triggered by viral infection or rotavirus vaccination, where mesenteric lymphadenopathy can act as a "lead point" for the intestinal telescoping. Underlying Meckel diverticulum is also a common lead point. - In ultra-rare scenarios, intussusception in elderly can occur due to colon cancer. - Classic presentation is intermittent/colicky abdominal pain, where the kid will have episodes of drawing the legs to the chest or squatting, with blood in the stool +/- bilious vomiting. - **An abdominal mass** (i.e., sausage-shaped mass) is a key feature, but not in all Qs. - Diagnosis is first done with ultrasound, which may show a sausage mass or target sign (I've never seen NBME show the image, so they don't care). After the ultrasound, **enema is both diagnostic and therapeutic.** USMLE doesn't care what kind of enema is used; air-contrast enema is classic, but you will see all different types of enemas as correct answers. - "Air enema with ultrasound" is the answer on one 2CK form.
Midgut volvulus	- Failure of rotation of proximal bowel. - Almost always under age 1. - Presents similarly to intussusception (often mistaken for it), but there is **no mass.** - Upper-GI series shows corkscrew appearance.  - If an abdominal x-ray is performed, it will show "dilated loops of small bowel with air fluid levels," since we have an obstruction. - Treatment is surgical.

Necrotizing enterocolitis	- Necrotic bowel occurring in premature neonates born <32 weeks' gestation. - Presents with pneumatosis intestinalis (air in bowel wall); resembles bubbles.  - USMLE will show you above image in kid born, e.g., at 26 weeks' gestation, and then answer is just necrotizing enterocolitis. Not hard. - It's to my observation necrotizing enterocolitis usually shows up as a *wrong* answer choice on NBMEs, where I see students pick it in, e.g., an adult, and I'm like, "Dude you see that in fucking hyper-preemies." And they're like Oh. So you need to remember this as specifically a premature neonate condition.
Hirschsprung	- Failure of migration of neural crest cells distally within the colon. - Causes an aganglionic distal segment that results in ↑ anal sphincter tone.  - Often associated with Down syndrome. - Presents usually as failure to pass stool at birth, as chronic constipation in an infant, depending on the severity. But do not confuse with "meconium ileus," which is a term that refers to failure to pass stool at birth specifically due to cystic fibrosis. - Diagnose with anal manometry showing ↑ anal sphincter tone, followed by biopsy.
Cystic fibrosis	- Discussed in extensive detail in the HY Pulm PDF. - Just a reiteration that meconium ileus + **exocrine pancreatic insufficiency** are exceedingly HY.

	- Secretions within the pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This means the enzymes can't make their way to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy). - D-xylose test is normal, since the intestinal lining/architecture is intact. Any malabsorption that occurs is due to mere paucity of enzymes in the small bowel. Recall that abnormal D-xylose test would be Celiac and Crohn on USMLE.
Biliary atresia	- As discussed earlier, ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - Ultrasound will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis. - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

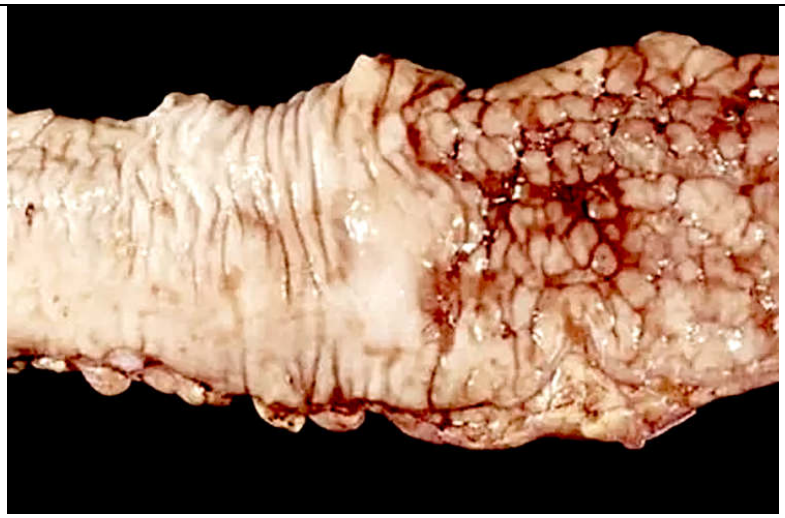
Autoimmune intestinal absorptive disorders	
Celiac disease	- Intolerance to gluten (i.e., to gliadin proteins found in wheat, oats, rye, and barley; but not rice). - Causes type-IV hypersensitivity response where T cells attack the small intestinal villi, resulting in **flattening of the villi** and malabsorption. The biopsy/image is pass-level. Below, we have normal villi on the left, and flattening of villi in Celiac on the right.  - Even though the disorder is T-cell-mediated destruction of the villi, patients still develop antibodies which are HY: anti-endomysial (aka anti-gliadin) and anti-tissue transglutaminase IgA. - Patients with concurrent IgA deficiency will have false (-) results for tissue transglutaminase IgA antibody. Since "autoimmune diseases go together," and "autoimmune and immunodeficiency syndromes go together" patients with IgA deficiency have 15x greater likelihood of developing Celiac. - Celiac presents as vague bloating/diarrhea in patients who might not be able to pinpoint what they're eating to cause the symptoms. - I'd say one of the highest yield points on USMLE regarding Celiac is that it can cause **iron deficiency anemia.** This is because iron absorption in the duodenum is impaired from the flattened villi. - For example, you might get a difficult/vague vignette where you're not sure if the diagnosis is Celiac or lactase deficiency, but you see that hemoglobin is low, and you're like "Boom. Celiac." - Associated with dermatitis herpetiformis, which presents as itchy, vesicular lesions on extensor areas. This causes "IgA deposition at dermal papillae."

	 - D-xylose test is abnormal because the intestinal lining/architecture is abnormal (i.e., we have flattening of villi). - Increased risk of gastrointestinal T-cell lymphoma in Celiac. This is called enteropathy-associated T-Cell lymphoma (EATL). - Diagnosis is made via antibody screening, followed by duodenal biopsy as confirmatory. - Treatment of Celiac is simply with gluten-free diet.
Lactose intolerance	- Deficiency of brush border disaccharidase (i.e., aka lactase deficiency). - Brush border is located at "tips of the villi." This is in contrast to stem cells of the GI tract, which are located at the "base of the crypts." USMLE asks the locations of these things. - Diarrhea/bloating in response to dairy. - In contrast to Celiac disease, there is **no iron deficiency anemia.** - ↑ incidence in Asians, but can also develop idiopathically starting in teenage years-onward. When a young adult has a new-onset diarrhea-related condition, it's usually lactose intolerance, as lactase production can ↓ with age. But once again, USMLE can trick you, so I'll be an asshole and reiterate that if you see ↓ Hb, choose Celiac instead. - Secondary lactose intolerance can occur following viral gastroenteritis (e.g., rotavirus or Norwalk virus) → sloughing of the tips of villi → vignette will say new-onset diarrhea/bloating following gastro illness 1-2 weeks ago. This is only transient and patient will recover. This is HY cause of lactose intolerance. - **Biopsy is normal.** This is in contrast to the flattened villi in Celiac. - Diagnosis is made using hydrogen breath test or detecting ↓ stool pH. - D-xylose test is normal, since the intestinal lining/architecture is intact. - Treatment is with avoidance of lactose-containing products, or with lactase pills.

Inflammatory bowel disease (IBD)
- Refers to both ulcerative colitis (UC) and Crohn. - Both can present with bloody, mucoid stools. - Both are associated with HLA-B27 → **PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis. - For example, if a patient has psoriasis + bloody stools, you say, "The bloody stools are probably IBD." Or likewise, if patient with known IBD has lower back pain that's worse in the morning, you say, "that's probably ankylosing spondylitis." - Both UC and Crohn can be associated with other autoimmune diseases unrelated to HLA-B27, like vitiligo. - Both can cause anterior uveitis (non-specific finding seen in many autoimmune diseases). - Both have ↑ risk of colon cancer if the colon is involved. But if you're forced to choose, UC has > risk than Crohn because the colon is always involved in UC but not always in Crohn.

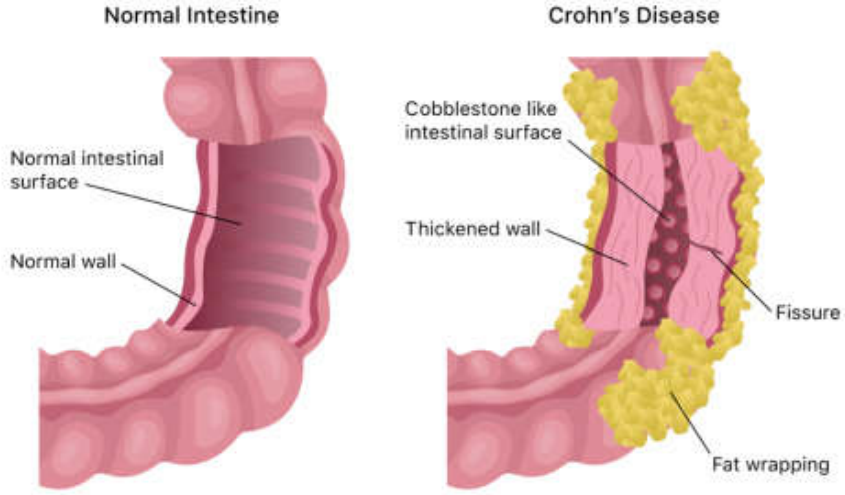
<table>
<tr><td colspan="2">- Both are treated with 5-ASA NSAID compounds (mesalamine / sulfasalazine) or steroids. If they ask first Tx, go with mesalamine or sulfasalazine, whichever they list (they won't list both) over steroids. Q on 2CK IM form 8 has "prednisone therapy" as answer for Crohn, but a 5-ASA isn't listed.
- USMLE wants you to know anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) are used in IBD in patients who fail initial Tx with 5-ASAs and steroids.
- If they ask you for which cytokine can be anti-inflammatory for IBD, the answer is IL-10. Sounds weird, but just know IL-10 and TGF-β are mostly anti-inflammatory mediators. Don't worry about Th1 vs Th2 nonsense.</td></tr>
<tr><td>Ulcerative colitis</td><td>- Rectum-ascending.
- Not transmural – i.e., only affects mucosa and submucosa. This means fistulae are not seen. If they say fistulae to the skin overlying the anus or to any organ, this means transmural involvement (Crohn).
- Colonoscopy will show pseudopolyps and crypt abscesses. You don't need to know what these look like. You just need to know they = UC.
- Barium enema shows "lead-pipe appearance" due to loss of haustra. This detail is very important for UC.
 
Descending colon resembles a "lead pipe"

- Can cause primary sclerosing cholangitis, as discussed earlier. This can be associated with pANCA antibodies.
- Pyoderma gangrenosum is a necrotic skin lesion seen rarely in patients with UC. There is an NBME Q that describes this as an "ulcer with necrotic debris."
- Can cause toxic megacolon, which presents as SIRS and sometimes low BP in a patient with UC. They might say abdominal x-ray shows a 12-cm cecum (NR 3-8).
- If the patient has normal BP, NBME for 2CK wants steroids first for toxic megacolon. If the patient is unstable, go straight to laparotomy.
- Colectomy is sometimes performed in patients with severe UC, but very rarely in Crohn.</td></tr>
<tr><td>Crohn disease</td><td>- "Mouth to anus" – i.e., can occur anywhere in GI tract. USMLE loves giving mouth ulcers in Crohn. But terminal ileum is highest yield location.
- Transmural – i.e., can cause anal fistulae + to other organs.
- Colonoscopy shows skip lesions, where there is alternating diseased vs normal bowel segments, with cobblestone ulcers.</td></tr>
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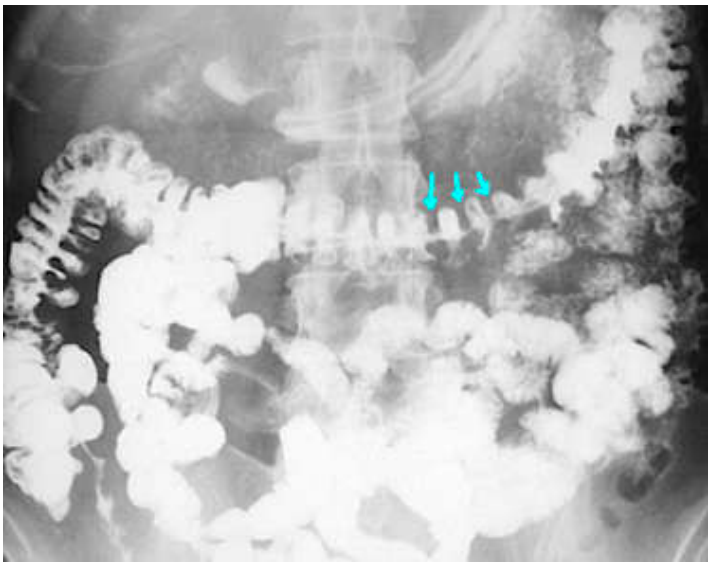


Skip lesion (left part normal + right part inflamed and cobblestoned)

- "Creeping fat" is buzzy term that can be seen on NBME, which refers to intestinal fat migration that wraps around the bowel.



- Barium enema shows "string sign," where inflamed segments are narrowed in comparison to normal bowel.



	- Biopsy shows **non-caseating granulomas.** Very HY for USMLE you know that Crohn + sarcoidosis both have non-caseating granulomas. - Sometimes associated with erythema nodosum. Not specific for Crohn in any regard, but tends to have ↑ association, whereas UC is pyoderma gangrenosum. - Can cause anti-saccharomyces cerevisiae (yeast) antibodies. This is on a 2CK NBME, where they say (-) for these Abs, but (+) for pANCA, where answer is UC. - Intestinal malabsorption can occur, resulting in B12 deficiency most commonly due to terminal ileum being classic inflammatory location. - Impaired fat absorption can result in ↑ calcium oxalate urolithiasis, as discussed in the HY Renal PDF.

Irritable bowel syndrome (IBS)
- Don't confuse with IBD. IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."

Other HY GI motility scenarios	
Diabetes	- Diabetic gastroparesis = ↓ peristalsis due to neuropathy of the GI tract. - Presents as GERD-like presentation in someone with Hx of advanced diabetes. - USMLE wants endoscopy first to rule-out physical obstruction. If endoscopy is negative, do gastric-emptying scintigraphy (aka gastric emptying scintigraphic assay) to confirm delayed gastric emptying. - Metoclopramide is used 1st for Tx. It is a prokinetic agent (i.e., ↑ peristalsis) and anti-emetic. It is a D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - Erythromycin can also be used to agonize motilin receptors. - "2-week trial of PPI" is wrong answer. This is the answer for "regular GERD." If the patient has GERD-like presentation with advanced diabetes, however, the Dx is diabetic gastroparesis, not GERD. - Neuropathy to the hypogastric nerves (sympathetic) causes severe diarrhea. This is because the sympathetic nerves are "anti-peristalsis," so if we knock them out, we get too much peristalsis. - Neuropathy to the pelvic splanchnic nerves (parasympathetic) causes severe constipation. This is because the parasympathetic nerves are pro-peristalsis, so if we knock them out we get unopposed hypogastric. - In other words, diabetes + diarrhea → answer = hypogastric nerves are fucked up. - Diabetes + constipation → answer = pelvic splanchnic nerves are fucked up.
Thyroid	- Hypothyroidism can cause constipation. - Hyperthyroidism can cause diarrhea. - USMLE will give vignette of a thyroid condition with GI disturbance, and then the answer will just be "motility disorder." This is same answer for diabetes.
Supplements	- Iron and Aluminum can cause constipation ("Aluminimum amount of feces.") - Magnesium can cause diarrhea.
Drugs	- Verapamil causes constipation. - Macrolides, orlistat, and α-glucosidase inhibitors (e.g., acarbose) cause diarrhea.

<table>
<tr><th colspan="2">Viral hepatitis</th></tr>
<tr><td colspan="2">- Hep A and E cause acute hepatitis only. Hep B, C, and D can cause chronic hepatitis.
- Hepatitis in general classically has ALT > AST, where #s can be in the hundreds to thousands, but I've seen plenty of variability on NBME Qs, which is why I don't consider this a foundational point.</td></tr>
<tr><td>Hep A</td><td>- The answer for acute hepatitis in the United States most of the time. The Q might say the patient had recent travel to Mexico.
- Fecal-oral; only causes acute hepatitis.
- IgM against HepA means acute infection.
- IgG against HepA means patient has cleared infection (because there is no chronic HepA).
- USMLE wants you to know HepA vaccine is indicated for IV drug users and MSM. There's a 2CK NBME Q where they just mention otherwise healthy MSM, and answer is Hep A vaccination.</td></tr>
<tr><td>Hep B</td><td>- Mandatory stuff for USMLE is the serology (I discuss below).
- Most common hepatitis infection in the worldwide. USMLE likes China for hepatitis B. Just a pattern I've noticed. Due to ↑↑ unvaccinated. In the USA, HepC is most common.
- Parenteral; can be acute or chronic.
- Transmitted vertically from mother to neonate, sex, via IV drugs, or blood exposure.
- Present in all body fluids, including breast milk.
- Serology very HY:

- (+) Surface antigen = patient currently has HepB.
- (+) Surface antibody = patient is immune to HepB.
- (+) Core antibody = patient has HepB now or did in the past.
- (+) Core antibody IgM = patient has acute infection.
- (+) Core antibody IgG = patient has chronic HepB, OR cleared HepB.
- (+) Surface antibody / (-) Core antibody = Immune → Vaccinated against HepB.
- (+) Surface antibody / (+) Core antibody = Immune → History of HepB / cleared it.
- (-) Surface antibody / (-) Core antibody = Not immune / Susceptible → need to vaccinate.
- (-) Surface antigen / (-) Surface antibody / (+) Core antibody IgM → window period (below).

- Vaccination against HepB is at birth, 2 months, and 6 months (no longer at 4 months).
- Only give HepB IVIG to neonate if mom is confirmed (+). A 2CK NBME Q gives mother's status as unknown when child is born → answer = "Give HepB vaccine now + only give IVIG if mother is positive."
- If patient has Hx of completed HepB vaccination but has titers that show susceptibility, the answer is just "give more vaccine." Sometimes people's immunity wanes.

- Once a susceptible patient is exposed to HepB and the immune system attempts to clear it, sometimes Surface antigen will decline to the point that it is no longer detectible. But at the same time, the Surface antibody might not be high enough / at detectable levels yet. This is called the "window period," where both Surface antigen and antibody are negative, so it can appear as though the patient doesn't have an infection. However, Core antibody IgM will be (+). So the key point is that 1) you know the double-negative Surface antibody/antigen combo is seen in the window period, and 2) that Core antibody IgM is most reliable during the window period.
- USMLE really doesn't give a fuck about HepB pharm (i.e., entecavir, tenofovir). Waste of time.
- You could be aware that interferon-α can be used for HepB.</td></tr>
<tr><td>Hep C</td><td>- Parenteral; can be acute or chronic.
- Transmitted almost exclusively from IV drugs/blood exposure. Not present in breastmilk and non-sanguineous body fluids (in contrast to HepB).
- In contrast to HepB, HepC is not considered sexually transmitted. Large longitudinal study of couples with one HepC(+) partner showed sexual transmission almost nil (possibly due to menses exposure). If you're forced to choose for behavioral science-type Qs, however, still inform that abstinence or barrier contraception minimizes risk.
- No vaccine due to ↑↑ antigenic variation (i.e., >7 genotypes and 80 subtypes of HepC exist).
- IgM against HepC means acute infection.
- IgG against HepC means usually means chronic HepC.</td></tr>
</table>

	- Many drugs can be used to treat. USMLE doesn't care. What you could be aware of is pegylated interferon-α.
Hep D	- Requires hepatitis B in order to infect, which can be due to co-infection (happening at the same time) or superinfection (occurs later in someone who already has HepB). - If USMLE asks how to prevent HepD infection, answer = vaccination against hepatitis B. There is no vaccine against HepD. - Apparently HepB antigen forms the envelope for HepD (i.e., forms a circle around HepD).
Hep E	- Causes fulminant hepatitis / ↑↑ risk of death in pregnant women. - Same as with HepA, only causes acute hepatitis. - Seen more in Asia, e.g., Tibet. But if USMLE says Mexico + pregnant woman + fast death from hepatitis, you still have to use your head and know that's HepE over HepA.

Other HY hepatitis causes	
NASH	- Non-alcoholic steatohepatitis. - Answer on USMLE for liver condition in an overweight patient with normal labs. - Underrated diagnosis in that students often have no idea about it but it shows up on both Steps (higher yield on 2CK IM). - Patient will have metabolic syndrome (i.e., ↑ BMI + ↑ lipids) + either mild transaminitis or completely normal labs. I'd say ~50% of Qs will give ALT and AST a very tiny bit elevated; the other ~50% of Qs will give you completely normal labs where you say WTF. - For example, they'll give you 40-year-old patient with BMI of 35 who comes in for routine health maintenance exam who has no complaints + lab studies are all completely normal; then the answer is just "fatty liver" or "non-alcoholic steatohepatitis." Student says, "Wait, but the labs are completely normal though." → Exactly. There's nothing wrong. But it's assumed most overweight people have some degree of fatty liver. - The condition starts out with completely normal labs, then progresses to very mild transaminitis. Over time, LFTs can worsen. Rarely it can cause cirrhosis and carcinoma. - Theoretical Tx is just hit the treadmill more and stop eating your dumb fast food.
Alcoholic	- Classically AST > ALT. - The ratio need not be 2:1. I've seen variations on NBME forms where there can be very mild transaminitis (i.e., both AST and ALT are <100, with AST only slightly higher than ALT). If the vignette is hyper-easy, they'll say something like AST is 400 and ALT is 200. Both should normally be <50 U/L. - Causes steatosis (fatty liver).
Acetaminophen	- Acetaminophen overdose most important for USMLE. *N*-acetylcysteine is antidote. - Breakdown of acetaminophen produces metabolite called NAPQI that depletes reduced glutathione. The latter normally mops up free radicals / prevents oxidation. - Patient will OD on acetaminophen and be asymptomatic for 24-28 hours, then will proceed to get ALT and AST in the tens of thousands + require liver transplant or death. - "*N*-acetylcysteine regenerates reduced glutathione." Memorize that. - Oxidized glutathione had disulfide bond (-S—S-); reduced glutathione has thiol (-SH). - Activated charcoal should apparently be administered to patients who present very acutely after massive ingestion, as this can ↓ acetaminophen absorption. But I've never seen this on NBME. It tends to be more Qbank that's gotchya-style this way. I've only ever seen *N*-acetylcysteine assessed across NBMEs.
Other drugs	- I'd say some important drugs that can cause transaminitis are: - Statins and fibrates, methotrexate, lithium, valproic acid, propylthiouracil. - Mild transaminitis is normal and expected with these agents; the answer is you do not need to decrease dose. - For statins, myopathy is more common than toxic hepatitis (offline NBME).

Heavy metal liver diseases	
Hereditary hemochromatosis	- Autosomal recessive; *HFE* gene; chromosome 6. - Iron overload. - **Mechanism USMLE wants is "increased intestinal iron absorption."** - Main iron regulation is via shutting off intestinal absorption; this is impaired in hemochromatosis. - Body has poor ability to excrete iron; otherwise there are minor losses via skin shedding. Since women excrete naturally via menses, presentation occurs earlier in men (who clearly don't have this mechanism of excretion). - Can present as "bronze diabetes" → hyperpigmentation due to hemosiderin deposition in skin + ↑ fasting sugars (iron deposition in tail of pancreas). - Miscellaneous other findings can be seen like infertility (iron deposition in hypothalamus, anterior pituitary, or gonads), cardiomyopathy, or arthritis (pseudogout). - Hereditary hemochromatosis, primary hyperparathyroidism, and hypothyroidism are 3 most important causes of pseudogout. I used to only discuss the former two with students over the years, but the latter shows up on a new 2CK NBME exam where they mention chondrocalcinosis (calcium deposition in cartilage). - USMLE wants you to know there is ↑ risk of hepatocellular carcinoma. There is easy NBME Q of hemochromatosis where they ask what patient is at increased risk for, and the answer is simply "hepatocellular carcinoma." - Diagnose with **ferritin >300 mg/dL.** Transferrin saturation will also clearly be ↑. - It is exceedingly rare that ferritin is >300 in other conditions, however this can occur in lymphoma and leukemia, where ↑ ferritin is a poor prognostic marker in non-Hodgkin lymphoma. There is one NBME Q on 2CK where ferritin is 300 where it's not hemochromatosis, but USMLE won't play gotchya. - Treat with **serial phlebotomy, not chelators.** - Chelators such as deferoxamine or deferasirox are for secondary hemochromatosis due to transfusional siderosis (i.e., repeated blood transfusions that contain iron, for e.g., β-thalassemia major).
Wilson disease	- Autosomal recessive. - Copper overload. - Inability to secrete copper into bile from the liver. Copper is normally excreted by the body via secretion into bile. - ↑ urinary copper + ↓ serum ceruloplasmin (copper-binding protein in the blood; in the case of copper overload, body tries to minimize amount carried in blood). - Buzzy / pass-level detail is Keiser-Fleischer rings, which is copper deposited in the cornea of the eye. Vignette can give you what sounds like Wilson disease, and then the answer is **"slit-lamp exam."** - Can cause ↑ LFTs with cirrhosis and Parkinsonism. - Copper deposits in basal ganglia, especially the putamen. - Parkinsonism in a young patient = Wilson until proven otherwise. - In old patient, Parkinsonism = Parkinson disease, normal pressure hydrocephalus, Lewy-body dementia, or progressive supranuclear palsy. - Treat with the copper chelator penicillamine.

Primary hepatocellular carcinoma (HCC) vs Metastases
Three main points: 1) Primary HCC is going to be one large lesion, sometimes with tiny satellite lesions. In contrast, metastases to the liver will present as multiple lesions everywhere. 2) USMLE wants **colon cancer** as the most common type that goes to liver. 3) Do CT scan of abdomen as first step in diagnosis of suspected liver cancer. Biopsy is confirmatory.

Liver in relation to heart/kidney for Surg	
Pulsatile liver	- Just know this is an examination finding seen in tricuscpid regurg. - As I talked about in the HY Cardio PDF, the highest yield cause of tricuspid regurg on USMLE is pulmonary hypertension / cor pulmonale.
Hepatorenal syndrome	- Just know that in chronic liver disease, sometimes the kidneys can also fail. - This is supposedly due to ↓ blood flow in patients with cirrhosis and ascites. - Biopsy shows no abnormalities. That is probably the only notable point. - Never seen this as correct answer on NBME. Only reason I'm including it is because I see it as a distractor on some 2CK Qs in particular, so you should at least know it exists and is likely always wrong.

Bowel ischemia	
Ischemic colitis	- Patient with cardiovascular disease (CVD) + blood in the stool. - Caused by bleeding at ischemic ulcers at watershed areas in the colon (i.e., splenic flexure and rectosigmoid junction). - Can occur randomly or due to inciting event like recent AAA surgery. - Diagnose with colonoscopy to visualize the ischemic ulcers.
Chronic mesenteric ischemia	- Patient with CVD + abdominal pain 1-2 hours after eating meals. - Caused by atherosclerosis of the SMA or IMA → consuming food ↑ oxygen demand of bowel → angina of bowel. - The timing of the pain in relation to meals sounds like duodenal ulcers, however instead of the vignette being a 29-year-old dude from Indonesia with *H. pylori*, it will be a 69-year-old dude with Hx of coronary artery bypass grafting, intermittent claudication, HTN, and diabetes. - Next best step is "mesenteric angiography" on USMLE.
Acute mesenteric ischemia	- Presents 3 different ways on USMLE: 1) Severe abdo pain in patient with AF → LA mural thrombus launches off to SMA or IMA. 2) Severe abdo pain in patient just cardioverted/defibrillated → can launch LA thrombus off to SMA or IMA. There's a Q on 2CK form where this is the case, where they don't mention AF in the stem. 3) Severe abdo pain in patient with Hx of chronic mesenteric ischemia (i.e., acute on chronic) → atheroma within SMA or IMA ruptures, effectively causing an "MI of the bowel." - USMLE wants "mesenteric angiography" as next best step. - Laparotomy is also answer on NBME form. - I've never seen medications or endarterectomy as answers.

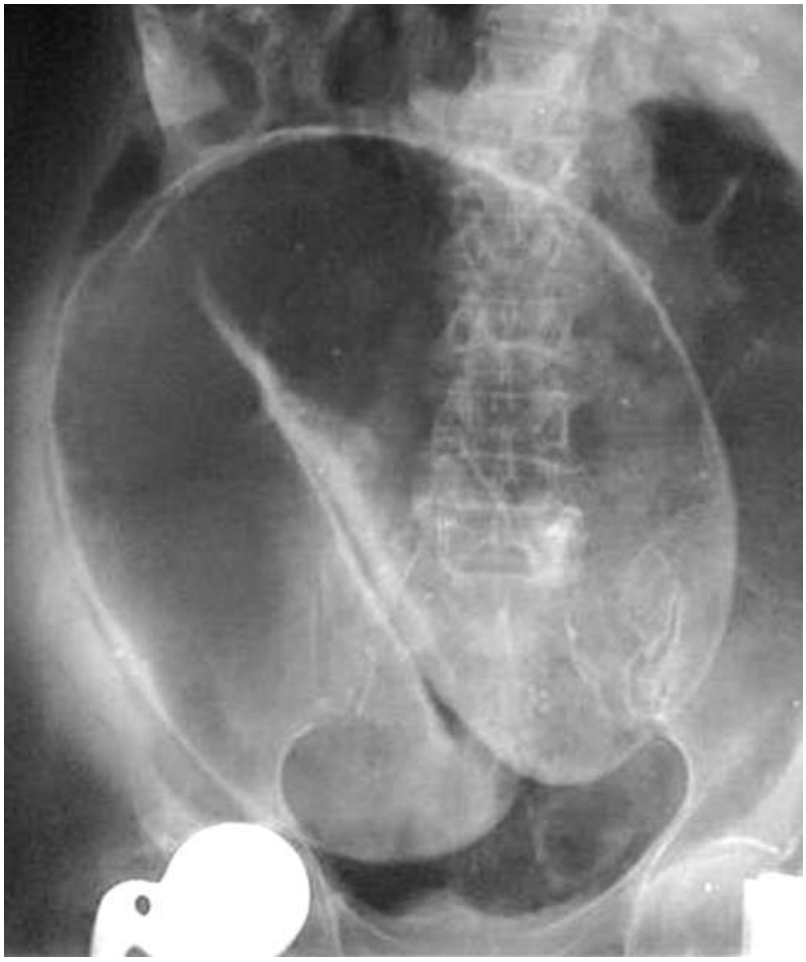
GI thromboses	
Mesenteric vein thrombosis	- The splenic vein and superior mesenteric vein (SMV) merge to form the portal. The inferior mesenteric vein (IMV) goes to the SMV. - Cirrhosis → ↑ portal venous pressure → ↑ SMV + IMA pressure → stasis. - In addition, any malignancy a patient might have → hypercoagulable state. - NBME Q gives patient with cirrhosis and lung cancer who has abdominal pain + dark, mottled small bowel on surgical examination → answer = "mesenteric venous thrombosis." Students ask why. I say, "Well there's ↑ mesenteric venous pressure from the cirrhosis + hypercoagulable state from the malignancy."
Splenic vein thrombosis	- Risk factors are same as above. - As I discussed earlier, USMLE wants you to know splenic vein thrombosis is a cause of esophageal varices due to collateral development with the left

	gastric vein (which drains the esophageal veins, so if ↑ left gastric venous pressure, then ↑ esophageal venous pressure). - USMLE wants you to know the short gastric veins feed into the splenic vein. So if we have ↑ splenic veinous or portal veinous pressure, then the short gastric veins also have ↑ pressure.
Budd-Chiari syndrome	- Hepatic vein thrombosis. The hepatic vein drains the liver. - Caused by polycythemia vera and pregnancy. - Triad = abdominal pain, ascites, and hepatomegaly.
Portal vein thrombosis	- Caused by portal venous stasis from cirrhosis (portal vein enters the liver).

Mandatory GI anatomy for Surg		
Structure	**Spinal level**	**HY points**
Celiac trunk	T12	- Supplies foregut (oral cavity until 1st part of duodenum). - Its 3 branches are: splenic artery, common hepatic artery, left gastric artery. - For whatever reason, the USMLE wants you to know the spleen is supplied by an artery of the foregut but is not itself derived from the foregut (i.e., it is derived from midgut). - The short gastric arteries come off the splenic artery and supply the superior greater curvature of the stomach.
SMA	L1	- Supplies the midgut (2nd part of duodenum until 2/3 distal transverse colon). - What the USMLE will do is give vignette of SMA thrombosis in peripheral vascular disease, causing acute mesenteric ischemia, and then ask which structure is involved → answer is anything that is midgut (e.g., **jejunum,** ascending colon, etc.)
IMA	L3	- Supplies the hindgut (distal 1/3 of transverse colon until the rectum). - Same as for SMA, they'll say there's acute mesenteric ischemia of IMA and then ask for involved structure (e.g., descending colon). - USMLE will also ask cancer lymphatic/venous drainage, which follows the same distribution as arterial supply. For instance, they'll say there's a cancer of the descending colon, and the answer is "inferior mesenteric lymph nodes" or "inferior mesenteric vein." Not complicated.
Renal/gonadal arteries	L2	- Gonadal arteries = testicular arteries in males and ovarian arteries in females.
Aortic bifurcation	L4	- Bifurcation of abdominal aorta into the common iliacs occurs as L4. - What USMLE will do is tell you a AAA repair is performed where a graft needs to be inserted between the aortic bifurcation and the renal arteries. Then they'll ask blood supply to which structure will get compromised. So you need to say, "Well the bifurcation is at L4, and the renal arteries are at L2, so anything that's IMA would get fucked up since IMA is L3." → answer = "descending colon."

Appendicitis
- Triad of RLQ pain, fever, and vomiting (Murphy's triad). - Pain starts at epigastrium (visceral pain) and migrates toward RLQ (parietal pain). USMLE will ask why there is movement of the pain → answer = "inflammation of the parietal peritoneum." - Pain at McBurney's point (1/3 of the way from the anterior superior iliac spine to the umbilicus). - (+) Rovsing sign (pain felt in RLQ upon palpation of LLQ). - USMLE wants you to know appendicitis pain can be **RUQ in pregnancy** due to displacement of bowel and a shift in appendiceal location. What they will do is tell you pregnant woman in 3rd trimester has RUQ pain, fever, and vomiting + negative abdo ultrasound (meaning not cholecystitis) → answer = "appendicitis." - "Inflammatory mass and fecalith in the cecum" is phrase used on 2CK NBME form to describe appendicitis.

- Diagnosis is done via ultrasound first, and sometimes CT. However, I have not seen USMLE assess either of these, likely because their utility is debated. What I have seen is **laparoscopic removal in the stable patient.**
- USMLE will give ethics/consent Q, where they say patient has operation performed for ovarian procedure but it is seen that she has an acutely inflamed appendix → answer = "remove due to necessity of medical emergency."
- However, if the scenario is reversed and, while performing an appendectomy, the surgeon notices a suspicious lesion on, e.g., one of the ovaries, the answer is "do not biopsy." Consent must first be obtained for all non-emergencies.

HY colorectal / proctocolonic diagnoses	
Angiodysplasia	- Tortuous, superficial vessels in the colonic wall that cause painless bleeding per rectum in elderly. - Classically associated with aortic stenosis (Heyde syndrome; possibly related to pressure backup to the colon). - NBME Q gives older guy arguing with his wife + gets chest pain and bleeding per rectum. Even though colonoscopy is used for diagnosis, they say in this Q that colonoscopy is negative, likely to tell you it's not cancer causing the bleeding.
Sigmoid volvulus	- **Patient over 75** + 2-3 days of constipation + abdo pain. - Rotation around its mesentery causes "dilation of sigmoid colon" (answer on NBME as what is most likely to be seen in patient). - **Abdominal x-ray** is used to diagnose, which shows a **coffee bean sign,** which is one of the **highest yield radiographic images on Surg** (hence I'm making it big AF):  - Tx on NBME is "sigmoidoscopy-guided insertion of rectal tube."

Diverticulosis	- Mere presence of diverticula in the colon; >50% of people over age 60 in the US. - ↑ straining throughout life leads to herniation of mucosa + submucosa through the muscularis propria of the colonic wall. - Usually asymptomatic, but can bleed. Diverticular bleed is most common cause of painless bleeding per rectum in elderly, followed by colorectal cancer, followed by angiodysplasia. - Can cause **colovesical fistula** (on new 2CK NBME), where a passageway between the colon and bladder forms, leading to UTI and mixed flora in the urine. - Do not confuse with diverticulitis, which is when a diverticulum becomes inflamed.
Diverticulitis	- **LLQ pain + fever in patient over 60.** One of the highest yield diagnoses on USMLE. - Inflamed diverticulum, usually in the sigmoid colon. - **CT of the abdomen with contrast** is how to diagnose. - Do not scope acutely, as this can cause perforation. - After patient is treated with antibiotics, a colonoscopy should be scheduled weeks to months later to rule out malignancy. But once again, never scope acutely. - Perforated diverticulitis can require colectomy.
Fecal impaction	- Patient over 70 + chronic constipation + **"hard stool palpated in the rectal vault."** - Can sometimes cause fecal incontinence and paradoxical overflow diarrhea leading to encopresis (i.e., word for shitting yo pants). - Idiopathic, but exacerbated by opioids. - Treatment is with enema and laxatives.
Hemorrhoids	- Bleeding per rectum that will be described as "blood on the toilet paper," or "blood that drips into the toilet bowl." - Hemorrhoids are technically normal vascular structures within the anal canal that facilitate/cushion the passage of stool. - Bleeding from internal hemorrhoids is painless. These are above the pectinate line in the anal canal. - Bleeding from external hemorrhoids is painful. These are below the pectinate line. - Pregnancy and cirrhosis are risk factors. - 2CK NBME wants rectal exam followed by **anoscopy** for diagnosis. - Hemorrhoids often self-resolve and can be managed conservatively with sitz bath, NSAIDs, and ↑ dietary fiber. However, if USMLE forces you to choose surgical management for more severe cases, the answer is **rubber band ligation.**
Anal fissure	- Young adult who has painful bowel movements +/- blood in the stool. - The key detail is **they refuse the rectal exam because the pain is so bad.** - For whatever reason, they can also say there's an associated skin tag. I've seen this more than once, where the student says, "What's with the skin tag?" No fucking idea. - If exam forces you to choose a location for anal fissure, the answer is "posterior in the midline." Stupid, but it's asked. - Mechanism can be "increased anal sphincter tone." - Tx on 2CK Surg is **sitz bath.** - If the USMLE forces you to choose a med, topical nitrates or diltiazem is used.
Pilonidal cyst/abscess	- Answer on USMLE for cystic mass at the superior aspect of the gluteal cleft. - Contains hair; often caused by ingrown hairs. - Tx on 2CK Surg = incision and drainage.
Perianal abscess	- Answer on USMLE for painful, erythematous mass near anal verge. - Increased risk in Crohn and diabetes. - Tx on 2CK Surg = incision and drainage.
Anal malignancy	- 2CK Surg Q gives cancer at anal verge + asks for next best step in management → answer = colonoscopy; excision is wrong answer. Presumably the scoping is done to first investigate the extent of the disease, as that might alter management.

HY GI bacterial infections	
Gram-negative rods	
ETEC	- Enterotoxigenic *E. coli* causes traveler's diarrhea. - Will present as brown/green diarrhea in person who's gone to Mexico or Middle East classically.
EHEC	- Enterohemorrhagic *E. coli* causes bloody diarrhea 1-3 days after consumption of beef. - Produces shiga-like toxin, which can cause hemolytic uremic syndrome (HUS; triad of renal dysfunction, schistocytosis, and thrombocytopenia).
Shigella	- Bloody diarrhea 1-3 days after consumption of beef. - Also can cause HUS via shiga toxin.
Salmonella	- Food poisoning is caused by *Salmonella typhimurium* and *Salmonella enteritidis*. The nomenclature has changed with time, but those names are acceptable. - Bloody diarrhea 1-3 days after infection from consuming poultry, or following exposure to eggs or reptiles (i.e., turtles, lizards, etc.). New NBME Q mentions bloody diarrhea in someone with a pet lizard. - *Salmonella typhi* causes typhoid fever, which will be rose spots on the abdomen in a patient who's "prostrated" (i.e., lying down in pain); can be either diarrhea or constipation. The reservoir for *S. typhi* is humans, not chickens/reptiles.
Vibrio	- *Vibrio cholerae* (cholera) presents as "liters and liters" of rice-water stool in someone who went traveling to, e.g., Mexico. The way you can differentiate this from ETEC traveler's diarrhea is that cholera is notably **profusely high-volume.** - Acquired fecal-oral (i.e., fecal-contaminated food/water). - Both ETEC and cholera vignettes can tell you the patient has 8-12 stools daily, so it's not the # of stools that matters; it's the emphasis on volume. Cholera causes death via severe dehydration and electrolyte disturbance. - Tx is **oral rehydration** on USMLE; if patient has low BP or altered mental status (i.e., confusion/coma), IV hydration is done. - *Vibrio parahemolyticus* doesn't cause profuse, watery diarrhea the same way cholera does. I've seen this organism asked once in an NBME vignette where the patient ate sushi (can be acquired from sushi and shellfish). - *Vibrio vulnificus* causes severe sepsis in half of patients. This is asked on offline NBME 19 where a dude went running on a beach and got sepsis, with no mention of consumption of food. But it's apparently acquired from shellfish.
Yersinia enterocolitica	- Causes bloody diarrhea + either **appendicitis-like (i.e., RLQ) pain** or **arthritis.** - The RLQ pain is from mesenteric adenitis or terminal ileitis.
Campylobacter jejuni	- Bloody diarrhea 1-3 days after consumption of **poultry.** - Can cause Guillain-Barre syndrome (ascending paralysis + ↓ tendon reflexes + albuminocytologic dissociation in the CSF → Tx with IVIG + plasmapheresis).
Gram-positive rods	
Bacillus cereus	- Watery diarrhea and/or vomiting in patient who's consumed reheated or fried rice. The process of heating/re-heating causes germination of spores. - Can also cause eye infections (weird, but on USMLE). They'll say patient who's had eye surgery + now has infection caused by gram(+) rod → will be only gram(+) rod listed.
Clostridium perfringens	- Watery/secretory diarrhea following consumption of poultry. - Causes gas gangrene (CO_2 gas) due to production of α-toxin/phospholipase; presents as black skin / crepitus. - Can also cause emphysematous cholecystitis (air in gall bladder wall).
Clostridium difficile	- Diarrhea (pseudomembranous colitis) ~7-10 days after commencing oral antibiotics. - Antibiotics kill off normal bowel flora, allowing *C. difficile* to overgrow. - *C. difficile* is not normal flora, however. It is acquired via **consumption of spores.** - Can be watery or bloody diarrhea on NBME. Can also cause LLQ cramping (not RLQ as with *Yersinia*).

	- There is NBME Q where they say 28-year-old with LLQ cramping and bloody diarrhea 7 days after starting oral antibiotics → answer = *C. diff*; wrong answer is *Yersinia.* - USMLE doesn't care which antibiotics cause it; can be any. - Diagnose with **stool AB toxin test; stool culture is wrong answer.** - Treat with **oral vancomycin.** - Vancomycin has poor oral bioavailability, so is usually given IV for things like endocarditis and meningitis. But for *C. diff* infection, *that's a good thing* because we want it to stay within the GI tract. - Other fancy Abx like fidaxomicin, rifaximin, etc., I've never seen on NBME. More just masturbation around dumb factoids.
Gram-positive cocci	
Staph aureus	- *S. aureus* pre-formed heat-stable toxin is acquired from two main sources on NBME: 1) various meats sitting under a heat lamp / out for long periods of time, e.g., at a buffet; 2) dairy products like creams, custards, potato salad. - Notably causes **vomiting 1-6 hours** after consumption. This is notable, as the symptoms occur rather quickly. Diarrhea, both watery and bloody, can occur, but is not mandatory. The main crux is the vomiting.
Other GI bacteria	
Tropheryma whipplei	- Causes Whipple disease, a GI malabsorptive syndrome where the patient can also get arthritis and renal and cardiac disease. - 100% of Qs will say "PAS-positive macrophages in the lamina propria." You don't need to know the image for this. But for whatever magical reason, the USMLE gives a fuck about this detail. Old-school pathology docs who reminisce maybe. - New highly pedantic Q on 2CK NBME wants "ceftriaxone + daily TMP/SMX for one year" as the treatment. Organism isn't even HY. No idea why they care.
Tropical sprue	- Causes flattening of villi similar to Celiac, but considered infective/bacterial in origin. Literature says exact etiology not certain. - Answer on USMLE for patient living in tropical area with unknown malabsorptive disease, where you can easily eliminate the other answers. - Tx = tetracycline.

HY GI viral infections	
Rotavirus	- Most common cause of watery diarrhea in unvaccinated children < 5 years. - Vaccine normally given orally at 2, 4, and 6 months of age.
Norwalk virus	- Most common cause of watery diarrhea in adults and rotavirus-vaccinated children. - Cruise ships and business conferences are buzzy places to acquire (fecal-oral); basically any place with high density of people. - If the Q says a young child + family all have watery diarrhea, the answer is Norwalk, not Rota, since only the young child would get Rota, not the family also.
Herpes	- HSV1/2 causes herpes esophagitis → odynophagia + punched-out ulcers in esophagus.
CMV	- CMV esophagitis →odynophagia + linear (confluent) ulcers in esophagus. - CMV colitis → bleeding per rectum in AIDS patient with CD4 count <50. - CMV retinitis → blurry vision in HIV (or immunosuppressed) patient. - Most common organism transmitted via blood transfusions and organ transplants. - USMLE likes "intranuclear inclusions" or "intranuclear inclusion bodies" for CMV. This refers to the "owl eyes" that can be seen on histo.

HY GI protozoal infections	
- A protozoan is a unicellular eukaryote. - **ECG** → ***E****ntamoeba histolytica,* ***C****ryptosporidium parvum,* and ***G****iardia lamblia* are all GI protozoa that are acquired via **cysts in water** (i.e., they are **water-borne**). If "water-borne" and "fecal-oral" are both listed as answers, USMLE wants "water-borne" as means of acquisition.	
Entamoeba histolytica	- Bloody diarrhea in person who went to Mexico. - Can cause "flask-shaped ulcers" in the small bowel and **liver abscess.** - Demonstrates "erythrophagocytosis," where RBCs can be seen within it on LM. - Treat with metronidazole. - Iodoquinol kills intraluminal parasite.
Cryptosporidium parvum	- Watery diarrhea in person who went to Mexico. - Appears as **acid-fast cysts** (same stain as TB). - Self-limiting in immunocompetent persons → Tx = supportive care. - Chronic diarrhea in HIV → Tx = nitazoxanide.
Giardia lamblia	- Steatorrhea in person who went to Mexico. - Steatorrhea = bloating + extremely foul-smelling stool that floats. - The steatorrhea is due to *Giardia* causing **malabsorptive** diarrhea. - There is one NBME Q for *Giardia* where they say "foul-smelling *watery* diarrhea with bloating," which is audacious, since 9 times out of 10 it is not described as watery. But the "bloating" and "foul-smelling" are consistent with steatorrhea in this context. Just letting you know it exists. - Acquired via fresh water lakes / scuba diving. There is a *Giardia* NBME Q where they say a woman goes scuba diving in Mexico and then gets foul-smelling stools with bloating. Student says, "Wait but I thought *Giardia* was fresh water. If she went scuba diving in the ocean then how is that *Giardia*?" → People can go scuba diving in fresh water lakes too bro. Don't know what to tell you. - Flagellated protozoan. - Tx = metronidazole.

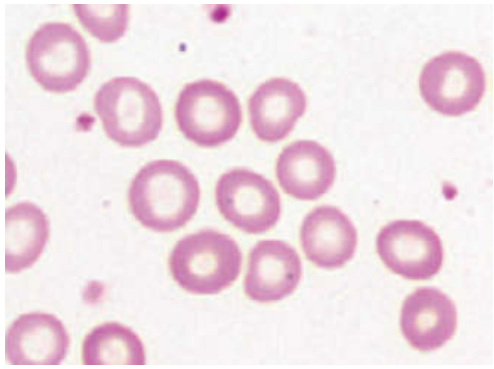
HY GI parasitic (helminth) infections	
- USMLE wants you to know **routes of acquisition** for the nematodes (i.e., ingested or through feet, etc.). - You do not need to obsess over exactly which anti-helminthic agent treats which organism. Waste of time. What USMLE wants you to know is the big-picture concept that -bendazoles (i.e., mebendazole) and pyrantel pamoate are the two agents that treat nematodes primarily. - Praziquantel is used for cestodes and trematodes. Albendazole is the odd one out that is used for neurocysticercosis (*Taenia solium*; cestode). - USMLE doesn't give a fuck about diethylcarbamazine.	
Nematodes (roundworms)	
Ascaris lumbricoides	- Ingested. Giant roundworm; causes intestinal obstruction. (Ascariasis)
Enterobius vermicularis	- Ingested. Causes perianal itching / (+) tape test in children. (Enterobiasis)
Angiostrongylus	- Ingested by eating raw slugs/snails. Causes eosinophilic meningitis.
Trichinella spiralis	- Ingested by eating pork/**bear** meat. Causes triad of fever, periorbital edema, and myalgias. (Trichinosis)
Strongyloides stercoralis	- Acquired through soil/one's feet. Causes intestinal obstruction + pulmonary symptoms (apparently travels to lungs via blood, then travels up respiratory tree into esophagus, where it is swallowed).
Hookworms	- Acquired through soil/one's feet. "Hookworms" refers to *Ancylostoma duodenale* and *Necator americanus*. - Cause **microcytic anemia / ↓ hemoglobin** due to sucking blood in GI tract.
Cestodes (tapeworms)	
Diphyllobothrium latum	- Fish tapeworm. Nothing else you need to know.
Taenia solium	- Pork tapeworm. Causes cysticercosis (muscle pain/cysts) + neurocysticercosis ("Swiss cheese" appearance of brain or soap bubbles in ventricles).
Echinococcus granulosus	- Aka hydatid worm, or dog tapeworm; causes hydatid cyst disease.

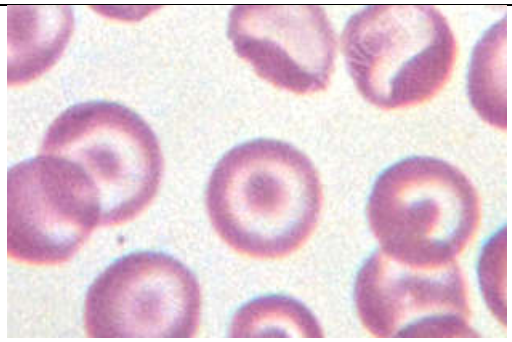
	- Acquired from dogs (clearly). - Causes liver cysts. Only point you need to know is that you do not biopsy because this can cause anaphylaxis. Tx = surgically remove.
Trematodes (flukes)	
Schistosoma	- Penetrates skin when swimming; snail is host. - *Schistosoma mansoni/japonicum* can cause portal hypertension. Not HY but shows up once on NBME. Not to be confused with *Schistoma hematobium*, which causes squamous cell carcinoma of the bladder.
Paragonimus westermani	- Ingested from crab meat or crayfish. - Aka lung fluke. Most common cause of hemoptysis in the world.
Clonorchis sinensis	- Ingested from fish. Causes cholangiocarcinoma.

HY GI pharm for Surg	
PPIs	- Omeprazole. - Shut of proton pumps on parietal cells → ↓ acid secretion. - Irreversible and non-competitive; more efficacious than H2 blockers, which are reversible and competitive. - Choose PPIs over H2 blockers for Tx of most things, like GERD, ulcers, and *H. pylori.* - Diagnosis of GERD is 2-week trial of PPIs. - PPIs cannot be given with sucralfate or -azole antifungals (on NBME).
H2-blockers	- Cimetidine, ranitidine. - Not used often. PPIs more efficacious. - One 2CK Surg form has "2-week trial of H2 blocker" as correct for diagnosis of GERD, but PPIs aren't listed. As I said above though, if you're forced to choose between a PPI and H2 blocker, the PPI is correct basically always.
Metoclopramide	- As discussed earlier, anti-emetic + prokinetic agent (means ↑ peristalsis). - D2 antagonist but also an antagonist of serotonin 5HT3 and agonist of 5HT4 receptors. The effects on serotonin receptors ↑ gut peristalsis. - HY use on USMLE is **diabetic gastroparesis** (i.e., sounds like GERD but patient has severe diabetes, where answer is metoclopramide, not PPI). - Because it's a D2-antagonist, adverse effects are similar to the anti-psychotics – i.e., prolong QT interval, hyperprolactinemia, anti-pyramidal effects (acute dystonia, parkinsonism, akathisia, tardive dyskinesia). - Metoclopramide as cause of parkinsonism, where the answer is "discontinue metoclopramide" as next best step before simply giving the patient Parkinson disease meds.
Ondansetron	- Powerful anti-emetic classically used for nausea/vomiting from chemoradiotherapy. - 5HT3 antagonist.
Erythromycin	- Motilin receptor agonist that can be used for diabetic gastroparesis. - Also a 50S ribosomal subunit inhibitor (antibiotic).
Misoprostol	- PGE_1 analogue. Used for **NSAID-induced ulcers** after PPIs. - NSAIDs ↓ prostaglandins. Therefore misoprostol is the replenished prostaglandin. - Prostaglandins ↓ acid production, ↑ mucous and bicarb production, and ↑ gastric mucosal blood flow.
Magnesium	- Antacid. Causes diarrhea.
Aluminum.	- Antacid. Causes constipation ("Aluminimum amount of feces.")
Calcium	- Antacid. Causes milk-alkali syndrome (↑ Ca^{2+} + ↑ $HCO3^-$+ urolithiasis). - Bisphosphonates (e.g., alendronate) and tetracyclines (e.g., doxycycline) cannot be taken with calcium or iron supplements (i.e., divalent cations), since the latter chelate the drugs and ↓ absorption / oral bioavailability. USMLE loves this. - USMLE will give you, e.g., prostatitis treated with ciprofloxacin, + list all sorts of other meds the patient is on, as well as a calcium supplement, and then tell you the patient's condition isn't improving + ask why → answer = calcium carbonate.
Bismuth	- Can be used as part of *H. pylori* regimens (but not first-line).

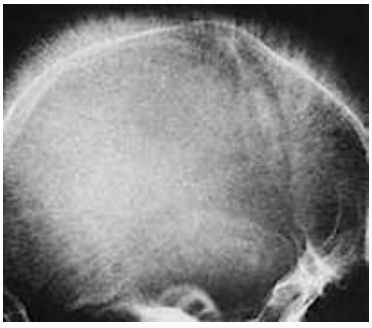
Sucralfate	- Can be used to coat ulcers / form a barrier to protect against acid and ↓ pain. - Cannot take with PPIs. The latter ↓ ability of sucralfate to crosslink/work correctly.
5-ASA compounds	- Mesalamine, sulfasalazine; NSAIDs used for IBD, as discussed earlier.
TNF-α blockers	- Infliximab, adalimumab, and etanercept. - USMLE likes these for IBD *after* 5-ASA compounds and steroids are attempted. - Infliximab and adalimumab are monoclonal antibodies against TNF-α. - Etanercept is a recombinant receptor that mops up soluble TNF-α.
Octreotide	- Somatostatin analogue used for esophageal varices Tx after banding. - Acts by ↓ portal blood flow.
Propranolol	- Beta-blocker that can be used for esophageal varices prophylaxis (not Tx).
Lactulose	- Used to Tx hyperammonemia in cirrhosis. - Carbohydrate that is metabolized by gut bacteria into acidic substrates that trap ammonia as ammonium. The latter is then excreted (ions aren't absorbed as easily). - The answer on USMLE is "acidification, $NH4^{+}$" for what lactulose causes in the gut.
Neomycin	- Used to Tx hyperammonemia in cirrhosis. - Antibiotic that kills ammonia-producing bacteria.
Loperamide	- Mu-opioid receptor agonist used to treat diarrhea-predominant IBS. - Opioids cause constipation, so "this is a good thing" in this case.
Ezetimibe	- Blocks cholesterol absorption in the small bowel.
Cholestyramine	- Bile acid sequestrant. Causes reduced enterohepatic circulation of bile acids at terminal ileum → liver must now convert more cholesterol into bile acids in order to replenish them → liver pulls cholesterol out of the blood to accomplish this.
Orlistat	- Pancreatic lipase inhibitor used for obesity; can cause steatorrhea.
Nystatin	- Mouth wash used for oropharyngeal candidiasis. - Forms pores in fungal ergosterol membrane.
Fluconazole	- Used for candidal esophagitis. - Inhibits ergosterol synthesis (i.e., lanosterol → ergosterol) by inhibiting 14α-demethylase. USMLE can write this as "P-450-mediated demethylation reaction."
Acyclovir	- HSV esophagitis; HSV1/2 causes "punched out" ulcers. - DNA polymerase inhibitor; causes chain termination; resistance = altered viral thymidine kinase.
Ganciclovir	- CMV esophagitis and colitis; causes linear/confluent ulcers. Same MOA as acyclovir.
CAP	- Clarithromycin, Amoxicillin, PPI. - First-line Tx for *H. pylori*. - If patient still has (+) urease breath test after 4 weeks, assume antibiotics resistance and switch out the CA and add tetracycline, metronidazole, and bismuth (keep PPI).
Vancomycin	- Given orally to treat *C. difficile.*
Metronidazole	- "Anaerobes below the diaphragm." - Used for diverticulitis in combo with a fluoroquinolone. - Tx for *Giardia* + *Entamoeba.*
Imipenem	- Carbapenem antibiotic that has fantastic penetration for pancreatitis.
Mebendazole	- Used for most nematodes (i.e., *Ascariasis, Enterobiasis,* hookworms, etc.). - Inhibits microtubules. An NBME Q has "tubulin" as the answer for protein it inhibits.
Albendazole	- Same as mebendazole, but for whatever reason it's known as a preferred agent for neurocysticercosis, which is caused by a cestode (*Taenia solium*).
Pyrantel pamoate	- Agent used against nematodes. Shows up on an offline NBME as answer. - Causes paralysis.
Praziquantel	- Used for cestodes and trematodes. - Causes paralysis.

Surg Heme/Onc

<table>
<tr><th colspan="4">Iron deficiency anemia vs thalassemia</th></tr>
<tr><th>Variable</th><th>IDA</th><th>Thalassemia</th><th>HY points</th></tr>
<tr><td>Hb</td><td>↓</td><td>↓</td><td rowspan="5">- Thalassemia means abnormal synthesis of either the α or β globin chain. It presents two ways on USMLE:
1) As a microcytic anemia despite normal iron + ferritin; or
2) As a patient who has a microcytic anemia despite iron supplementation.
- Most IDA is seen in menstruating women, even if their menses aren't heavy. NBMEs will actually explicitly say menses are normal/not heavy, but Dx is still IDA.
- NBME will give you microcytic anemia in a woman who's 32, and then the answer is just "check serum iron and ferritin." This is because thalassemia is often misdiagnosed as IDA, where if we get the iron studies back and see iron + ferritin are normal, we say, "Ok that's thalassemia, not IDA." → next best step = hemoglobin electrophoresis.
- Obgyn forms like to give pregnant women at first antenatal screening with microcytic anemia who are given iron for 3 weeks but show no improvement. Q won't mention anything about patient's iron or ferritin levels. So you say, "Ok, this is a microcytic anemia despite iron supplementation, meaning the iron + ferritin are actually normal, so this is thalassemia." → next best step = hemoglobin electrophoresis.</td></tr>
<tr><td>MCV</td><td>↓</td><td>↓</td></tr>
<tr><td>Serum iron</td><td>↓</td><td>Normal</td></tr>
<tr><td>Ferritin</td><td>↓</td><td>Normal</td></tr>
<tr><td>Hb electrophoresis</td><td>Normal</td><td>α: normal;
β: ↑ HbA2 + HbF
(See discussion at bottom of table)</td></tr>
<tr><td>What happens if we give iron?</td><td>Improvement</td><td>No improvement (HY)</td></tr>
<tr><td>RDW</td><td>↑ (HY)</td><td>↓/Normal (HY)</td><td>- Red cell distribution width.
- All you need to know is, basically always, ↑ RDW means IDA straight-up 9/10 Qs.
- The only Q where ↑ RDW is seen in thalassemia is on CMS FM form 4, where IDA is wrong, but they give ferritin within the normal range, so IDA isn't possible. But the ↑ RDW is essentially an erratum, as this is rare as fuck.
- Therefore we can essentially say that ↑ RDW is sensitive but not specific for IDA, where if RDW isn't ↑, we can rule-out IDA, but be careful about automatically saying "must be IDA" if RDW is ↑.</td></tr>
<tr><td>Blood smear</td><td>Pale RBCs</td><td>Target cells</td><td>- IDA causes pale RBCs (i.e., ↑ central pallor). Not dramatic.
 
- Target cells are highly buzzy and pass-level for thalassemia. Probably 14/15 Qs on USMLE mentioning target cells are thalassemia (α or β).</td></tr>
</table>

			 - There is one NBME Q with target cells in a splenectomy patient, but they're a background finding and don't facilitate answering the Q in any way.
Tx	Iron	Transfusions if severe	- Oral iron is ideal over parenteral (IV) or intramuscular iron always. I mention this because answers such as "parenteral iron" are wrong/distractors basically always. I'll say to a student, "You want to give this patient IV iron?" - Oral iron can cause constipation and black stools. USMLE wants you to know iron, aluminum (antacid), and verapamil all cause constipation. - Transfusional siderosis (secondary hemochromatosis) is sophisticated way of saying iron overload due to repeated blood transfusions, e.g., in β-thalassemia major. Each RBC transfusion contains iron. Use chelators (e.g., deferoxamine) to treat.

- Thalassemia is impaired synthesis of either the α- or β-globin chain in hemoglobin.
- α-thalassemia can have 1-4 alleles mutated; β has 2. Don't worry about obscure intermediate types, etc.
- USMLE overwhelmingly focuses on adults for both α- and β-thalassemia, since these patients are usually misdiagnosed as having iron deficiency anemia. That's why you need to know about thalassemia, so you don't misdiagnose patients with IDA.
- Adults with α-thalassemia will be asymptomatic (1 mutation), where they have an incidental microcytic anemia picked up on antenatal screening (hence Obgyn forms love thalassemia). Adults with 2 mutations have symptoms akin to IDA, but they'll fail to improve with iron supplementation. 3 α mutations (HbH disease; β4 tetramer) will present as a sick kid; 4 α mutations (Hb Barts; γ4 tetramer) will be fatal *in utero*.
- β-thalassemia minor will be 1 mutation in an adult who presents similar to 2 α-thalassemia mutations (i.e., similar to IDA).
- β-thalassemia major is 2 mutations and will present as a sick kid similar to 3 α-thalassemia mutations.
- It is exceedingly HY you know that β-thalassemia has ↑ HbA2 (α2/δ2) and HbF (α2/γ2); α-thalassemia (1 or 2 mutations) has a normal hemoglobin electrophoresis.
- An NBME Q gives two asymptomatic Vietnamese parents and then asks what their child is at increased risk for → answer = α thalassemia trait (1 mutation); wrong answer is β-thalassemia minor. Students think this Q is a guessing-game about whether α or β is more likely in Vietnamese; it's not. The reason it's α thalassemia trait is because both parents are asymptomatic, which is possible with 1 α mutation. If either of the parents had a β mutation, he or she would be symptomatic.
- Sick children with thalassemia are known to get "chipmunk facies/skull" and hepatosplenomegaly occur due to ↑↑ extramedullary hematopoiesis.

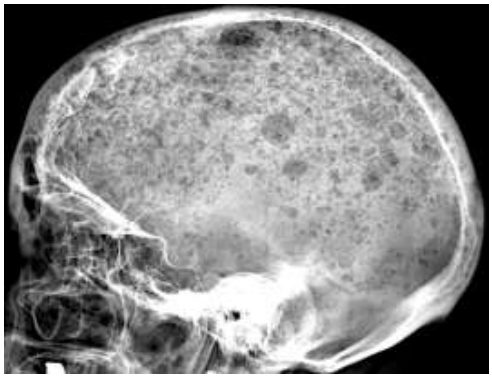


Anemia of chronic disease (AoCD)

- The answer on USMLE for anemia (↓ Hb) in patient who has **renal failure,** an autoimmune disease (i.e., RA, JRA, IBD, SLE, etc.), hepatitis B/C, HIV, or malignancy.
- Patients will have low iron but **normal ferritin** (or sometimes slightly high). Ferritin is the most sensitive marker of iron stores, which means these patients are not iron deficient.
- There's two main reasons iron is low in AoCD: 1) RBC production is decreased because cytokines (namely IL-6 and TNF-α) block iron release from storage sites, and 2) hepcidin, a molecule produced by the liver, decreases gut absorption of iron and promotes its storage in cells instead.
- It's to my observation that **renal failure** is the highest yield cause of AoCD on USMLE. This is due to "cytokine-mediated EPO deficiency." This is an answer on an NBME form. EPO is the treatment for AoCD only if renal failure is the etiology. Otherwise the Tx is merely addressing the underlying condition (e.g., the RA), and EPO is a wrong answer.
- There is an AoCD NBME Q for RA where they ask how to treat and answer is "no specific measures indicated." Student says, "Wait, I thought we were supposed to address the underlying condition, so I was confused by this answer choice." I agree. So don't take it up with me. Take it up with NBME.
- One of the highest yield points I can communicate is that even though classic MCV is normal in AoCD, it will be ↓ in ~50% of NBME Qs. I've seen students say, "Oh it can't be AoCD cuz MCV is low." No. USMLE loves giving low MCV in AoCD, especially on 2CK forms. I've seen MCVs of 72 and 75 in JRA Qs.

Multiple myeloma

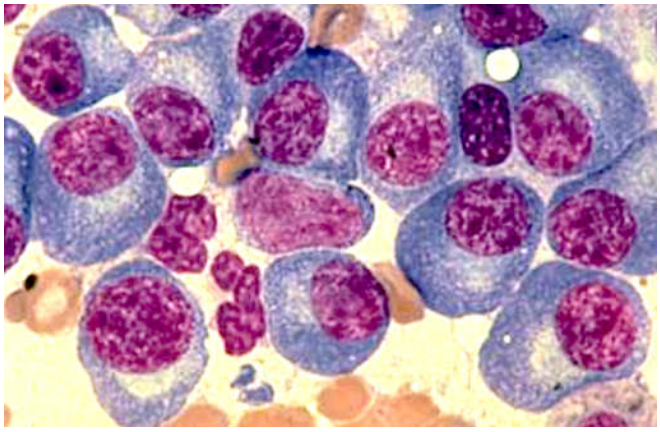
- Cancer of plasma cells.
- Plasma cells normally secrete immunoglobulins, so we have ↑↑ serum immunoglobulins.
- Multiple myeloma buzzy presentation is patient over 50 with mid-back pain + hypercalcemia.
- ↑ serum Ca^{2+} occurs due to lytic lesions of bone caused by the proliferating plasma cells. This can present as pathologic rib fractures or "pepper pot skull."



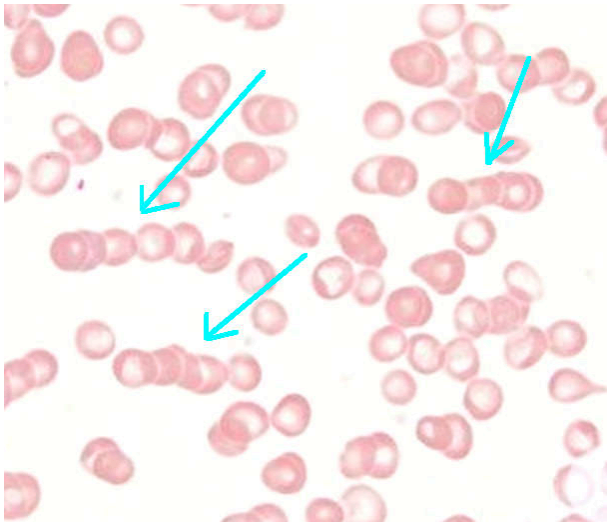
- There is NBME Q floating around where they give lytic lesions of the humerus as well; not buzzy/textbook in comparison to rib/skull lesions, but they show image similar to following:



- First step in diagnosis is **serum protein electrophoresis (SPEP),** which shows ↑↑ serum IgG kappa or lambda light chains. This is known as an M-spike (monoclonal spike; this does not mean IgM). Then **bone marrow biopsy** confirms the diagnosis, showing >10% plasma cells.
- Smear in multiple myeloma will show plasma cells (blue cells below) with "clockface chromatin," which is the appearance ascribed to the nuclei (purple below).



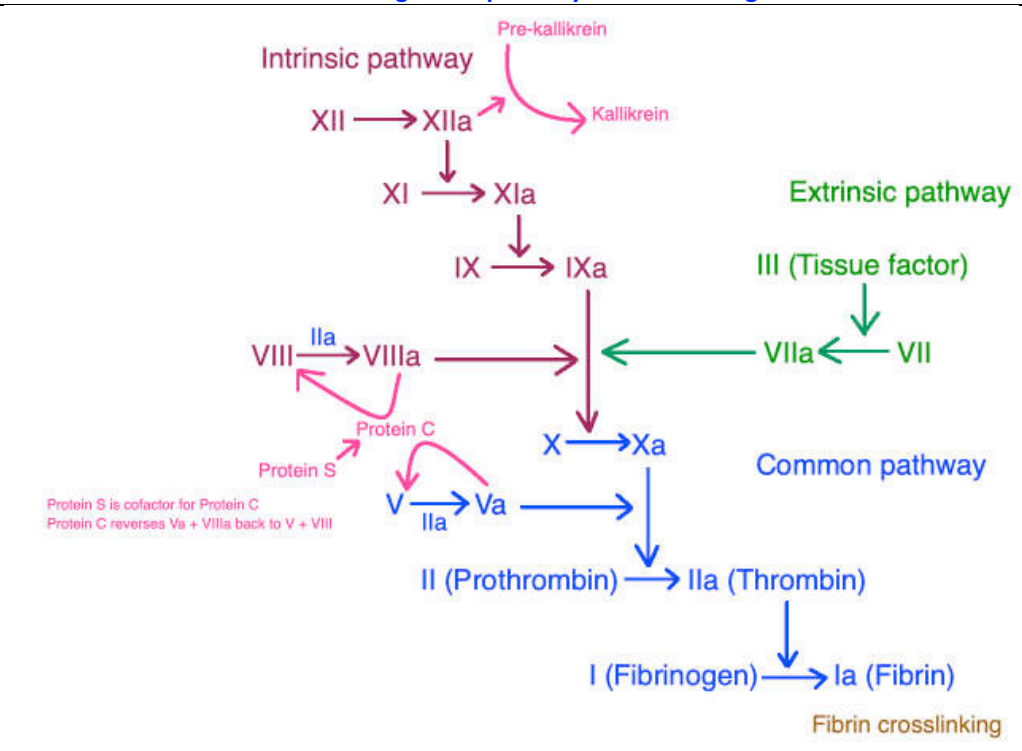
- Urinalysis will show Bence Jones proteinuria, which is simply the IgG light chains in the urine.
- Amyloidosis is proteins depositing where they shouldn't be depositing. Immunoglobulins are proteins. Since these are flying around in the blood, they deposit in the heart (**cardiac amyloidosis;** S4 heart sound with diastolic dysfunction) and kidney (**renal amyloidosis;** nephrotic syndrome).
- The IgG light chains stick to RBCs, causing the RBCs to stick to each other as stacks (i.e., Rouleaux). These are heavy and sediment quickly on centrifugation, hence ↑ ESR. USMLE can show you a picture of the Rouleaux rather than the clockface chromatin, where you need to know instantly the diagnosis is MM.



- MCV can sometimes be ↑ in MM (e.g., 108; NR 80-100). Don't be confused by this. It is common. This is caused by 1) ↑ immunoglobulins being transported by RBCs (thereby ↑ RBC size); 2) ↓ RBC production within bone marrow due to infiltrating plasma cells, causing ↑ RBC size to compensate; and 3) B9 and B12 deficiencies can be caused by MM.
- Tx is initially IV bisphosphonate to ↓ risk of bone lysis.
- Should be noted that patients with bone marrow biopsy showing <10% plasma cells have a condition known as MGUS (monoclonal gammopathy of undetermined significance). There is small annual risk of progression to multiple myeloma. These patients might have fatigue and/or ↑ MCV but no hypercalcemia or renal dysfunction.

Waldenstrom macroglobulinemia
- Cancer of plasmacytoid cells. - "-Oid" means looks like but ain't. So these are cells that look like plasma cells but they ain't plasma cells. - Causes IgM M-protein spike on SPEP (MM is IgG M-protein spike). - In contrast to MM, does not cause hypercalcemia, lytic lesions, Bence-Jones proteinuria, or amyloidosis. - Waldenstrom causes **hyperviscocity syndrome,** which will present as headache, blurry vision, tinnitus, or Raynaud phenomenon.

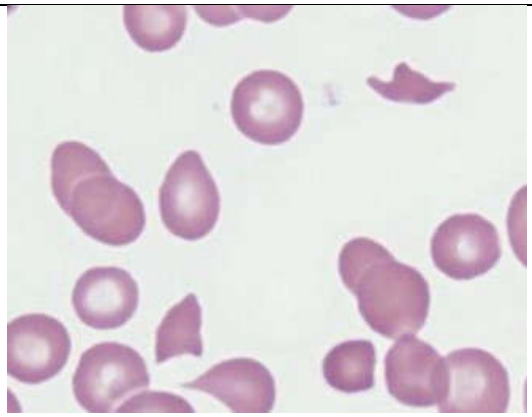
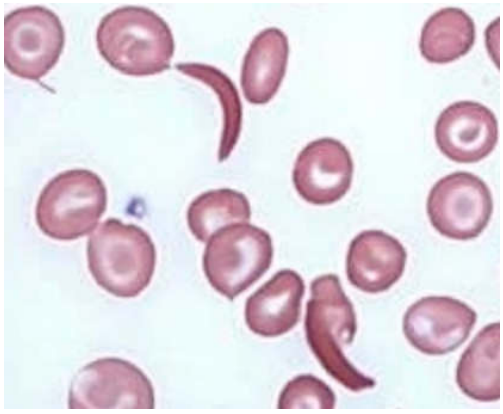
Polycythemia vera vs secondary polycythemia	
Polycythemia vera	- JAK2 mutation causing "proliferation of hematopoietic stem cells" (answer on NBME). - Qs will give ↑ Hb at 18+ (NR 12-17.5 g/dL) *and* ↑ WBCs and/or platelets. - In other words, at least 2/3 cell lines will be ↑, with RBCs always ↑. - ↑ Hb can cause hyperviscosity syndrome. - Basophilia can cause pruritis after a shower. - EPO is suppressed due to ↑ Hb. - USMLE wants **phlebotomy** as the answer to ↓ acute symptoms associated with hyperviscosity syndrome (i.e., "What is next best step to ↓ patient's headache / blurry vision?"). - Serial phlebotomy and hydroxyurea can be answers to ↓ recurrent symptoms; the USMLE will not give both at the same time.
Secondary polycythemia	- Isolated ↑ in RBCs due to ↑ EPO – i.e., WBCs and platelets are normal. - Seen in those with lung disease where arterial O2 tension is low → kidneys sense ↓ O2 and secrete EPO → acts on bone marrow to stimulate erythroid precursors. - Also seen in **renal cell carcinoma,** which can secrete EPO (and PTHrp causing hypercalcemia). - Exogenous administration (e.g., cyclists).

Brief clotting factor pathway review for Surg


- Prothrombin time (PT) reflects the functioning of the extrinsic pathway; activated partial thromboplastin time (PTT; aPTT) reflects intrinsic pathway.
- In other words, if PT alone is high, then the extrinsic pathway is fucked up.
- If PTT alone is high, then the intrinsic pathway has a problem.
- If both PT and PTT are elevated, then the common pathway has an issue.
- Normal PT is 10-15 seconds.
- Normal PTT is 25-40 seconds.
- Normal bleeding time is 2-7 *minutes*. This relates to **platelets** and has nothing to do with clotting factors.

HY Core DDx	
ITP	- Idiopathic/immune thrombocytopenic purpura. **- Antibodies against glycoproteins IIb/IIIa on platelets.** **- Causes ↓ low platelets; ↑ bleeding time; no change PT and PTT.** - Bleeding time is high because platelets are fucked up. PT and PTT and normal because these refer to clotting factors, which have nothing to do with ITP. - Normal platelet count is 150-450,000/μL. Platelets in ITP Qs are usually <100,000. - Precipitated by viral infection, where antibodies against viral proteins cross-react with platelets (disrupts platelet aggregation); type II hypersensitivity. - Viral infection **will be asymptomatic in ~50% of vignettes.** In other words, the Q need not mention the viral infection, so don't get confused if you don't see it. - Will present two ways on USMLE: 1) School-age kid (i.e., 12) who has coryza for a few days followed by petechiae and/or nosebleeds. 2) Woman 30s-40s + no mention of viral infection + random bruising + bleeding time 9 minutes. - For the latter vignette, if they say exact same Q but bleeding time 6 minutes (normal), then answer is domestic abuse, not ITP. - There is an unusual 2CK Q on new NBME where they say girl only has heavy menses (classically a clotting factor presentation for vWD) and the answer is ITP, but vWD isn't listed and the other answer choices are clearly wrong. Just letting you know this exists. - Tx = steroids, then IVIG, then splenectomy, in that order. - If splenectomy is performed and platelets return to normal range, but then 6 months later they fall again, the answer is "accessory spleen." Sounds weird, but a small % of people have a second spleen the size of a pea that can grow in size if the main spleen is removed.
vWD	- von Willebrand disease; **autosomal dominant.** - von Willebrand factor (vWF) normally bridges glycoprotein Ib on platelets to vascular endothelium / underlying collagen (platelet adhesion). **- ↑ bleeding time; ↑ PTT (only half the time); no change PT; normal platelets.** - ↓ vWF means ↓ platelet function (and hence ↑ BT), but platelet count is unchanged. - vWF also has secondary role of stabilizing factor VIII in plasma; since we have ↓ vWF, the intrinsic pathway gets fucked up, so we have ↑ PTT. But bear in mind, this effect on factor VIII *is only a mere secondary* role, so PTT isn't always ↑. It's to my observation that only ~50% of vWD vignettes give ↑ PTT. PT is normal because the extrinsic pathway is fine. - 8/10 Qs presents as a teenager or young adult with a mix of one platelet problem and one clotting factor problem. - Platelet problem = mild, cutaneous findings (i.e., petechiae, bruising) + epistaxis. - Clotting factor problem = excessive bleeding with tooth extraction + heavy menses. (Hemarthroses are also clotting factor but are seen in hemophilia, not vWD, as I explain below). - For example, the Q can give 17-year-old girl who has nosebleeds + heavy periods; or 17-year-old girl who has bruising + Hx of excessive bleeding with tooth extraction (i.e., we have one platelet problem + one clotting factor problem). This is 8/10 Qs I'd say.

	- 1/10 Qs might give only isolated platelet or clotting factor problem (i.e., only say the patient has Hx of excessive bleeding with tooth extraction), but they'll also say *one of the parents* has Hx of nosebleeds → so you say, "Oh ok. vWD is AD, so that's your combo of platelet problem + clotting factor problem linked via autosomal dominant inheritance." - 1/10 Qs will say the patient has a cut on the finger that takes longer to heal than normal + has heavy menses + normal platelets + has normal PT and PTT + they don't mention BT. The cut on the finger is their way of saying BT is elevated. Despite only an isolated ↑ in BT here, since platelets are normal, we know it's not ITP and is instead vWD. - Ristocetin cofactor assay is abnormal. All you need to know is that this is some test that is abnormal if platelet adhesion is fucked up. Tx = desmopressin (DDAVP), which can in theory help boost functional vWD levels.
Hemophilia A+B	- Deficiency of factors VIII and IX, respectively. - **Both X-linked recessive.** - Isolated ↑ PTT; BT and PT are normal. - Factors VIII and IX are in the intrinsic pathway, so PTT is ↑. Extrinsic pathway is unaffected so PT is normal. Platelets have nothing to do with this condition, which is why bleeding time is normal. - Will present two ways on USMLE: 1) School-age boy with hemarthrosis who has isolated ↑ PTT. They can also say he has a maternal uncle who died years ago from minor head trauma. 2) Neonate who has excessive bleeding with circumcision. - Hemophilia A is treated with DDAVP and factor VIII replacement. - Hemophilia B is treated with factor IX replacement. DDAVP doesn't work apparently. - If the Q tells you a patient with hemophilia A has persistently elevated PTT despite factor VIII supplementation, the answer is "check for antibodies against factor VIII." Sometimes patients can develop resistance against exogenous factor replacements.
Vitamin K deficiency	- **Causes ↑ PT, ↑ PTT, no change bleeding time.** - Vitamin K is cofactor for gamma-glutamyl carboxylase, which is an enzyme that activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. - Bleeding time is normal because platelets aren't affected. - Will present two ways on USMLE: 1) Neonate who has bleeding from umbilical stump +/- retinal hemorrhages. The latter can sound like shaken-baby syndrome, but the bleeding from the umbilical stump is highly buzzy for vitamin K deficiency. 2) Adult who's been on broad-spectrum antibiotics for 6 weeks for, e.g., endocarditis, who now requires a lower dose of warfarin → answer = "depletion of bowel normal flora" (Vitamin K is normally synthesized by colonic normal flora). - Warfarin inhibits vitamin K function, so if Vit K is ↓, we don't need as much warfarin.
DIC	- Disseminated intravascular coagulation. - Runaway effect of clotting factor/platelet consumption, precipitated by multifarious etiologies such as trauma, sepsis, amniotic fluid embolism, and treatment of AML. **- ↑ BT, ↑ PT, ↑ PTT, ↑ D-dimer, ↑ plasmin activity.** **- ↓ fibrinogen, ↓ platelets, ↓ clotting factors.** - Fibrinogen decreased because it is converted to fibrin. - Plasmin breaks down fibrin, so more fibrin means more plasmin is upregulated in an attempt to dissolve it. - D-dimer = fibrin degradation products; since more fibrin is being broken down, fibrin degradation products (D-dimer) ↑. - "Bleeding from catheter/IV sites" is 9 times out of 10 synonymous with DIC. However on one 2CK Surg Q, dilutional thrombocytopenia secondary to ↑ blood transfusions presents the same. - Blood smear shows **schistocytes.**

	
HUS	- Hemolytic uremic syndrome. - Presents as triad of: 1) thrombocytopenia; 2) hemolytic anemia with **schistocytes;** and 3) renal insufficiency with or without hematuria. - The combination of thrombocytopenia + schistocytosis = microangiopathic hemolytic anemia (MAHA). - Mechanism is: E. coli (EHEC O157:H7) and Shigella both secrete toxins (Shiga-like toxin and Shiga toxin, respectively) that cause inflammation and damage of renal microvasculature → platelets adhere to plug damaged endothelium → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells).
TTP	- Thrombotic thrombocytopenic purpura. - Presents as HUS triad + fever and neurologic signs. In other words: - Presents as pentad of 1) thrombocytopenia; 2) **schistocytes;** 3) renal insufficiency; 4) fever; 5) neurologic signs. - TTP is caused by antibodies against, or a mutation in, a protein called ADAMTS13, which is a metalloproteinase that breaks down vWF multimers → unable to break down platelet aggregates that plug renal microvasculature → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells). - Can present as a stroke-like presentation in young woman, where they just want "schistocytes" as the answer (Step 1 NBME). - 2CK NBME has "plasmapheresis" as Tx.
Sickle cell	 - **Autosomal recessive;** glutamic acid → valine missense mutation on the beta-chain. - Carrier status (one mutation) is referred to as sickle cell trait, which is less severe than sickle cell anemia (two mutations). - NBME wants "heterozygote advantage" as the basis for the existence of sickle cell – i.e., confers resistance to malaria.

	- Sickle cell crises can present as **abdominal or chest pain,** as well as dactylitis (inflammation of the fingers). - The abdominal pain, in particular, as the initial presentation for sickle crisis has become HY on NBME exams. A new 2CK form gives RLQ pain + low Hb in child from Libya, where "blood smear" is the answer for what will confirm the diagnosis. The Q is tricky because it almost sounds like appendicitis. - Diagnosed with blood smear, followed by hemoglobin electrophoresis to detect HbS. - Offline NBME Q asks which of the following best describes the molecular basis for sickling in a patient → answer = "gain of stabilizing hydrophobic interactions in the deoxygenated form of hemoglobin S." This is because valine is more hydrophobic than glutamic acid, so we have hydrophobic interactions enabling sickling. - Another NBME has "beta chain slips into a complimentary hydrophobic pocket on the alpha chain" as answer for sickling. - Sickling notably occurs with dehydration + increased acidity. This makes sense, since the surrounding environment would become less lipophilic, meaning the HbS would consolidate/collapse unto itself to minimize surrounding surface area contact. - **Hydroxyurea ↑ HbF** and can ↓ recurrence of sickle crises. Sounds weird, but the mere presence of more HbF simply means there's fractionally less HbS floating around.
Myelofibrosis	- Usually JAK2 mutation (same as polycythemia vera). - The answer on USMLE if they say "tear-drop-shaped RBCs" or "dry tap on bone marrow aspiration." - Massive splenomegaly seen in 100% of questions. - I've seen NBME also write "tear-drop shaped poikilocytes" (fancy word that means abnormal RBCs seen as >10% on a smear). - In myelofibrosis, the proliferation of abnormal cells in the bone marrow, notably megakaryocytes and granulocytes, leads to the release of growth factors and cytokines that stimulate deposition of collagen, resulting in bone marrow fibrosis.
Essential thrombocytosis (ET)	- Aka essential thrombocythemia. - Usually JAK2 mutation (same as myelofibrosis and polycythemia vera) causing over-proliferation of platelets. - USMLE Q will tell you patient has pain in fingertips, Raynaud phenomenon, or headache (hyperviscosity symptoms) + platelet count is, e.g., 1.4 million/μL (NR 150,000-450,000/μL). - ET is a myeloproliferative neoplasm where platelets **can be >1,000,000/μL.** - The patient will present with no specific triggering event. Compare this to reactive thrombocytosis below. - Tx = aspirin + plateletpheresis (removing platelets from blood).
Reactive thrombocytosis (RT)	- Aka reactive thrombocythemia. - Triggering event (i.e., inflammation, infection, surgery) can stimulate thrombopoietin production and increased synthesis and release of platelets. - It is to my observation the USMLE loves to give reactive thrombocytosis in the setting of **infective endocarditis,** where they'll tell you platelet count is 900,000, and the student is like, "why the fuck are the platelets 900,000." It's just RT. - ET is usually women >50 and is symptomatic; RT is anyone and is usually asymptomatic. When RT occurs, it will be a super-high platelet count you incidentally notice as part of the stem, but the focus of the Q isn't the patient's symptoms related to the ↑ platelets. - For example, there is a difficult offline NBME Q where they give neutrophilic shift with platelets of 1.4 million, where it looks like it could be RT due to infection (i.e., neutrophilic shift usually indicates bacterial infection), but the answer is ET on this form, not RT, presumably because the platelet count is >1 million (rare in RT to be this high) and the patient has symptoms (pain in fingertips). - Tx = address underlying infection + give aspirin. - I haven't seen RT as an actual correct answer on an NBME form; I've only seen it baked into questions as an incidental finding where the labs show high platelets in the setting of infection, and you, as the student, need to not be thrown off by that or think it's weird. You just say, "Oh that's probably just RT because we have infection/sepsis here."

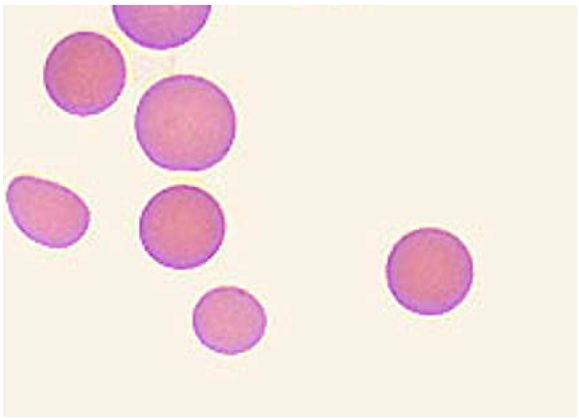
Glanzmann	- Aka Glanzmann thrombasthenia. I've seen this on NBME as just "thrombasthenia." - Deficiency of glycoproteins IIb/IIIa on platelets → defective platelet aggregation.
Bernard-Soulier	- Deficiency of glycoprotein Ib on platelets → defective platelet adhesion. - Same as with vWD, has abnormal ristocetin cofactor assay.
SLE	- Systemic lupus erythematosus; classically women 20-40s. - Most common presenting feature is arthritis (>90%). - Malar rash too easy for most Qs, although you should know this is a type III HS. - **Thrombocytopenia** is mega-HY for Step – i.e., 32-year-old woman with arthritis + low platelets = SLE. - RBCs and WBCs can also be ↓, but it's the ↓ platelets that's highest yield. The cell lines are down due to antibodies. In other words, if you get a patient with SLE where all cell lines are down, this is due to "increased peripheral destruction," not "deficient bone marrow production." - Discoid lupus = skin lesions. Mucositis = mouth ulcers. - ↑ risk for malignancy (i.e., non-Hodgkin lymphoma) → i.e., 44-year-old woman with SLE has seizure + ring-enhancing lesion seen on CT of head → answer = primary CNS lymphoma; Toxo is wrong answer. - Causes diffuse proliferative glomerulonephritis, as discussed in HY Renal PDF. - SLE in pregnant women can cause congenital heart block in neonates. Sounds weird, but shows up on NBME somewhere. - Treat standard flares with steroids (i.e., prednisone).

Thrombotic disorders	
Factor V Leiden	- Just be aware that there is a condition called FVL, which means the patient will have ↑ clots / DVTs. - Normally factor Va and VIIIa are reversed to V and VIII by protein C (protein S is a cofactor for protein C). In FVL, we have "activated protein C resistance," which means Factor V is resistant to cleavage by protein C. - Conditions such as Prothrombin mutation and MTHFR also cause ↑ clots, but are lower yield.
Protein C deficiency	- Causes clots/DVT, same as FVL, but is notably associated with skin necrosis if the patient receives warfarin.
Antithrombin III deficiency	- AT III is required to reverse IIa (thrombin), Xa, and IXa back to inactive II, X, IX. - Presents two ways on USMLE: 1) Young, otherwise healthy adult who has thromboses + the Q mentions nothing about PT or PTT. 2) Patient with nephrotic syndrome who has DVT, superficial thrombophlebitis, renal vein thrombosis, or varicocele. AT III is a protein that can be lost in the urine.
Antiphospholipid syndrome (APS)	- Antibodies against phospholipids that cause *in vivo* thromboses despite a paradoxical *in vitro* ↑ PTT (i.e., if PTT is ↑, you'd think we'd have bleeding diathesis, not thromboses). - PTT is fucked up because we need phospholipids for the *in vitro* assay to work. - *In vivo*, we have thromboses however because 1) antibodies against phospholipids in platelet membranes activate platelets and cause them to secrete procoagulant molecules; 2) these antibodies also bind to vascular endothelial cells, causing endothelial dysfunction and prothrombotic state; 3) the antibodies are pro-inflammatory, where ↑ activity of IL-6 and TNF-α activates tissue factor (factor III). **USMLE WILL NOT** assess you on these three mechanisms; I merely write them here because this diagnosis is exceedingly HY but remains perennially elusive and confusing as fuck for students. - Can be idiopathic (i.e., caused by antibodies against β2-microglobulin or cardiolipin), or **secondary to SLE.** - If APS is due to SLE, we merely call these antibodies "lupus anticoagulant," but they're the same antibodies.

	- For whatever magical reason, APS can cause **false(+) VDRL syphilis** tests.

Neutropenic fever (Febrile neutropenia)
- Neutropenia + fever, LOL! - Can present as **mouth ulcers, fever, and sore throat.** - Can occur in the setting of patient who has chemo- or radiotherapy-induced pancytopenia (i.e., RBCs, WBCs, and platelets are all ↓), or viral-induced neutropenia (i.e., Parvovirus B19), where you see that the patient has fever and neutrophils are low (+/- RBCs and platelets). - Questions will not usually come out and say, "Yes, the kid has neutropenia." The situation might be, e.g., a kid who's undergoing chemotherapy for AML who now has mouth ulcers and/or sore throat. - Medical emergency (i.e., the patient has infection but no ability to fight it). USMLE wants "IV broad-spectrum antibiotics" as the next best step. - After Abx, the USMLE wants granulocyte colony-stimulating factor (i.e., G-CSF, or filgrastim) as the answer. - I've never seen granulocyte transfusion as a correct answer on USMLE.

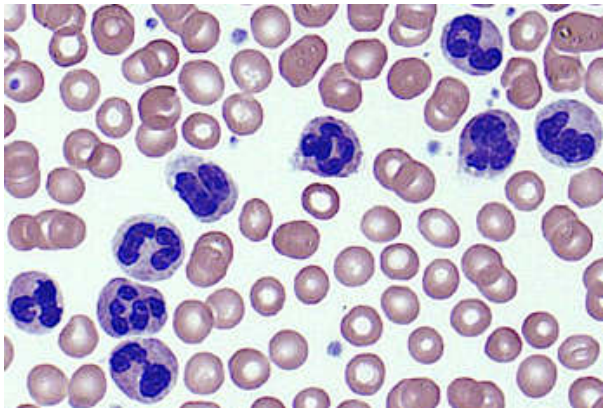
Aplastic anemia
- Means all cell lines (i.e., RBCs, WBCs, platelets) are ↓. - Can be due to Parvo B19, where it infects not just myeloid progenitors, causing neutropenia, but also erythroid progenitors and megakaryocytes, causing ↓ RBCs and platelets, respectively. - Can also be due to other viruses, such as HepA (weird, but on CMS Peds form 5). - Can be chemo- and/or radiotherapy-induced. - Fanconi anemia is rare, AR aplastic anemia that, for whatever reason, also presents with hypo-/aplastic thumbs/radii.

Spherocytosis	
Hereditary	- **Autosomal dominant** (one of the highest yield inheritance patterns on USMLE). - Heterozygous mutation in ankyrin, spectrin, or band proteins, causing an RBC cell membrane/cytoskeletal defect. This causes the RBC to deform away from the normal biconcave disc and into a spherical shape. - The USMLE might give you hereditary spherocytosis and then the answer is just "cell membrane," or "cytoskeleton," or "cytoskeletal defect." - Can be described as RBCs lacking central pallor. If they show you a smear, they look full and red.  - Causes a normochromic, normocytic anemia (asked on an NBME directly). - Can cause **pigment stone cholelithiasis** due to the spleen identifying the spherocytes as abnormal and increasing RBC turnover.

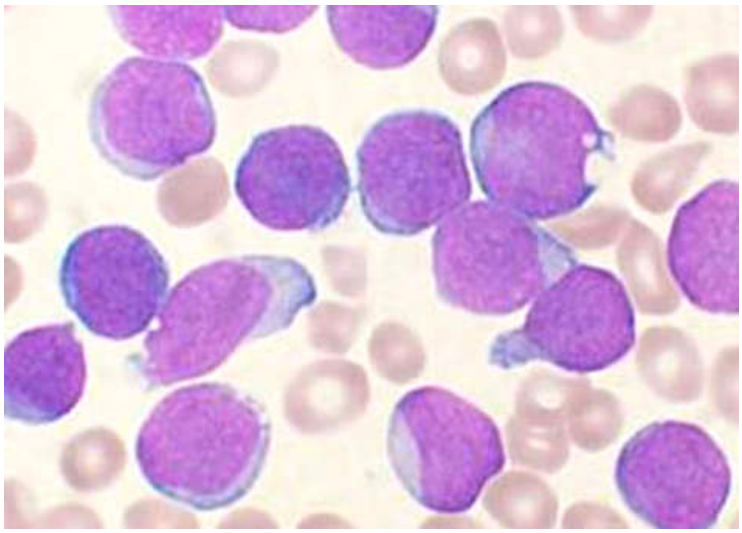
	- RBC mean corpuscular hemoglobin concentration (MCHC) can be ↑. This is not specific to spherocytosis but is nevertheless a rare lab parameter to see in Qs. It's to my observation that if they mention it, 4/5 times the Dx is spherocytosis. - Diagnose via osmotic fragility test and eosin-5-maleimide. - Coombs test is negative (meaning we don't have antibodies against RBCs). This is important in terms of contrasting with spherocytosis due to hemolysis (as I discuss below). - Treatment is splenectomy (to ↓ RBC turnover). The answer can appear on USMLE as "splenectomy and cholecystectomy" together. - USMLE can mention a child who has a low Hb + one of the parents had splenectomy +/- cholecystectomy performed in the past for "an anemia" → answer = hereditary spherocytosis. I also point out to the student the AD inheritance pattern based on one of the parents having the condition as well. - Q can also just ask straight-up which heme condition is often treated with splenectomy, and the answer is hereditary spherocytosis; students sometimes erroneously choose sickle cell. Remember, sickle cells causes autosplenectomy due to repeated microinfarcts within splenic microvasculature; it is not treated with splenectomy.
Hemolytic anemia	- Spherocytosis is not always due to hereditary spherocytosis and can sometimes be seen with drug- or infection-induced hemolysis. - In this case, the Coombs test is (+), meaning we have antibodies against RBCs. In contrast, with hereditary spherocytosis, the mechanism has nothing to do with antibodies, so Coombs test is (-). - Antibodies binding to RBCs can trigger complement-mediated damage, which can lead to deformation of the RBC membrane and cytoskeleton. - USMLE will give kid with recent viral infection who has low Hb and spherocytes on a smear + positive Coombs; answer = hemolytic anemia, not hereditary spherocytosis (i.e., answer = "no specific mutation").

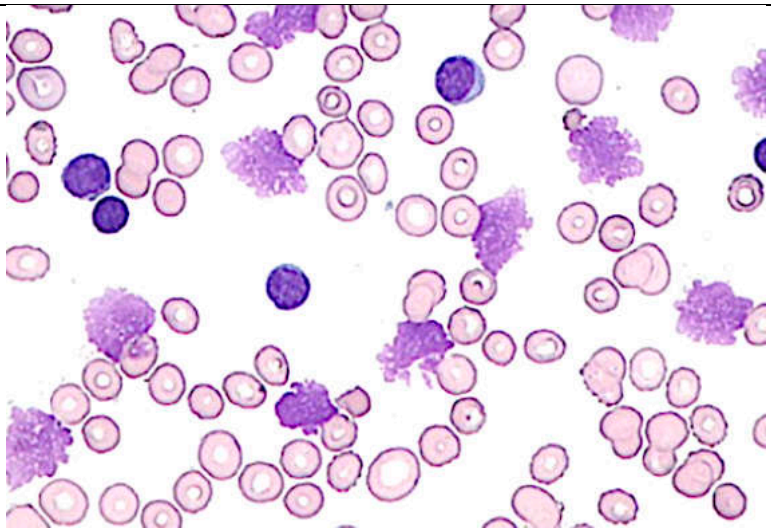
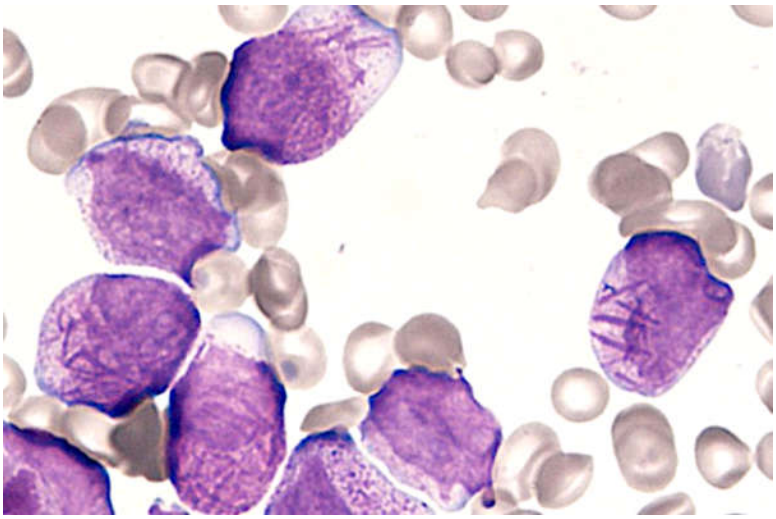
Leukemoid reaction

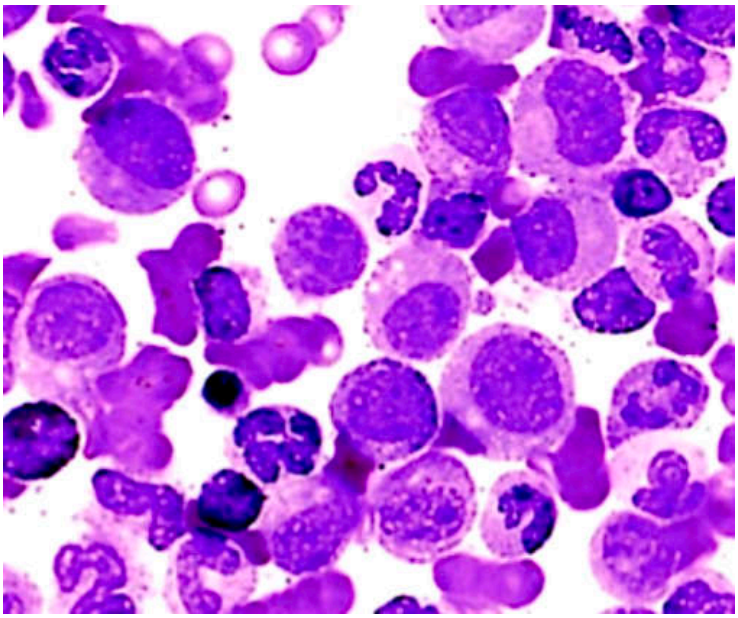
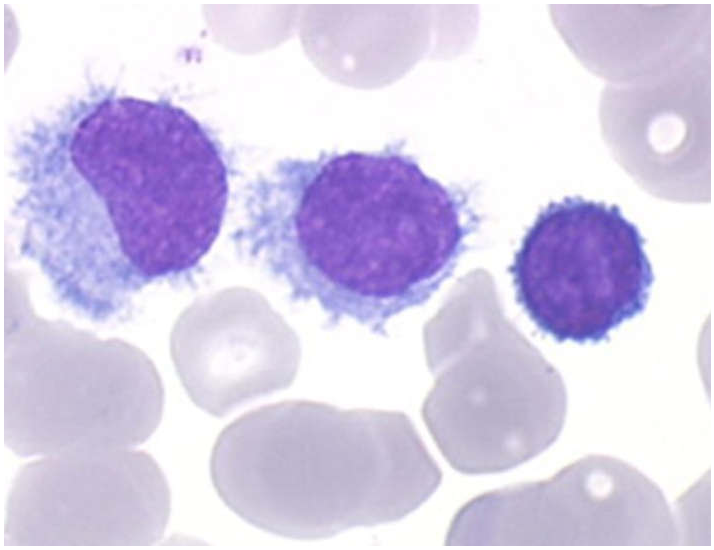
- Exaggerated immune response to infection in which WBCs go ↑↑ and can hence resemble leukemia.
- As I mentioned before, "-oid" means "looks like but ain't," so the patient's lab can look like leukemia (because the WBCs are so high), but it's not leukemia.
- Normal leukocyte count is 4-11,000/μL. Typical infections might cause WBCs to go ↑ to the teens or 20,000s/μL. But when WBC levels creep into the 30s, this becomes increasingly rare, even for patients who are septic, so we want to start thinking about leukemia. This is why if the patient doesn't turn out to have leukemia we call it leukemoid reaction.
- This will be the answer on USMLE for patient who has **infection + WBCs >30,000/μL, PLUS** they give you a blood smear that shows neutrophils or they say leukocyte ALP is elevated (in CML it's low).



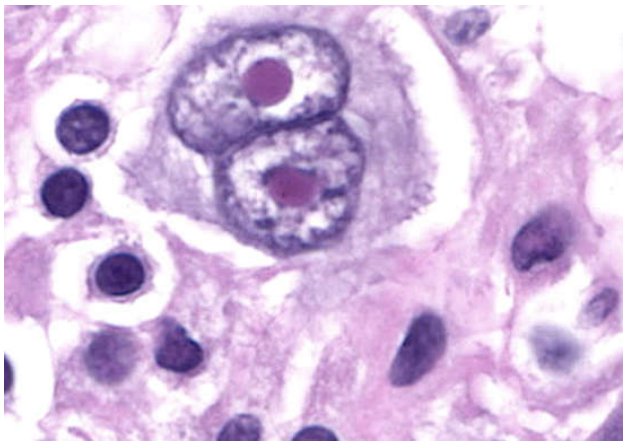
- This is the smear the USMLE will show you if they want leukemoid reaction. This is showing you **neutrophilia** (i.e., just lots of neutrophils). They can also write the answer as **reactive granulocytosis,** or as "increased release of leukocytes from bone marrow post-mitotic reserve pool."
- A typical leukemoid reaction Q might give you a UTI + WBCs 32,000/μL + the above smear. Not hard.

HY Leukemias	
- A leukemia is a cancer of white blood cells in the blood. - They are almost always B-cell in origin. - Apart from Sezary syndrome and mycosis fungoides (discussed below), if one of the HY lymphocytic leukemias (i.e., ALL, CLL) is T-cell in origin, USMLE will give an SVC-like syndrome (i.e., positive Pemberton sign with congested neck veins and/or facial erythema/swelling) due to a thymic lesion.	
ALL	- Acute lymphoblastic leukemia. - The answer on USMLE for a kid with leukemia (i.e., pre-adolescent). - Q will give you kid who's, e.g., 6-years-old, and has leukocyte count of 60,000/μL, where it's 90% lymphocytes. - **An important DDx of ALL is pertussis,** that for whatever reason, can cause an ALL-like picture where WBCs can be 30-40,000+, where it's 90% lymphocytes. This is called reactive lymphocytosis. The distinction with ALL is that, in pertussis, they'll say there's a cough and/or post-tussive emesis (vomiting after cough) and/or hypoglycemia (HY finding in pertussis). - ALL can be CD10 and TdT positive. Dumb, but I've received feedback of them showing up. - ↑ Risk with Down syndrome. The Q can give you standard vignette of Downs and then ask for what child is at greatest risk of developing → answer = "excess lymphoblasts."  ALL; lymphoblasts appear smooth and relatively uniform.
CLL	- Chronic lymphocytic leukemia. - Vignette will sound just like ALL but not in a kid (usually older adults). - The Q might say 70-year-old + WBCs 87,000/μL (90% lymphocytes), where they ask for the next best step → answer = "quantitative immunoglobulin assay," which can show ↑ monoclonal immunoglobulin sometimes in CLL. Sounds nitpicky, but it's an answer on an NBME. - CLL smear can show smudge cells, which are fragile cells on LM.

	 - Leukemic cells are CD5 and CD23 positive (asked on an NBME). - Can be associated with warm autoimmune hemolytic anemia (IgG against RBCs).
AML	- Acute myeloblastic leukemia. - Will show Auer rods on a smear.  - The image of Auer rods is exceedingly HY for the Step. They are composed of myeloperoxidase, which is a blue-green heme-containing pigment. - If an AML Q doesn't give you the buzzy image of Auer rods, they'll say there's "30% blasts" or "50% blasts." - Acute promyelocytic leukemia (APL; aka AML type M3) has a t(15;17) translocation. - Tx is all-*trans* retinoic acid (vitamin A). This causes the maturation of leukemic cells into mature granulocytes. - Tx of AML also leads to leukemic cell lysis, where release of Auer rods into the circulation can precipitate DIC.
CML	- Chronic myelogenous leukemia. - Caused by t(9;22) translocation (Philadelphia chromosome). - Results in formation of bcr/abl tyrosine oncogenic tyrosine kinase, which is a "fusion protein." - The stem will say there's all sorts of myelo-sounding cells – i.e., myelocytes, promyelocytes, metamyelocytes. 9/10 Qs that mention these cells are CML. - Leukocyte ALP is low. - Smear shows what I refer to as a "motley mix" or "soup," which shows all different types of cells.

	 - This is HY smear that differentiates it from leukemoid reaction (which instead shows neutrophilia). - Tx is with imatinib, which targets the bcr/abl tyrosine kinase.
Hairy cell	- Q will merely say patient has leukemia characterized by "cytoplasmic projections" that stains positive for tartrate-resistant acid phosphatase (TRAP).  - The stem can give low WBCs, which is unusual for leukemia. 9 out of 10 questions for leukemia will give ultra-HY WBCs. The NBME Q I've seen on hairy cell gives image similar to above + gives WBCs 3,500 + they say WBCs with cytoplasmic projections" + stains positive for TRAP.
Sezary Syndrome	- Cutaneous T-cell lymphoma that extends to the blood as a T-cell leukemia. - Caused by human T-cell lymphotropic virus (HTLV). - Has cerebriform-shaped T cells in blood stream.

	 - Presents classically as erythroderma (red skin).

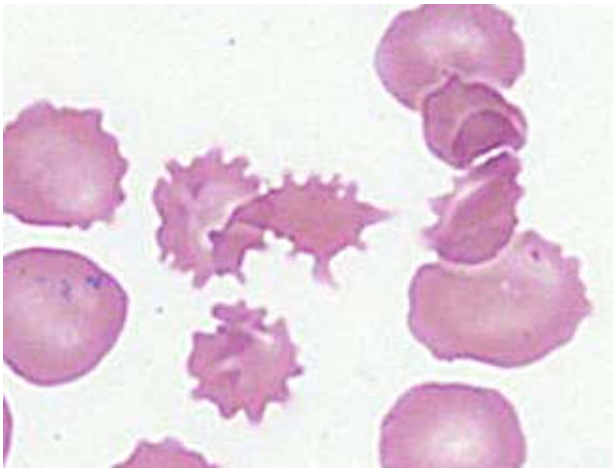
Lymphomas	
- A lymphoma is a cancer of the lymphatic system. - Usually starts in lymph nodes, but can also be present in other areas of the lymphatic system, such as the spleen and bone marrow. - Almost always B-cell in origin. - Lymphomas are categorized as Hodgkin or non-Hodgkin (NHL). - Epstein-Barr virus, HIV, general immunosuppression (i.e., chronic corticosteroids or immunosuppressant drugs), and autoimmune disease (e.g., SLE, Hashimoto) increase the risk for lymphomas.	
Hodgkin	- Characterized by Reed-Sternberg cells, which are B cells with a characteristic “owl eye” appearance.  - 100% of questions give a painless lateral neck mass or facial swelling (lymphadenopathy). - The USMLE Q will then give at least one additional finding of: 1) hepatomegaly; 2) mediastinal mass (not thymoma; this is mediastinal lymphadenopathy); 3) Virchow node (Troisier sign of malignancy; palpable left supraclavicular lymph node); this is not limited to gastric cancers; I’ve seen an NBME Q where they give this latter finding for Hodgkin. - B symptoms (i.e., fever, night sweats, weight loss) are common but not mandatory in vignettes. - Question can give normal leukocyte count and differential. An NBME Q gives WBCs of 10,000/μL (NR 4-11,000) and lymphocytes of 33% (NR 25-33). This is because lymphomas are cancers of lymphatic tissue. They’re not leukemias, which are cancerous WBCs floating around the blood.
Burkitt	- t(8;14) translocation of c-myc gene. - Presents classically as jaw lesion in African boy, or as abdominal lesion.

	 - Histo is buzzy "starry sky" appearance, which is a background of basophilic (purple) B cells with translucent macrophages.
Follicular	- t(14;18) translocation of *Bcl-2* gene. - Most common indolent NHL (which means less aggressive). - Presents as waxing/waning painless neck mass over 1-2 years.
Mycosis fungoides	- Cutaneous T-cell lymphoma characterized by cerebriform-shaped cells. - Less aggressive than the T-cell lymphoma of Sezary syndrome and does not extend to the blood as a T-cell leukemia.  - An important differential for "skin rash."

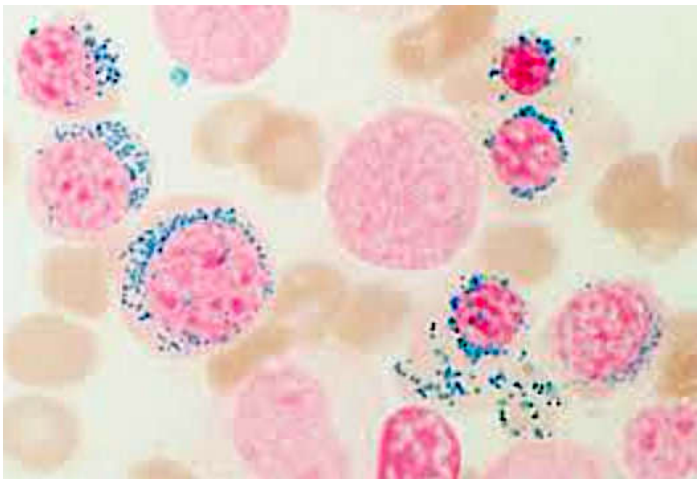
Warm vs cold autoimmune hemolytic anemia (AIHA)	
- Both present with (+) Coombs test, which just means antibodies against RBCs. - In other words, if a Q says Coombs is (+), we have Abs against RBCs; if it's (-), we don't.	
Warm	- IgG antibodies against RBCs. - "Warm" because agglutination of RBCs occurs at warmer temperatures. The stem can say agglutination occurs when the nurse holds the tube of blood by hand. - Seen on USMLE for autoimmune diseases like SLE or RA; CLL; or can be drug- or infection-induced.
Cold	- IgM antibodies against RBCs.

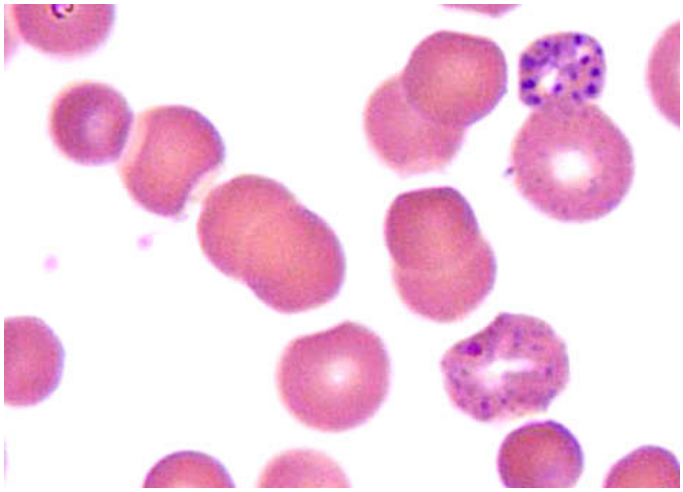
	- "Cold" because agglutination of RBCs occurs at cooler temperatures. The stem can say agglutination occurs while the tubes are transported en route to the laboratory. - Seen on USMLE for *Mycoplasma* pneumonia and CMV mononucleosis.

Infectious mononucleosis
- Usually caused by EBV, but can also be CMV. - The virus invades B cells, which then stimulates the immune system to produce CD8+ T cells attacking the viral-infected B cells. In mono, these CD8+ T cells are called "atypical lymphocytes." The USMLE wants you to know these are **reactive CD8+ T cells.** They are "reactive" because they are responding to the viral-infected B cells. - Primary infection presents usually in teenager or young adult with fever >38 C, lymphadenopathy, tonsillar exudates, and lack of cough, making the presentation appear bacterial (in the HY Pulmonary PDF, I talk about CENTOR criteria for differentiating bacterial from viral URTIs). As a result, it is often misdiagnosed as *Strep* pharyngitis. - If amoxicillin or penicillin is given to treat EBV, this can cause a rash. This is not to be confused with a rash caused by allergy to beta-lactams. If pre-adolescent receives beta-lactam and gets a rash, that is likely beta-lactam allergy. If a patient adolescent or older gets a rash, we do a heterophile antibody (Monospot) test as next best step in diagnosis, as EBV mono is more likely. - Following the primary infection, mono can present as recurrent episodes of extreme fatigue that arise at interval-separations of months to years.

Acanthocytes (spur cells)
- Spiky RBCs that show up notably in **liver failure due to heat stroke.** - I've probably only seen around 4 questions with acanthocytes on NBME material ever. 3 out of 4 are liver failure; 1 out of 4 is abetalipoproteinemia. Other resources get the yieldness reversed, where they mention abetalipoproteinemia as though it's this HY association. Nonsense. USMLE cares about liver failure. - **Heat stroke** is end-organ damage due to hyperthermia (>104 F); heat exhaustion is mere fatigue with no end-organ damage due to hyperthermia. Heat stroke presents on Qs as abnormal liver and/or renal function tests. - Q will say old lady found in her house in the summer has body temp of 106 F and acanthocytes on a blood smear; what's the diagnosis? → answer = liver failure (due to heat stroke). - Another NBME Q gives hepatitis C patient with acanthocytes. The latter aren't critical to answer the Q, but I noted that the NBME mentioned them for hepatitis (i.e., not just heat stroke). 

Macrocytosis causes	
- Normal RBC mean corpuscular volume (MCV) is 80-100 fL (femtoliters). - Macrocytosis refers to large RBCs (>100 fL).	
B9 (folate) / B12 deficiencies	- Deficiency of either vitamin leads to impaired DNA synthesis, where RBCs remain in the bone marrow for a longer time period, leading to RBCs that are larger and less mature. - High MCV as a result of impaired DNA synthesis is called megaloblastic anemia. - Impaired DNA synthesis also causes hypersegmented neutrophils, where longer time spent in the bone marrow results in an increased number of nuclear lobes and segments. 
Alcohol	- An important cause of high MCV on USMLE in the setting of normal B9/B12 levels. - Alcohol can also cause other types of anemia (i.e., sideroblastic).

Heme synthesis disorders	
Sideroblastic anemia	- Impaired first step of heme synthesis. - Can be X-linked recessive, but also can be caused by alcohol. - Failure to incorporate iron into heme. This leads to “ringed sideroblasts,” which are erythroblasts (RBC precursors) where iron accumulates inside the mitochondria, where the latter form a ring around the nucleus. - The iron-laden macrophages stain blue with Prussian blue stain, making this a very buzzy and pass-level image for USMLE. 
Lead poisoning	- The stereotypical vignette of a 2-year-old eating paint chips in an old house leading to mental deterioration is too easy and maybe only ~1/3 of lead questions.

	- The other 2/3 will be an adult who has **microcytic anemia** and mental decline. Buzzy hobbies can be hunting (bullets containing lead) or home-brewing of alcohol (casks made of lead).  - Can cause basophilic stippling of RBCs (rRNA precipitates). - Tx is calcium EDTA, dimercaprol, or succimer.
AIP	- Acute intermittent porphyria. - Answer on USMLE for the heme synthesis disorder where patient has red/pink/port wine colored urine and abdominal pain. - Patients can sometimes have miscellaneous neurologic findings, but not mandatory. - Treatment is hematin and glucose.
PCT	- Porphyria cutanea tarda. - Has increased urinary uroporphyrins. - Answer for the heme synthesis disorder where patient has photosensitivity / blistering. - Urine can appear “tea-colored” / dark. Don’t confuse this with the red/pink/port wine colored urine in AIP.

Intra- vs extra-vascular hemolysis	
Intravascular	- Lysis of RBCs within the bloodstream. - Causes release of bilirubin directly into the blood, leading to **hemoglobinuria.** - **Serum haptoglobin is decreased.** Haptoglobin is a protein produced by the liver that binds to and removes free hemoglobin released from damaged or lysed RBCs. This mopping up of hemoglobin by haptoglobin prevents renal damage and helps to recycle iron. Low serum haptoglobin levels can be used to differentiate intra- from extra-vascular hemolysis). - A few HY examples of intravascular hemolysis are: - a) G6PD deficiency (oxidative damage to RBCs). - 2) PNH, due to complement-mediated hemolysis; - 3) HUS, TTP, HELLP syndrome, DIC, mechanical hemolysis by prosthetic valves due to shearing of RBCs (all schistocytes).
Extravascular	- Phagocytosis and lysis of RBCs within the spleen (and also liver). - Hemoglobin from lysed RBCs is not released directly into the bloodstream; it is broken down within macrophages, forming biliverdin, which is converted to bilirubin. Bilirubin then goes to the liver for excretion in bile. - There is no hemoglobinuria. - Haptoglobin is not decreased. - HY examples are: 1) Hereditary spherocytosis (abnormal shape leads to clearance by spleen).

	2) Thalassemias (imbalanced Hb chains leads to RBC structural imbalance, which is recognized by the spleen as defective RBCs). 3) Warm and cold AIHA, where Abs opsonize RBCs, leading to their removal in red pulp of the spleen. (It should be noted that Abs in AIHA can also lead to intra-vascular hemolysis if complement is recruited, but most destruction of RBCs in AIHA is extra-vascular.)

Transfusion reactions	
Febrile non-hemolytic transfusion reaction (FNHTR)	- Caused by activation of donor WBCs when they come into contact with antibodies in the recipient's plasma. - This interaction triggers the release of cytokines from donor WBCs. - These cytokines then cause symptoms of fever and chills. - Coombs test is **negative,** since the process doesn't involve antibodies against donor RBCs. A (+) Coombs test means there are Abs against RBCs. - The vignette will be someone who gets a blood transfusion, and then within 2 hours gets fever + chills. They will then ask for the mechanism as the answer. I've seen them write it one of two ways: 1) Preformed antibodies against donor leukocyte antigens; and 2) Cytokine release from transfused blood. - Leukoreduction, a process that removes or reduces the number of WBCs from blood components before transfusion, can be used to mitigate risk of FNHTR by reducing probability of donor WBCs coming into contact with recipient antibodies.
Hemolytic (ABO mismatch)	- Caused by recipient antibodies against A and/or B antigens on donor RBCs. - This causes RBC agglutination and intravascular hemolysis, causing the release of hemoglobin into the bloodstream. - Binding of recipient antibodies to donor RBCs can also lead to complement activation and lysis of RBCs. - LDH can be increased in setting of hemolysis (RBCs are packed with LDH). - The vignette will be someone who gets fever, chills, and often flank pain within minutes of a blood transfusion. - Vignette can say, "10 minutes after the transfusion is started, the patient feels an 'impending sense of doom' and the transfusion is stopped" (sounds like MI, but NBME writes this). They then ask for next best step, which is antiglobulin test (another way of writing Coombs test).
Delayed transfusion reaction	- Caused by recipient antibodies against minor antigens on donor RBCs, such as Rh, Kell, Duffy, Kidd. - Coombs test is (+). - Vignette will be someone who had surgery + received RBCs intraoperatively + over the course of a week following the surgery, hemoglobin gradually decreases. The answer is then "Coombs (+); unconjugated bilirubin" as the combo that would be seen.
Transfusion-associated lung injury (TRALI)	- The answer on USMLE if a patient has an ARDS-like presentation with bilateral crackles and low O2 sats **<6 hours** following a transfusion. - Mechanism is abnormal priming of neutrophils in the lung that react to cytokines within transfused blood products (this is straight from a CMS Surg form 8 explanation). - Technically a type of non-cardiogenic pulmonary edema.
Transfusion-associated circulatory overload (TACO)	- Aka transfusion-induced hypervolemia. - Differs from TRALI in that this is a type of cardiogenic pulmonary edema (i.e., the left heart can't handle the ↑ hydrostatic pressure from ↑ volume, so transudation into the alveoli occurs).

	- The answer on USMLE if a patient has an ARDS-like presentation **>6 hours** following a transfusion, OR gets ARDS-like presentation following transfusion in the setting of cardiovascular disease. In other words, presents one of two ways: 1) They don't mention Hx of cardiovascular disease + the vignette will sound exactly like TRALI **> 6 hours** following a transfusion. - Surgery form 7 for 2CK gives an elderly dude who received only 3 packs of RBCs + he develops bilateral crackles and low O2 sats, but they say this occurs 12 hours after admission (not <6 hours as with TRALI), where the answer is "X-ray of the chest" as the next best step in diagnosis. The Q doesn't rely on you discerning TACO vs TRALI to get it right, but the explanation says it's TACO. 2) If a patient *with Hx of heart failure or MI* develops respiratory distress following transfusion of repeated blood products. - A 2CK NBME Q gives a 72-yr-old with Hx of MI ten years ago who gets shortness of breath and bilateral crackles 30 minutes after transfusion with crystalloid solution and 4 packs of RBCs (they don't specify the volume of crystalloid in the Q). In summary, if ARDS-like picture after transfusion: If <6 hours → TRALI. If >6 hours → TACO. If heart disease → TACO regardless of time frame.

HY Heme/Onc Pharm for Surg	
Aspirin	- Irreversible cyclooxygenase (COX) 1&2 inhibitor. - Non-steroidal anti-inflammatory drug (NSAID). - Aka salicylic acid (salicylate). - Blocks synthesis of prostaglandins and platelet thromboxane A2. - ↓ prostaglandin means ↓ inflammation, pain, and body temperature. - ↓ thromboxane A2 means ↓ platelet clumping. - Many HY use cases on USMLE: - Atrial fibrillation (with low CHADS2 score – i.e., Congestive heart failure, Hypertension, Age >75, Diabetes, Stroke/TIA/Hx of emboli, with the S being worth 2 points). If 0 or 1 points, give aspirin to AF patient; if 2+ points, give warfarin. USMLE doesn't obsess over valvular vs non-valvular AF in terms of giving warfarin vs NOACs (e.g., dabigatran, apixaban); they will either give aspirin or warfarin as the answer choices. - Peripheral vascular disease and carotid stenosis → USMLE wants triad of anti-platelet therapy (aspirin alone sufficient on USMLE), statin, and ACEi or ARB. - Ischemic heart disease (i.e., patients who have angina pectoris) → nitrates are used for acute angina, but daily aspirin is also given to these patients. - Ischemic strokes after 4.5 hours has transpired; if <4.5 hours, give tPA. - Aspirin can cause GI bleeds (↓ prostaglandin → ↓ gastric mucosal barrier mucous production and protection), nephropathy (pre-renal, interstitial nephropathy, nephrogenic diabetes insipidus), asthma (inhibition of COX causes shunting of arachidonic acid down the lipoxygenase pathway → leukotriene synthesis → bronchoconstriction). - Never give aspirin to children due to Reye syndrome risk (defect in beta-oxidation, with cerebral edema and hepatitis).

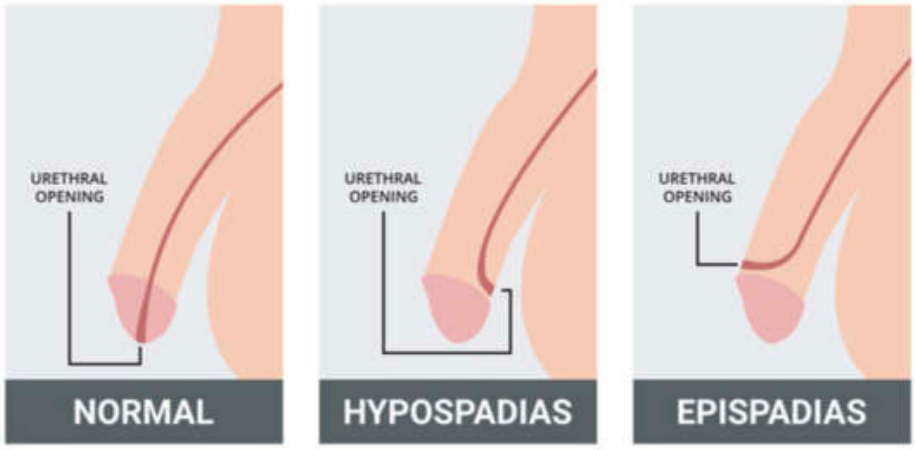
	- Misoprostol is given to patients following PPIs if they have NSAID-induced gastric ulcers (misoprostol is PGE1 analogue). - Can cause high-anion gap mixed metabolic acidosis-respiratory alkalosis (S in MUDPILES).
Other NSAIDs	- "Other NSAIDs" = reversible COX inhibitors, in contrast to aspirin (irreversible). - Indomethacin → used for acute gout and pericarditis. - Ibuprofen → general NSAID used broadly for pain, fever, and autoimmune diseases (i.e., RA, IBD, SLE, etc.). - Naproxen → USMLE-favorite drug for causing GI bleeds and nephropathy; shows up on 2CK forms a lot. - Diclofenac, ketorolac (just know these names = NSAIDs). - 5-ASA compounds (mesalamine, sulfasalazine) used for IBD.
Acetaminophen	- Central-acting COX inhibitor. - Not considered an NSAID because it doesn't act peripherally. - Decreases pain and fever, but is not anti-inflammatory in the periphery. - First-line med for osteoarthritis (weight loss first Tx). NSAIDs not ideal first-line because they kill the kidneys + OA is non-inflammatory, so NSAIDs don't provide added benefit over acetaminophen. Patients with chronic pain who take NSAIDs frequently are at risk of nephropathy and GI bleeds. - Metabolite NAPQI causes massive hepatotoxicity. Activated charcoal can be given immediately after consumption, otherwise *N*-acetylcysteine HY as Tx. Latter regenerates reduced-glutathione, which neutralizes NAPQI.
Celecoxib	- Selective COX-2 inhibitor that does not cause GI bleeds. - Still can cause nephropathy. - NBME asks why celecoxib can increase the risk of MI → answer = decreases prostaglandin production without inhibiting platelets. - Can in theory be used for pain and inflammation.
Clopidogrel	- Anti-platelet agent; inhibits ADP P2Y12 receptor on platelets. MOA is HY. - Can be used to fulfill the anti-platelet component of the peripheral vascular disease and carotid stenosis triad (as mentioned before: 1) anti-platelet therapy; 2) statin; 3) ACEi or ARB). - In theory can be used for stenting procedures, but I haven't seen USMLE give a fuck. It's more that this use case is perpetuated in resources. - Can also be used in patients with ischemic heart disease who can't tolerate aspirin. - Prasugrel, ticlopidine, and ticagrelor drugs with same MOA, but not assessed.
Dipyridamole	- Phosphodiesterase-3 inhibitor; prevents breakdown of cAMP. - Causes arteriolar dilation + also an anti-platelet agent. USMLE really likes this – i.e., they will ask, "which of the following is both a vasodilator and anti-platelet agent?" → answer = dipyridamole or cilostazol (whichever is listed). - Dipyridamole/thallium is combo used frequently for pharmacologic stress testing. Arteriolar vasodilation causes HR to go up, where thallium uptake into myocardium can help visualize areas of ischemia.
Cilostazol	- Phosphodiesterase-3 inhibitor; prevents breakdown of cAMP. - Causes arteriolar dilation + also an anti-platelet agent, same as dipyridamole. - Used for intermittent claudication / peripheral vascular disease **after a walking/exercise program is already prescribed.** Do not choose cilostazol as a first treatment for peripheral vascular disease. Choose walking/exercise program first. Then, if they force you to choose a drug for additional management, choose cilostazol.
Abciximab	- GpIIb/IIIa inhibitor on platelets. - Prevents platelet aggregation (i.e,. platelet-platelet interaction). - Don't confuse with GpIb, which is for platelet adhesion. vWF normally bridges GpIb on platelets to vascular endothelium / underlying collagen, and is unrelated. - Since physiologically, fibrinogen is normally what bridges GpIIb/IIIa on adjacent platelets, abciximab is considered to be a fibrinogen analogue. - Drugs such as tirofiban and eptifibatide have same MOA but aren't assessed.

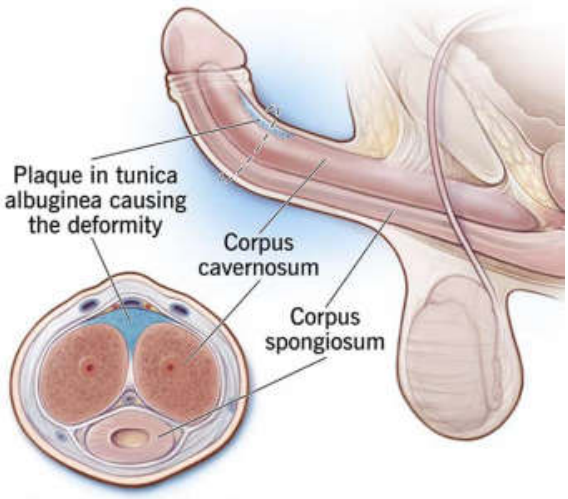
tPA (alteplase), streptokinase	- Tissue plasminogen-activator actives plasminogen, which breaks down fibrin clots. - I've only seen this used twice on NBME forms: 1) for ischemic strokes within 4.5 hours; and 2) for pulmonary embolism where there is obstructive shock (i.e., PE with low BP).
Heparin	- Activates anti-thrombin III, leading to inactivation of thrombin (factor II) and factors IXa, Xa, XIa, and XIIa. - Used as a bridging agent to warfarin when patient needs anticoagulation. This is because warfarin, by inhibiting protein C as part of its function, temporarily makes the patient hypercoagulable. - Causes both an ↑ in PT and aPTT, but aPTT is used to monitor heparin over PT because aPTT is more accurate/sensitive to the effects of heparin. This is because heparin inhibits intrinsic pathway factors (i.e., IXa, Xia, and XIIa) *in addition to* the common pathway factors (i.e., thrombin and Xa). In other words, 5 clotting factors within the intrinsic pathway are affected by heparin, but only 2 within the extrinsic pathway are (i.e., the common pathway ones). - Subcutaneous enoxaparin (LMWH) used first-line for DVT and superficial thrombophlebitis. Heparin is also used for acute limb ischemia and vertebral artery dissection (clot can form within false lumen; asked on 2CK NBME). - USMLE won't ask about unfractionated heparin vs LMWH, but apparently unfractionated heparin is used in patients with 1) severe renal impairment and 2) in those who might need rapid-reversal due to uncertainty about dosing. In other words, unfractionated heparin can be reversed with protamine sulfate much faster than LMWH. - Heparin-induced thrombocytopenia (HIT) is exceedingly HY for USMLE. Mechanism is autoantibodies against heparin-platelet factor 4 (PF4) complex, where administration can lead to ↓ platelets. This is a type II hypersensitivity (platelets are cells). NBME can write the answer simply as "anti-platelet antibodies." Treatment is cessation of heparin + give direct-thrombin inhibitor (e.g., dabigatran). Warfarin is wrong answer. A 2CK NBME Q simply has "direct-thrombin inhibitor" as the answer, without listing specific drug name.
Warfarin, dicoumarol	- Vitamin K epoxide reductase inhibitors, which prevent vitamin K from being recycled back to its active form. - Vitamin K normally is a cofactor for an enzyme called γ-glutamyl carboxylase, which activates clotting factors II (prothrombin), VII, IX, and X, as well as anti-clotting proteins C and S. - PT is used for monitoring because 3 clotting factors are in the extrinsic pathway (II, VII, X), whereas only 2 are in the intrinsic pathway (i.e., the common pathway ones – II and X), so PT is more sensitive than aPTT for changes in warfarin. - International normalized ratio (INR) is the ratio of the patient's PT to a control standard's PT, where 2.0-3.0 is the usual target range for warfarin. - Highest yield use cases on USMLE are for maintenance anticoagulation therapy in patients who have DVT, prosthetic valves, or atrial fibrillation with elevated CHADS2 score. - As discussed earlier, heparin must be used as bridging agent initially with warfarin because warfarin's inhibition of Protein C, which has a very short half-life, makes the drug initially pro-coagulable. - Warfarin levels are sensitive to drugs that inhibit or stimulate P-450. For example, P-450 inhibitors can ↑ INR and cause bleeding diathesis; P-450 inducers can ↓ INR and cause thromboses. - As discussed earlier, vitamin K deficiency due to broad-spectrum antibiotics causing depletion of bowel normal flora can ↑ INR (i.e., ↓ warfarin is required to inhibit clotting, but we haven't changed dose). - Teratogenic; must be avoided in pregnancy. Warfarin is a small, lipophilic molecule that readily crosses the placenta. In contrast, heparin is a large, acidic polymer that does not cross the placenta. Warfarin can cause bleeding diathesis and **bone abnormalities** in the fetus.

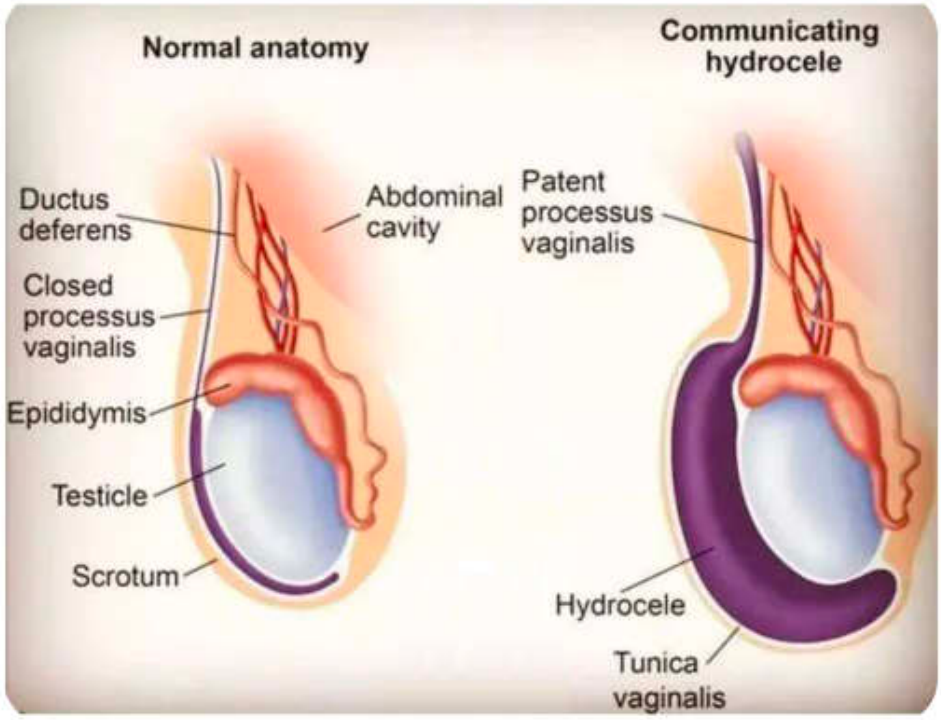
	- Can cause skin necrosis in patients with protein C deficiency. - Slowly reversed with vitamin K (requires re-synthesis of clotting factors by the liver); quickly reversed by fresh frozen plasma (supplies clotting factors).
Dabigatran	- Direct-thrombin inhibitor. - Used for HIT. - Sometimes in patients with non-valvular AF with high CHADS2 score.
Apixaban	- Factor Xa inhibitor. Just know MOA. - USMLE doesn't assess use cases, but can be given for non-valvular AF with high CHADS2 score.
Fondaparinux	- Factor Xa inhibitor. Just know MOA. - USMLE doesn't assess use cases, but can be given for DVT and PE in place of heparin in some patients.
Cisplatin	- Crosslinking agent. - Can cause nephro-, oto-, and neurotoxicity. - Chloride diuresis (i.e., normal saline) and amifostine can mitigate toxicity.
Doxorubicin (Adriamycin)	- Intercalating agent + produces free radicals. - Can cause dilated cardiomyopathy (HY). - Dexrazoxane mitigates toxicity by mopping up free radicals.
Bleomycin	- Induces DNA breaks. - Causes pulmonary fibrosis.
Cyclophosphamide	- Guanine N7 alkylating agent. - Causes hemorrhagic cystitis (red urine). - Mesna mitigates toxicity by using a thiol (-SH) group to mop up acrolein, which is the toxic metabolite of cyclophosphamide that is toxic to uroepithelium.
Azathioprine	- Pro-drug that is metabolized into 6-MP.
Imatinib	- Bcr/able tyrosine kinase inhibitor used to treat CML. - Can cause fluid retention / peripheral edema.
Erlotinib	- EGFR tyrosine kinase inhibitor used for non-small cell lung cancer.
Rituximab	- CD20 inhibitor on B cells. - Can ↑ risk of bacterial pneumonia (asked on NBME). This is because B cells mature into plasma cells, which make immunoglobulins, which are crucial for humoral immunity to fight bacterial infections.
Methotrexate	- Dihydrofolate reductase inhibitor (enzyme required in folate pathway). - First-line disease-modifying anti-rheumatic drug (DMARD) in rheumatoid arthritis. - First-line oral agent in plaque psoriasis that is non-responsive to topical agents, or for systemic psoriasis (arthritis). - Causes pulmonary fibrosis, mucositis (mouth ulcers due to neutropenia), and hepatotoxicity. - Toxicity mitigated with folinic acid (not folic acid), which is aka "leucovorin rescue."

Surg Repro/Obgyn

Pelvic floor stuff	
Kegel exercises	- Aka pelvic floor exercises. - Used to strengthen the pelvic floor muscles in women who have stress incontinence. - USMLE will ask which muscle is *not* strengthened by Kegel exercises. The student says, "Huh? That could be any muscle. The deltoid isn't for instance." Answer = internal anal sphincter. The big picture concept is: internal sphincters of the anus and bladder are under autonomic (sympathetic) control, not somatic (voluntary). So you say, "Even though I'm not an obstetrics expert, clearly you can't voluntarily do an exercises to strengthen a muscle that's not under voluntary control." The external anal and bladder sphincters, in contrast, are under voluntary control.
Episiotomy	- Procedure performed prior to Stage 2 of labor (i.e., delivery of the fetus) to cut the vagina posteriorly in the midline. The purpose is to prevent vaginal laceration during delivery of a macrosomic (large) fetus. - USMLE will ask, "If an episiotomy is performed posteriorly in the midline and the doctor cuts too far, what will he/she cut into?" → answer = external anal sphincter.

Male repro DDx	
Hypospadias	- Urethral meatus opens on underside of penis (ventral aspect). 
Epispadias	- Urethral meatus opens on top of penis (dorsal aspect).
Phimosis	- Cannot retract the prepuce (foreskin). - In other words, cannot pull it back to expose the glans.
Paraphimosis	- Cannot reduce the prepuce. - In other words, once it is retracted, cannot replace it back over the glans. - Just remember that paraphimosis is the bad one / emergency.
Cryptorchidism	- Undescended testis. - Increased risk of infertility (↑ temperature within abdomen → ↓ spermatogenesis). - Increased risk of testicular carcinoma. - USMLE wants no treatment prior to age 6 months; after 6 months, perform orchidopexy.
Testicular torsion	- Acutely painful testis; medical emergency; can lead to ischemia and loss of testis. Can be caused by abnormality of the gubernaculum, which is a ligamentous cord that anchors the testis within the scrotum. This leads to "bell-clapper deformity," where improper anchoring = ↑ risk of rotation. - Negative Prehn sign, which is failure of relief of pain upon lifting of the scrotum. Apparently this is an unreliable test in real life, but NBME vignettes can still mention it. In contrast, a (+) Prehn (relief of pain upon lifting the scrotum) is more indicative of epididymitis.

	- Negative cremasteric reflex, which is failure of the scrotal skin to retract upon palpation of the medial thigh. In contrast, a (+) cremasteric reflex (scrotal skin retracts) is more indicative of epididymitis. Normally, when the inner thigh is stroked, the cremaster muscle contracts and pulls up on the ipsilateral testis. In torsion, the twisted spermatic cord can impair the nerve fibers responsible for triggering the reflex. - USMLE wants Doppler ultrasonography as next best step, followed by surgery.
Epididymitis	- Inflammation/infection of epididymis. - (+) Prehn sign and (+) cremasteric reflex. - Chlamydia and gonorrhea are most common causes in males <35. - Over 35, *E. coli* is most likely. - The USMLE will not play games where they give you a borderline case and force you to guess. They will give you either a 25- or 45-year-old male. This also goes for prostatitis (same organisms). There is an NBME Q of a 45-year-old where the answer is *E. coli*, and chlamydia is wrong.
Torsion of appendix testis	- Not the same as torsion of testis. - Asked on a 2CK Peds form. - Q will tell you there's a kid with a painful testis + the superior pole is blue + there's an intact cremasteric reflex ("regular torsion" it's not intact). - "Blue dot" sign = blue superior pole of testis. - Not my opinion if you think it's nitpicky or weird. Take it up with the NBME exam.
Peyronie disease	- Curvature deformity of penis caused by fibrous plaque within tunica albuginea.  - Doesn't have to be treated surgically unless causing functional impairment.
Orchitis	- Inflammation/infection of the testis. - Mumps is HY cause in unvaccinated / immigrants (sometimes implies unvaccinated). - If not caused by mumps virus, then the bacterial causes are same as epididymitis and prostatitis for young vs old (i.e., STIs vs *E. coli*, respectively).
Priapism	- Erection lasting longer than 4 hours. - Can be caused by sickle cell or by PDE-5 inhibitors (i.e., tadalafil/sildenafil). - Injection of alpha-1 agonist (i.e., phenylephrine) into the erectile tissue is treatment → decreases blood flow due to blood vessel constriction.
Hydrocele	- Serous fluid collection within the scrotum between the layers of the tunica vaginalis surrounding the testis. - Mechanism is patent processes vaginalis. - The processus vaginalis is an embryonic outpouching of peritoneum that descends into the scrotum. It normally obliterates shortly after birth, but if it remains open, can cause hydrocele or indirect inguinal hernia.

	 - Transilluminates when a light is shone to it. - USMLE wants no treatment before the age of 1; most spontaneously close.
Spermatocele	- Obscure serous fluid collection within the epididymis containing sperm. - Occurs usually in adult men (rather than neonates as with hydrocele). - Transilluminates similar to hydrocele. But once again, it's an older adult, not neonate.
Varicocele	- Dilation/congestion of pampiniform venous plexus draining the scrotum. - Does not transilluminate (in contrast to hydro- and spermatoceles). - Can be described as buzzy "bag of worms." Sounds too easy, but an NBME Q literally uses this colloquialism. - Can also be described as a heavy, dragging scrotum lower on the left. - Occurs almost always on the left side due to the anatomy of venous drainage of the testis. - On the left, we have left testicular vein → left renal vein → IVC. - On the right, we have right testicular vein → IVC ("right to IVC"). - The left testicular vein drains at a 90-degree angle into the left renal vein, creating a proclivity for hydrostatic pressure backup ipsilaterally. - There is a 2CK NBME Q where bilateral varicocele is the answer. Highly obscure, but just be aware it's somehow possible/exists. - Can occur due to left renal vein thrombosis in setting of nephrotic syndrome (loss of antithrombin III in the urine → hypercoagulable state).

HY incontinences for USMLE	
Stress	- The answer for loss of urine with ↑ intra-abdominal pressure from laughing, sneezing, coughing. - Stereotypical risk factor is grand multiparity (i.e., Hx of many childbirths) leading to weakened pelvic floor muscles. - I'd say only ~50% of Qs will mention Hx of pregnancy. The other ~50% are idiopathic. - Buzzy vignette descriptors and answer choices are: "downward mobility of the vesicourethral junction," "urethral hypermobility," and "urethral atrophy with loss of urethrovesical angle." - Treatment is pelvic floor (Kegel) exercises. These notably strengthen **levator ani, pubococcygeus,** and the **external urethral sphincter.**

	- USMLE is known to ask which muscle is *not* strengthened by Kegel exercises, which sounds obscure, since any muscle could theoretically be the answer ("Well the deltoid isn't strengthened."). But a favorite answer here is **internal urethral sphincter.** The way you know this is the answer is because internal sphincters are under sympathetic (i.e., involuntary; autonomic) control, which means it's impossible to strengthen it via a voluntary (i.e., somatic) exercise. USMLE doesn't expect you to be an obstetrician. The bigger picture concept is simply knowing internal sphincter control is involuntary. It is external sphincter control that is voluntary (somatic). - Do **not** give medications for stress incontinence on USMLE. - If Kegel exercises fail, patients can get a mid-urethral sling (LY; asked once).
Urge	- The answer on USMLE for patient who has an "urge" (NBME will literally say that word and I've seen students get the Q wrong) to void 6-12+ times daily **unrelated** to sneezing, coughing, laughing, etc. (otherwise stress incontinence). - Ultra-HY for **multiple sclerosis.** I've had students ask whether MS is urge or overflow. It shows up repeatedly on the NBMEs as urge; I've never seen it associated with overflow. I'd say ~1/3 of urge incontinence vignettes on NBME forms are MS. - Other vignettes will be peri-menopausal women, or idiopathic in old women. - Mechanism is "detrusor hyperactivity," or "detrusor instability." - Vignette can mention woman has urge to void when stepping out of her car, or when sticking her key in the car/front door of her house. Sounds weird, but these are important Qs to ask when attempting to diagnose urge incontinence. - UTIs can present similarly to urge incontinence. Some students have asked, "Well isn't that because UTIs are a cause of urge incontinence?" Not really. It just happens to be that UTIs can sometimes cause transient urinary urgency. For example, if they give a Q where they say patient had Hx of urinary catheter + now has dysuria and urinary urgency, answer = "urinary tract infection" on NBME; "detrusor hyperactivity" is wrong answer. - Treatment is **oxybutynin** (muscarinic receptor antagonist); this ↓ activity of the detrusor muscle of the bladder. - Some students get hysterical about mirabegron (β3-agonist), but I've never seen NBME forms assess this.
Overflow	- Will be due to either BPH or diabetes on USMLE. - Will have ↑ **post-void volume.** Normal is < ~50 mL. On USMLE for overflow, they'll give you 300-400 mL as post-void volume. - As I talked about earlier for BPH, they will give old dude + high creatinine (post-renal azotemia). Next best step is "insertion of catheter" to relieve the obstruction. If they don't have this listed, "measurement of post-void volume" can be an answer. We then treat the BPH with finasteride (5α-reductase inhibitor) or an α1-blocker (tamsulosin, terazosin). - For diabetes, the mechanism is neuropathy to the bladder causing "neurogenic bladder," or "hypotonic bladder," or "hypocontractile bladder/detrusor muscle." - For neurogenic bladder causing overflow incontinence +↑ post-void volume, remember that USMLE is first obsessed with "measure post-void volume" and "insertion of catheter" if they are listed. They will not force you to choose between the two. But they like these answers prior to giving medications. - Give **bethanechol** (muscarinic receptor agonist); this stimulates the detrusor muscle. - Making sure you don't confuse oxybutynin and bethanechol is pass-level for USMLE.

The "celes"	
Urethrocele	- Herniation of urethra into the vagina. - **Anterior inferior vaginal wall.** - Can present with stress incontinence, recurrent UTIs, or a palpable bulge in the anterior inferior vaginal wall.

	- There is one 2CK Obgyn Q where they say the vaginal wall appears well-supported (i.e., doesn't sound like a urethrocele), but then they say a cotton-tipped applicator placed into the urethra arcs at a 45-degree angle when the woman coughs → answer = urethrocele. Call it weird all you want, but it's on the CMS form.
Cystocele	- Herniation of bladder into the vagina. - **Anterior superior vaginal wall.** - Similar to urethrocele, can present with stress incontinence, recurrent UTIs, or a palpable bulge on the vaginal wall (anterior superior).
Rectocele	- Herniation of the rectum into the vagina. - **Posterior inferior vaginal wall.** - Can present as difficulty with bowel motions.
Enterocele	- Herniation of small bowel into the vagina. - **Posterior superior vaginal wall.** - NBME Q will tell you a woman feels "movement" within her vagina, which refers to the peristalsis of the bowel. - I've seen NBME write "superior" wall as "high on the vaginal wall."
Cystovaginal fistula	- Including this here because I've seen it as a distractor in quite a few "cele" Qs. - Abnormal communication between the bladder and the vagina, leading to continuous, involuntary leakage of urine into the vagina. - Can be caused by prior obstetric trauma (e.g., prolonged labor) or surgery. - I've seen it once as a correct answer. NBME will say there is "pooling of urine" in the vagina.

Endometrial cancer
- Highest yield point on USMLE is that it is caused by "unopposed estrogen," usually due to anovulation/PCOS. - Estrogen normally stimulates growth of endometrium; progesterone prevents overgrowth. - In anovulation/PCOS, no corpus luteum is formed, since the latter is the follicular remnant. - The corpus luteum normally secretes progesterone to maintain the endometrial lining if fertilization occurs. If no corpus luteum is formed, there's no progesterone produced to balance estrogen. This leads to endometrial hyperplasia, which increases the risk of endometrial adenocarcinoma. - USMLE can give an overweight female who's perimenopausal who has breakthrough bleeding (i.e., mid-cycle bleeding), or a post-menopausal woman with any vaginal bleeding, and they want endometrial biopsy as the answer. The fact that the patient has high BMI insinuates that anovulation/PCOS may have been a part of her past, where unopposed estrogen is a part of her history. - Can occur secondarily as a result of granulosa cell tumor of the ovary (as discussed above). This is a rare cause of unopposed estrogen. - Can occur in women who take hormone-replacement therapy (HRT) who stop taking the progesterone component (mentioned in one NBME Q). - Can be associated with hereditary non-polyposis colorectal cancer (HNPCC; Lynch syndrome).

Fibroid points
- Fibroids are aka leiomyomata uteri and are most common benign tumor in women. - They are smooth muscle tumors of the myometrium; often found incidentally on ultrasound or autopsy. - If fibroids are picked up incidentally on ultrasound, the answer is observe / follow-up. The highest yield point for USMLE is that these are not managed almost always. - There is negligible risk of progression into leiomyosarcoma (malignant version that doesn't exist on USMLE). In other words, do not choose answers like myomectomy. - If a woman has bleeding due to fibroids, OCPs and NSAIDs can be attempted in theory, although I have not seen USMLE assess these. Other treatments like leuprolide are also nonexistent on USMLE.

Vulvovaginal lesions for Surg	
Condylomata acuminata	- Aka warts; caused by HPV 6/11. - Present as painless, skin-colored or slightly hyperpigmented cauliflower-like popular lesions. - I discuss all of the STDs in more detail later.
Lichen sclerosus	- Presents as whitish-grey, rough, irritated or scratchy patch on the vulva or perineum. - Thought to be caused by a mix of chronic irritation and autoimmunity; not HPV-related. - Characterized by atrophy, hyperkeratosis, and a band of lymphocytes in the dermis. It is not neoplasia, dysplasia, or metaplasia. - USMLE wants biopsy as next best step to rule out SCC (on an old Step 1 Free 120 even though NBS Qs are classically 2CK). If SCC is negative and the diagnosis is LS, USMLE wants topical steroid as treatment.
Vulvovaginal carcinoma	- Cancer of the vagina and/or vulva. - Almost always squamous cell carcinoma due to HPV 16 or 18. - Can occur rarely due to lichen sclerosus.
Bartholin gland cyst	- Shows up as a tender/painful bump at a 4- or 8-o'clock position on the vulva. - If the cyst becomes infected (i.e., warm and red), we call it Bartholin gland abscess. - USMLE wants "polymicrobial" as the most likely organism. - For cysts, sitz bath + warm compresses are first Tx. - For overt abscesses, drainage is the answer.

Cervical cancer + Pap smear screening (2CK/3 only)
- Squamous cell carcinoma; occurs at the transitional zone between the stratified squamous ectocervix and columnar endocervix. - Caused by HPV 16/18. There are other strains students get pedantic about, but USMLE doesn't give a fuck. - Can cause vaginal bleeding + uni- or bilateral hydroureter / -nephrosis. - Pap smears should be started at age 21. If the vignette gives you a teenager who's sexually active, the answer is you test for STDs but do not do a Pap smear. - Perform Pap smear age 21-65. - From age 21-30, they are done every 3 years without HPV co-testing. - From age 30-65, the female has the option of continuing Paps every 3 years, or doing Pap-HPV co-testing every 5 years. - Pap smears are stopped at age 65. - If the female has a history of a high-grade squamous intra-epithelial lesion (HSIL) or cervical intra-epithelial neoplasia (CIN), Pap smears are continued after age 65. - Patients who have had hysterectomy due to cervical cancer should have continued Pap smears of the vaginal cuff. Patients with hysterectomy due to endometrial cancer do not. - Patients with HIV require two Pap smears in the first year following diagnosis (6 months apart), followed by annual Pap smears thereafter. The vignette will tell you HIV patient had normal Pap smear last year, and student is like, "Cool, she doesn't need one for another 2 years." No. This is important. You need to know HIV patients need once/year Paps. - Low-grade squamous intra-epithelial lesions (LSIL) and atypical squamous cells of undetermined significance (ASC-US) are managed with either repeat Pap smear or HPV co-testing, where (+) vs (-) HPV test determines whether colposcopy + biopsy is performed. **USMLE will not** assess the algorithmic specifics. If you memorize the algorithm based on the female's age, you're wasting your time for USMLE purposes. - HSIL on Pap smear is managed with immediate colposcopy + biopsy. - CIN I is what we call LSIL that is confirmed on colposcopy + biopsy. - CIN II/III is what we call HSIL that is confirmed on colposcopy + biopsy. - CIN I has high rate of spontaneous regression. Repeat Paps are done in a year. - CIN II/III requires excisional or ablational treatment due to higher risk of progression to invasive cervical cancer. Such treatments include LEEP, coning, cryotherapy, or laser ablation.

- Cone biopsy is also an answer on NBME for next best step in patient who has colposcopy performed for HSIL on Pap + "the entire squamocolumnar junction cannot be visualized." In other words, do a cone biopsy if colposcopy is insufficient.
- CIN III is **not** excised/ablated during pregnancy. HPV lesions (including warts) are known to get temporarily worse during pregnancy due to relative immunosuppression (biologic attempt to minimize attack against fetal antigens). In the post-partum period, CIN III lesions are evaluated for signs of regression over the course of weeks; if they do not regress, they are excised same as non-pregnant women with CIN II or III.

Breast neoplasia + HY points	
- Specific breast cancer types are higher yield than ovarian cancer types for Surg shelf.	
DCIS	- Most common malignant (i.e., has metastatic potential) breast cancer. - "Microcalcifications" on mammography = DCIS till proven otherwise. - 2CK wants "needle-guided open biopsy" as next best step; FNA is wrong. - Paget disease of breast (looks like a nipple with eczema in patient over 50) is often the cutaneous extension of an underlying DCIS.
Invasive ductal	- Invasive form of ductal carcinoma. - Cells on biopsy have stellate morphology.
LCIS	- Same as DCIS, malignant potential but not yet invaded. - Appears as linear rows of cells ("Indian file") on histo. This is supposedly due to ↓ E-cadherin expression, where the cells don't clump together the same way they do as with ductal carcinomas. I haven't seen USMLE give a fuck about this detail, but I'm mentioning it because if you ever talk to an attending about breast cancers, they become like hysterical over this detail.
Invasive lobular	- Invasive form of lobular carcinoma. - 20-30% bilateral.
Intraductal papilloma	- Unilateral bloody/rusty nipple discharge = intraductal papilloma till proven otherwise.
Inflammatory carcinoma	- Can appear as a red/inflamed breast, as though there's an infection. - Rubor (redness), dolor (pain), tumor (swelling), calor (heat). - Peau d'orange is an orange peel-appearing texture of the breast due to tethering of the skin by Coopers ligaments in the setting of blockage of lymphatic drainage by tumor cells.
Fibroadenoma	- Most common breast tumor overall; benign. - Presents as rubbery, mobile, non-tender breast lump in woman 20s-30s. - Will appear as solid breast mass without calcification on ultrasound. - FNA is next best step to confirm Dx. If confirmed fibroadenoma, they do not need to be excised.
Cystosarcoma phyllodes	- Fast-growing breast lesion that appears "leaf-like" on light microscopy. - Can be benign or malignant based on histo.
BRCA points	- *BRCA1/2* are tumor suppressor genes but are inherited in an autosomal dominant fashion with incomplete penetrance (i.e., sometimes can skip a generation). A pedigree requiring you know this pattern is asked on NBME. - NBME also wants you to know that, with *BRCA* mutations, the molecular process that's fucked up is "recombinational double-stranded DNA repair."
Screening/Mx points	- For USMLE, mammography is done every two years between ages 50-74. - Since committees / cancer task forces have varying recommendations, the USMLE doesn't play trivia as far as "oh em gee do we start at 45 vs 50, etc." They won't be borderline about it. They will just give you a patient who's, e.g., 56 who needs a mammogram since she hasn't had one for two years. - If first degree family member (i.e., parent or sibling) has Hx of breast cancer, then start mammography at age 40, not 50. - Any breast screening done for a women under 30, do not do mammography on USMLE. Ultrasound is typically done. This is because younger women have

	denser breast tissue, making mammography less specific (i.e., more false-positives) and hence less reliable of a diagnostic modality. - Estrogen and progesterone receptor (ER/PR) positivity is correlated with better prognosis for breast cancers. This is partially because SERMs and aromatase inhibitors can be used as Tx. - HER2/neu positivity is correlated with worse prognosis. Trastuzumab (Herceptin) can be used to target HER2/neu. - Triple-negative ER/PR/HER2/neu is associated with bad prognosis due to ↑ aggressiveness, ↑ rates of recurrence, and ↑ risk of metastasis to lungs/brain. - 2CK NBME wants "bilateral mastectomy + oophorectomy" in patients who have confirmed *BRCA* mutation. - Breast cancer risk is ↑ as a result of HRT, even when progesterone is given as part of it. Breast cancer risk is ↑ as a result of ↑ absolute estrogen exposure the female has in her life. This has nothing to do with unopposed estrogen, which is the major risk factor for endometrial cancer. - Combined OCPs have been suggested in some literature as ↑ the risk of breast cancer, but there is no significance. They won't assess this on USMLE either way. But they do assess ↑ breast cancer risk as a result of HRT. - As I talked about before, the reason the only approved indication of HRT is severe vasomotor Sx (i.e., and not to maintain bone density, etc.) is because HRT ↑ the risk of breast cancer, thromboembolic events, MI, and stroke. This is because estrogen upregulates fibrinogen and factors V and VIII.

Other breast conditions	
Fibrocystic change	- Benign cystic alteration in breast tissue in women 20s-50s. - Shows up three ways on NBMEs: 1) Highly buzzy/classic presentation is a woman 20s-30s who has bilateral, waxing/waning breast tenderness over the course of her menstrual cycle; 2) Unilateral tenderness or sharp pain in one breast (students are often surprised that it's not always bilateral). 3) Painless, unilateral breast cyst in woman in her 40s that drains brown/greenish fluid → answer = fibrocystic change straight-up on the NBME. Call it weird all you want. Take it up with NBME, not me. - You do not need to biopsy or do ultrasound. It is a clinical diagnosis and Tx is symptomatic only (i.e., warm showers, evening primrose oil). - In theory, biopsy of fibrocystic change will show some buzzy findings: "sclerosing adenosis," apocrine metaplasia," and/or "blue dome cysts."
Simple breast cyst	- "Simple" means hypoechoic, or black on ultrasound. It reflects a fluid-filled cyst. - "Complex" means at least in some part hyperechoic, or white on ultrasound. It reflects a solid component to the cyst. - Simple cysts are observed and do not require FNA drainage unless they cause discomfort to the patient (e.g., large cysts). There is no specific size that requires drainage on USMLE. - There is an NBME Q where they say a woman was started on HRT three months ago and now has a simple cyst → answer = "biopsy of the lesion." So you should be aware that sudden growth in size can be indication for biopsy. - All complex cysts require FNA biopsy, similar to fibroadenoma.
Mastitis	- Classically presents as red, cracked, fissured nipple in breastfeeding woman. - Usually caused by *S. aureus*, but Group A Strep also possible. - USMLE wants oral dicloxacillin (not doxycycline) + continue breastfeeding through the affected breast as treatment. There is no risk of harm to the neonate. Methicillin-class beta-lactams (e.g., dicloxacillin) and first-generation cephalosporins (e.g., cephalexin) are equivalent on USMLE for all intents and purposes. They both treat *S. aureus* skin infections.

	- High-yield point is that mastitis need not affect the nipple and can present as a red, warm, tender, ***non***-fluctuant mass of the breast. When we talk about infection of the breast, non-fluctuant = mastitis; fluctuant = abscess. This is asked insidiously on a 2CK form and students get it wrong all the time thinking it's abscess – i.e., "They say it's a warm, red, tender mass of the breast though!" Yeah, but they say non-fluctuant, not fluctuant. Not my issue if you get emotional over dumb shit.
Abscess	- Warm, red, tender *fluctuant* mass of the breast. - Drain as next best step.
Galactocele	- Aka milk retention cyst; usually seen in recently breastfeeding women. - Often confused with breast abscess. - Can present as painless or tender fluctuant mass of the breast that is peri-areolar. - USMLE will say the patient is afebrile and that the lesion is not warm or red. - Tx is warm compresses followed by drainage.
Ductal ectasia	- As the name implies, this is dilation of lactiferous ducts. - Diagnosis of exclusion on USMLE (i.e., we eliminate to get there). - Presents 2/3 Qs as an inverted nipple. - Presents 1/3 Qs as some random breast lesion where they say biopsy shows "dilation of the ducts," which is what the name means.
Virginal breast hypertrophy	- Virginal breast hypertrophy is benign asymmetric growth of the breasts that can occur in young women. USMLE will try to trick you into thinking it's sinister by saying a first-degree relative had breast cancer. - They will say 16-year-old girl has large breast lump under the nipple on one side. Answer is just follow-up / observe.
Gynecomastia	- Physiologic gynecomastia can occur in teenage boys on USMLE, where there can be ↑ breast tissue, either tender or not, occurring uni- or bilaterally. Just observe. - Can occur as a result of ↑ estrogen in liver failure, or due to certain drugs like cimetidine, spironolactone, ketoconazole, or even marijuana. - Caused by direct estrogen effect on breast. It is not caused by prolactin. - As I talked about earlier, an NBME Q gives gynecomastia in a male patient taking hCG. They ask for the mechanism → answer = "testis producing estrogen." They can also have as answer "estrogen; direct effect on breast." The reason this makes sense is because, since hCG shares the same alpha-subunit with LH and FSH (and TSH), giving hCG is as though we are giving LH and FSH at the same time. So LH effect → androgen production; FSH effect → aromatase production → therefore androgens are converted to estrogens in the testis → mechanism via which we could get gynecomastia.
Necrosis	- Non-enzymatic fat necrosis of breast due to trauma; can calcify. - Don't confuse with enzymatic fat necrosis, which is acute pancreatitis. - Can occur due to overt trauma (e.g., getting hit by softball) or sometimes by tight-fitting sports bras.

Ovarian neoplasms	
- Only mentioning the ones I see as relevant here to the 2CK Surg shelf. - There is no formal screening for ovarian cancer (unless family Hx of BRCA). - OCPs ↓ risk of ovarian cancer overall in population due to synchronization of cycles.	
Sertoli-Leydig cell tumor	- Produces androgens; can cause virilization in females.
Granulosa cell tumor	- Produces estrogen; can cause endometrial hyperplasia and ↑ risk of endometrial cancer due to unopposed estrogen. - NBME Q gives female with ovarian neoplasm + ↑ thickness of endometrial stripe + vaginal bleeding → answer for ovarian lesion = granulosa cell tumor.
Krukenberg tumor	- Gastric metastases to the ovaries bilaterally. - Has mucin-containing signet-ring cells on biopsy. - Spread to ovaries from stomach is usually hematogenous.

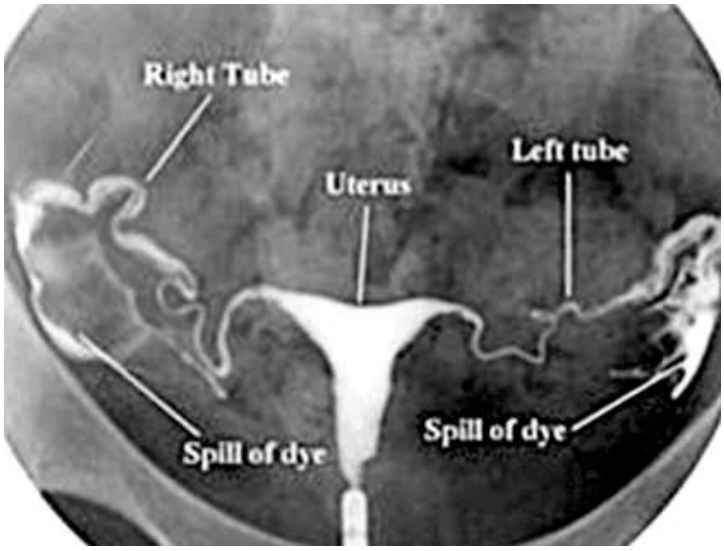
Other ovarian conditions	
Ruptured ovarian cyst	- Presents as sudden, sharp pain in an adnexa (adnexa = combo of Fallopian tube + ovary). - The Q will often say there is 10-15 mL of serosanguineous fluid in pouch of Douglas on culdocentesis (means aspiration of pouch of Douglas). - The Q will say there is no mass / will not mention mass. - Treated symptomatically (i.e., NSAID). Do not do laparoscopy.
Adnexal/ovarian torsion	- 3/4 Qs will give intermittent adnexal pain that becomes constant over hours to weeks (I've seen NBME give both time frames). - 1/4 Qs will just say few-hour Hx of constant pain without the intermittent detail first. - Presents as "large" adnexal mass (i.e., 8 x 12-cm). - There will be no mention of fluid in pouch of Douglas. - Treatment is laparoscopic detorsion. - Tricky detail is that the biggest risk factor is a pre-existing anatomic abnormality (e.g., cyst). So if they say Hx of cyst + now has large adnexal mass and pain, that is torsion, not ruptured cyst.
Theca-lutein cyst	- Arises from theca cells of the ovary; usually bilateral. - Frequently associated with conditions where hCG is very high, such as moles, multiple gestation pregnancies, or fertility drug use. - Can occasionally bleed, where the answer is "hemorrhagic theca-lutein cyst" as a diagnosis of exclusion (i.e., we eliminate to get there), where endometrioid cyst can be eliminated because the Q won't mention anything about dysmenorrhea or signs of endometriosis.
Endometroid cyst	- Aka endometrioma, or "chocolate cyst." - Cyst filled with endometrial tissue that bleeds. - Occurs in endometriosis (endometrial tissue growing outside the myometrium, usually on the ovary, as with this case; discussed more later).
Dermoid cyst	- Aka mature cystic teratoma. - Contains tissue from multiple germ layers – i.e., hair, skin, teeth, etc. - Can calcify.

Testicular cancer	
- Just listing what's relevant for Surg shelf.	
Seminoma	- Most common testicular cancer; occurs teenage years and older. - Large, uniform cells with watery cytoplasm. - Chemo- and radiosensitive. - Placental ALP can be a tumor marker.
Sertoli-Leydig cell tumor	- Secrete androgens. - Can cause gynecomastia (shows up on NBME). This is because ↑ androgens can be aromatized into estrogens.
Granulosa cell tumor	- Secretes estrogen. - Can also cause gynecomastia. If NBME gives you Q with Sertoli-Leydig cell tumor causing gynecomastia, they won't simultaneously list granulosa cell tumor as an answer.

Prostate cancer vs BPH points	
Prostate cancer	- Adenocarcinoma (i.e., glandular); most common cancer in men. - Stimulated by DHT. - Causes osteo**blastic** metastases (highest yield point on USMLE). - Can present as random guy over 50 with no PMHx who just suddenly has neurologic symptoms in the legs and two blastic lesions in his spine picked up on MRI. - If neuro Sx are present due to spinal mets, give IV methylprednisolone. - If no neuro Sx are present and they just say vertebral blastic lesions, answer will be radiotherapy. I've seen distractor answers in this case such as "morphine + corticosteroids," where the student says, "I thought we give steroids though." Yeah, if the patient has neuro Sx, which he doesn't here, plus jumping to morphine before trying other pain management first is wrong. - Orchiectomy will ↓ androgen production → leads to atrophy and shrinkage of prostatic cancer cells. - Flutamide blocks androgen receptor. It is given prior to leuprolide, which is a GnRH receptor agonist that, when given continuously, causes desensitization at GnRH receptor. Since leuprolide can ↑ LH and FSH prior to desensitization, the flutamide is given first. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA (i.e., not bound to protein) is low compared to bound-PSA in prostate cancer. - Gleason scoring is used to grade prostate cancer. - The parasympathetic nerves (for erection) are at greatest risk of injury during radical prostatectomy (asked somewhere on NBMEs).
BPH	- Benign prostatic hyperplasia. - All old dudes have enlarged prostates due to lifetime exposure to DHT, which causes prostatic enlargement. - Presents as dribbling, hesitancy commencing urination, or interrupted stream in a male over 60 (but usually 70s+). - "Old dude + high creatinine = BPH till proven otherwise." - Discontinue anti-cholinergic meds because they cause urinary retention. This concept shows up a lot on NBMEs. Three classes of drugs in particular: 1) First-gen H1 blockers (i.e., diphenhydramine, chlorpheniramine); 2) TCAs (i.e., amitriptyline). 3) Anti-psychotics. - You'll get a Q where an older male has high Cr and then the answer is just "discontinue anticholinergic meds," or "discontinue the diphenhydramine." - Can be described as "nodular" on palpation. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA is high compared to bound-PSA in BPH. - Tx is alpha-1 blocker, such as tamsulosin/terazosin; or the 5-alpha-reductase inhibitor, finasteride. - Transurethral resection of prostate (TURP) for refractory cases.

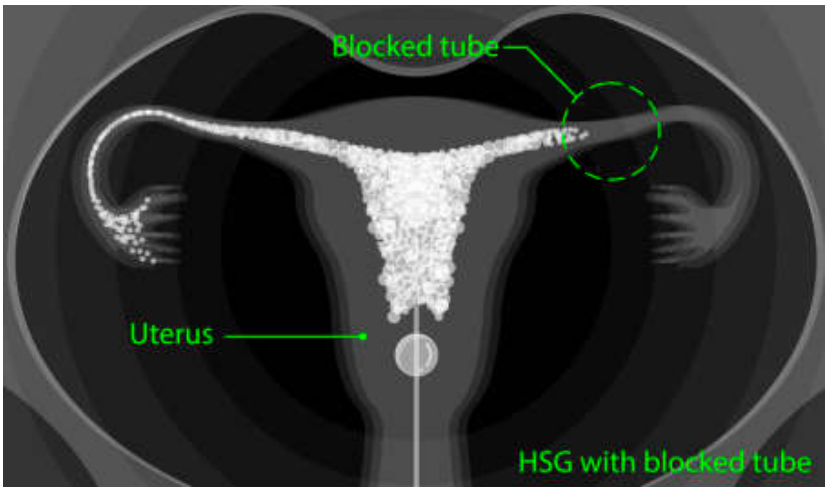
Pelvic inflammatory disease (PID)
- As discussed earlier, is caused by either *Chlamydia* or *Gonorrhea*. - When the infection ascends the uterus and Fallopian tubes, it can lead to inflammation and scarring, thereby increasing the risk for ectopic pregnancy (i.e., loss of Fallopian tube cilia leads to premature implantation usually in the ampulla of the Fallopian tube). - Adnexal and **cervical motion tenderness** are buzzy. But particularly the latter. If they say cervical motion tenderness, you know right away they're talking about PID. - USMLE wants you to know that the Fallopian tubes are normally open on both ends, where the release of the ovum from the ruptured Graafian follicle is "caught" by the fimbriae of the Fallopian tube, prior to its

transit to the uterus. Because the Fallopian tubes are open on both ends, when a hysterosalpingogram is performed, spillage of dye bilaterally into the peritoneal cavity *is normal.*



Normal hysterosalpingogram

- If there is Hx of PID and the Fallopian tube is scarred, it may be sealed, where the hysterosalpingogram shows failure of spillage.



- If the Q tells you a girl has PID and is treated with antibiotics but has persistent fever and adnexal pain, next best step = ultrasound to look for **tubo-ovarian abscess,** which is a potential sequela of PID.
- Extension of PID to the liver capsule is called Fitz-Hugh-Curtis syndrome.

Imperforate hymen	
- The hymen is a thin membrane that partially covers the vaginal opening in females. It normally has a small hole that allows the passage of menstrual blood out of the vagina. The size and shape of the hymenal opening can vary between females. - Imperforate hymen is when the hymen completely covers the vaginal opening, which can lead to hematocolpos and hematometra.	
Hematocolpos	- Blood accumulation within the vaginal canal due to imperforate hymen. - USMLE Q will give a teenage girl who's never had a menstrual period. Physical exam shows a "blueish bulge" behind the hymen. The Q might say the girl has had monthly pain

	(i.e., she has in fact had menses, but blood is blocked from leaving the vagina due to the imperforate hymen). - Answer is either hematocolpos or "cruciate incision of the hymen."
Hematometra	- Blood backing up all the way to the uterus causing uterine distention. - Q will give what sounds like imperforate hymen + hematocolpos but you'll notice the blood pressure is low → distention of the uterus can trigger a vagal response leading to low BP.

Hemolytic disease of newborn types	
Rhesus (Rh) type	- ~85% of people have Rh antigen on their RBCs – i.e., Rh(+). Women who are in the ~15% that are Rh(-) are at risk of developing IgG antibodies against Rh antigen if the fetus is Rh(+) and the circulations mix inadvertently. - When the mother develops antibodies against Rh (i.e., has become sensitized to it), we call this "Rh isoimmunization." NBME will throw this word around a bit, so you need to know it means "she's developed IgG antibodies against Rh." - Mechanism: Rh(-) mom in first pregnancy is exposed to fetal blood that is Rh(+); mom makes Abs against Rh; subsequent pregnancy results in IgG against Rh crossing placenta and targeting fetal Rh(+) RBCs, leading to hemolysis in the fetus. - As mentioned earlier, Rh status is checked at the first antenatal screening at 8-10 weeks. If the mother is Rh(-), then give RhoGAM at 28 weeks as well as at parturition. It must also be given if there are any interventions (e.g., amniocentesis), or if there's complications like spontaneous abortion or abruptio placentae. - If 2nd pregnancy onward, if Rh (-) woman is found to have titers against Rh, do not give RhoGAM during the pregnancy, since it's too late. - Kleihauer-Betke (KB) test is done in setting of hemorrhage or suspected mixing of maternal and fetal circulations in Rh(-) women. The mother's blood is drawn, and the test works by exploiting the resistance of HbF to acid, allowing fetal RBCs to be distinguished from maternal cells on a blood smear. Depending on the fraction of fetal RBCs present, this determines the dose of RhoGAM necessary.
ABO type	- Most anti-A and -B Abs people have to opposing blood types are IgM, but people with O blood can have fractionally greater IgG; if mother with O blood has higher % of anti-A and -B Abs that are IgG, fetus can be symptomatic. - USMLE will give you O+ mom usually in first pregnancy and A or B fetus (can occur in Rh- women in second pregnancies, but the USMLE wants to assess you specifically know the ABO type of HDN, so they'll give O+ mom in first pregnancy).
Kell, Duffy, Kidd type	- Minor antigens beyond Rh (i.e., Kell, Duffy, Kidd) can sometimes cause HDN. - The process for isoimmunization is same as Rh. - What will happen is, an A+/B+/AB+ mother will have a second pregnancy where fetal hemolysis is occurring, but based on her blood type, you know it's not ABO or Rh type HDN. - The Q will say her blood titer for, e.g., Kell is high. The next best step is "check Kell status of the father." - In other words, since the mother has titers against Kell, you know she's Kell negative. So if the father's RBCs are Kell(+), the fetus is likely Kell(+), and that would explain the HDN.

Ectopic pregnancy points
- Implantation of conceptus outside the uterus, including the parametrium of the uterus and cervix. - β-hCG levels will be much lower than expected for the gestational age. You don't have to be an obstetrician and know exact values. - β-hCG should double approximately every 2-3 days in early pregnancy, so the Q can also mention something about poor rate of change of increase. - Highest yield point for USMLE is that ectopic risk is ↑↑ in women who have Hx of PID. As discussed earlier, this is due to scarring of the Fallopian tubes and disruption of the cilia. Ectopic implantation most frequently occurs in the ampulla (70-80%). - Methotrexate is given for Tx (asked once on 2CK form) for "small, stable ectopics" – i.e., mother is hemodynamically stable, there is no evidence of tubal rupture / fluid in the peritoneal cavity, the ectopic is <3.5 cm, and β-hCG is <5000 mIU/mL. For your Obgyn rotation, I would know those criteria. - Laparoscopic salpingostomy/salpingectomy is done if methotrexate cannot be given. - If the female is hemodynamically unstable (i.e., low BP), the answer is laparotomy, not laparoscopy.

HY Obgyn infections	
Chorioamnionitis	- Infection of the uterus during pregnancy. - Greatest risk factor is ROM >18 hours. - Cause is usually polymicrobial. - Tx = ampicillin + gentamicin +/- clindamycin.
Endometritis	- Infection of the uterus *after* pregnancy. - Endometritis = "Postpartum endometritis" for all intents and purposes. - Can present as fever, uterine tenderness, and foul-smelling lochia within a few days of delivery. - Cause is usually polymicrobial. - Q will say woman had C-section two days ago + now has fever + lower abdo pain +/- foul-smelling lochia → answer = postpartum endometritis; chorioamnionitis is wrong answer because she's not pregnant anymore. - Tx = ampicillin + gentamicin +/- clindamycin. - USMLE also wants you to be aware of what is "normal post-operative course" versus endometritis. For example, the NBME Q can say something like, woman had Caesar two days ago + temp 99.3 F + incision site mildly erythematous + two inguinal lymph nodes are palpable and tender → answer = "normal post-operative course." Surgery is trauma, and the immune system will generate a natural response to it. Palpable lymph nodes can be a normal response to inflammation and surgery.
Septic pelvic thrombophlebitis	- Formation of infected blood clots in the pelvic veins (umbrella term for uterine and ovarian veins). - Endometritis is big risk factor. - Differentiated from endometritis in that it causes persistent pelvic pain and fever despite administration of antibiotics. - Answer choice shows up quite a bit on Obgyn forms, so know it exists.
Puerperal sepsis	- Generalized term for sepsis occurring after childbirth. - Presentation is variable; causes high fever, chills, tachycardia, and low BP. - Q might list both puerperal sepsis and septic pelvic thrombophlebitis as answer choices. Remember that SPT is pelvic pain unresponsive to antibiotics; puerperal sepsis is actual sepsis due to a number of causes, but often endometritis.
Toxic shock syndrome	- Caused by TSST superantigen of *S. aureus*. - Tampon use, or vulvovaginal/perineal skin infections are HY etiologies. For non-Obgyn Qs, cotton nasal packing also HY. - Toxin bridges MHC-II on macrophages with T cell receptor on CD4+ T cells, causing macrophages to release cytokines leading to shock. - Shows up on Obgyn material where they ask most likely cause of death in TSS, and the answer is ARDS. Slightly unusual answer, but I don't know what to tell ya.

Peripartum bleeding	
Uterine atony	- Most common cause of postpartum bleeding (70-80%). - Presents as vaginal bleeding in the context of large, boggy uterus. - Uterine massage immediately following Stage 2 delivery ↓ risk. - Refractory bleeding can be treated oxytocin to enhance uterine tone and contractions. If ineffective, ergonovine/methylergonovine can be administered. - Surgery such as uterine artery embolization or hysterectomy is last resort. - Ergonovine is contraindicated in HTN, preeclampsia, migraine with aura, smokers, and cardiovascular disease.
Vaginal laceration	- Usually seen in the setting of fetal macrosomia (e.g., maternal diabetes), as discussed earlier. - Episiotomy can be performed to ↓ risk.
Placenta previa	- Placenta implants over the internal cervical os. - Causes **painless 3rd trimester bleeding.** - C-section recommended 37 weeks onward. - Prior to 36 weeks, placental implantation site can occasionally move off the cervical os as the uterus continues to grow. - Hx of prior interventions to the uterus (i.e., myomectomy, previous Caesars) ↑ risk of placenta previa due to ↑ probability of abnormal implantation (i.e., if the endometrial lining is disrupted in any form, then chance of normal implantation is ↓).
Abruptio placentae	- Placenta detachment from the uterine wall prior to parturition. - Causes **painful 3rd trimester bleeding.** - Can present as intense cramping. - Classic causes are cocaine and deceleration injury (i.e., car accident).
Placenta accreta	- Placental attachment to the surface of the myometrium. - Causes postpartum bleeding.
Placenta increta	- Placental attachment into the myometrium. - Causes postpartum bleeding.
Placenta percreta	- Placental attachment through the myometrium, sometimes onto adjacent structures such as the bladder (percreta = perforates). - Causes postpartum bleeding.
Vasa previa	- Fetal vessels overly the internal cervical os and are prone to shearing forces and bleeding. - Presents as triad of 1) painless third-trimester bleeding, 2) rupture of membranes, 3) fetal bradycardia. - Caused by conditions such as velamentous cord insertion or succenturiate lobe of placenta. - Velamentous cord insertion = portion of fetal vessels normally protected by Wharton jelly within umbilical cord are exposed within the fetal membranes before they insert onto the placenta → ↑ risk of shearing/trauma → ↑ risk of bleeding. Sometimes the vessels overly the internal cervical os, ↑ risk of vasa previa. - A succenturiate lobe is an accessory/extra lobe of the placenta that is separate from the main body of the placenta but remains connected to it by blood vessels. When vessels connecting the main placenta and the succenturiate lobe run across the cervical os, this can cause vasa previa.
Retained placental tissue	- Fragments of the placenta or membranes remaining in the uterus post-delivery. - Just be aware it's a cause of postpartum bleeding.
Heme disorders	- Conditions such as DIC, ITP, and vWD ↑ risk of postpartum bleeding.
Uterine inversion	- Uterus turns inside out following the delivery of the placenta. - Typically presents as postpartum bleeding and abdominal pain following/during delivery of the placenta + the Q will say the uterus is not palpable (abdominally). - The Q does not need to say vigorous cord traction was applied. There is a 2CK Obgyn Q where they literally say "gentle cord traction," and the answer is still uterine inversion.
Uterine rupture	- Wall of uterus tears; often associated with labor or previous uterine surgery like Caesars (especially classic vertical incision, rather than low, horizontal incision).

	- USMLE Q will tell you oxytocin was given + there were strong and persistent contractions + now fetal parts are palpable in the RUQ. Student says, "What do they mean there's fetal parts in RUQ?" Exactly. They shouldn't be there, which indicates there was a rupture of the uterus. - Oxytocin can cause tachysystole and hypertonus, which ↑ risk of rupture. - Tachysystole = abnormally high frequency of uterine contractions, usually more than five contractions in 10 minutes, averaged over a 30-minute window. - Hypertonus = prolonged contraction of the uterus > 2 minutes. - Contractions should normally occur once every 2-5 minutes and last 30-60 seconds. - I explain contractions more below with respect to Montevideo units.

HY Obgyn cardiovascular DDx	
Amniotic fluid embolism	- The answer on USMLE for acute-onset shortness of breath and tachycardia 30 seconds to 2 minutes after delivery of the placenta. - Amniotic fluid leaks into maternal circulation + goes to pulmonary arterioles. - Can cause **disseminated intravascular coagulation** with bleeding from IV lines / catheter sites. - Tx = supportive.
Pulmonary embolism	- The answer on USMLE for acute-onset shortness and tachycardia two days postpartum when the mother gets up to go to the bathroom (presumably a DVT that formed while in hospital bed launched off to lungs). - Tx = heparin first, followed by CT of the chest (aka spiral CT or CT angio). - *If woman is pregnant,* then do V/Q scan before CT. This is in order to decrease potential radiation exposure to the fetus. - If V/Q scan is performed during pregnancy that shows "segmental perfusion defects" (i.e., suggestive of PE), the next best step in diagnosis is CT of the chest if it's listed. Student says, "Wait, but I thought you just said we don't do that in pregnancy because of radiation." I agree with you. That's why we did V/Q scan first. But if they force you to choose a NBS in Dx following the V/Q scan, the answer is still CT. Take it up with USMLE, not me.
Peripartum cardiomyopathy	- Dilated cardiomyopathy almost always during 3rd trimester or in the first few months postpartum. - Cause is multifactorial, but some studies have identified an autoantibody-mediated process. - Will present as gradually worsening shortness of breath on exertion (i.e., LHF at a minimum). Depending on severity, can also present with JVD and significant pitting edema (RHF findings). - Peripartum cardiomyopathy tends to get worse with each subsequent pregnancy. - Diagnosis is made with transthoracic echo looking for low ejection fraction. - Qs like to ask how to determine degree of maternal/fetal risk in subsequent pregnancies → answer = transthoracic echo (looking for EF). In other words, the next pregnancy is going to have an even lower EF than the current, so if the EF is already significantly low (NR 55-70%), we know it'll only be worse subsequently.
Normal peripheral edema	- Peripheral edema is common in pregnancy due to ↑ plasma volume + the uterus compressing the pelvic veins and IVC; both cause ↑ venous hydrostatic pressure.
IVC compression syndrome	- The answer on USMLE for 3rd trimester female who gets lightheadedness, nausea, palpitations, shortness of breath, and sometimes a drop in blood pressure while supine. - Due to compression of uterus on IVC, ↓ venous return. - Ameliorated with positioning/sleeping left-lateral decubitus.
Atelectasis	- Confusing word that refers to "lung collapse," or "collapse of alveoli." - Highest yield point for USMLE is that it is the most common cause of fever within 24 hours of post-surgery. If this is the first time you're reading this, that might sound weird, but this is pass-level and extremely important for 2CK.

	- There is one 2CK Q where they say a woman had a C-section two days ago and the answer was still atelectasis, so even though it's most common <24 hours, just be aware one Q exists where, oh em gee, it's 2 days later. - The mechanism is related to combo of pain meds + sedentation, where breathing becomes slower + shallower in hospital bed, leading to mild collapsing of some alveoli. This is why breathing exercises can be important post-surgery. - Will often present as **bibasilar shadows or opacities.** In other words, patient had surgery yesterday + now has fever + CXR shows mild opacity at the lung bases → answer = atelectasis. - NBME assesses obstructive (aka resorptive) atelectasis. This is when an area of lung distal to an obstruction from, e.g., a tumor, can cause alveoli to collapse. This then increases the chance for pneumonia distal to the obstruction. - 2CK IM form has "endobronchial obstruction" as answer for distal area of lung collapse (i.e., atelectasis) in patient with lung cancer; "vascular occlusion by tumor" is wrong answer (makes sense, as the tumor obstructs the respiratory tree, not blood vessel, in this case, but I've seen students accidentally choose the latter).
Budd-Chiari syndrome	- Hepatic vein thrombosis due to hypercoagulable state in pregnancy. - Presents as abdominal pain + hepatomegaly +/- ascites (latter indicates hydrostatic pressure backup to the portal vein). - Placenta produces plasminogen-activator inhibitor (PAI-2), which ↓ plasmin activity and fibrinonlysis → hypercoagulable state. In addition, ↑ estrogen and progesterone contribute to hypercoagulable state by ↑ fibrinogen, clotting factor, and vWF synthesis. - Hypercoagulable state in pregnancy is an evolutionary mechanism to ↓ hemorrhage risk at parturition.
Hemorrhoids	- Bleeding or protrusion from rectal veins. - Common during pregnancy, particularly in the second and third trimesters. - ↑ Pelvic blood flow and pressure from uterus on the pelvic and rectal veins. - ↑ Progesterone during pregnancy relaxes the walls of the veins, allowing them to swell more easily. - Constipation, which is also common during pregnancy, can cause straining during defecation, further contributing to the development of hemorrhoids. - Tx = dietary modifications to prevent constipation, using cushions or pillows to relieve pressure when sitting, and avoiding prolonged periods of standing or sitting.

Polycystic ovarian syndrome (PCOS)
- High BMI female → insulin resistance. - Insulin resistance causes abnormal GnRH pulsation → leads to ↑ LH and ↓ FSH. This is often truncated as just saying there's an ↑ LH/FSH ratio. Some students think FSH is also ↑, but it's just the ratio that's ↑. That's wrong. FSH is low. - LH normally acts on the theca lutein cells of the ovaries to make androgens. Since LH is ↑, we get hirsutism. - FSH normally stimulates follicular development. Since FSH is ↓, we have poor follicular development, leading to failure of a Graafian follicle to rupture during ovulation. The unruptured follicle is retained as a follicular cyst. - Failure of ovulation is called anovulation. This term is exceedingly HY on USMLE. It presents as a female with irregular periods. - Anovulation is a broader term than PCOS, as it can be due to other conditions as well, such as hypothyroidism and Cushing syndrome. But when you hear the word "anovulation" alone, it is usually used as synonymous for the same mechanism as PCOS – i.e., high-BMI female who has irregular periods due to abnormal GnRH pulsation causing an ↑ LH/FSH ratio, with the ↓ FSH causing failure of follicular rupture. - So high-BMI female + irregular periods = anovulation till proven otherwise. - Anovulation + hirsutism = PCOS.

- In clinical practice, part of the PCOS diagnosis requires ultrasounds showing 11+ cysts bilaterally (Amsterdam criteria). But USMLE doesn't assess this. You just need to know anovulation + hirsutism = PCOS.
- Normally when a follicle ruptures, the remnant is called the corpus luteum, which secretes progesterone.
- Progesterone inhibits growth of endometrium; estrogen stimulates growth of endometrium.
- Women who have anovulation have ↓ progesterone production because they don't form a corpus luteum.
- This means they have ↑ estrogen in comparison to progesterone. We call this **unopposed estrogen.** This is one of the highest yield phrases for USMLE.
- Unopposed estrogen means ↑ risk of endometrial hyperplasia and, in turn, ↑ risk of endometrial carcinoma.
- The Q can give you a high-BMI female who's post-menopausal + has vaginal bleeding. Answer is just straight-up endometrial biopsy. Student asks how we know it's endometrial cancer. My response is, if she's overweight, this implies she was probably overweight in the past, which implies she's had history of anovulatory cycles and endometrial hyperplasia, leading to ↑ endometrial cancer risk.
- Because insulin resistance is the basis for PCOS, patients are at ↑ risk of developing type II diabetes.

- As mentioned above, hypothyroidism and Cushing syndrome are also HY causes of anovulation on USMLE.
- ↑ Glucocorticoids in Cushing syndrome can cause insulin resistance and anovulation, where the diagnosis can appear like PCOS. The difference is PCOS is idiopathic in response to high BMI – i.e., it is not caused by a known secondary etiology like Cushing syndrome, even though the presentations can be similar.
- Hypothyroidism leads to ↑ thyroid-releasing hormone (TRH), which stimulates prolactin, which causes abnormal GnRH pulsation. Even though this is the mechanism, the caveat I issue is that USMLE does not directly assess this, and it is infinitely more important you know that the mechanism for prolactin secretion is ↓ dopamine, or overt dopamine-2 receptor antagonism (i.e., for severance of pituitary stalk Qs).

Progesterone withdrawal test

- Progesterone withdrawal test is the process of giving progesterone followed by seeing if bleeding occurs following its withdrawal. It can be used to help diagnose the cause of oligo- or amenorrhea.
- As discussed earlier, menses are caused by the presence of progesterone followed by its withdrawal.
- If bleeding occurs (i.e., positive progesterone withdrawal test), it means estrogen levels are normal and the endometrial lining builds up just fine, and that anovulation is the cause of the irregular menses.
- In PCOS, for instance, since we don't have a corpus luteum and progesterone secretion, we don't have the sequence of "progesterone present followed by withdrawal," so we can't get regular menses. However, since estrogen levels are normal in PCOS, the endometrial lining can grow without a problem, so if progesterone is given exogenously, followed by its withdrawal, we will see bleeding.
- In conditions like Turner (where estrogen is low) or Asherman (scarred uterus that can't grow), giving progesterone won't induce bleeding since the endometrial lining hasn't developed. The bleeding will only occur if estrogen is present in normal amounts to allow for endometrial growth, where the administration, followed by withdrawal, of exogenous progesterone essentially induces forced menses.

Primary dysmenorrhea vs endometriosis vs adenomyosis

Primary dysmenorrhea	- Dysmenorrhea = period pain. - Primary dysmenorrhea = "normal period pain"; benign. - Mechanism is prostaglandin PGF2α hypersecretion. - During menstruation, the endometrial cells release high levels of prostaglandins, which cause uterine contractions and pain. The prostaglandins can also make their way into the systemic circulation, leading to headache and nausea. - This is the answer on USMLE for period pain under the age of 20. The Q can say, girl who is 17 has menstrual pain so bad she has to miss class + physical exam shows no abnormalities → answer = primary dysmenorrhea. - The answer can also be written simply as "prostaglandin secretion." - The key detail regarding primary dysmenorrhea is that **the physical exam is normal.** - In endometriosis, the physical exam is abnormal + is always over age 20. - Treatment is NSAID. OCP is wrong answer if listed alongside NSAID (on NBME).

Endometriosis	- Growth of endometrial tissue outside the uterus. Most common location is the ovary, but can also grow in other locations like the pouch of Douglas. - Mechanism is thought to be retrograde menstrual flow through the Fallopian tubes + seeding onto the ovaries or peritoneum. - Causes severe menstrual pain over age 20 + **an abnormal physical exam.** - The abnormal physical exam can be any miscellaneous finding – i.e., there is no specific finding you have to memorize. They can say findings such as nodularity of the uterosacral ligaments, a fixed retroverted uterus (due to adhesions from lesions), or pelvic tenderness. - In contrast, severe period pain + normal physical exam = primary dysmenorrhea. - In other words: - 23F + period pain so bad she has to miss work + normal physical exam → answer = prostaglandin secretion / primary dysmenorrhea. - 23F + period pain so bad she has to miss work + physical exam shows nodularity of the uterosacral ligaments → answer = endometriosis. - Descriptors such as pain with defecation (due to pouch of Douglas lesions) or dyspareunia (pain during sex) are highly buzzy and pass-level but too easy, so are often omitted from NBME questions, as per my observation. - USMLE wants diagnostic laparoscopy as next best step. - NSAIDs and OCPs are short-term measures, but definitive Tx is laparoscopic removal of lesions.
Adenomyosis	- Growth of endometrial tissue within the myometrium. - Presents as a diffusely enlarged uterus + vaginal bleeding in woman 30s-50s. - May or may not be painful. - Q will say woman had tubule ligation two years ago + now has uterus that is 8 weeks' gestation in size + vaginal bleeding → answer = adenomyosis. Student is confused and says "how can she be pregnant if she had tubule ligation?" She's not. Use your head. It's just how the Q can describe the ↑ size. - Tx = NSAIDs + OCPs. - Leuprolide can be used in theory, but I haven't seen it assessed.

Miscellaneous repro/obgyn DDx for Surg	
Asherman syndrome	- Scarring of the uterus by fibrous adhesions, **aka uterine synechiae.** - Caused by Hx of instrumentation – i.e., D&C. - **Can cause amenorrhea.**
Cervical stenosis	- Q will say female has a Hx of LEEP for cervical CIN2 or 3 six months ago + now has pelvic pain with menses + examination shows a small, scarred cervical os. - Pain can occur in theory if outflow of menses is ↓, similar to imperforate hymen.
Chemo- / radiotherapy-induced amenorrhea	- As the name implies. You just need to be aware that chemo- and radiotherapy can cause amenorrhea. - Shows up on NBME as "primary hypogonadism" as the answer – i.e., the ovaries themselves are hypo-secreting hormones as reason for amenorrhea.
Chronic interstitial cystitis	- Unexplained pelvic pain, bladder pain, and/or dysuria >6 weeks, where all laboratory studies are normal. - Can present with anterior vaginal wall pain (bladder is anterosuperior to vagina). - Q will say 25F + 6 months of dysuria + pubic pressure + pregnancy test negative + labs all normal → Dx = interstitial cystitis straight-up. Not hard. - Treatment is mundane, soft porn stuff like lifestyle modification and dietary changes. Steroids are a **wrong** answer.

Repro/obgyn drugs	
- Not meant to be exhaustive list; just HY ones for Surg.	
Aromatase inhibitors	- ↓ Conversion of androgens to estrogens. - Letrozole is used to stimulate ovulation by ↓ negative-feedback at hypothalamus / anterior pituitary. Some students ask about this agent in comparison to clomiphene (discussed below). The literature seems to be split on it, and I haven't seen NBME give a fuck. - Anastrozole and exemestane are used for breast cancer in post-menopausal women.
BPH meds	- Finasteride is 5α reductase inhibitor used for BPH. ↓ conversion of testosterone to DHT. Since DHT causes prostatic growth, ↓ DHT means ↓ prostatic growth. - Tamsulosin/terazosin are α1 blockers that ↓ constriction of internal urethral sphincter of the bladder and help promote urinary outflow.
Clomiphene	- Selective-estrogen receptor modulator (SERM) used to stimulate ovulation in those with irregular cycles (in particular those PCOS). - Has partial-agonist effects at the hypothalamus that are weaker than endogenous estrogen, so the hypothalamus interprets this as ↓ estrogen is present, so negative-feedback also ↓ → GnRH ↑.
GnRH receptor agonists	- Leuprolide, goserelin, and nafarelin are all GnRH receptor agonists. When administered continuously, they cause desensitization of the GnRH receptor at the anterior pituitary, leading to ↓ LH and FSH secretion (i.e., even though the drugs are pharmacologic agonists, they function clinically as antagonists).
Methotrexate	- Given for small, stable ectopic pregnancies – i.e., mother is hemodynamically stable, there is no evidence of tubal rupture / fluid in the peritoneal cavity, the ectopic is <3.5 cm, and β-hCG is <5000 mIU/mL. - If the above criteria are not fulfilled, do salpingostomy/salpingectomy for ectopic (or laparotomy if unstable).
SERMs	- SERMs are agents that have different effects depending on the tissue. - Tamoxifen and raloxifene are SERMs used for estrogen-receptor (+) breast cancer, where they both are antagonistic at breast and agonistic at bone. - Tamoxifen causes ↑ risk of endometrial cancer due to agonistic effects at endometrium.

Surg Derm

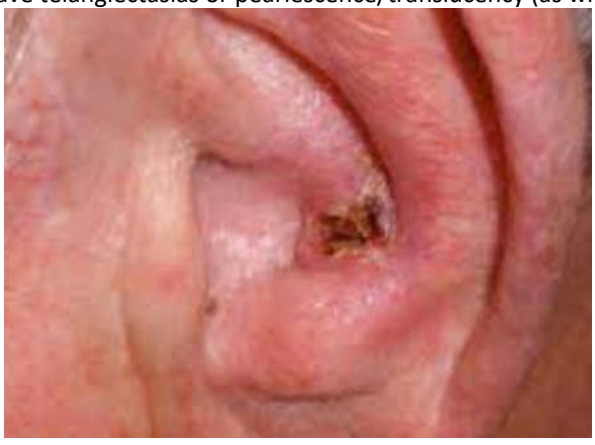
HY Derm conditions for Surg	
Cellulitis	 - *Staph aureus* exceeds *Strep pyogenes* (Group A Strep) as causal organism; must give beta-lactamase-resistant beta-lactam in the methicillin class (i.e., dicloxacillin, flucloxacillin) or first-generation cephalosporin (i.e., cephalexin, cephazolin), or Augmentin (amoxicillin-**clavulanate**). - Amoxicillin and penicillin alone are wrong answers. - 90% of community *Staph* (i.e., MSSA) produces beta-lactamase, so amoxicillin and penicillin alone will not work if *Staph* is the cause.
Erysipelas	 - Infection of upper dermis and superficial lymphatics. - Caused by *Strep pyogenes* (Group A Strep), which eclipses *Staph aureus* for erysipelas. - Looks worse than cellulitis but is more superficial / “not as bad”; has characteristic “fiery red” appearance and may appear well-demarcated with raised edges. - Although Group A Strep > *Staph* for erysipelas, Tx is same as cellulitis (oral dicloxacillin, cephalexin, or Augmentin) because Staph can still cause it. Penicillin alone can be used for *Strep* pharyngitis.
Meningococcus	

	 - *Neisseria meningitidis;* gram-negative diplococcus. - Causes meningitis + characteristic non-blanching rash. - Low BP can be endotoxic shock, but student should bear in mind Waterhouse-Friderichsen syndrome is often asked; give hydrocortisone to increase BP after normal saline is administered.
Parvovirus B19	 - Fifth disease = “slapped cheek” facial erythema caused by Parvo B19 in Peds. - Next best step = check serum IgM titers; if IgM titers are not listed, choose bone marrow biopsy. - Increased risk of aplastic anemia in sickle cell. - Once child has developed the red cheeks, he/she has immunologically cleared the illness (i.e., if they turn it into a behavioral science Q, tell parents to chill the fuck out / relax because the child has cleared the virus).  - Can cause body rash +/- arthritis in adults; same as with Fifth disease, next best step is check serum IgM titers.

Dermatitis herpetiformis	 - Extra-intestinal cutaneous eruption that can occur in patients with Celiac disease. - Despite the name, this is not actual herpes. It is IgA deposition at dermal papillae. - Don't confuse with eczema herpeticum.
Hidradenitis suppurativa	- Skin condition characterized by painful abscesses and sinus tracts, most commonly in the groin, axilla, and under the breasts; frequently affects apocrine glands. - Cause is a combination of genetic and environmental factors (idiopathic). - NBME answer is "surgical excision of lesions." 
Erythema nodosum	 - Panniculitis (inflammation of subcutaneous fat) that resembles a rash. - Seen with autoimmune diseases like sarcoidosis and Crohn, as well as part of serum sickness due to medications (e.g., sulfa).

Dermatomyositis	 - Image shows heliotrope rash (left) and shawl rash (right) seen in dermatomyositis. - Anti-Jo1 antibodies and electromyography + nerve conduction studies = diagnostic. - Do not confuse heliotrope rash of dermatomyositis with malar rash of SLE. - Give steroids for flares.  - Patients can also get Gottron papules (above image) and “mechanics’ hands” (i.e., rough-surfaced / scaly hands).
Latex allergy	 - Q will give you a nurse or laboratory technician. The implication is that he/she has been using gloves resulting in contact dermatitis due to latex. - Can also occur due to adhesives (i.e., from bandages applied).

NF1	 - Neurofibromatosis type I; autosomal dominant; one of the phakomatoses (neurocutaneous disorders → NF1/2, VHL, TSC, Sturge-Weber). - Image shows café au lait spot. - NF1 → café au lait spots, neurofibromas, axillary/groin freckling, optic glioma, pheochromocytoma.
Tuberous sclerosis	- Autosomal dominant; one of the phakomatoses. - Image shows adenoma sebaceum (angiofibromas). - Intracranial/periventricular nodules ("tubers" – i.e,. hamartomas), adenoma sebaceum (angiofibromas), cardiac rhabdomyoma, lymphangeoleiomyomatosis, subungual fibromas, renal angiomyolipoma.  Image shows adenoma sebaceum
Acanthosis nigricans	 - Brown/black velvety skin, usually along nape of neck. - Associated with hyperinsulinemia and type II diabetes almost always. - Can also very rarely be seen in patients with underlying, visceral malignancy.

Basal cell carcinoma (BCC)	 - Classically pearlescent / slightly translucent; talengiectasias common; borders can sometimes be described as “heaped up” or “rolled”; may or may not be ulcerated. - Treatment for skin cancers on cosmetically sensitive areas such as the eyelid or nose can be managed with Mohs micrographic surgery (on NBME).
Actinic keratoses	  - Aka solar keratoses; precursor to squamous cell carcinoma (SCC); classically described as red/scaly lesions on forehead, ear, or arms of fisherman, farmers, or construction workers. Cryotherapy is usual treatment.
Squamous cell carcinoma (SCC)	- Ulcerated lesion in the setting of a patient who’s had actinic keratoses – i.e., Q says patient has had scaly red areas on forehead for many years + now has recent ulceration → answer = SCC. - SCC will not have telangiectasias or pearlescence/translucency (as with BCC). 

Keratoacanthoma	 - Can be confused with SCC; described as dome-shaped with a hyperkeratotic core, surrounded by a wall of inflamed skin. - Tx is surgical excision.
Melanoma	 - Depth is most important for prognosis. - Answer on NBME = excisional biopsy; suspected malignant lesions in non-cosmetically sensitive areas (i.e., not on the head/neck) can simply be excised with narrow margins; excision of additional tissue is important if margins are positive (i.e., entire lesion wasn't excised). - "Full-thickness biopsy" answer on USMLE for suspected melanoma on back of neck (NBME Q gives lesion similar to above on neck, with full-thickness biopsy as answer). Excision of lesion is management for neck lesions if full-thickness biopsy confirms melanoma; do Mohs micrographic surgery for facial lesions. - Punch biopsy is a type of full-thickness biopsy; if Q asks, choose side of lesion for biopsy, rather than middle of lesion. - Students will sometimes ask about shave biopsy. I have never seen this assessed on NBME, but literature says it can sometimes be used to remove superficial non-pigmented lesions where the clinician does not suspect melanoma (performing shave biopsy on melanoma can create problems for assessing depth, prognosis, and therapy). 

	- Acral lentiginous melanoma occurs on the palms/soles (areas not usually exposed to sun); more common in persons of African and Asian descent.  - Can also occur under the finger nail.
Seborrheic keratoses	 - Sun exposure and old age are biggest risk factors; also seen in smokers. - Described as “greasy,” or waxy, skin growths that appear like they can be “peeled off”; not malignant. - Sign of Leser-Trélat = sudden, eruptive seborrheic keratoses secondary to underlying, visceral malignancy (i.e., unrelated to age or sun).
Blue nevus	- Aka Mongolian spot. - Type of benign birthmark where dermal melanocytes fail to migrate superficially to stratum basale. - More common in Asians and blacks. - Often mistaken for child abuse. - Schedule routine follow-up; wrong answer is contacting child protective services. 

Strawberry hemangioma	 - Benign capillary tumor that will grow slightly then regress spontaneously within a few years. - No Tx necessary unless causing functional impairment. - Don't confuse with cherry hemangioma, which is due to sun exposure in elderly and presents as a tiny cherry-colored papule.
Cherry hemangioma	 - Benign capillary tumors caused by sunlight; benign; greater prevalence as age ↑. Students confuse with strawberry hemangioma in peds.
Kasabach-Merritt syndrome	 - Aka infantile hemangioma with thrombocytopenia; asked several times on 2CK Peds assessment. - Child will have thrombocytopenia due to platelet sequestration within the lesion. - Large and deforming; don't confuse with strawberry hemangioma. - Tx is surgical.

Cushing syndrome	 - USMLE wants you to know purple striae are HY physical exam finding. - Weakening of dermal collagen + capillary walls → micro bleeding into skin.
Keloids	- Disorganized growth of dermal collagen. - Tx is surgical excision, although recurrence is common. Benign. 
Varicose veins	- Q will ask next best step in diagnosis → answer = duplex venous ultrasonography for venous disease / valvular insufficiency. - Tx is compression stockings. - Do not choose surgical interventions for varicose veins on USMLE. 
Superficial thrombophlebitis	- Similar to DVT, but in more superficial vein; presents as painful, warm, “palpable cord” at the ankle that may or may not track up to the knee. - Treated with subcutaneous enoxaparin (heparin); wrong answer is compression stockings; very difficult Q since compression stockings are common answer for venous disease, but if patient has active venous occlusion (i.e., DVT or superficial thrombophlebitis), give heparin.

	- In other words, for simple venous disease / varicose veins, choose compression stockings, but if the patient has an active clot presenting as a palpable vein with pain, give heparin instead. 
Buerger disease	 - Aka thromboangiitis obliterans; condition is digital gangrene in males who are heavy smokers; contrasts with gangrene due to diabetes, which is pedal. - Tx = smoking cessation. - Do surgical excision / debridement of necrotic tissue.
Charcot joint	- Aka neurogenic joint; mechanism is “lack of appropriate joint sensation” – i.e., patient can feel his or her feet due to neuropathy → damage → can’t heal the area due to vasculopathy. - Q will be diabetic with poor HbA1c + foot ulcer + x-ray of the shows disorganization of the tarsals/metatarsals. 

Cryoglobulinemia	- Cryoglobulins are immune complexes that precipitate at cold temperatures.  - Often presents as livedo reticularis, which is a mottled, reticulated vascular pattern that has many etiologies; associated with hepatitis C. - Cryoglobulinemia is associated with decreased serum complement protein C4 (in contrast, C3 is sometimes down in SLE flares and Group A Strep infections).
SVC syndrome	 - Image shows Pemberton sign (facial erythema due to diminished venous return when arms are raised above head). - Seen classically in Pancoast (i.e., apical) tumors of the lung.

Surg Neuro

Stroke risk factors
- The two most important risk factors for stroke (cerebral infarction) on USMLE are hypertension and atrial fibrillation. - Regarding HTN, a strong systolic impulse pounds the carotid arteries, leading to endothelial damage and increased development of atherosclerotic plaques (carotid stenosis), which then launch off to the brain/eye. - Atrial fibrillation results in turbulence and stasis, leading to left atrial mural thrombus, which launches off. - Hypertension is most common in the population for causing stroke. Blood pressure control is more important than smoking cessation for decreasing stroke risk. - If the USMLE Q gives you a patient who has both AF and HTN, they want AF as the most important risk factor. In other words, even though HTN is the most common risk factor in the population for stroke, AF is still more likely to cause stroke if the patient has both. - There are other details regarding stroke I will discuss in this PDF, but for starters, these risk factors are pass-level points for understanding how/why most strokes occur – i.e., "How do most strokes occur?" → "Oh, well the patient will usually have carotid stenosis from HTN, where an atheromatous plaque has launched off, or will have AF where an LA mural thrombus has launched off." Simple. - I discuss carotid stenosis and AF management for stroke prevention in detail in the HY Cardio PDF.

Circle of Willis HY strokes	
Affected vessel	**HY Points**
ACA	- Anterior cerebral artery. - Motor/sensory abnormalities of contralateral leg.
MCA	- Middle cerebral artery. - Motor/sensory abnormalities of contralateral arm + face. - Dominant MCA stroke (usually left) can lead to Wernicke or Broca aphasia (discussed below). - Non-dominant MCA stroke (usually right) can cause hemispatial neglect (inability to draw clockface).
PCA	- Posterior cerebral artery. - Contralateral homonymous hemianopsia with macular sparing. - Prosopagnosia (inability to recognize faces).

Other stroke syndromes	
Lateral medullary syndrome	- Aka Wallenberg syndrome. - The answer for **dysphagia + ipsilateral Horner syndrome** after a stroke. - Due to stroke of posterior inferior cerebellar artery (PICA) or vertebral artery. - PICAchew (Pikachu) → PICA stroke causes dysphagia. - Students might know Horner syndrome (ipsilateral miosis, partial ptosis, anhidrosis) can be caused by Pancoast tumor, but another HY cause is lateral medullary syndrome.
Medial medullary syndrome	- The answer for **ipsilateral tongue deviation** after a stroke. - Due to stroke of anterior spinal artery.
Lateral pontine syndrome	- The answer for **ipsilateral Bells palsy** after a stroke. - Due to stroke of anterior inferior cerebellar artery (AICA). - FACIAL (contains AICA backwards) → "That's the one that's Bells palsy."
Weber syndrome	- Midbrain stroke. - The answer for **ipsilateral CN III palsy** (i.e., down and out eye) + contralateral spastic hemiparesis (weakness).
Locked-in syndrome	- Basilar artery stroke. - The answer for **inability to move entire body** except for eyes.

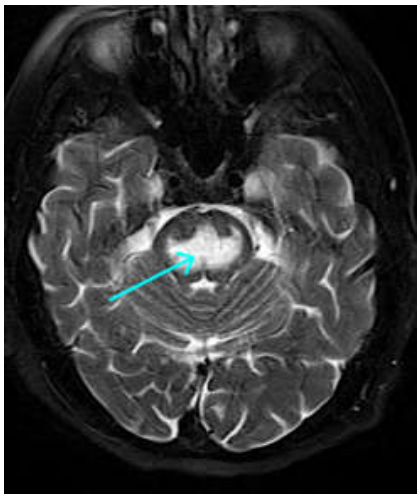
Gerstmann syndrome	- Stroke of angular gyrus of parietal lobe. - Tetrad of 1) agraphia (inability to write); 2) acalculia (cannot do math); 3) finger agnosia (can't identify fingers); 4) left-right disassociation (cannot differentiate between left and right sides of body).
Hemiballismus	- Stroke of subthalamic nucleus. - Causes "ballistic" flailing of contralateral arm and/or leg. - Weird condition, but for some reason asked. For example, if they say patient has random flailing of left arm following a stroke, answer = right subthalamic nucleus.

Lenticulostriate strokes	
Lacunar infarcts	- HTN can cause lipohyalinosis (fancy word for microatheroma formation) of small lenticulostriate arteries deep within the brain. - Ischemia within these small vessels leads to necrosis and reabsorption of tissue, leading to tiny cavities called lacunae. - Unlike larger strokes, lacunar infarcts often do not present with cortical deficits like aphasia, neglect, or visual field losses. Instead, they manifest as specific syndromes based on their location, such as pure motor hemiparesis, pure sensory stroke, ataxic hemiparesis. The USMLE doesn't give a fuck about the specific types of lacunar deficits/strokes. They just what you to know they exist / are possible, and that the mechanism is HTN causing lipohyalinosis of these small, penetrating vessels, as mentioned above.
Charcot-Bouchard microaneurysms	- Charcot-Bouchard microaneurysms are tiny (<1mm) aneurysms that can form within lenticulostriate arteries that can bleed an cause hemorrhagic strokes / intraparenchymal (intracerebral) bleeds.

Stroke management / Tx	
Ischemic stroke	- Non-contrast CT of the head is done to look for ischemic vs hemorrhagic stroke. If blood is present, it will appear as bright (hyperdense) areas on CT. - If stroke is ischemic (i.e., no blood seen on non-contrast CT), tPA is given if symptom onset has been within the past 4.5 hours and there are no overt contraindications to tPA (discussed below). - The start of the 4.5-hour window refers to when the patient was last confirmed/observed to be normal. So if the patient was sleeping prior to the stroke, it must be known exactly when the patient started sleeping, and if he/she was normal prior to going to bed. - The USMLE can be tricky about this point, where they might say a guy awoke this morning from sleep 2 hours ago + has facial drooping, where tPA is wrong because the stroke could have occurred at any point while he was asleep + the last time he was observed to have no stroke signs was the night before. - If the 4.5-hour window has elapsed, aspirin is the answer for ischemic stroke. - BP should be brought down rapidly until below 185/110 so that tPA can be given. Below this threshold, cautious/slower BP control should be done, since ↑ BP can actually facilitate perfusion of penumbric areas of ischemia (i.e., areas of ischemia around the stroke).
Hemorrhagic stroke	- Do not give tPA. First step is lowering BP (labetalol, nicardipine, or hydralazine). - Reduce BP rapidly to systolic <140 mm Hg. - Reverse anticoagulation if the patient is on it (i.e., FFP for patients on warfarin).

Hypercalcemic crisis
- Refers to cognitive dysfunction / a delirium-like state in the setting of severe hypercalcemia, often due to malignancy or primary hyperparathyroidism. - USMLE wants you to know that high calcium, as well as any sodium disturbance, can cause delirium.

- First step on USMLE for Tx of hypercalcemia is normal saline.
- After normal saline, USMLE wants bisphosphonate therapy (I've seen pamidronate listed on NBME).
- I've never seen calcitonin or loop diuretics as correct answers for hypercalcemia Tx. They're always wrong.

Sodium correction	
Cerebral edema	- Correction of hypernatremia too quickly with hypotonic saline can cause cerebral edema.
Central pontine myelinolysis	- Correcting hyponatremia too quickly with hypertonic saline causes central pontine myelinolysis. - Can be described as "osmotic demyelination." - Causes locked-in syndrome.  - USMLE can show MRI of the pons similar to the above.

Pseudotumor cerebri
- Term that refers to increased intracranial pressure in the setting of no clear structural cause. - Some notable causes are obesity (idiopathic intracranial hypertension), OCPs, isotretinoin, and danazol.

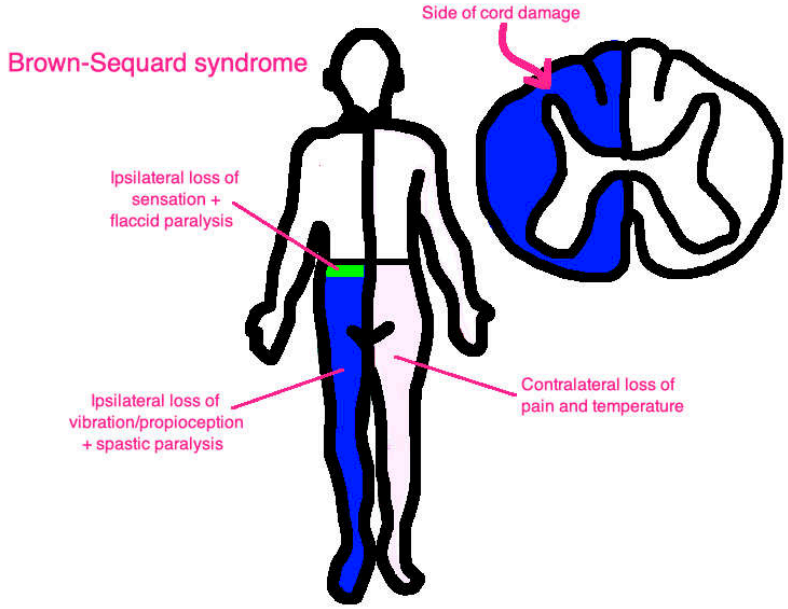
Reye syndrome
- Cerebral edema and hepatotoxicity seen with administration of aspirin to children (and sometimes adolescents) during a viral infection. - Thought to be due to impairment of beta-oxidation, but USMLE doesn't care.

Diffuse axonal injury
- Deceleration injury (e.g., car accident) followed by severe cognitive and/or motor/sensory dysfunction (i.e., following an MVA, the patient has to relearn tying his/her shoes, etc.). - USMLE can show MRI showing scattered hyperintense (white) lesions.



Pass-level spinal tract functions	
Spinothalamic tract	- **Pain and temperature sensation from contralateral body.** - Cross-over (decussation) point is in spinal cord, meaning if, e.g., the left spinothalamic tract in the spinal cord is fucked up, we lose pain and temperature on the right side of the body below the lesion. - Decussation point is called anterior white commissure. A lesion at this point is called syringomyelia, where there is bilateral loss of pain and temperature below the lesion.
Corticospinal tract	- **Motor function from ipsilateral body.** - In other words, if left-sided corticospinal tract is fucked up, we lose motor function on the left side below the lesion.
Dorsal columns	- **Carry vibration + proprioception from ipsilateral body.** - Tabes dorsalis is dorsal column lesion caused by neurosyphilis. - If the dorsal columns are disrupted, patient will have (+) Romberg test, which is where he/she falls over when standing with eyes closed. I discuss this more below.

Upper- vs lower motor neuron findings	
UMN	- Lesion of the corticospinal tract (i.e., lesion from the cerebral cortex down until the anterior horn in the spinal cord). - Present as hyperreflexia, hypertonia, clonus, Babinski sign.
LMN	- Lesion involving the motor neuron from the anterior horn until the muscle itself. If the USMLE gives you an anterior horn lesion, this will be LMN, not UMN. - Present as hyporeflexia, hypotonia/flaccidity, fasciculations.

HY Spinal conditions	
Tabes dorsalis	- Loss of dorsal columns due to neurosyphilis. - Bilateral loss of vibration + proprioception. - Ataxia with (+) Romberg test → indicates dorsal columns are fucked up as the cause of the ataxia, since proprioception is lost. In contrast, if the cause of ataxia is cerebellar, for instance, Romberg test is (-).
Syringomyelia	- 9/10 Qs will give bilateral loss of pain and temperature due to lesion of anterior white commissure (decussation point for spinothalamic tract). - Maybe 1/3 of Qs will also involve corticospinal tract, where you'll see some impaired motor function in addition to the bilateral loss of pain/temperature. - 1/10 Qs will not mention pain/temperature loss but instead just give impaired motor function, where it sounds nothing like syringomyelia, but you're forced to eliminate to get there (on a 2CK Neuro CMS form).
Brown-Sequard syndrome	- Hemisection of spinal cord. - Ipsilateral loss of vibration + proprioception (dorsal columns). - Ipsilateral loss of motor function (corticospinal tract).  - Contralateral loss of pain + temperature (spinothalamic). - Not typically caused by stab wound where half the spinal cord is perfectly transected. It's to my observation on 2CK NBME forms that they like viral infection or SLE causing transverse myelitis.
Transverse myelitis	- Inflammation of spinal cord resulting in variable deficits. - As mentioned above, USMLE likes viral infection and SLE (autoimmune). - Brown-Sequard syndrome is one way transverse myelitis presents.
Central cord syndrome	- The answer when weakness in the upper limbs is greater than weakness in the lower limbs. - Typically follows hyperextension injury of the neck during whiplash in a rear-end MVA.
Anterior cord syndrome	- The answer when there's preservation of vibration/proprioception but loss of motor + pain/temperature, classically in patient with arterial disease. - In other words, everything but the dorsal columns is fucked up. Essentially the opposite of tabes dorsalis.

Neurogenic joint
- Aka Charcot joint, where patient injures joint due to lack of joint sensation from peripheral neuropathy. - Usually seen in diabetes; can also be seen in neurosyphilis. - The answer on USMLE when they say diabetic patient has “disorganization of the tarsometatarsal joints” on foot x-ray.

Hydrocephalus	
Communicating	- CSF can flow freely between the ventricles and subarachnoid space around the brain and spinal cord. - The answer for meningitis causing hydrocephalus, where the mechanism is “failure of CSF reabsorption by the arachnoid granulations.”
Non-communicating	- Aka obstructive hydrocephalus. - When there is a blockage preventing flow of CSF between the ventricles or subarachnoid space. - There is an NBME Q where they give a tumor of the spinal cord in the setting of hydrocephalus, and the answer is obstructive hydrocephalus. - Can be caused by aqueductal stenosis (narrowing of the cerebral aqueduct of Sylvius between the 3rd and 4th ventricles), which is the most common cause of congenital hydrocephalus. - Can be caused by **colloid cyst (HY for Surg),** which is a benign fluid-filled growth at the anterior roof of the third ventricle, just posterior to the foramen of Monro.
NPH	- Normal pressure hydrocephalus. - “Wet, wobbly, wacky” +/- Parkinsonism. - In other words: urinary incontinence, ataxia, and cognitive dysfunction. - Due to impingement on the corona radiata (asked on an NBME), which is a white-matter bundle that connects the motor cortex superiorly to the internal capsule inferiorly. - Mechanism for urinary incontinence = “failure to inhibit the voiding reflex.”
Congenital	- Aqueductal stenosis is most common cause. - Most important cause on NBME, in my opinion however, is congenital toxoplasmosis, which presents as a triad of 1) hydrocephalus, 2) chorioretinitis, and 3) intracranial calcifications.
Ex-vacuo	- Appearance of enlarged lateral ventricles on imaging without true hydrocephalus. - Can be caused by loss of surrounding brain matter. - Alzheimer and schizophrenia are important causes on USMLE, where enlargement of the lateral ventricles is a finding that can sometimes show up in question stems.

Autoimmune neuropathies	
Multiple sclerosis	- T-cell-mediated attack against oligodendrocytes and myelin-basic protein. - Idiopathic autoimmune disorder classically in white women 20s-30s who live far from the equator. - Relapsing-remitting type is most common, where the patient will have repeated episodes of neurologic disturbances over many years at varying intervals. - Highest yield findings are optic neuritis and internuclear ophthalmoplegia. - Urge incontinence is exceedingly HY. Some students ask about overflow incontinence due to MS; I’ve never seen this on NBME. Urge incontinence is what shows up all over the NBME forms for MS. I discuss the incontinences in extensive detail in my HY Renal and Repro/Obgyn PDFs. - MRI is gold standard for diagnosis, showing scattered white matter lesions within the CNS (i.e., both brain and spinal cord). - CSF analysis shows IgG oligoclonal bands. - Clonus and (+) Babinski sign (UMN findings) also HY.

	- Tx is steroids for acute flares (IV methylprednisolone). - β-interferon is what USMLE wants between flares to decrease recurrences. - Baclofen (GABA-B receptor agonist) is used for spasticity.
Neuromyelitis optica	- Aka Devic disease. - Sounds exactly like MS but they will say CSF oligoclonal bands are negative, and then the answer is just "antibodies against aquaporin-4." - You say Wtf? Not my opinion. It's asked on a 2CK form.
ALS	- Amyotrophic lateral sclerosis. - In ALS, the lateral corticospinal tracts (UMN) and anterior horns (LMN) degenerate. - There are two things you will see in all ALS Qs, making the Dx very easy: 1) The vignette will always have combination of UMN and LMN findings. Fasciculations, decreased reflexes, decreased tone, and muscle atrophy are classic for LMN. Babinski reflex, clonus, increased (brisk) reflexes, increased tone are classic for UMN. Serum CK can be elevated due to increased tone. 2) Must have **no** sensory findings. If any sensory abnormalities (i.e., paresthesias, numbness, etc.) are present in the vignette, ALS is the wrong answer. USMLE will often give vignette of ALS and then the answer is simply "motor neurons." - Knowing this second point in particular will help on difficult 2CK Qs, where you'll get a big paragraph + they say somewhere in the stem something about a sensory abnormality, and you can say, "Cool, not ALS." For 2CK, next best step in Dx = "electromyography and nerve conduction studies."
GBS	- Guillain-Barre syndrome (aka acute inflammatory demyelinating polyneuropathy). - T-cell- and antibody-mediated destruction of Schwann cells and myelin of peripheral nerves. - In other words, MS = CNS/oligodendrocytes; GBS = PNS/Schwann cells. - Can be described as "segmental and inflammatory demyelination." - Presents as ascending paralysis and loss of deep tendon reflexes. - Can affect both upper and lower limbs. - Symptoms can start as tingling in the hands and/or feet. - USMLE can say "weakness of proximal and distal muscles in lower limbs and weakness of distal muscles in upper limbs" → the implication is that the weakness has already ascended in the legs but not yet in the arms. - Textbook cause is *Campylobacter jejuni*, but in almost all Qs they will not give a patient who has bloody diarrhea after a BBQ. In other words, you can't use this as a crutch. They will just give you the symptoms alone and you have to know it's GBS.
CIDP	- Chronic inflammatory demyelinating polyneuropathy. - Vignette will sound exactly like GBS, but symptoms will be present for months and progress slowly. - GBS, in contrast, develops quickly over days and peaks within weeks. - In other words, if you get a vignette where it sounds like GBS, but they say the symptoms have been present for 4 months, that's CIDP, not GBS. There's one Q on a 2CK form on it.
CMTD	- Charcot-Marie-Tooth disease. - Obscure peripheral nerve autoimmune disease that is the answer on USMLE if they give you high-arched feet (pes cavus), hammer toes, and foot drop.
Bell's palsy	- LMN lesion of CN VII (facial nerve). - Paralysis of ipsilateral facial muscles. - Usually idiopathic/autoimmune following viral infection, shingles (i.e., Ramsay-Hunt syndrome II), or Lyme disease. - If the Q gives you Bell's palsy and asks for next best step, the answer is do serology for Lyme disease. If this isn't listed (or the patient doesn't live in endemic area, such as Africa), choose "no further diagnostic studies indicated." Nerve conduction studies are wrong (this is on NBME). Bell's palsy is a clinical diagnosis.

	- IV steroids, followed by a 10-21-day taper of oral steroids is indicated to minimize the immune response. - Ipsilateral hyperacusis (due to paralysis of stapedius muscle) and/or loss of taste to the anterior 2/3 of the tongue are basically nonexistent in questions. - There is a Q on NBME where they say patient has loss of taste following trauma + neurologic exam shows no abnormalities → answer = olfactory nerve (CN I) palsy, not CN VII, due to cribriform plate fracture (loss of smell means loss of most taste). If the Q wanted CN VII lesion, they would give a concurrent Bell's palsy presumably, since loss of taste wouldn't happen in isolation with CN VII palsy. - In contrast to CN VII lesion that is 19 times out of 20 LMN causing Bell's palsy, you should be aware that 1/20 are UMN, which results in loss of motor function of the contralateral middle and lower face, with the forehead spared.
PANDAS	- Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococci. - In rare cases, Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask what will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Sydenham chorea	- Autoimmune movement disorder seen as part of rheumatic fever. - Chorea = fast, purposeless, jerky movements. - Vignette will give child who has rheumatic fever and then ask about the cause of the movement disorder, and the answer is just "autoimmune." - I discuss rheumatic heart disease in detail in the HY Cardio PDF.
Myoneural junction disorders	
Myasthenia gravis	- Autoantibodies against post-synaptic nicotinic acetylcholine receptors. - Classic vignette is female office worker in her 40s who has triad of 1) diplopia, 2) dysphagia, and 3) ptosis that worsens throughout the day. - Gets worse with recurrent stimulation of muscle (vignette might say the patient cannot perform upward gaze for 60 seconds). - Tensilon test is administration of edrophonium (short-acting acetylcholinesterase inhibitor) → patient experiences significant improvement in symptoms. - Tx = pyridostigmine (longer-acting acetylcholinesterase inhibitor). - MG can be a paraneoplastic syndrome of thymoma. - 10-15% of patients with MG have thymoma. - If MG is diagnosed, a chest x-ray should be performed to look for thymoma. If the x-ray is abnormal, a CT scan is ordered. - Removal of the thymoma in these patients can significantly improve/cure the MG.
Lambert-Eaton	- Autoantibodies against pre-synaptic voltage-gated calcium channels. - Presents as proximal muscle weakness that improves with activity (vignette might say the patient tries to get up from a chair a few times before finally being able to). - Tensilon test does not significantly improve Sx in comparison to MG. - Highest-yield point is that it is a paraneoplastic syndrome of small cell lung cancer.

Vertigo	
BPPV	- Benign paroxysmal positional vertigo. - Vertigo is the feeling of the room spinning. - Brief episodes of dizziness, usually 30-60 seconds, sometimes with vomiting. - Caused by a semicircular canal otolith (ear stone; aka otoconia). - Otoconia made of calcium carbonate are normally found within the utricle and saccule of the inner ear and play a role in detecting acceleration and motion. If one becomes dislodged and enters the semicircular canals, BPPV results. - Diagnosis is made via Dix-Hallpike maneuver. In this test, nystagmus is induced when the patient is rapidly moved from a seated position to lying down with the head tilted backward and turned to one side. If nystagmus occurs, Dx = BPPV.

	- Tx is Epley maneuver, which is a series of head movements aimed at moving the semicircular canal otolith back to the utricle, thereby curing the BPPV.
Vestibular neuritis	- Viral infection + vertigo. - Inflammation of the vestibular nerve (branch of CN VIII, vestibulocochlear nerve).
Labyrinthitis	- Viral infection + vertigo +/- tinnitus. - Inflammation of both the vestibular nerve and cochlear nerve. - This distinction of "VN is just vertigo whereas labyrinthitis is also tinnitus" is perpetuated on the internet, but I can tell you there is an NBME Q that is repeated across forms) where they don't mention anything about tinnitus and the answer is labyrinthitis (where vestibular neuritis isn't listed). This is why I write "+/-" for tinnitus above. - The tympanic light reflex can be abnormal, which can reflect increased middle ear pressure (sometimes seen with viral infection). I've seen the NBME write in the stem "abnormal light reflex," which is a confusing point, since this could be misconstrued as referring to the eyes, not the ear. The vignette in which they mention this they also have multiple sclerosis as a wrong answer + mention the patient has a sibling with MS. So the Q is dumb/gotchya-style overall. The key sentence in this NBME Q is that the patient had nausea and reduced ability to eat foods past few days, which implies recent viral infection.
Meniere	- Waxing and waning, asymmetric tinnitus and vertigo that has a slowly progressive course over many years. - Has a familial component but no strict inheritance (i.e., the vignette can sometimes mention a relative with similar findings). - Due to defective endolymphatic drainage. - Can cause low-frequency hearing loss (i.e., patient says it is difficult to hear conversations at the dinner table with many people). This is in contrast to presbycusis, which is high-frequency hearing loss in elderly.
Drug-induced	- Aminoglycosides (i.e., gentamicin, amikacin), chemo agents like cisplatin, and loop diuretics (furosemide, ethacrynic acid) are known for ototoxicity. - Keep point is that this ototoxicity need not be hearing loss and can be vertigo – i.e., the patient is receiving IV Abx for endocarditis treatment and feels like the room is spinning (ototoxicity due to gentamicin).

HY headache types	
Trigeminal neuralgia	- 11/10 lancinating, knife-like pain that occurs as episodes lasting usually <1 minute. - Classically brought on by minor stimuli, such as brushing one's hair/teeth, or a gust of wind. - Thought to be caused by trigeminal nerve irritation or impingement (i.e., at the exit points from the skull). - There is an NBME Q where they ask what part of the brain is fucked up (i.e., they don't list any nerves), and the answer is pons, since CN V originates from the pons. - Prophylaxis is carbamazepine. - There is no Tx for acute episodes since pain lasts such short duration. - Often confused with cluster headache if occurring in the V1 (ophthalmic branch of trigeminal nerve) distribution.
Cluster headache	- Male 20s-40s who wakes up from sleep with 11/10 lancinating, knife-like headache; often associated with ipsilateral lacrimation, rhinorrhea, and sometimes pupillary changes (I've seen NBME say ipsilateral miotic pupil). - Episodes last minutes to half hour (longer than trigeminal neuralgia), where the Q will say the guy wakes up from sleep + paces around his room until pain eventually goes away. - Prophylaxis is verapamil. - Tx for acute episodes is oxygen (bizarre, unless patient has O2 tank lying around).
Migraine	- Unilateral throbbing/pounding headache that lasts hours.

	- Can be associated with prodromal visual aura or sounds. - Prophylaxis is propranolol. - Tx is NSAIDs first, followed by triptans (e.g., sumatriptan). - Triptans are serotonin (5HT) receptor agonists. - Triptans are not prophylactic meds; they are only used as abortive therapy. - USMLE will give 30s female with hypertension who has migraine Hx, and the treatment is "beta blockade" for her HTN management. - Estrogen-containing contraceptives are contraindicated in patients who have migraine with aura.
Tension headache	- Bilateral, dull, band-like headache. - Treated with sleep and acetaminophen.
Temporal arteritis	- Aka giant cell arteritis. - 9/10 Qs will be painful unilateral headache in patient over 50. I've seen one Q on NBME where it's bilateral. - Flares can be associated with low-grade fever and high ESR. - Patients can get proximal muscle pain and stiffness. This is polymyalgia rheumatica (PMR). The two do not always go together, but the association is HY. (Do not confuse PMR with polymyositis. The latter will present with ↑ CK and/or proximal muscle weakness *on physical exam*. PMR won't have either of these findings. I talk about this stuff in detail my MSK notes.) - Patients can get pain with chewing. This is jaw claudication (pain with chewing). - Highest yield point is we give **steroids before biopsy** in order to prevent blindness. - An NBME has "ischemic optic neuropathy" as the answer for what complication we're trying to prevent by giving steroids in temporal arteritis. - IV methylprednisolone is typically the steroid given, since it's faster than oral prednisone. - It's to my observation many 2CK NBME Qs will give the answer as something like, "Steroids now and then biopsy within 3 days," or "IV methylprednisolone and biopsy within a week." Students ask about the time frames, but for whatever reason USMLE will give scattered/varied answers like that. - Another 2CK Neuro CMS Q gives easy vignette of temporal arteritis and then asks next best step in diagnosis → answer = biopsy. Steroids aren't part of the answer. Makes sense, since they're asking for a diagnostic step.

WKS and ataxia	
- Ataxia is a neurological disorder characterized by a lack of coordination of voluntary muscle movements, leading to unsteady gait and difficulty with balance.	
WKS	- Wernicke-Korsakoff syndrome. - Damage to primarily the mamillary bodies due to thiamine (B1) deficiency. - Wernicke encephalopathy = **A COW** = **A**taxia, **C**onfusion, **O**phthalmoplegia, **W**ernicke. - Korsakoff psychosis = retrograde amnesia; causes confabulations, which means making up stories about the past because of loss of memory regarding prior events. - Therefore, WKS = A COW + confabulations. - Application to USMLE is that alcoholics presenting to hospital with confusion, ataxia, or eye findings require thiamine. - One NBME Q asks what giving thiamine will help reduce risk of in alcoholic → answer = anterograde amnesia (apparently by reducing risk of confusion due to Wernicke), where retrograde amnesia isn't listed as answer.
General Ataxia	- USMLE wants you to be able to differentiate ataxia caused by cerebellar vs dorsal column lesions. - Dorsal column lesions cause a (+) Romberg test, meaning the patient falls over when standing with the eyes closed due to loss of proprioception. - Cerebellar lesions won't cause a (-) Romberg test almost always due to maintenance of proprioceptive capacity.

Cerebellar lesion signs	
Asterixis	- Aka “hepatic flap.” - Motor disturbance characterized by brief, sudden lapses of muscle tone, presenting as the patient demonstrating a “flapping” motion of the hands when he or she extends the arms and hands anteriorly.
Dysdiadokokinesia	- Impaired ability to perform rapid alternating movements; presents as difficulty rapidly tapping a finger or quickly pronating and supinating the hand.
Intention tremor	- Kinetic tremor that worsens as the patient’s hand approaches a target.
Truncal ataxia	- Tends to occur with more midline cerebellar lesions (i.e., of the vermis). - Can be seen, e.g., in pediatrics with medulloblastoma, where a kid has morning vomiting (often indicates brain tumor) and truncal ataxia.
Limb ataxia	- Tends to occur with more lateral cerebellar lesions. - An important rule is that cerebellar lesions cause ipsilateral deficits, unlike many brain lesions that cause contralateral limb findings.

Neuro conditions caused by vitamin deficiencies	
Vit B12	- As discussed earlier, causes subacute combined degeneration, with reversible dementia and/or systemic neurologic findings. - Hx of gastrectomy, pernicious anemia, veganism, or strict vegetarianism is buzzy. - If patient has B12 deficiency + an unrelated autoimmune disorder, e.g., vitiligo, Hashimoto, etc., the cause is probably pernicious anemia, since “autoimmune diseases go together,” where having one autoimmune disease ↑ risk of others. - Patients need not have neurologic dysfunction in B12 deficiency. This is a common misconception. - Causes ↑ MCV (megaloblastic anemia) and hypersegmented neutrophils. - If Q gives you older patient on tea and toast diet + high MCV + no other information, folate (B9) deficiency is more common than B12 deficiency.
Vit B1	- Can cause WKS, dry beri beri (neuropathy), and wet beri beri (dilated cardiomyopathy). - Two new 2CK NBME Qs assess thiamine deficiency causing neuropathy (dry beri beri) *post-gastrectomy*, which is bizarre, since we classically think of this as causing B12 deficiency. - A helpful detail that B1 deficiency is the answer post-gastrectomy instead of B12 is if they give *confusion*, which is characteristic of Wernicke encephalopathy, not subacute combined degeneration. - One of the Qs also mentions a (+) Romberg test for B1 deficiency as well. Everyone gets these Qs wrong because the notion of B1 deficiency as correct post-gastrectomy, as opposed to B12, is recondite and obscure.
Vit E	- Can cause peripheral neuropathy. - Weirdly an extremely HY vitamin deficiency on USMLE, despite it not seeming that common. Shows up in **cystic fibrosis questions.** - Patients with CF have exocrine pancreatic insufficiency leading to fat-soluble vitamin malabsorption. So CF patient + neuropathy = vitamin E deficiency. - Can in theory be seen with any cause of exocrine pancreatic insufficiency (i.e., chronic pancreatitis due to alcoholism, pancreatectomy, severe diabetes), but it’s HY for CF Qs as I said. - Likewise, bowing of tibias / low Ca^{2+} in CF patients would be vitamin D deficiency.
Vit B6	- Neuropathy and/or seizures in patients taking isoniazid for tuberculosis.

Tremor types	
Essential	- Autosomal dominant intention tremor. - Can occur at any age, but usually middle age and older.

	- Patients self-medicate with alcohol, which ↓ tremor. - Propranolol (beta-blockade) is treatment.
Resting	- Parkinson disease (or Parkinson-plus disorder) in older patients. - Wilson disease till proven otherwise in younger patients.
Intention	- Cerebellar lesions or essential tremor.
Drug-induced	- Lithium and valproic acid can both cause tremor. - Lithium is classic for tremor, but an NBME Q gives valproic acid as correct answer (and lithium wrong answer) for patient who has tremor + abnormal liver function tests, since valproic acid is known for hepatotoxicity but lithium is not. - Albuterol (β2-agonist for asthma) shows up in an NBME Q as causing tremor. - Alcohol withdrawal (delirium tremens). Treat with benzo.

Meningitis	
- Presents as high fever + **stiff neck** and photophobia.	
Bacterial	- Usually Group B Strep in neonates (gram-positive cocci). - *S. pneumo* and *N. meningitides* most common otherwise. - CSF shows ↑ Protein, ↓ glucose, ↑ neutrophils (polymorphonuclear cells).
Fungal	- *Cryptococcus neoformans* in immunocompromised patients (i.e., HIV). - CSF shows ↑ Protein, ↓ glucose, ↑ lymphocytes.
Viral	- Aka aseptic meningitis. - Commonly echovirus (if vaccinated); can also be mumps (if unvaccinated). - CSF shows normal protein (or a tad elevated), normal glucose, ↑ lymphocytes.
Encephalitis	
- Presents as high fever + **confusion.**	
HSV ½	- Herpes encephalitis can occur following vertical transmission. - **RBCs in the CSF** due to temporal lobe hemorrhage. - Non-contrast CT of the head is often negative, so don't be confused by this. However they can say EEG shows abnormal spikes over the temporal region. - Other CSF findings same as viral meningitis. - I've only ever seen one NBME Q on 2CK Neuro CMS form where RBCs were absent in CSF in herpes encephalitis, but you could still get the answer because other CSF findings were viral + no other virus listed.
Management points	
- Do lumbar puncture before antibiotics, as the latter first can interfere with culture results. - Four indications for doing CT before LP in suspected meningitis / encephalitis: 1) Focal neurologic signs. 2) Seizure. 3) Papilledema, or the optic fundi cannot be visualized. 4) Confusion that interferes with neurologic exam / reduced Glasgow score. - The reason for CT before LP in the above scenarios is because of the risk of intracranial mass lesion, where if LP is performed, which acutely ↓ intracranial pressure, herniation and death might occur. - 2CK NBME Q gives point #3 above, where answer is CT, not LP, and many students get it wrong. - Empiric Tx for meningitis is ceftriaxone + vancomycin. Steroids can be added if confirmed *S. pneumo*. - For Group B Strep only, choose ampicillin + gentamicin, or ampicillin + cefotaxime; the combo of vancomycin + ceftriaxone is wrong on the NBME for GBS. - 2CK NBME Q has amp + gent as correct over vanc + ceftriaxone for GBS. - We normally avoid ceftriaxone in the first month of life and prefer cefotaxime as the 3rd-gen cephalosporin. The exact age of the transition is not so important. What you need to know is, if you get a Peds Q and both cefotaxime and ceftriaxone are listed, cefotaxime is what we generally prefer. - Ceftriaxone, compared to cefotaxime, displaces bilirubin from albumin, thereby ↑ risk of neonatal hyperbilirubinemia. It can also cause calcium deposits in the lungs and kidney. - HY that you know meningitis can cause sensorineural hearing loss. If they tell you, e.g., 6-year-old boy isn't paying attention in class + isn't doing well in school + has Hx of meningitis; next best step = air-conduction audiometry. Do not confuse with tympanometry, which evaluates mobility of the tympanic membrane.	

Other CNS infections for Surg	
Mastoiditis	- Complication of untreated or severe otitis media. - Q will tell you a child has tenderness behind the ear + the pinna is displaced upward and outward. - Next best step is CT or MRI of the temporal bone to visualize a potential fluid collection, since if this is not drained it can lead to brain abscess. - Peds CMS form has CT of temporal bone as answer for mastoiditis; sounds wrong to do head CT in a kid, but it's what they want on the form. - X-ray is wrong answer, as it doesn't adequately visualize a potential fluid collection.
Brain abscess	- Abscess in the brain. Not rocket science. - Can be caused by septic embolus due to endocarditis. - Q will give you murmur + fever + neurologic findings → vegetation from heart valve has launched off to the brain, causing stroke-like presentation. - Tx = drainage of abscess + IV antibiotics.
Epidural abscess	- Usually presents as "point tenderness over a vertebra," which on USMLE means epidural abscess, compression fracture (osteoporosis), or metastases. - Patient may or may not have neurologic findings. Fever is not mandatory. - Three demographics I've seen on NBME exams for epidural abscess: 1) IV drug user who now has point tenderness over vertebra; 2) Patient who's had some sort of cellulitis of the legs / lower body that has spread to the spine, where you've eliminated the other answers to get there; 3) Patient with active endocarditis (i.e., murmur + fever) who has neurologic findings in the legs (septic embolus also possible, but this tends to produce stroke-like findings rather than spinal pain + lower leg findings). - Tx = drainage of abscess + IV antibiotics.
Pott disease	- Infection / granulomatous destruction of the vertebral column by tuberculosis. - Just know that this is possible, the same was psoas abscess is also possible.

Cavernous sinus thrombosis
- Severe headache, diplopia, proptosis, and/or ophthalmoplegia in patient with recent sinus infection. - Bacterial infection can ascend the valveless facial veins to the cavernous sinus, leading to clotting of blood. - Usually *S. aureus*. - Tx = IV antibiotics + heparin. - CN VI palsy can occur with pathologies of the cavernous sinus.

Primary CNS lymphoma
- HY and underrated diagnosis for USMLE. - Causes ring-enhancing lesion on head CT that can be mistaken for Toxo. As discussed above, if AIDS patient has CD4 count <100 and is already on TMP-SMX, the Dx is primary CNS lymphoma, not Toxo. - Can be seen in non-AIDS patients who have immunosuppression due to other causes (i.e., organ transplant recipients on immunosuppressant meds; patients undergoing radio- and/or chemotherapy). - Can be caused by autoimmune diseases. For example, if they give you a 1-liner where they say, "44-year-old woman with SLE has ring-enhancing lesion on CT" → answer = CNS lymphoma, not Toxo. - Many autoimmune diseases (e.g., SLE, dermatomyositis, Hashimoto) ↑ the risk of non-Hodgkin lymphoma.

Encephalopathy	
- General term that refers to altered brain function (usually presenting as confusion) due to various causes.	
Hepatic	- Seen in alcoholism due to hyperammonemia. - Classically occurs on USMLE when there is an ↑ in blood in the GI tract, such as following acute variceal bleed. Blood is broken down in the GIT into proteins that liberate amino acids, which are converted by gut bacteria into ammonia that is absorbed. - Lactulose is important drug for USMLE for treatment of hepatic encephalopathy. It is a sugar that is broken down by gut bacteria into acidic products. This leads to the conversion of ammonia into ammonium. The latter is ionic and not readily absorbed. The USMLE wants "acidification; ammonium" as the answer for what happens when we give lactulose. - Neomycin is another drug for hepatic encephalopathy. It is an aminoglycoside antibiotic that kills ammonia-producing gut bacteria.
Uremic	- Confusion in the setting of renal failure due to elevated blood urea nitrogen.
Metabolic	- Confusion due to electrolyte imbalances, hypoglycemia, and thyroid storm.
Toxic	- Mental status changes due to toxins. Reye syndrome and lead poisoning are HY.
Inflammatory	- Herpes encephalitis (infectious). - Encephalitis is infection of the substance of the brain, causing confusion (encephalopathy). - As discussed earlier, infection + confusion = encephalitis; infection + stiff neck and/or photophobia = meningitis. - Lupus cerebritis (autoimmune).
Anoxic	- Confusion caused by diminished oxygen delivery to the brain.

Neuropathic pain disorders	
Diabetic neuropathy	- Presents as pain, paresthesia, and/or numbness in diabetic. - USMLE likes TCAs (i.e., amitriptyline) or gabapentin as first-line for Tx. - Use nortriptyline in elderly if a TCA is given (fewer side-effects).
Herpetic neuralgia	- Pain, burning, tingling experienced prior to, during, or after herpes or varicella (shingles) episode. - Acyclovir is used to treat the virus; gabapentin used classically for the pain.
Post-trauma	- Patients who have injuries resulting in nerve trauma can sustain chronic neuropathic pain. Treat with gabapentin or TCAs. - Trauma can result in complex regional pain syndrome, which is neuropathic pain that occurs as sequela of injury that is disproportionate to the injury itself.
Thalamic pain syndrome	- Damage to the thalamus due to a stroke that results in body pain, usually contralateral to the side of the infarction, occurring many months post-stroke. - The answer on USMLE for severe limb pain many months following a stroke.

Brain bleeds	
Epidural hematoma	- Rupture of middle meningeal artery (branch of external carotid); classically occurs following head trauma to the temple. **- Lucid interval** – i.e., patient loses consciousness briefly, followed by arousing and returning to normal (i.e., has a period of lucidity), then goes home thinking he/she is okay, then goes to sleep, then dies. - Bleed appears as lens (biconvex) shape; fast accumulating. - If low GCS, do "intubation + hyperventilation," *then* do craniotomy. If normal GCS, just do craniotomy. - Hyperventilation mechanism: decreased CO_2 → decreased cerebral perfusion → reduces intracranial pressure.

	
Subdural hematoma	- Rupture of bridging (superior cerebral) veins. - **No lucid interval** (i.e., patient did not lose consciousness). - Crescent-shaped; can be slowly accumulating and heterogenous in appearance due to combination of dried + fresh blood. - Increased risk in elderly/dementia and alcoholics; answer in acceleration-deceleration injuries (shaken baby syndrome; motor vehicle accidents). - NBME wants craniotomy as the answer for Tx (no evidence for mere observation as the answer on any of the USMLE material). - If the patient has decreased Glasgow score, do “intubation + hyperventilation” as answer before craniotomy. 
Subarachnoid hemorrhage	- Rupture of anterior communicating artery (AcoM) or posterior communicating artery (PcoM) saccular/berry aneurysm. - AcoM > PcoM in terms of location. - Patient often reports “worst headache of his/her life.” - **Can present with stiff neck** (meningism) the same way meningitis does, due to irritation of the meninges by the hemorrhage. This often confuses students and is a very HY detail. - PcoM aneurysm classically associated with ipsilateral blown pupil (mydriasis) due to CN III impingement. - HTN is most common risk factor in population and accounts for most cases. - Ehlers-Danlos and autosomal dominant polycystic kidney disease (ADPKD) are HY non-HTN-associated specific causes of saccular aneurysms. - A patient with HTN is still more likely to die from myocardial infarction than berry aneurysm (asked on NBME). But if patient has berry aneurysm, it is most likely from HTN.

	- Bleed appears like a “star fish” or “sand dollar.” Q can say there is blood visualized in the basal cisterns.  - Treatment is usually supportive. HTN must be controlled. Anticoagulants must be reversed (i.e., FFP for patients on warfarin). Nimodipine can prevent vasospasm and neurologic sequelae during convalescence.
Intracerebral hemorrhage	 - Rupture of Charcot-Bouchard microaneurysms within the lenticulostriate arteries in patients who have chronic HTN. - Can present on USMLE as a 1-2-line stem, without any image or patient info, where they just say, “X person has bleed + decorticate posturing; what’s the diagnosis?” → answer = intracerebral hemorrhage. - In other words, USMLE wants you to know intracerebral hemorrhage has high likelihood of causing brainstem compression. - Associated with Alzheimer (i.e., amyloid angiopathy, which presents in Alzheimer as recurrent hemorrhagic stroke). - Can be associated with brain cancer as well (e.g., glioblastoma multiforme).
Pituitary apoplexy	- Bleeding into the pituitary gland; can cause bitemporal hemianopsia. - USMLE wants you to know this can occur puerperium and also due to pituitary tumors. - 2CK form gives bitemporal hemianopsia in woman who just gave birth, where pituitary apoplexy is only answer listed that could cause this visual pattern.

Cushing reflex

- **Cushing reflex** is a triad seen with ↑ ICP, where the patient will have: 1) HTN, 2) bradycardia, and 3) respiratory disturbance (classically ↓ RR, but there's an NBME Q where they don't give RR as ↓).
- ↑ ICP causes compression of intracerebral vessels → chemoreceptors sense ↓ oxygen delivery + ↑ CO_2 → this causes sympathetic activation and vasoconstriction of peripheral arterioles → ↑ BP → ↑ stretch of carotid sinus baroreceptors → ↑ CN IX afferent firing to solitary nucleus of medulla → ↑ CN X efferent firing to nodal tissue → ↓ HR.
- The ↓ RR is due to ↑ ICP causing compression of brainstem.
- NBME will give you a big 15-line paragraph with patient who's been in an accident + ask you for the reason for his or her HTN → answer = "increased intracranial pressure."
- Don't confuse Cushing reflex with Cushing ulcer (completely unrelated), which is head trauma causing vagal outflow → increased acid production + peptic ulcers.

Glasgow coma scale (GCS)

Glasgow Coma Scale		
Response	Scale	Score
Eye Opening Response	Eyes open spontaneously	4 Points
	Eyes open to verbal command, speech, or shout	3 Points
	Eyes open to pain (not applied to face)	2 Points
	No eye opening	1 Point
Verbal Response	Oriented	5 Points
	Confused conversation, but able to answer questions	4 Points
	Inappropriate responses, words discernible	3 Points
	Incomprehensible sounds or speech	2 Points
	No verbal response	1 Point
Motor Response	Obeys commands for movement	6 Points
	Purposeful movement to painful stimulus	5 Points
	Withdraws from pain	4 Points
	Abnormal (spastic) flexion, decorticate posture	3 Points
	Extensor (rigid) response, decerebrate posture	2 Points
	No motor response	1 Point
Minor Brain Injury = 13-15 points; **Moderate Brain Injury** = 9-12 points; **Severe Brain Injury** = 3-8 points		

- It's to my observation across NBME exams (in particular 2CK Neuro CMS forms) that they will never ask you to formally calculate GCS score.
- What will occasionally surface in NBME Qs is "decorticate" or "decerebrate" posturing, where you have to know, "Oh that means GCS is low; patient has compromised brain function, possibly herniation / ↑ intracranial pressure (ICP)."
- And as I mentioned earlier regarding suspected meningitis diagnostic workup, if GCS is not 15, then head CT should be done before LP.

Post-concussion syndrome

- A concussion is a type of traumatic brain injury caused by a blow to the head or by a sudden movement that results in the brain moving rapidly inside the skull. The rapid movement can cause brain cells to stretch and become damaged.
- Post-concussion syndrome (aka post-concussive disorder) comprises a range of symptoms that can persist for weeks, months, or even years after a concussion.
- Such symptoms include cognitive disturbance, memory loss, headache, and sensitivity to light or sound.
- USMLE will ask for the diagnosis straight-up on Neuro forms.

Base of skull fracture
- Fracture at base of the skull (you'd never have guessed it). Classic tetrad of signs is: 1) Battle sign, which is bruising over the mastoid process behind the ear. 2) Raccoon eyes, which refers to bilateral periorbital ecchymoses (bruising around both eyes). 3) Otorrhea (CSF leaking out of the ear). 4) Rhinorrhea (CSF leaking from the nose). - Hemotympanum can also be seen, which is blood visualized behind the tympanic membrane.

CNS malignancy	
- Can present with seizure and/or morning vomiting (a sign of ↑ ICP). - Can be described as ring-enhancing lesions (i.e., this descriptor is not limited to Toxo; it can also be tumors/metastases and abscesses).	
Peds brain tumors	
Pilocytic astrocytoma	- Most common brain tumor in kids. - Has solid (white) and cystic (black) components on imaging. - Not aggressive.
Medulloblastoma	- Aggressive brain tumor of cerebellum. - Can cause truncal ataxia if located at vermis, and ipsilateral limb ataxia if lateral.
Craniopharyngioma	- Most common pituitary tumor in children and teenagers. - Bitemporal hemianopsia (can occur with any pituitary lesion). - Embryo is that it originates from the "roof of the primitive oropharynx," or "roof of the primitive oral cavity." - Can calcify and contain cholesterol crystals; high rate of recurrence. - Don't confuse that embryo with "endoderm of foramen cecum," which refers to thyroglossal duct cyst.
Pinealoma	- Can cause vertical gaze palsy in kids (Parinaud syndrome).
Hamartomas	- The "tubers" in tuberous sclerosis are hamartomas, which are described as periventricular or cortical nodules on imaging.
Adult brain tumors	
Glioblastoma multiforme	- Most common adult brain tumor; aggressive. - Aka "butterfly glioma," since it can cross through the corpus collosum. - Heterogenous in appearance due to areas of necrosis and hemorrhage. - USMLE wants you to know GM is a grade IV fibrillary astrocytoma.
Meningioma	- Benign tumor that grows from the meninges and appears to sit on top of the brain. - NBME Q floating around gives one that they describe as adjacent to falx cerebri. - *Can also occur within the spinal canal,* overlying the spinal cord (on NBME).
Schwannoma	- Arises from Schwann cells, which are responsible for myelinating peripheral nerves. - Vestibular schwannoma arises from CN VIII (vestibulocochlear nerve), often at the cerebellopontine junction, and can cause tinnitus and/or vertigo. - Can be idiopathic or as part of neurofibromatosis type II (NF2). - USMLE wants you to be able to identify its growth as causing sensorineural hearing loss (Rinne: AC > BC; Weber: lateralizes contralaterally).
Prolactinoma	- Most common pituitary tumor in adults. - Can cause amenorrhea in women and impotence in men. - Galactorrhea can occur in everyone. - I discuss other pituitary tumors and hormones in detail in the HY Endocrine PDF.
Optic glioma	- Seen in neurofibromatosis type I (NF1).
Oligodendroglioma	- Tumor of oligodendrocytes (the cells that normally myelinate CNS neurons). - Histo has fried-egg appearance and chicken-wire capillary pattern.
Ependymoma	- Tumor of ependymal cells, which normally produce CSF. - Rosenthal fibers on histo (eosinophilic corkscrew filaments).
Tumor of foramen magnum	- Can cause hypoglossal nerve (CN XII) palsy, leading to ipsilateral tongue deviation. - Shows up on a 2CK Neuro form.

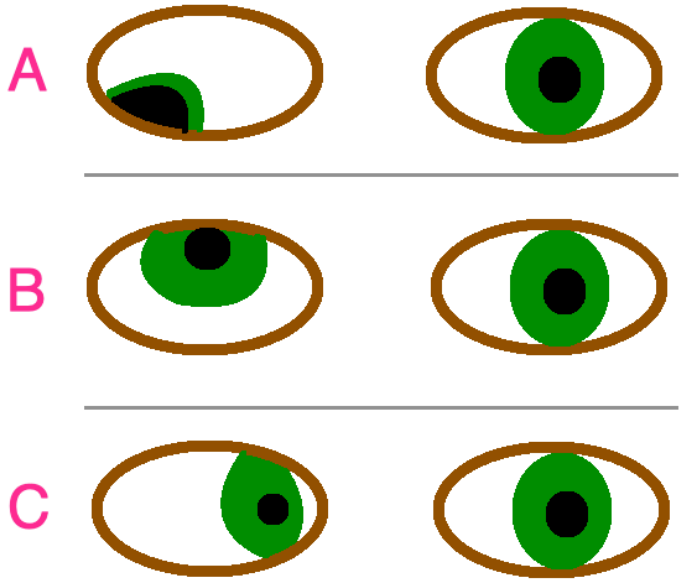
	- So you should know that CN XII lesions occur from medial medullary syndrome as well as tumors of the foramen magnum.
Metastases	- Can be described as lesions at the grey-white junction. - Choriocarcinoma and lung cancer love to metastasize to brain.

Phakomatoses (neurocutaneous disorders)		
- I discussed NF1 and TSC a briefly earlier, but these conditions are exceedingly HY and need to be reinforced.		
Condition	**Genetics**	**HY points**
Neurofibromatosis type I (NF1)	*NF1*; AD; chromosome 17	- Neurofibromas, café au lait spots (hyperpigmented macules), axillary/groin freckling, Lisch nodules (iris hamartomas), pheochromocytoma, **optic nerve glioma,** oligodendroglioma, ependymoma.
Neurofibromatosis type II (NF2)	*NF2*; AD; chromosome 22	- In kids, sometimes causes bilateral cataracts. - Bilateral acoustic schwannomas + meningioma in adults.
Tuberous sclerosis (TSC)	*TSC1*; AD; chromosome 9; hamartin *TSC2* ; AD; chromosome 16; tuberin	- Periventricular nodules (tubers) on MRI of head, seizures, adenoma sebaceum (angiofibromas), subungual fibromas, renal angiomyolipoma, cardiac rhabdomyoma, hypopigmented macules ("ashleaf spots"), hyperpigmented velvety lesions ("shagreen patches").
Von Hippel-Lindau	*VHL*; AD; chromosome 3	- Cerebellar/retinal hemangioblastomas, bilateral renal cell carcinoma, pancreatic cysts. - Mutation causes constitutive activation of hypoxia-inducible factor.
Sturge-Weber	Not inherited; somatic mosaicism of *GNAQ* gene	- Port wine stain birth mark (nevus flammeus; may also present as violaceous papules in temporal distribution), leptomeningeal angioma (presenting as seizure); glaucoma.

Cranial nerve exit points		
- You should know this stuff for Surg because occasionally they're known to give trauma or tumors of the cranial nerves, where they expect you to know functions / locations.		
Cranial nerve	**Exit point**	**Extra points**
Olfactory (CN I)	Cribriform plate	- Only CN that doesn't relay through thalamus.
Optic (CN II)	Optic canal	**- Afferent component of light reflex.**
Oculomotor (CN III)	Superior orbital fissure	**- Efferent component of light reflex (constricts pupil via parasympathetic innervation).**
Trochlear (CN IV)	Superior orbital fissure	- Innervates superior oblique (eye looks inferomedially).
Trigeminal ophthalmic branch (V1)	**S**uperior orbital fissure	- Afferent component of corneal reflex.
Trigeminal maxillary branch (V2)	Foramen **R**otundum	- Trigeminal mnemonic: "**S**tanding **R**oom **O**nly."
Trigeminal mandibular branch (V3)	Foramen **O**vale	- Muscles of mastication.
Abducens (VI)	Superior orbital fissure	- Abducts eye (patient gazes laterally).
Facial (VII)	Auditory canal	- Efferent component of corneal reflex. - LMN results in ipsilateral Bells palsy. - UMN results in contralateral mid + lower facial paralysis (forehead spared).

Vestibulocochlear (VIII)	Auditory canal	- Can give rise to acoustic schwannoma.
Glossopharyngeal (IX)	Jugular foramen	- Afferent component of gag reflex + carotid sinus baroreceptors.
Vagus (X)	Jugular foramen	- Efferent component of gag reflex + parasympathetic to slow HR; lesion causes **contralateral** uvula deviation.
Accessory (XI)	Jugular foramen	- Innervates sternocleidomastoid + trapezius.
Hypoglossal (XII)	Hypoglossal canal	- Lesion causes **ipsilateral** tongue deviation. - Tumor of the foramen magnum can cause CN XII palsy. - Medial medullary syndrome can cause CN XII palsy.

Extraocular nerve palsies



- A = CN III (oculomotor) lesion → presents as down and out.
- B = CN IV (trochlear) lesion → presents as upward gaze + patient will tilt head toward unaffected side.
- C = CN VI (abducens) lesion → medially directed eye.

Rinne + Weber tests	
Rinne test	- A vibrating tuning fork is placed against the mastoid bone (bone conduction; BC). Once the patient can no longer hear it, it is removed from the bone and placed in front of the ear canal (air conduction; AC). - For both normal hearing and sensorineural hearing loss, AC > BC. - For conductive hearing loss, BC > AC.
Weber test	- A vibrating tuning fork is placed on the center of the forehead or top of the head, and the patient is asked if they hear the sound equally in both ears or if it's louder in one ear. - If it is louder in one ear, this is called “lateralization” to that ear.
Normal hearing	- Rinne shows air > bone conduction; Weber does not lateralize.
Conductive on left	- Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive on right	- Rinne shows bone > air conduction; Weber lateralizes to right.
Sensorineural on left	- Rinne shows air > bone conduction; Weber lateralizes to right.
Sensorineural on right	- Rinne shows air > bone conduction; Weber lateralizes to left.

Pro-cholinergic meds	
Muscarinic receptor agonists	
Bethanechol	- Stimulates detrusor muscle. - Used for overflow incontinence in diabetic patients with neurogenic (i.e., hypo-contractile) bladder.
Carbachol	- Constricts pupil and lowers intraocular pressure. - Used for glaucoma. - See my HY Ophthalmology PDF for more detail on that stuff.
Pilocarpine	- Constricts pupil and lowers intraocular pressure. - Powerful agent used for closed-angle glaucoma.
Methacholine	- Causes bronchoconstriction. - Used to help diagnose asthma. If administered to patient (not during acute attacks), spirometry will worsen and asthma symptoms may be replicated.
Acetylcholinesterase inhibitors	
Donepezil	- Increases acetylcholine available in synaptic cleft (i.e., pro-cholinergic). - Used 1st-line for Alzheimer disease. - Galantamine and rivastigmine have same MOA and use-case, but LY.
Edrophonium	- Aka Tensilon (the brand name). - Short-acting acetylcholinesterase inhibitor. - Administration to patients with myasthenia gravis causes marked improvement in symptoms and muscle function. - Administration to patients with Lambert-Eaton syndrome does not result in as marked improvement of symptoms.
Pyridostigmine	- Treatment for myasthenia gravis.
Physostigmine	- Antidote for Jimson weed poisoning (asked on NBME). - Jimson weed contains atropine-like compounds that are anti-cholinergic.
Neostigmine	- Used post-surgically to reverse the effects of neuromuscular blockade. - Toward the end of surgery, administration helps increase availability of ACh so that it can bind to its receptors and displace the nicotinic receptor antagonists used for neuromuscular paralysis.
Organophosphates	- Vignette will either be backpacker in his/her 20s picking fruit on a farm, or someone who drank fluid in a suicide attempt. - The phosphate group binds to acetylcholinesterase, inhibiting it. - NBME Q asks how toxicity could best be prevented → answer = "wearing gloves," where "wearing mask" is wrong answer. So acquisition occurs mostly through skin. - USMLE wants atropine first, followed by pralidoxime as Tx (discussed below).

Anti-cholinergic meds	
Muscarinic receptor antagonists	
Atropine	- Classic textbook drug that blocks muscarinic receptors. - Can be used to increase heart rate during surgery. - Given as 1st-line Tx for organophosphate poisoning before pralidoxime. - Pralidoxime is a special anti-cholinergic that works by regenerating functional acetylcholinesterase by kicking out the phosphate group that inhibited it.
Benztropine	- Used for acute dystonia due to anti-psychotics. - In other words, if patient starts anti-psychotic and then within hours to days gets stiff neck (torticollis), muscle rigidity (without fever), and/or abnormal eye movements (oculogyric crisis), this is 1st-line Tx. - Muscle rigidity with fever, in contrast, suggest neuroleptic malignant syndrome.

Diphenhydramine	- 1st generation H1-blocker that has strong anti-cholinergic side-effects to the point that it is often just characterized straight-up as a primary anti-cholinergic med. - Can be used same as benztropine for acute dystonia (USMLE won't list both). - Can be used for motion sickness (anti-cholinergic effects treat motion sickness). If USMLE tells you it is used in this setting + they ask you antagonism of which receptor enables its anti-motion sickness effect, the answer is M3. The wrong answer is H2, since this refers to stomach acid secretion. Diphenhydramine is an H1 blocker, not H2. - Most Qs on USMLE regarding diphenhydramine revolve around you knowing it should be discontinued from a patient's regimen, rather than added.
Prochlorperazine	- Anti-psychotic used as an anti-nausea med. - Even though it is an D2-receptor antagonist, as I mentioned before, anti-psychotics have anti-cholinergic side-effects. In this case, the anti-muscarinic side-effects are actually a "good" thing since these treat nausea and motion sickness.
Ipratropium	- One of the 1st-line agents used for COPD. - COPD patients can be started on either a muscarinic receptor antagonist, such as ipratropium, or a beta-2 agonist, such as olodaterol. - New literature gets pedantic about the duration (i.e., long- vs short-acting) anti-muscarinics and beta-2 agonists, but USMLE doesn't (i.e., some sources say tiotropium is recommended over ipratropium 1st-line now, but I've had students come out of the exam saying ipratropium showed up, but that it didn't matter since it was the only drug listed that worked anyway).
Oxybutynin	- Used for urge incontinence.
Scopolamine	- Scopolamine patches are used for motion sickness (i.e., put it on the arm before getting on a boat or plane). This drug is an anti-cholinergic straight-up. - As mentioned above, oral diphenhydramine can also be used for motion sickness, but scopolamine is the classic treatment.
Hyoscyamine	- Used sometimes for irritable bowel syndrome.
Non-depolarizing nicotinic receptor antagonists	
Tubocurarine	- "Non-depolarizing" means the agent does not cause any action potentials. - NBME Q mentions a non-depolarizing agent used for neuromuscular blockade during surgery, and the answer is just "tubocurarine" straight-up.
Vecuronium	- Used for paralysis during surgery. - Cisatracurium and rocuronium are other important agents.
Depolarizing nicotinic receptor agonist (acts as antagonist)	
Succinylcholine	- Functions as antagonist due to desensitization of receptors. - Binds to nicotinic receptors and agonizes them, causing initial action potentials and muscle twitching (fasciculations), prior to desensitization and muscle paralysis. - Akin to leuprolide causing desensitization of GnRH receptors with continued binding, where despite being a GnRH receptor agonist, it acts functionally as an antagonist. - Can cause malignant hyperthermia (high fever + muscular rigidity; treat with dantrolene, which inhibits the ryanodine Ca^{2+} channel).
Smoking cessation agents	
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Varenicline	- Used for smoking cessation. - Partial agonist at nicotinic receptors, but since this effect is less than endogenous ACh, it functions like an antagonist.

Drugs for neuropathic pain

TCAs	- Prevent reuptake of both serotonin and norepinephrine. - Amitriptyline, nortriptyline, imipramine. - Neuro use-case is neuropathic pain, i.e., diabetic peripheral neuropathy. - Causes CCC → Coma, Convulsions, Cardiotoxicity (QT prolongation on ECG). - Have nasty anti-cholinergic side-effects (cognitive dysfunction and urinary retention, especially in elderly).
Gabapentin	- Blocks calcium channels, leading to neuroinhibition. - Used for neuropathic pain, e.g. , diabetic peripheral neuropathy.

Headache drugs	
Migraine	- Abortive therapy: NSAIDs first, then sumatriptan (serotonin receptor agonist). - Prevention: propranolol (beta-blockade).
Cluster	- Abortive therapy: oxygen. - Prevention: verapamil (non-dihydropyridine calcium channel blocker).
Tension	- Rest + acetaminophen.
Trigeminal neuralgia	- Abortive: goes away on its own because episodes last maximum 30-60 seconds. - Prevention: carbamazepine (sodium channel blocker).

Quick anesthetics garbage	
MAC	- Mean alveolar concentration. - Used to determine the potency of inhaled anesthetics. - Dose of inhaled anesthetic required to prevent movement in 50% of patients to external noxious stimuli. - Less drug needed to achieve effect = more potent = lower MAC. - More drug needed to achieve effect = less potent = higher MAC. - Drugs that are more lipophilic (i.e., have high blood:lipid coefficient) are more potent and have lower MAC. This is because they dissolve well into the membranes of neurons and are able to achieve faster anesthesia. - Halothane = high potency = low MAC + high blood:lipid coefficient. - Nitrous oxide = low potency = high MAC + low blood:lipid coefficient.
Blood:lipid coefficient	- Reflects how lipophilic / how well the drug dissolves in adipose tissue. - As discussed above, high blood:lipid coefficient = lower MAC = higher potency. - Halothane is an example of a drug with high blood:lipid coefficient and high potency. - Nitrous oxide is an example of a drug with low blood:lipid coefficient and low potency. - USMLE wants you to know that coming out / waking up from anesthesia is due to “redistribution of drug into adipose tissue.
Blood:gas coefficient	- Reflects solubility in blood. - Low blood:gas coefficient = fast onset of action (nitrous oxide). - High blood:gas coefficient = slow onset of action (halothane). - Blood:gas and blood:lipid coefficients are not necessarily inversely related – i.e., drugs that have higher potency (high B:L coefficient) don’t necessarily have slow onset of action (high B:G coefficient).

Inhaled anesthetics	
- All used for general anesthesia (i.e., to put the patient to sleep). - All potentiate GABA + inhibit NMDA glutamate receptors.	
Halothane	- Can cause malignant hyperthermia (same as succinylcholine).

	- Rarely used today due to severe liver toxicity, but you need to know it as classic agent. - High potency due to high blood:lipid coefficient.
Nitrous oxide	- Fast-onset of action (low blood:gas coefficient).
Sevoflurane	- Fast-onset of action (low blood:gas coefficient).

Intravenous anesthetics	
Ketamine	- NMDA glutamate receptor antagonist. - Often used in trauma situations for procedural sedation.
Propofol	- Classic agent used to maintain general anesthesia. - White in color (looks like milk). - Has calories. (Won't get asked on USMLE, but weird factoid anesthetists sometimes ask). - The drug that killed Michael Jackson.
Benzos	- Midazolam is classic agent used for adjunct to anesthesia as an anxiolytic.
Barbiturates	- Thiopental used classically for induction of anesthesia but has been replaced by propofol.

Local anesthetics	
- Block sodium channels.	
Lidocaine	- Very common local anesthetic. - Also can be used as a type Ib anti-arrhythmic in theory.
Procaine	- Short duration of action (i.e., effect doesn't last long).
Bupivacaine	- Long duration of action (i.e., effect lasts long).

Neurotoxic drugs	
Vincristine	- Prevents microtubule assembly; used as general chemotherapy. - USMLE can ask what drug inhibits and answer is just "tubulin." - Peripheral / autonomic neuropathy. - Answer shows up as "toxic neuropathy" on a 2CK Neuro CMS form.
Cisplatin	- Crosslinking agent used as general chemotherapy. - Causes neuro- and ototoxicity.
Aminoglycosides	- 30S ribosomal subunit inhibitors used for gram-negative rods. Highest yield use-case is in combination with vancomycin as empiric Tx for endocarditis. - Gentamicin, tobramycin, amikacin (latter sounds unusual but shows up on an NBME). - Cause ototoxicity (sensorineural hearing loss and/or vertigo) + nephrotoxicity.
Loop diuretics	- Block the $1Na^+/1K^+/2Cl^-$ symporter on the apical membrane of the thick ascending limb. - Furosemide, ethacrynic acid, etc., can cause sensorineural hearing loss and/or vertigo.

Surg Psych

Depressive disorders	
Major depressive disorder (MDD)	- At least 5 out of 9 SIGECAPS must be present for at least a 2-week period. S – Sleep disturbance (insomnia or hypersomnia). I – Interest loss. G – Guilt or feelings of worthlessness. E – Energy loss. C – Concentration problems. A – Appetite changes (usually causes weight loss, but sometimes weight gain). P – Psychomotor agitation (restlessness; slowed speech or movements). S – Suicidal ideation. - The symptoms must cause socio-occupational impairment and must not be attributed to a substance or another medication condition. - As far as NBME Qs go, I can say that USMLE doesn't actually give a fuck about fulfilling the 5/9 criteria. They'll give you an elderly male who's a bit quiet + teary-eyed + who's had weight loss, and the answer is just depression. Students get confused because they don't count 5/9, but as I said, USMLE doesn't care. - It's to my observation that *weight loss* is one of the most buzzy indicators in a psych vignette that MDD is likely the answer. - Terminal insomnia (patient wakes up too early in the morning) can be seen. - Can cause ↓ libido, but patient still has nocturnal erections (asked on NBME). - Can be caused by beta-blockers (propranolol); asked on NBME. - Tx is with drugs such as SSRIs, CBT, and occasionally ECT.
Depression due to a medical condition	- Hypothyroidism is important cause of low mood and apathy. - USMLE likes post-MI and cancer diagnoses as causes of depression. - Q might give guy who's had weight loss + teary-eyed post-MI; answer on an NBME form is just sertraline (an SSRI).

Eating disorders	
Anorexia	- Intense preoccupation with maintaining low body weight and an intense fear of gaining weight. BMI must be low (i.e., <18.5) for diagnosis. - Purging can be seen (not limited to bulimia). - Vignette can give girl who runs 12 miles per day and barely eats. - USMLE likes metatarsal stress fractures due to decrease bone density. - The Q can ask what the patient is at risk for later in life → answer = osteoporosis (reduced adipose → reduced estrogen). - Amenorrhea in females can be seen but is no longer mandatory for diagnosis. USMLE wants "abnormal GnRH pulsation" as the mechanism for amenorrhea in anorexia, where LH and FSH are both decreased. This is called hypogonadotropic, or central, amenorrhea. - BMI <15 or overt malnutrition is indication for hospital admission. - Tx is CBT and/or SSRI. Mirtazapine is an α2-antagonist that stimulates appetite and is used for patients with depression and anorexia. - Avoid bupropion in patients with eating disorders (lowers seizure threshold). - Most common cause of death is ventricular fibrillation due to hypokalemia. - Olanzapine can be used in theory (anti-psychotic that ↑ appetite).
Bulimia	- Binge eating followed by compensatory behaviors such as vomiting, excessive exercise, or the use of laxatives. - BMI is normal or increased (unlike anorexia, where it must be decreased). - Tx is CBT and/or SSRI.

Somatic symptom and related disorders	
- Symptom generation and motive are both unconscious.	
Somatic symptom disorder	- One or more chronic somatic symptoms (e.g., GI disturbance/pain, adnexal pain, shortness of breath, neuropathic pain) that are distressing or result in socio-occupational dysfunction, but in which medical examination and diagnostic interventions cannot find any abnormalities.
Conversion disorder	- Neurologic deficits (i.e., paralysis, loss of function, numbness) usually following a stressor, e.g., loss of job, argument with spouse, etc. - Patient has La belle indifference, which refers to apparent lack of care or concern about the deficit (i.e., the patient supposedly can't move his arm but doesn't seem fazed by it). - Example is a child being dropped off at summer camp + says he can't move his finger → child is likely stressed by being separated from parents. - Hoarseness of voice shows up on NBME form where answer is conversion disorder. - NBME also gives guy with blindness in one eye, but there's no relative afferent pupillary defect (indicating CN II isn't fucked up), so Dx = conversion disorder. If you're confused about RAPD, go to HY Neuro PDF.
Illness anxiety disorder	- Formerly known as hypochondriasis. - Preoccupation with fear of having or acquiring a serious illness; worries persist even after medical reassurances of no significant health concern/risk.

Factitious disorders	
- Symptom generation is conscious/intentional; motive is unconscious. - Production of symptoms in factitious disorders is driven by a need to be seen as ill or injured, rather than for a tangible benefit such as drugs, money, or time out of prison. - Seeking medical attention/care is known as attempting to seek "primary gain," whereas something concrete like cash or time out of prison is "secondary gain."	
Factitious disorder imposed on self	- Formerly known as Munchausen syndrome. - Intentional production / feigning of physical symptoms (e.g., GI pain). - The primary motivation is to assume the sick role and receive medical attention, care, or sympathy. - Willing to undergo invasive procedures. The Q might say patient has history of cholecystectomy or appendectomy.
Factitious disorder imposed on another	- Formerly known as Munchausen syndrome by proxy. - Feigning symptoms in another person in order to achieve medical attention. - Considered a form of abuse. Can occur with parents or caretakers toward their children, elderly, or those in their care. - Needs to be reported.
Factitious thyrotoxicosis	- Injection of thyroid hormone. Aka surreptitious thyrotoxicosis. - USMLE vignette will usually make this an endocrine-type Q (see HY Endocrine PDF), where it need not be attached to the patient's desire to seek medical care (i.e., the patient could just be attempting to lose weight), however some patients will inject as a means to acquire medical attention. - Q need not say patient is a pharmacist (students use that as crutch).
Factitious use of glycemic meds	- Use of insulin or hypoglycemic meds (e.g., sulfonylureas) to achieve hypoglycemia or weight loss. - USMLE will usually ask this in the setting of C-peptide levels (i.e., low C-peptide indicates exogenous insulin; high C-peptide means either secretagogue abuse like sulfonylureas, or insulinoma). - Some patients use these meds to achieve medical attention.

Malingering
- Symptom generation and motive both intentional. - Aim is secondary gain – i.e., money, drugs, time out of prison/school. - Patient is not willing to undergo invasive procedures. For example, a prisoner complains of severe RLQ pain and feigns a rigid abdomen, but once at hospital, refuses to undergo laparoscopic appendectomy.

Sleep disorders	
Sleep apnea	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose. - When obstructive sleep apnea progresses to the point that the patient is a chronic CO_2 retainer with pulmonary hypertension and/or cor pulmonale, we call it obesity hypoventilation syndrome (Pickwickian syndrome). If you're confused about the cardio, go to the HY Cardio PDF.
Restless leg syndrome	- Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used. - USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.

Important child abuse findings
- Spiral fractures (rotational/twisting force applied to bone). - Posterior rib fractures (squeezing). - Circular burns (cigarettes). - Burns sparing flexor regions (from being dipped in hot water [child flexes limbs to ↓ exposed surface area). - Retinal detachment / hemorrhages. - Subdural hematoma (shaken baby syndrome). - **Avoidance of eye contact** (if they say Hx of meningitis, think instead neurosensory hearing loss).

Peds bladder/bowel control	
Enuresis	- Nocturnal enuresis (i.e., wetting the bed) is normal on occasion up through age 5. - First step in ↓ recurrence is behavioral science-type answers (i.e., ↓ stressors + ↑ time spent with child). Q might say child wets the bed after a sibling is diagnosed with ALL (i.e., stressor). - After behavioral-type answers, the next step is star chart (positive reinforcement), where the child receives a star on a chart every time he/she doesn't wet the bed, where after, e.g., 10 stars, he/she gets an extra dessert. - If star chart is not listed, enuresis alarm is the next best step (i.e., literally an alarm that detects moisture in a child's underwear, typically using a sensor, where if moisture is detected, indicating urination, the alarm sounds or vibrates child awake, thereby encouraging him or her to stop urinating and go to instead use the toilet). - Answers such as imipramine, desmopressin, and water deprivation are wrong on USMLE.
Encopresis	- Shitting your pants. Normal on occasion up through age 5.

Psychosomatic disorders	
Irritable bowel syndrome (IBS)	- Don't confuse with inflammatory bowel disease (IBD). - IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."
Chronic interstitial cystitis	- Not an infection. - This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care" and "physiotherapy."
Fibromyalgia	- This is a psych condition, not an actual muscle disorder, but is often confused with polymyositis and polymyalgia rheumatica (see HY Anatomy/MSK PDF). - Labs will be normal. ESR will not be elevated. Patient will not have fever. - Will be described as woman 20s-50s with multiple (and often symmetric) muscle tenderness points. - Treatment is SSRIs. USMLE can write this as "anti-depressant therapy." This confuses students ("But she doesn't have depression though.") → Right. But SSRIs are still anti-depressant medication.

Delirium	
- Acute disturbance in attention and cognition, usually over hours to days. - Most commonly presents as hyperactive delirium, which is where the patient appears confused, restless, agitated, and combative. - Can also present as hypoactive delirium, where the patient mute, lethargic, or slow to respond to stimuli. - Important causes are infection, electrolyte disturbance (e.g., hypercalcemic crisis), medications (especially first-gen H1 blockers like diphenhydramine), and metabolic causes (e.g., hypercarbia in COPD exacerbation).	
Hypercalcemic crisis	- Refers to cognitive dysfunction / a delirium-like state in the setting of severe hypercalcemia, often due to malignancy or primary hyperparathyroidism. - I've also seen this once in a patient on a thiazide (can cause hypercalcemia). - USMLE wants you to know that high calcium, as well as any sodium disturbance, can cause delirium. - First step on USMLE for Tx of hypercalcemia is normal saline. - After normal saline, USMLE wants bisphosphonate therapy (I've seen pamidronate listed on NBME). - I've never seen calcitonin or loop diuretics as correct answers for hypercalcemia Tx. They're always wrong.
Medication-induced delirium	
Reye syndrome	- Acute encephalopathy and hepatic dysfunction caused by giving aspirin in the setting of a viral illness and/or fever. - Aspirin should be avoided under age 12, and some sources say avoid in teens for that matter. - Mechanism is obscure and thought to be related to impairment of β-oxidation.
Anti-cholinergic delirium	- Anti-cholinergic medications can cause confusion and a delirium-like presentation. Children have ↑ susceptibility. - Q can mention a kid was given an over-the-counter medication and now has low-grade fever and confusion.

	- Details such as flushed, dry, warm skin, or enlarged pupils, suggest anti-cholinergic delirium over Reye. - Can be confused with Reye syndrome if they just say "over-the-counter" med. - 1st-gen H1-blockers (i.e., diphenhydramine) are notoriously anti-cholinergic.
Dextromethorphan	- Can cause delirium and psychosis in children when taken in excessive amounts. - An opioid that is an anti-tussive (cough suppressant).
Alcohol withdrawal	
Alcoholic hallucinosis	- Presents 12-48 hours after the last drink. - Patient can have hallucinations, usually tactile (i.e., bugs crawling on skin) or visual (bugs crawling up wall). - Treat with benzo + thiamine.
Delirium tremens	- Presents 2-4 days following last drink. - Patient will have tremulousness, tachycardia, diaphoresis, and restlessness. - Seizures can also sometimes occur. - USMLE-favorite vignette is 40s male who gets tremulousness and tachycardia while in hospital 2ish days after surgery. Answer is just benzo. - Can also show up on NBME as a guy who goes from drinking 12 beers a day to suddenly only 2 beers a day. - Treat with benzo + thiamine.

Depression meds	
SSRIs	- Selective-serotonin reuptake inhibitors. - Fluoxetine, escitalopram, sertraline, etc. - Cause sexual dysfunction (anorgasmia) and sleep disturbance. - The fact that they cause anorgasmia actually makes them the Tx for premature ejaculation. - Do not combine with drugs such as monoamine oxidase inhibitors or St John wort, as this can cause serotonin syndrome (discussed below). - There are unique side-effects of various SSRIs – e.g., sertraline is more likely to cause diarrhea; fluoxetine has a stimulating effect and is more likely to cause insomnia; citalopram can prolong QT interval at higher doses. But USMLE doesn't give a fuck. - Used for a variety of psych conditions external to depression, e.g., fibromyalgia, OCD, etc. - It can take 4-6 weeks for an SSRI to achieve desired effect. If after this time point the drug isn't working, the first step is increasing the dose. If this doesn't work, the next step is switching to a different SSRI, followed by switching to a different class agent.
TCAs	- Block reuptake of both serotonin and norepinephrine. - Amitriptyline, nortriptyline, clomipramine, doxepin, etc. - High-yield on USMLE as 1st-line for diabetic neuropathic pain; can also be used for neuropathic pain in general (e.g., from trauma). Gabapentin is otherwise frequently used. - Have nasty anti-cholinergic side-effects (anti-DUMBBELSS; see HY Neuro PDF for discussion of anti-cholinergic vs pro-cholinergic effects if you're confused). - Three HY anti-cholinergic side-effect vignettes are 1) palpable suprapubic mass in an older male → full bladder as a result of anti-cholinergic med + BPH. 2) Hot, red, dry patient (as a result of anhidrosis). 3) Confusion + dilated pupils (anti-cholinergic delirium + mydriasis). - Coma, convulsions, cardiotoxicity. - Sometimes the Q can just mention prolonged QT interval in patient on an anti-depressant, and the answer is the TCA, since they're cardiotoxic. - If elderly, use nortriptyline, since ↓ BBB penetration and anti-cholinergic side-effects. - TCAs, such as imipramine, for nocturnal enuresis are wrong on USMLE. - Doxepin is asked on 2CK Psych form, where the Q rides on you knowing it's a TCA to get it right (i.e., the only drug listed that causes anti-cholinergic effects). I mention this because students get the Q wrong and then are like wtf is doxepin.
MAOIs	- Monoamine oxidase inhibitors.

	- Phenelzine, tranylcypromine. - Avoided 1st-line overwhelming majority of the time because of ↑ risk of serotonin syndrome. This can be in isolation, but also when commenced too soon following discontinuation of SSRI. - Selegiline is MAO-B inhibitor used for Parkinson disease. Answer on NBME where they ask which Parkinson med caused serotonin syndrome in a patient.
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.

Serotonin syndrome
- Flushing, tachycardia, diarrhea following drug-drug interactions (i.e., MAOI + SSRI; SSRI + St John wort) or high-risk drugs in isolation (e.g., MAOI, trazodone). - Can cause high fever, i.e., 105F+. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Can be treated with cyproheptadine (blocks serotonin receptors). - Often confused with carcinoid syndrome, which is due to carcinoid tumors (serotonin-secreting tumors) of the lung, small bowel, or appendix. Carcinoid syndrome isn't due to drugs. Additionally, it can cause tricuspid vegetations, whereas serotonin syndrome does not. It is still diagnosed with urinary 5-HIAA.

Neuroleptic malignant syndrome (NMS)
- Muscle rigidity and fever following commencement of anti-psychotic. - The fever will usually be 103+ F, as per my observation on NBME exams. This is because mere anti-cholinergic side-effects can sometimes give low-grade fever (i.e., hot, red, dry patient). - Mechanism for NMS is: the ryanodine channel, which allows calcium to move from the sarcoplasmic reticulum into the cytosol, gets stuck open, so ↑ calcium moves into the cytoplasm. The cell then needs to use a lot of ATP to pump the calcium back into the sarcoplasmic reticulum. This generates heat → fever. - This mechanism for NSM secondary to anti-psychotic administration is the same as malignant hyperthermia (MH) due to succinylcholine (nicotinic neuromuscular blocking agent used during surgery). - Tx is **dantrolene,** which closes the ryanodine channel. - NBME will sometimes give vignette of NMS or MH, and then the answer for Tx is "decreases sarcoplasmic calcium release." - In theory, bromocriptine (D2 agonist) can be used in the setting of NMS only as an alternative to dantrolene, but I haven't seen NBME assess this. This is more a pedantic Q students will ask sometimes.

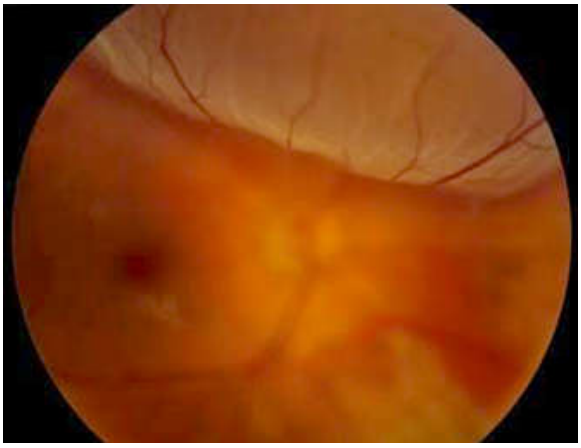
Epilepsy/hypnotic agents	
Benzodiazepines	- Diazepam, lorazepam, midazolam, chlordiazepoxide, etc. - Potentiate GABA-A receptor, which ↑ frequency of Cl^- channel opening, thereby allowing Cl^- to flow into the neuron, hyperpolarizing it and inhibiting it (i.e., ↓ firing). - Hypnotic (induces sleep); anxiolytic (reduces anxiety); anti-convulsant. - Given 1st-line for status epilepticus. - Used for alcohol withdrawal (delirium tremens / alcoholic hallucinosis). - Can cause respiratory depression. - Treat overdose with flumazenil (antagonist of benzo receptor).
Phenytoin	- Blocks sodium channels. - Generic anti-epileptic - Used for status epilepticus if benzo fails.

Barbiturates	- Phenobarbital, thiopental, etc. - Potentiate GABA-A receptor; ↑ duration of chloride channel opening, making them more efficacious (and dangerous) than benzos. - Used for general anesthesia. - Used for unremitting status epilepticus if benzos and phenytoin fail.

Opioids for USMLE	
- Mu (μ) and kappa (κ) are most well-studied, although delta (δ) and NOP forms exist. - Mu agonism is responsible for most of the classic effects of opioids – i.e., analgesia, euphoria, respiratory depression, and dependence. Agonism results in inhibition of neuronal firing and neurotransmitter release via opening of K^+ channels and closing of Ca^{2+} channels. - Kappa has analgesic effects but can also produce dysphoria. - Opioids can cause respiratory depression, miosis, and constipation. - Opioid withdrawal can cause flu-like symptoms, irritability, piloerection, and yawning.	
Opioid receptor agonists	
Morphine	- Gold standard opioid; used for moderate to severe pain. - USMLE wants you to know that morphine is "metabolized into active metabolites that accumulate." This is NBME answer for how overdose can occur in patients who use self-controlled pumps, since it can take time to work, so patient uses too much. - Patients who are in severe pain can sometimes have hypertension as a result. Surgery Q has morphine as answer for how to control BP post-op. - If patient has history of opioid abuse but has severe pain, e.g., due to trauma, do not withhold using opioids. Always treat pain fully.
Codeine	- Less potent than morphine; used for mild to moderate pain; in many antitussives.
Oxycodone	- Potent opioid often combined with acetaminophen. - USMLE wants you to know that oxycodone/acetaminophen combo is preferred initially in patients who are post-op. Do not just jump to morphine.
Fentanyl	- Potent synthetic opioid with high abuse potential.
Meperidine	- Potent opioid with high abuse potential. - I've noticed NBME likes this opioid for malingering Qs, where they'll say, e.g., a 47-year-old man comes into the ER in severe pain requesting meperidine.
Dextromethorphan	- Opioid used in antitussives. - Common cause of delirium from over-the-counter cold meds.
Loperamide, eluxadoline	- Opioids used for diarrhea-predominant irritable bowel syndrome. - Low addictive potential. - The fact that opioids cause constipation as an adverse effect "is a good thing" in the setting of wanting to help limit diarrhea in IBS.
Buprenorphine	- Mu partial agonist and kappa antagonist. - The kappa antagonism can help cause less dysphoria. - NBME Q floating around somewhere wants you to know it has the kappa effects.
Tramadol	- Unique MOA where it is weak mu opioid agonist + also an SNRI.
Methadone	- Long-acting full mu opioid agonist that is used for opioid/heroin withdrawal. - Reduces opioid cravings.
Opioid receptor antagonists	
Naloxone	- Opioid receptor antagonist used for acute overdoses.
Naltrexone	- Opioid receptor antagonist used for alcohol dependence (i.e., ↓ alcohol cravings). - Can also be used for opioid dependence in patients who are already detoxified in order to prevent relapse (i.e., by blocking opioid receptors, if the patient abuses any opioids, he/she won't feel the euphoric effects).

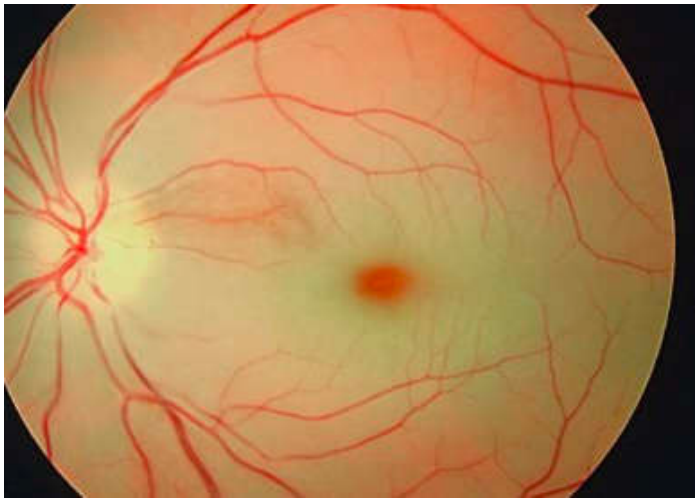
Surg Ophthal

Glaucoma	
Open-angle	- Chronic eye condition characterized by increased intraocular pressure due to impaired drainage of aqueous humor. - Unlike acute angle-closure (closed-angle) glaucoma, the drainage angle (canal of Schlemm) between the cornea and iris remains open. - Progresses slowly and is usually painless. - As it advances, causes peripheral vision loss (tunnel vision) and results in optic nerve damage characterized by "cupping of the optic disc" (i.e., an increased cup-to-disc ratio). - The USMLE wants **tonometry** as the first step in management (measures intraocular pressure). - Treatment for open-angle glaucoma for USMLE is not going to be a trivia game of "which drug first." They just want you to know that topical prostaglandins, such as **latanoprost,** can increase outflow of aqueous humor, thereby reducing IOP. - Topical beta-blockers, such as **timolol,** decrease production of aqueous humor. **- Carbonic anhydrase inhibitors,** such as dorzolamide, decrease production of aqueous humor. - **Pilocarpine** is a muscarinic receptor agonist that constricts the pupil and increases outflow of aqueous humor.
Closed-angle	- Presents on USMLE as a very buzzy red, teary, fixed mid-dilated pupil. - Due to closure of the drainage angle (canal of Schlemm) between the cornea and the iris, leading to a rapid increase in IOP. - Can lead to compression of the optic nerve and blood vessels, causing severe symptoms such as intense eye pain, headache, nausea, vomiting, blurred vision, and seeing colored rings or halos. - Similar to open-angle glaucoma, the USMLE wants **tonometry** as the first step to confirm increased IOP. - Treatment is IV mannitol (helps draw excess aqueous humor out of the eye), timolol, pilocarpine (muscarinic receptor agonist), or alpha-2 agonists such as brimonidine.

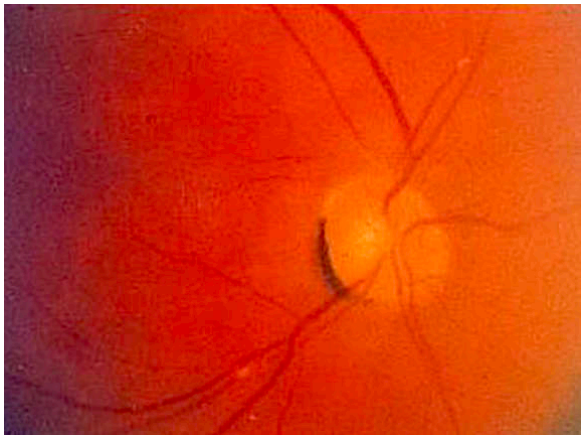
Retinal detachment
- Occurs classically in those with sudden trauma to the eye or severe myopia. - The fundoscopy is HY and important for USMLE, especially 2CK.  - Treatment is surgical and includes pneumatic retinopexy (i.e., gas bubble is injected into the eye to push the retina back into place; scleral buckling (piece of silicone is attached to the sclera to exert posteriorly-directed pressure so the retina reattaches); and vitrectomy (vitreous humor is removed and replaced with gas or silicone oil to flatten the retina).

Central retinal artery occlusion

- Presents as sudden, painless vision loss in a patient where a "curtain comes down."
- Due to a carotid atheroma (in setting of HTN) or left atrial mural thrombus (in setting of atrial fibrillation) that's launched off to the retinal vessels.
- A pale retina will be seen on fundoscopy with a cherry red spot due to maintenance of perfusion from the choroidal artery.



- Ocular massage can be attempted as Tx, where the embolus may be dislodged, restoring perfusion.

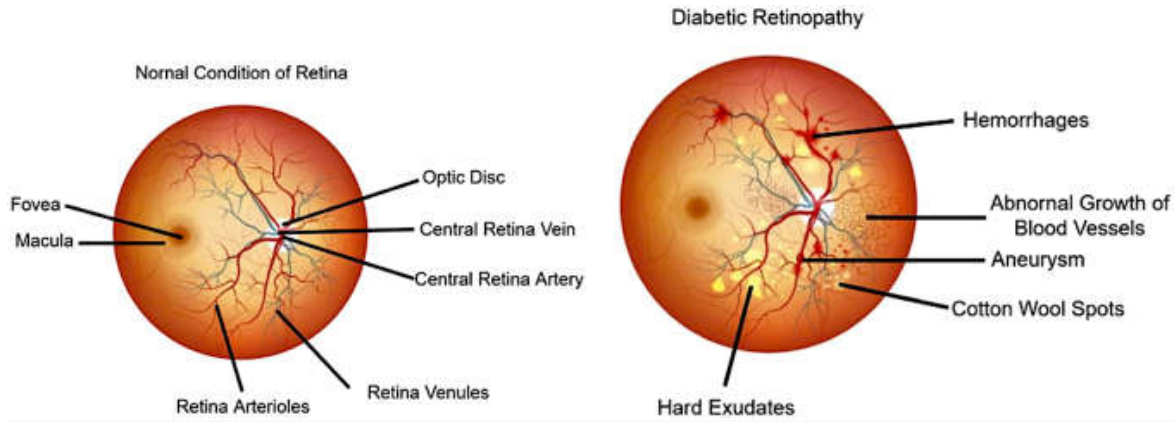


- USMLE can show image that looks similar to this one. They want you to know that weird crescent-shaped finding is called a "refractile body," and refers to an embolus.

Diabetic retinopathy

- Can present in numerous ways, such as with blurred or fluctuating vision, and difficulty recognizing colors.
- The USMLE vignette will usually say things like cotton wool spots (soft exudates), hard exudates, and retinal hemorrhages.
- Soft exudates are small, whitish lesions on the retina caused by microinfarctions and are axoplasmic material; hard exudates are lipoproteinaceous deposits due to leaky retinal vessels.
- Soft/hard exudates and retinal hemorrhages are seen classically in both diabetic and hypertensive retinopathies.

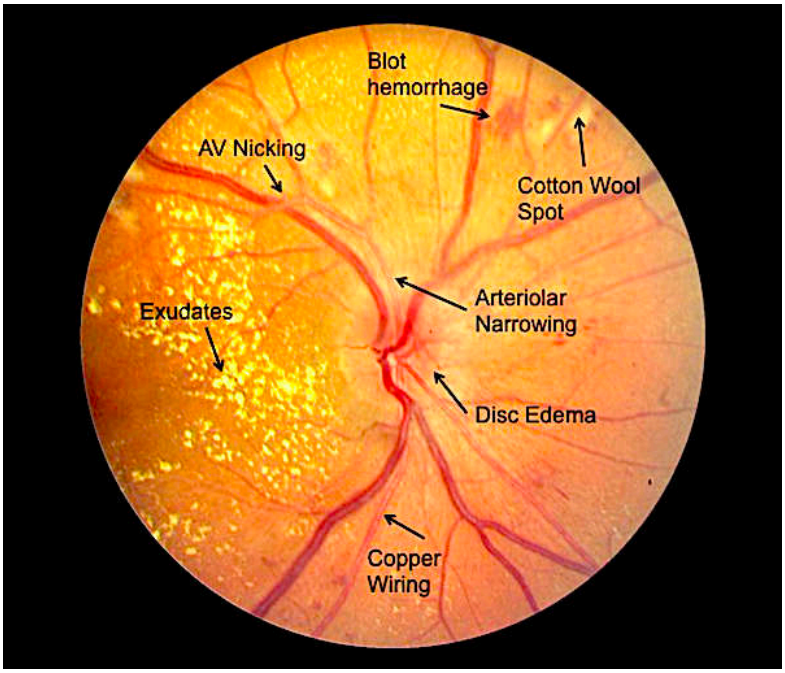




- Can be dry or wet (neovascular). Wet type can be treated with anti-VEGF agents.

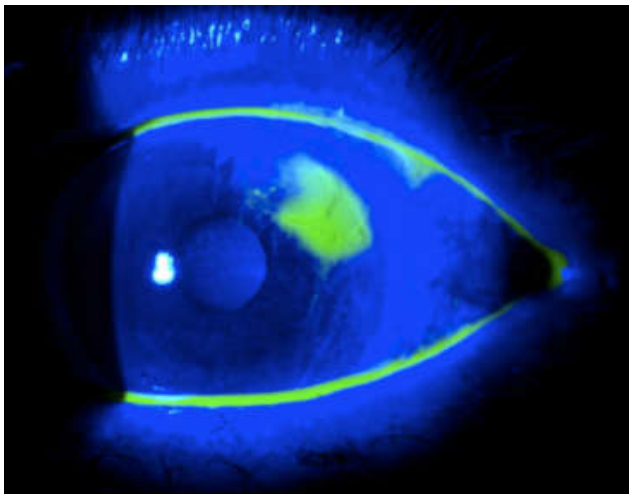
Hypertensive retinopathy

- Buzzy findings like AV nicking and arteriolar narrowing.
- Q can mention “sausage-like” appearance of retinal arterioles.



Corneal abrasion

- Scratchy or tearing sensation of the eye. Can be caused by contact lens-use or coming into contact with irritants such as sand (i.e., playing in sandbox with son/daughter).
- USMLE will either ask for the next best step, where the answer is “instillation of fluorescein into the eye,” or they will show you the following image (which is a fluorescein instillation) and then just ask for the diagnosis (i.e., corneal abrasion).



- Fluorescein is an orange dye that when dripped into the eye appears blue-green under violet light.
- Anything blue is normal. Anything green is abnormal.

Subconjunctival hemorrhage

- Buzzy image / spot-diagnosis for USMLE.
- Caused by bursting of superficial conjunctival vessels. Looks sinister but self-resolves uneventfully.
- Can be caused by trauma or brief periods of increased pressure (i.e., severe vomiting or coughing).

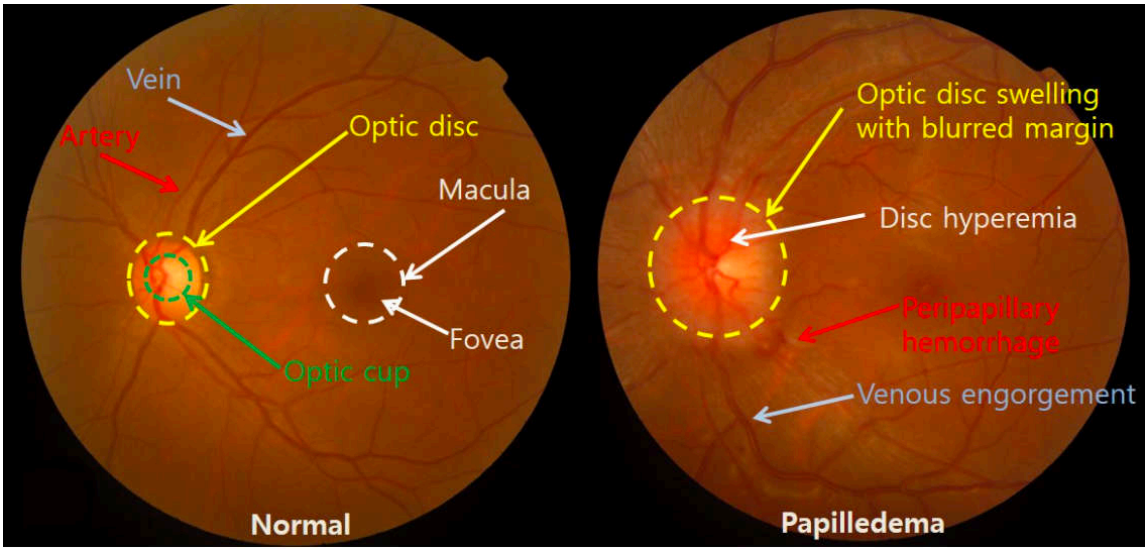


HY Peds Ophthal DDx for Surg	
Orbital cellulitis	 - Infection involving tissues posterior to orbital septum; *Staph aureus* most common cause; rare sequela of adjacent infection, e.g., from the paranasal sinus; medical emergency; requires IV antibiotics; preseptal cellulitis is a less severe version of orbital cellulitis.
Hordeolum (stye)	- Painful bump on the eyelid. - *Staph aureus* infection of sweat gland or oil duct; treat with warm compresses. - Can occur any age. 
Chalazion	 - Painless bump on the eyelid. - Blocked oil duct; not an infection; treat with warm compresses.

Retinoblastoma	 - The image shows leukocoria, which is an abnormal white appearance of the eye on light reflex (should normally appear red). - USMLE wants you to know that *RB* gene mutations not only cause congenital retinoblastoma, but also ↑ risk of osteosarcoma. - In other words, classic Q is they show you image above + ask what child is at risk for later → answer = osteosarcoma.

Papilledema

- Means increased intracranial pressure on USMLE.
- Papilledema refers to blurring of the optic disc margins due inflammation of the optic nerve head in the setting of increased intracranial pressure.



Anterior uveitis

- WBCs in the anterior chamber sometimes seen in flares of autoimmune diseases, especially sarcoidosis and ankylosing spondylitis.
- USMLE might sometimes just give you patient with sarcoidosis and then ask what needs to be done in terms of follow-up for the patient → answer = "slit-lamp examination" to check for anterior uveitis.

Surg MSK

Rotator cuff injuries		
Rotator cuff muscles (SITS)	Function	HY Points
Supraspinatus	Abduction of arm **(first 15 degrees)**	- Answer if patient has difficulty abducting arm first 15 degrees. - Empty-can (thumb-down) / full-can (thumb-up) tests for diagnosis → patient abducts arm to 90 degrees with thumb up or down → downward pressure applied to arm → if elicits pain, answer = supraspinatus injury. - There is Q on new 2CK CMS IM form 7 where answer is supraspinatus tendonitis, and they say patient has reduced ability to abduct the first 60 degrees.
Infraspinatus	Lateral (external) rotation	- Notion of "pitcher injury" = infraspinatus does more harm than good for USMLE; vignette can by all means give a pitcher with (+) full-can test above and answer is supra-, not infra-, spinatus injury. - Q might say patient simply cannot externally rotate arm, nothing more, where teres minor isn't listed as another answer, so infraspinatus is only one that could be right.
Teres minor	Adduction; lateral rotation	- Same as infraspinatus, just know it externally rotates the arm. - Also adducts arm, so if patient has issues with both lateral rotation **and** adduction, answer is teres minor over infraspinatus. - I've never seen NBME material assess the diagnostic tests for infraspinatus or teres minor.
Subscapularis	Adduction; medial (internal) rotation	- Can medially rotate and adduct the arm. - I've had students get asked Gerber lift-off test on both Steps, where the answer = subscapularis. - Gerber lift-off test = patient places dorsal aspect of hand on lower back, with palm facing posteriorly → examiner applies pressure into the patient's palm against his/her lower back → patient is then asked to move hand away under the pressure → if elicits pain / difficult to do, answer = subscapularis injury.

Upper limb nerve HY Points for Surg	
Axillary	- Main innervation of the deltoid, allowing for abduction of arm 15-90 degrees. USMLE wants you to know deltoid has an origin on the lateral clavicle and axillary nerve innervating it is at C5/C6. - Palsy caused by **surgical neck of humerus fracture.** - USMLE vignette will often say "flattened deltoid" or "loss of sensation over lateral upper arm / deltoid."
Median	- Main innervation is lateral "3 and a half" fingers / thenar pad, and lateral forearm. - Does thumb abduction. (In contrast, ulnar nerve does thumb adduction) - NBME wants "palmar cutaneous branch of median nerve" as answer for sensation over thenar region. - Palsy caused by **supracondylar fracture of humerus**, or "**distal shaft fracture."** Former is buzzy; latter sounds non-specific, but I've seen it this way on NBME. - Entrapment of median nerve causes carpal tunnel syndrome; will present as paresthesia/numbness of lateral hand / thenar region; can be caused by hypothyroidism (GAG deposition), acromegaly (growth of tendons), and pregnancy (edema); can occur bilaterally in construction workers using jackhammer. - Tx for carpal tunnel ultra-HY on 2CK. "Use of wrist pad when using computer" is answer on new 2CK NBME. If not listed, "wrist splint" is answer. If vignette says wrist splint fails, NSAIDs are wrong answer and not proven. USMLE wants "triamcinolone

	injection into carpal tunnel" (not IV steroids) as next answer. Surgery is always wrong answer for carpal tunnel on USMLE. - 2CK wants "electrophysiological testing" and "electromyography and nerve conduction studies" as next best step in diagnosis for carpal tunnel.
Ulnar	- Main innervation of medial "1 and a half" fingers, and medial forearm. - Ulnar nerve also does **finger abduction and adduction** (i.e., interosseous muscles). - USMLE loves **Froment sign** for ulnar nerve injury, which is inability to pinch a piece of paper between the thumb and index finger (ulnar nerve needed for thumb adduction against index finger, despite thumb being most lateral digit). - Distal compression (i.e., of wrist and hand only, not forearm) is aka **Guyon canal syndrome** and is caused by **hook of hamate fracture;** this can sometimes be seen in cyclists due to handlebar compression; presents as paresthesias / numbness of 4th and 5th fingers + hypothenar eminence. - Proximal compression (i.e., medial forearm + wrist/hand) is aka **cubital tunnel syndrome** and is one of the most underrated diagnoses on USMLE, since its yieldness, especially on 2CK, is comparable to carpal tunnel syndrome, but students often haven't heard of it. Essentially, patient will get paresthesias of medial forearm + hand, where it "sounds like carpal tunnel but on the ulnar side instead" → answer = cubital tunnel syndrome. - Tx for cubital tunnel syndrome is "overnight elbow splint." Surgery is wrong answer on USMLE.
Radial	- Main innervation for finger, wrist, and elbow extension. - Innervates **BEST** → **B**rachioradialis, **E**xtensors, **S**upinator, **T**riceps. - Palsy occurs as a result of **midshaft fracture of the humerus,** or as a result of fracture at the **radial groove** (latter is obvious). - Retired NBME Q says construction worker sustains "comminuted spiral fracture of humerus" (unusual, since spiral fracture classically = child abuse), and they ask for the resulting defect → answer = "loss of radial nerve function." - Highest yield point is that injury results in pronated forearm + wrist drop.
Musculocutaneous	- Main innervation of the biceps. - Just need to know injury results in loss of sensation over lateral forearm + decreased biceps function. - USMLE doesn't give a fuck about what kind of injury causes palsy.

Frequently confused shoulder conditions	
Subacromial bursitis vs rotator cuff tendonitis	- In NBME vignettes, I've seen both can give Hx of patient doing frequent overhead movement (i.e., painting a fence) + pain with palpation + pain that's worse when lying on one's shoulder in bed at night, making differentiating these difficult. - Subacromial bursitis will only present with above findings, which collectively are known as impingement syndrome. - Rotator cuff tendonitis will present with weakness when performing exam maneuvers (as described in prior table).
Biceps tendonitis	- Presents as anterior shoulder pain with focal tenderness over the biceps tendon (i.e., when pressing on anterior shoulder).
Adhesive capsulitis	- Aka "frozen shoulder," or arthrofibrosis. - Decreased passive and active motion of shoulder in all directions. - Idiopathic, but increased risk in diabetes. - Tx = range of motion exercises / physiotherapy. - This is a Dx that is LY on Step 1, but for whatever reason, becomes HY on 2CK.

Winged scapula

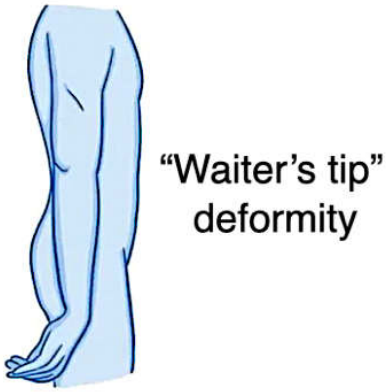


Winged scapula

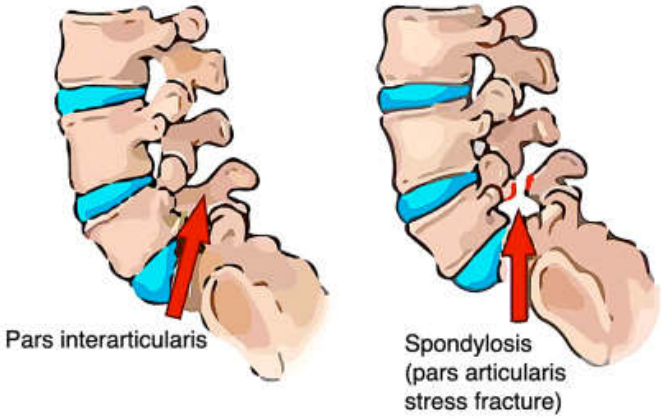
- Long thoracic nerve injury, which innervates serratus anterior.
- Can occur post-mastectomy.

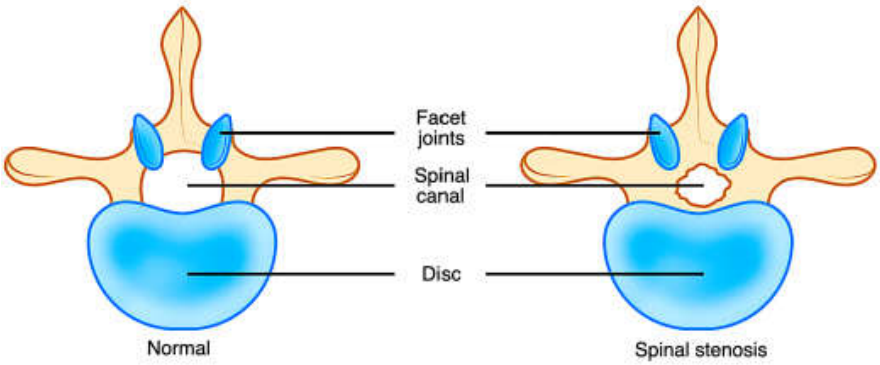
Brachial plexus injuries

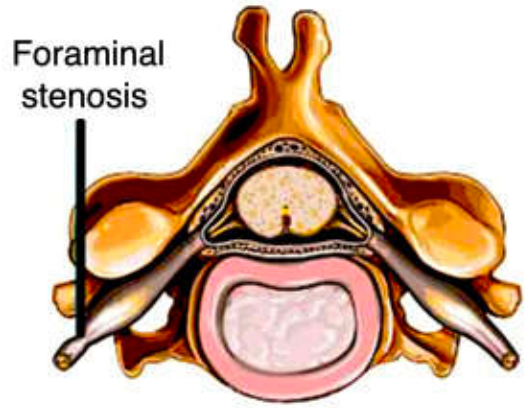
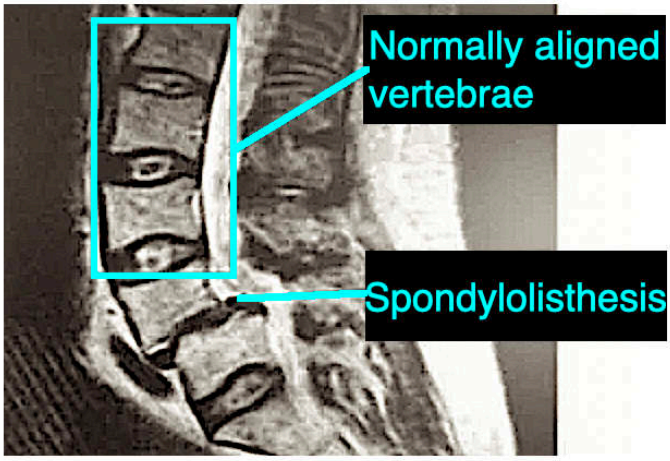
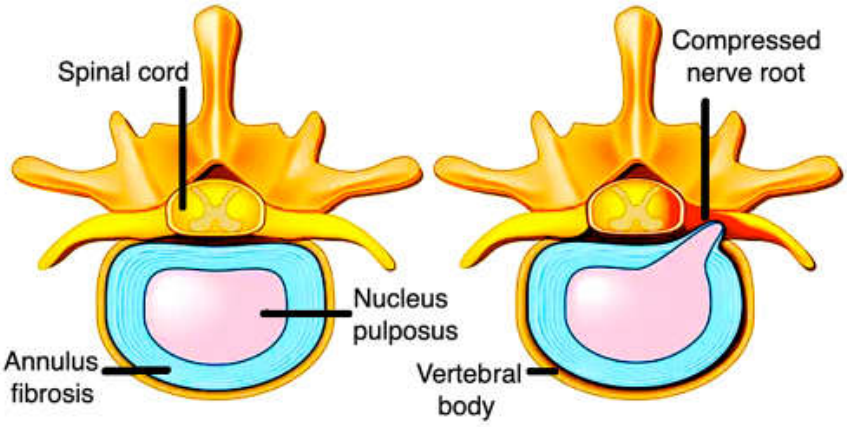
- Both can be caused in neonates by breech / traumatic labor.
- Can also be caused by injuries such as grabbing onto tree branch while falling.

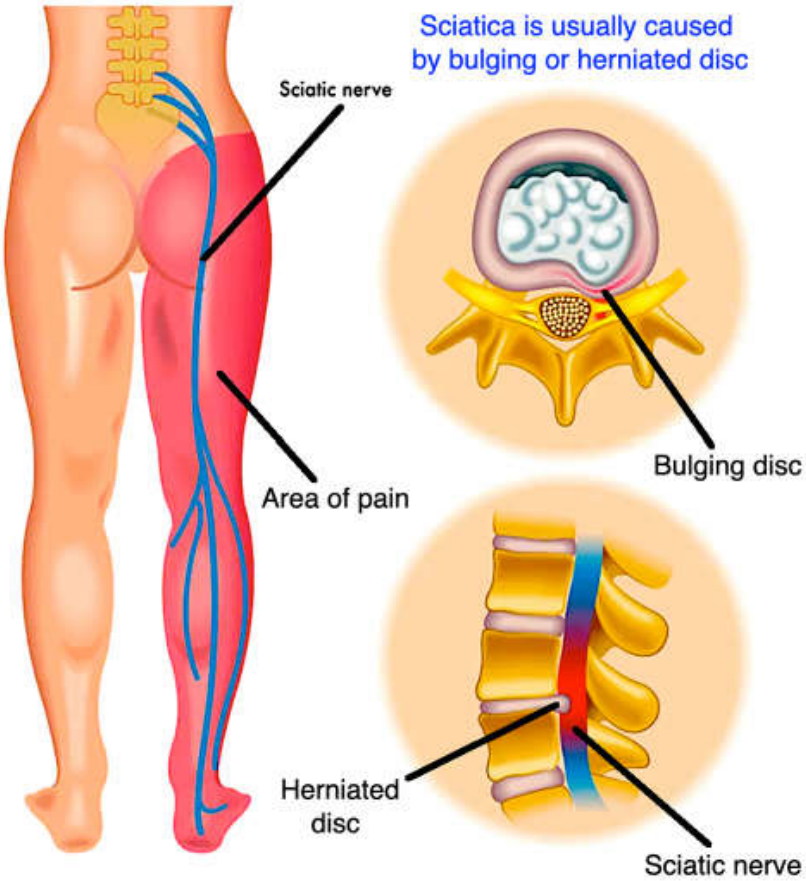
Erb-Duchenne	- Upper brachial plexus injury (C5-C6). - Sometimes rather than Erb-Duchenne written as the diagnosis, the answer will just be "upper brachial plexus" when they ask for what's injured.  "Waiter's tip" deformity - Arm is adducted, pronated, and wrist flexed.
Klumpke	- Lower brachial plexus injury (C8-T1). - Sometimes rather than Klumpke written as the diagnosis, the answer will just be "lower brachial plexus" when they ask for what's injured.  Klumpke "claw-hand"

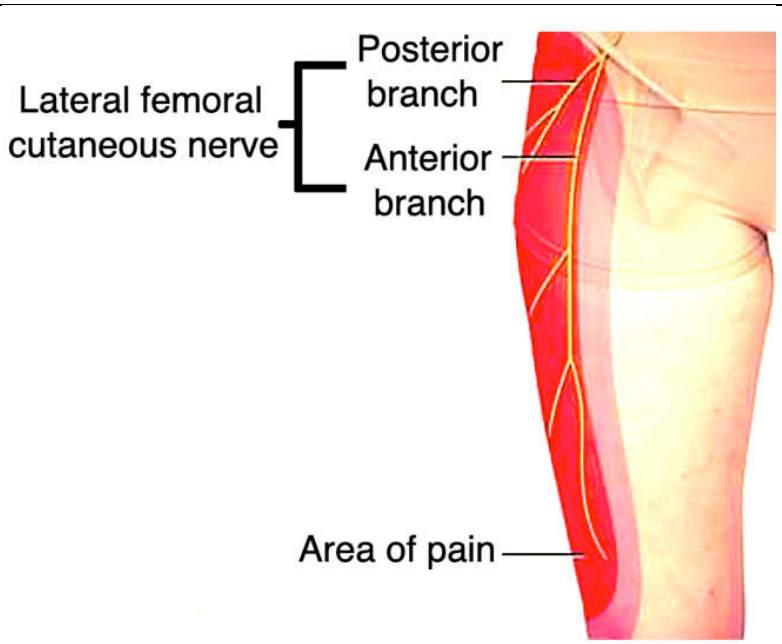
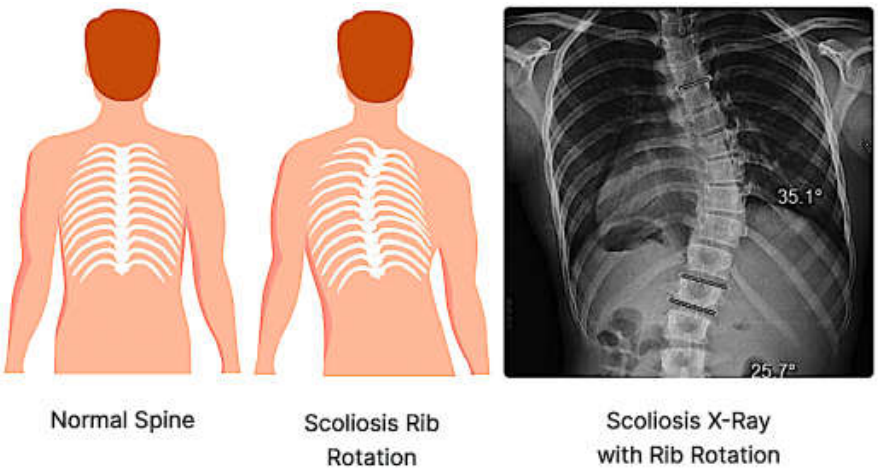
Other upper limb DDx for Surg	
deQuervain tenosynovitis	- Tenosynovitis means inflammation of tendon sheaths. - deQuervain is classic in breastfeeding women and is worsened with Finkelstein test (shown below) → 1) thumb is placed in palm; 2) 2nd-5th fingers are wrapped over the palm; 3) patient ulnar deviates the wrist → this causes pain.  - Patient should avoid offending activity, but since this is often breastfeeding, steroid injection can provide immediate relief.
Ganglion cyst	- Gelatinous collection of joint fluid; can occur on ankles and flexor areas as well, but classic location is dorsum of hand/wrist.  - If Q asks you most likely trajectory if untreated, answer = spontaneous regression. - Tx is needle drainage if disturbing to the patient; recurrence common because the root opening to the tendon sheath is not removed.
Lateral epicondylitis	- Tennis elbow. - Lateral elbow pain worsened when patient extends wrist against resistance. - NBME asks "extensor carpi radialis brevis" as answer for site of inflammation. - Tx = "forearm strap" on 2CK forms.
Medial epicondylitis	- Golfer elbow. - Medial elbow pain worsened when patient flexes wrist against resistance. - Inflammation at flexor carpi radialis and pronator teres. - Tx = forearm strap.
Radial head subluxation	- Aka "nursemaid elbow." - Child stops using arm + arm pronated and partially flexed. - Hx of child having arm pulled/yanked, or child holding hands and running with older sibling + the child falls, resulting in elbow pull. - Tx = hyper-pronation, OR supination when arm partially flexed. Either is correct. Both will not be listed at the same time as answers.
Olecranon bursitis	- Elbow pain, usually following contact injury. - Tx = compression bandage + NSAIDs. Steroid injection is wrong answer on USMLE.

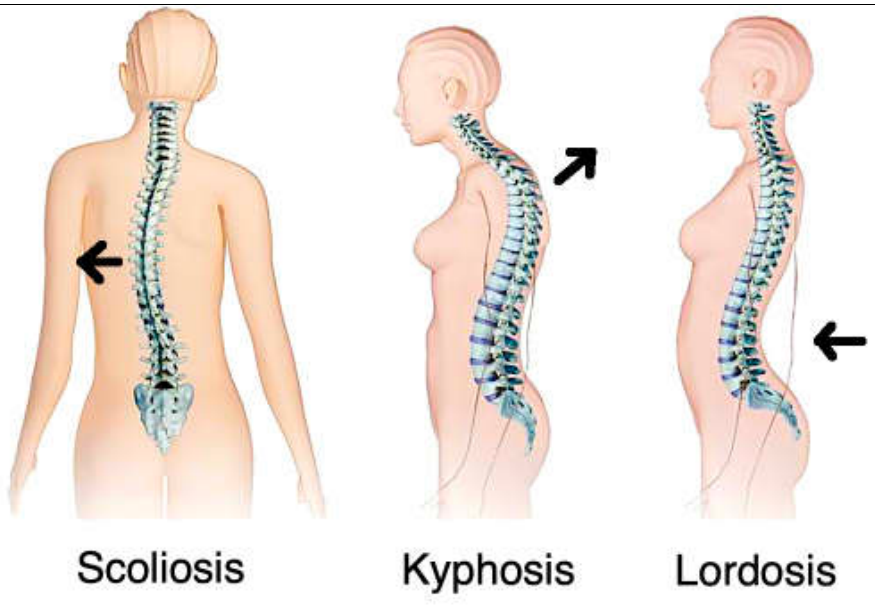
<table>
<tr><th colspan="2">HY MSK spinal conditions for Surg</th></tr>
<tr><td>Cervical spondylosis</td><td> 

- The answer on USMLE if they say patient over 50 has neck pain + MRI shows degenerative changes of cervical spine.
- Can occur in lumbar spine, but USMLE likes cervical spine for this.
- Technically defined as degeneration of pars interarticularis component of vertebral body.</td></tr>
<tr><td>Atlantoaxial subluxation</td><td>- Increased mobility between the first (atlas) and second (axis) vertebrae.
- Really HY on 2CK Surg and Neuro forms in patients who have rheumatoid arthritis.
- Must do CT or flexion/extension x-rays of cervical spine prior to surgery when a patient will be intubated; I’ve seen both of these as answers for different Surg Qs.
- Q on one of the Neuro CMS forms gives patient with RA not undergoing surgery who has paresthesias of upper limbs → answer is just MRI of cervical spine (implying atlantoaxial subluxation has already occurred).</td></tr>
<tr><td>Lumbar spinal stenosis</td><td>- Narrowing of the spinal canal.
- The answer on USMLE if they mention a patient over 50 who has lower back pain that’s worse when walking down a hill (i.e., relieved when leaning forward), or when standing/walking for extended periods of time.
- Can cause “neurogenic claudication,” where the vignette sounds like the patient has intermittent claudication, but they’ll make it clear the peripheral pulses are normal and that the patient doesn’t have cardiovascular disease; this shows up in particular on 2CK Neuro CMS forms.
- Technically an osteoarthritic change of the spine; therefore increased risk in obesity (but not mandatory for questions).
 
</td></tr>
<tr><td>Cervical foraminal stenosis</td><td>- The answer on USMLE if they say old woman has difficulty fastening buttons + weakness of hand muscles + loss of sensation of little finger → answer = “C7-T1 foraminal stenosis” (offline NBME 20).
- Not stenosis of cervical spinal canal, but stenosis of foramen where nerve exits.</td></tr>
</table>

	
Spondylolisthesis	 - The answer on USMLE if they a “step-off” between infra-/suprajacent vertebrae. In other words, they’ll say one vertebra “juts out” or has a “step-off” compared to those above/below it. - Can be due to trauma or idiopathic development.
Disc herniation	 - Herniation of nucleus pulposus through a tear in annulus fibrosis. - The answer on USMLE if they mention **radiculopathy** (i.e., shooting pain down a leg) after lifting a heavy weight or bending over (e.g., while gardening). They can write the answer as “herniated nucleus pulposus.” - As I mentioned with the radiculopathies above, be aware of the L4, L5, and S1 differences. - Be aware that cervical disc herniation “is a thing,” meaning it’s possible and also assessed on USMLE. 2CK neuro forms ask this a couple times, where patient has shooting pain down an arm, and answer is “C8 disc herniation.”

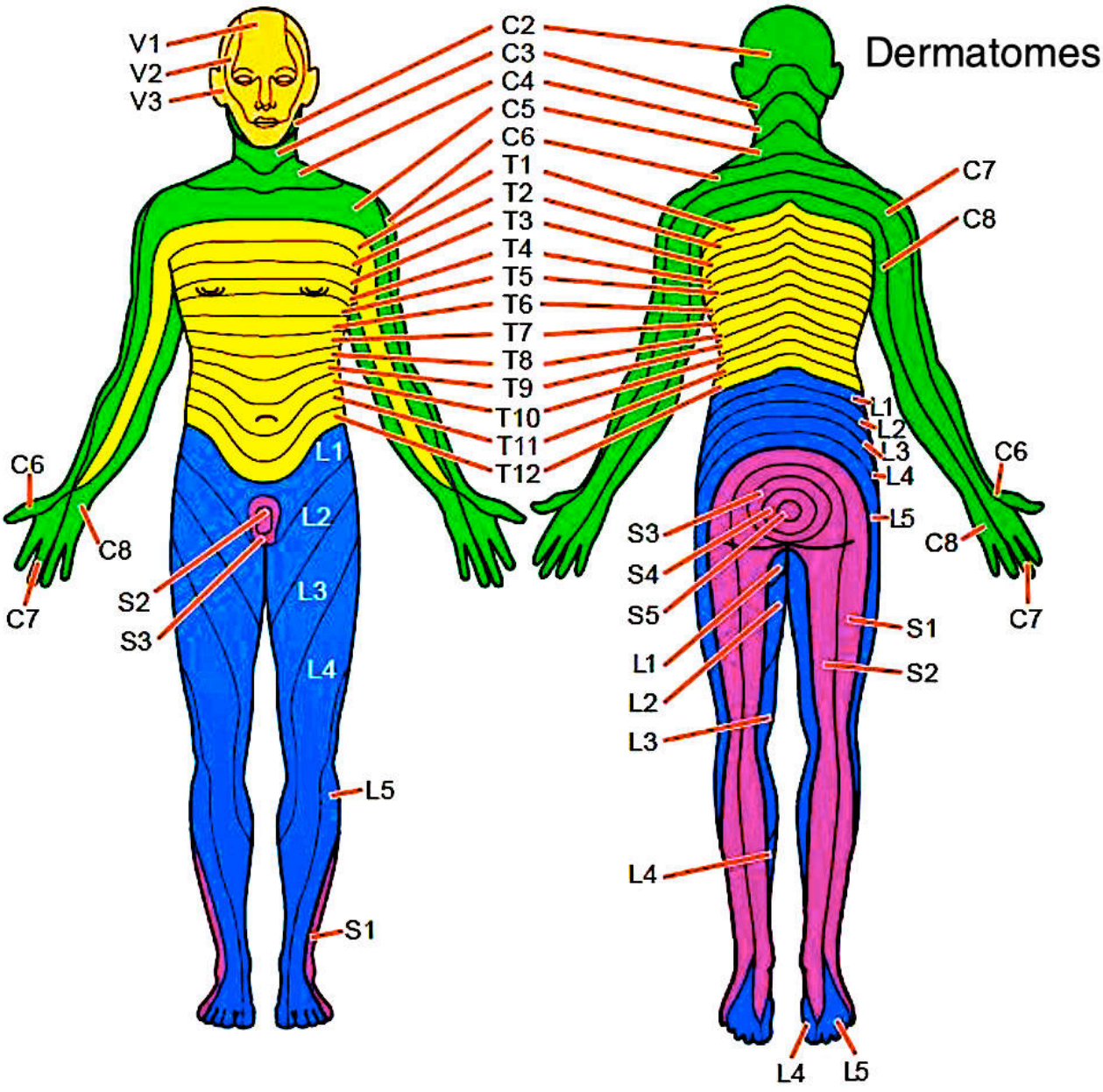
	- If suspected, newest NBMEs want "no diagnostic studies indicated." X-ray and MRI are not indicated unless there is motor/sensory abnormality (i.e., weakness or numbness). But for mere radiculopathy (i.e., radiating pain), no imaging necessary on new NBME content. - Straight-leg raise test is **not** reliable. Mere pain alone is a **negative** test. The test is only positive when they say it reproduces **radiculopathy/radiating pain.** There is a 2CK Q where they say straight-leg test causes pain (i.e., negative test) and answer is "no further management indicated" (i.e., Dx is only lumbosacral strain). - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Lumbosacral strain	- The answer on USMLE if they say patient has paraspinal muscle spasm following lifting of heavy box **without** radiculopathy. If they say radiculopathy, the answer is disc herniation instead. - Straight-leg test can cause pain (i.e., **negative** test). The test is only positive if they it reproduces radiating pain. - Do **not** x-ray. This is really HY for 2CK. Apparently lumbar spinal x-rays are one of the most frivolously ordered tests, and USMLE wants you to know that you do not order one for simple lumbosacral strain. - Tx is NSAIDs + light exercise as tolerated. Bed rest is wrong answer on USMLE.
Sciatica	- 90% of the time is due to disc herniation.  - Straight-leg test classically (+) – i.e., reproduces radiating pain. - Tx = Light exercise as tolerated + NSAIDs. Bed rest is wrong answer on USMLE. - On one of the 2CK CMS forms, ibuprofen straight-up is listed as the answer.
Meralgia paresthetica	- The answer on USMLE if they say patient has pain or paresthesias running down the lateral thigh. - Due to entrapment of lateral femoral cutaneous nerve. - Often seen as incorrect answer choice on Step, so at least be aware of it.

	
Scoliosis	- Sideways curvature of spine, creating an S- or C-shaped curve. - Usually idiopathic; affects 3% of population; girls 4:1.  Normal Spine Scoliosis Rib Rotation Scoliosis X-Ray with Rib Rotation - Can be associated with Marfan syndrome, Friedreich ataxia, NF1. - Adams forward bend test used to diagnose. - USMLE wants you to know most children do not need treatment, but that curvatures will remain throughout life. - Answer = bracing if curvature is >25 degrees and child is still growing. - I've never seen surgery as answer for scoliosis on NBME; literature says recommended only when curvature >40 degrees.
Kyphosis	- Abnormal convex curvature of thoracic spine. - Usually idiopathic due to old age; can be due to degenerative disc disease and compression fractures (osteoporosis). - If severe, can in theory cause restrictive lung disease due to impaired chest wall expansion.

Lower limb nerve HY Points for Surg	
Common peroneal (fibular) nerve	- The answer on USMLE if patient loses both eversion **and** dorsiflexion of the foot. - Sensation to upper third of lateral leg (around and below lateral knee). - Splits into superficial and deep peroneal (fibular) nerves.
Superficial peroneal nerve	- The answer on USMLE if patient loses only eversion of the foot, but dorsiflexion stays intact. - Sensation to lower lateral leg and dorsum of foot.
Deep peroneal nerve	- The answer on USMLE if patient only loses dorsiflexion of the foot, but eversion stays intact. - **D**eep for **D**orsiflexion, which means superficial is the one that does eversion instead. - Loss of dorsiflexion causes a high-steppage gait (patient has to lift foot high into the air with each step). - Also does sensation to webbing between 1st and 2nd toes. I've never seen NBME Qs ask or give a fuck about this sensation detail, but students get fanatical about it as if it's supposed to be high-yield.
Tibial nerve	- The answer on USMLE if patient loses plantarflexion of the foot (can't stand on tippytoes). - Sensation to bottom of foot / heel.
Sciatic nerve	- The answer on USMLE if patient has motor dysfunction of tibial and common peroneal nerves at the same time, or has sciatica (shooting pain down leg). - Splits into the common peroneal nerve and tibial nerve. - Does not supply sensation to thigh; sensation encompasses that supplied by the combination of the common peroneal nerve and tibial nerves. - Supplies some motor function to muscles of thigh but USMLE doesn't care. - Sciatica = shooting pain from the lower back down the leg usually as the result of disc herniation; 2CK Neuro forms simply want NSAIDs as treatment; straight-leg test is classically used in part to diagnose, but I've seen this test show up on NBME material for simple lumbosacral strain (i.e., the test is non-specific and not reliable).
Obturator nerve	- The answer on USMLE if patient has inability to adduct the hip with loss of sensation to medial thigh.
Femoral nerve	- The answer on USMLE if patient cannot extend knee and/or has buckling at the knee.

	- Also does sensation to anterior thigh + medial leg (not thigh), although I haven't seen sensation specifically asked for femoral nerve.
Saphenous nerve	- The answer on USMLE if patient loses sensation to medial leg. - Pure sensory branch of the femoral nerve.
Sural nerve	- The answer on USMLE if patient loses sensation to lower lateral leg. In contrast, if sensation loss is upper lateral leg, that's common peroneal nerve instead. - Often confused with saphenous. Good way to remember is: suraL is Lateral, therefore saphenous must be the one that's medial.
Superior gluteal nerve	- The answer on USMLE if patient has Trendelenburg gait → opposite side of pelvis will fall while walking, so patient will tilt trunk toward side of lesion while walking to maintain level pelvis. - Innervates gluteus medius and minimus.
Inferior gluteal nerve	- The answer on USMLE if patient cannot squat, stand up from a chair, or go up/down stairs. - Innervates gluteus maximus.



Lower limb reflexes / radiculopathies	
L4 radiculopathy	- The answer on USMLE if patient loses knee (patellar) reflex + has weakened knee extension. - Pain / paresthesias / numbness in L4 distribution (anterior thigh + medial leg). - Disc herniation of L3-4. - Just remember that L4 is the one where the knee reflex is fucked up.
L5 radiculopathy	- The answer on USMLE if patient loses dorsiflexion. - Pain / paresthesias / numbness in L5 distribution (lateral + anterior leg). - Disc herniation of L4-L5.
S1 radiculopathy	- The answer on USMLE if patient loses ankle (Achilles) reflex + has weakened plantar flexion. - Pain / paresthesias / numbness in S1 distribution (sole of foot + lower leg). - Disc herniation of L5-S1. - Just remember that S1 is the one where the ankle reflex is fucked up. - **SALT** → **S**1, **A**chilles, **L**ateral leg dermatome, **T**ibial motor issue (plantar flexion).

Lower limb DDx for Surg	
Trochanteric bursitis	- Vignette will be lateral hip pain that is worsened with palpation + lying on one's side in bed. - Tx = NSAIDs.
Septic bursitis	- USMLE will give inflammation of knee joint that sounds like septic arthritis, but they will say there's **no** joint effusion. This is how it presents in a 2CK NBME vignette, where students constantly ask "why not septic arthritis?" And they say in the vignette there's **no** joint effusion. - Tx = antibiotics.
Prepatellar bursitis	- Aka housemaid's knee; presents as anterior knee pain in people who are frequently on their knees (painters, plumbers, etc.). - Tx = NSAIDs.
Patellar tendonitis ("Jumper's knee")	- Inflammation of patellar tendon. - The answer on USMLE if they describe anterior knee pain that initially occurs only after finishing sports (e.g., basketball game), then progresses to more chronic pain. - Tx on NBME = "quadricep strengthening exercises."
Patellofemoral instability	- Presentation is annoyingly similar to patellar tendonitis, but do not confuse. - Patellofemoral instability is misalignment of the patella at the trochlear groove of the femur. - Q can mention crepitus. - Shows up on 2CK Peds CMS form 6 as teenage girl who has knee pain worse after jumping or running + has crepitus → answer = "patellofemoral instability" (patellar tendonitis / "jumper's knee" not listed as answer).
Patellofemoral pain syndrome	- Aka chondromalacia patellae; name implies softening of cartilage in the knee. - The answer on NBME if they say pain that worsens when sitting for long periods of time, or when going up or down stairs. - Classic in obesity or those who squat heavy weight (knees think you're obese). - Tx = quadriceps strengthening exercises.
Anterior cruciate ligament injury	- ACL is answer if (+) anterior drawer test or Lachman test → excessive anterior displacement of tibia relative to femur. - Classically injured when knee is hyper-extended, or with a rotational force on a fixed, planted knee.
Posterior cruciate ligament injury	- PCL is answer if (+) posterior drawer test → excessive posterior displacement of tibia relative to femur. - Classically injured when knee hits the dashboard in car accident.
Lateral collateral ligament injury	- LCL is the answer if varus test induces excessive lateral motion of the knee compared to the unaffected side.

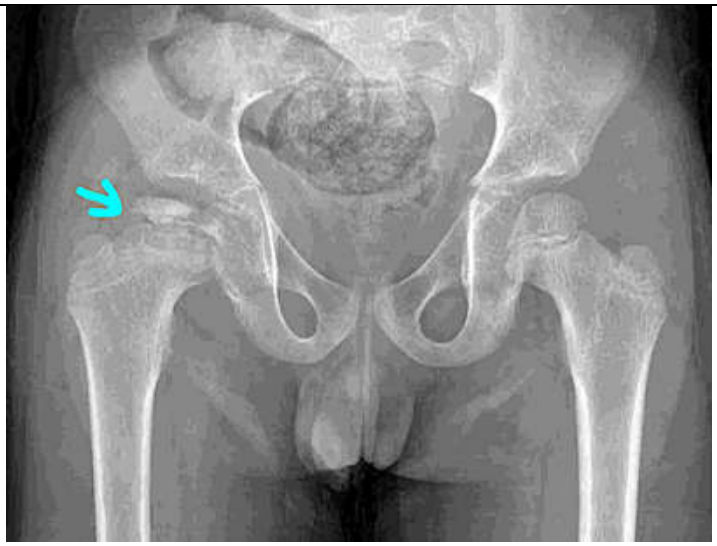
	- Varus test = hand placed on medial knee and pushing outward + other hand placed on lateral ankle and pushing inward.
Medial collateral ligament injury	- MCL is the answer if valgus test induces excessive medial motion of the knee compared to the unaffected side. - Valgus test = hand placed on lateral knee and pushing inward + other hand placed on medial ankle and pulling outward.
Lateral meniscal tear	- Lateral knee pain where patient experiences "locking" or "catching" of the knee in partial flexion. - Diagnosed with McMurray test → internal rotation of leg with concurrent knee extension causes lateral knee pain / "catching."
Medial meniscal tear	- Medial knee pain where patient experiences "locking" or "catching" of the knee in partial flexion. - Diagnosed with McMurray test → external rotation of leg with concurrent knee extension causes medial knee pain / "catching."
"Unhappy triad"	- Refers to a trio injury of the ACL, medial collateral ligament, and either the medial or lateral meniscus. - Students are sometimes fanatical about this triad as though it has yieldness. USMLE doesn't give a fuck. I cannot recall a single NBME question that has ever assessed this. This Dx primarily resides within the domain of Qbank, not the NBME.
Pes anserine bursitis	- The answer on USMLE if patient has inferomedial knee pain.
Iliotibial band syndrome	- The answer on USMLE if they say lateral knee pain, usually in a runner. - Iliotibial band runs from the hip to the knee. Pain may occur anywhere along the hip, lateral thigh, and lateral knee, but is worst in the latter. Tx = conservative with physiotherapy; NSAIDs for pain.
Osgood-Schlatter disease	- Buzzy vignette is knee pain in fast-growing teenage male who plays soccer. Don't pigeon-hole things, but that's classic vignette. - Inflammation of the patellar ligament at the tibial tuberosity. - Mechanism is repeated stress on the growth plate of the superior tibia.
Plantar fasciitis	- The answer on USMLE if they give severe heel pain that is worst when first getting out of bed in the morning.
Morton neuroma	- The answer on USMLE if patient has pain + abnormal growth occurring between the 2nd and 3rd metatarsals, usually worsened with high-heel shoes. - Benign growth/tumor of intertarsal nerve; cause is idiopathic. - First step in diagnosis is x-ray to rule out arthritis + fractures. Ultrasound is then done to confirm Dx, which will show thickening of interdigital/intertarsal nerve. - Tx = orthotics (comfortable shoes) + steroid injection.
Neurogenic joint	- As discussed already, aka Charcot joint, where patient injures joint due to lack of joint sensation from peripheral neuropathy. - Usually seen in diabetes; can also be seen in neurosyphilis. - The answer on USMLE when they say diabetic patient has "disorganization of the tarsometatarsal joints" on foot x-ray.

HY Fracture points for Surg	
Spiral fracture	- Pathognomonic for child abuse. - Caused by rotational/twisting force applied to a limb. - USMLE doesn't expect you to diagnose based on imaging.
Simple fracture	- Aka closed fracture – i.e., the skin is not broken and the underlying bone does not pierce the skin.
Compound fracture	- Aka open fracture – i.e., the skin is broken and the underlying bone pierces the skin.
Comminuted fracture	- Fracture where the bone is broken in at least two pieces.
Greenstick fracture	- Bone bends and cracks instead of breaking completely into two pieces. - More common in Peds than adults.

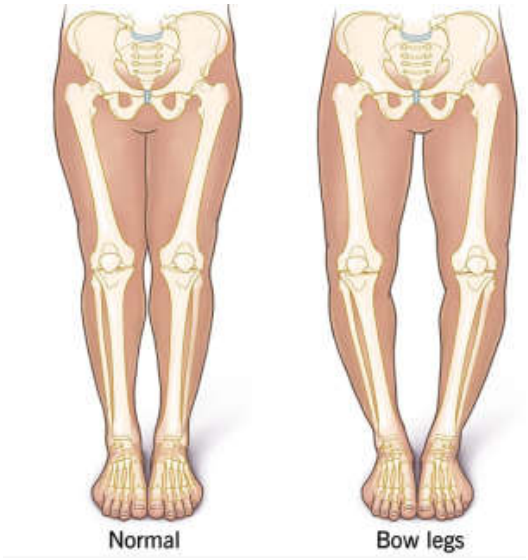
	- Children's bones are more flexible than adult bones adults' bones because they have ↑ collagen content and ↓ mineral content, allowing them to bend more. - When a force that would cause a complete fracture in an adult bone is applied to a child's bone, it might only cause a greenstick fracture, which is an incomplete fracture with the bone bending and breaking only on one side, resembling a green branch of a tree that bends and splinters on one side without breaking completely.
Linear skull fracture	- Most common type of skull fracture, where there is a break in the skull but the bone has not moved. - USMLE wants you to know this is classically associated with **epidural hematoma** (asked on 2CK CMS form).
Base of skull fracture	- Presents with tetrad of Battle sign (bruising over mastoid process), raccoon eyes (bruising around the eyes), rhinorrhea, and otorrhea.
Metatarsal stress fracture	- The answer on USMLE for pain in the metatarsal area of the foot in long-distance runners with low BMI. - Rather than asking diagnosis directly, USMLE will often give a vignette where a female long-distance runner with low BMI already has a metatarsal stress fracture, and then they'll ask what she's at greatest risk of developing → answer = osteoporosis.
Clavicular fracture	- Occurs with fall on outstretched hand (FOSH), or occasionally as a result of handlebar injury / impaction, with force transferred up to clavicle. - Most common site of break is the middle-third of the clavicle. - Tx = Figure-of-8 sling.
Scaphoid fracture	- As discussed earlier, will present as pain over anatomic snuffbox in patient with FOSH. - X-ray will usually be negative acutely. Must do thumb-spica cast to prevent avascular necrosis of scaphoid, followed by repeat x-ray in 2-3 weeks.
Lunate fracture	- The answer on USMLE if FOSH with pain in central palm + **no** pain over anatomic snuffbox.
Hook of hamate fracture	- Cause of distal ulnar nerve injury / Guyon canal syndrome. - Often from handlebar injury / impaction.
Surgical neck of humerus fracture	- Causes axillary nerve injury → loss of deltoid function + sensation over deltoid.
Midshaft fracture of humerus	- Causes radial nerve injury → wrist-drop + pronated arm.
Supracondylar fracture of humerus	- Aka "distal shaft fracture." - Causes median nerve injury → motor/sensory dysfunction of forearm muscles, first three fingers and thenar region.
Vertebral compression fracture	- Synonymous with osteoporosis on USMLE (i.e., post-menopausal, corticosteroid-use, Cushing syndrome). - Will often give point tenderness over a vertebra.
Pseudofracture	- Band of low-density bone that looks like fracture on x-ray but not actual fracture. - Synonymous with vitamin D deficiency (osteomalacia/rickets) on USMLE. - Can be seen in renal failure, since 1,25-D3 is low. Osteomalacia due to renal failure is called renal osteodystrophy.
Orbital floor fracture	- USMLE wants you to know this can cause entrapment of **inferior rectus and inferior oblique** muscles. - Vignette will say guy got hit in eye by baseball + has impaired upward gaze. - I talk about extraocular muscles and lesions in my neuroanatomy document, but this is one notable point you should be aware of here.

Ottawa criteria
- Used to assess probability of ankle fracture and tells us whether we need to order an X-ray or vs supportive care, e.g., RICE (rest, ice, compression, elevation). Before development of this criteria, x-rays for the ankle used to be ordered frivolously, with most showing no fracture. - Only order an x-ray for the ankle if: - Pain in the malleolar zone, AND any of the following: - Tenderness **posterior** to the lateral or medial malleolus; OR - Tenderness on the tip of the lateral or medial malleolus; OR - Patient cannot bear weight when walking four steps. - The above might seem nitpicky and pedantic, but this is HY for 2CK. Examples: - 25M + twisted ankle yesterday + moderate edema of lateral side of ankle with ecchymoses + tenderness to palpation lateral and anterior to lateral malleolus + patient can weight-bear; Q asks, in addition to 2-day ice pack application, what is next best step? → answer on shelf = "use a soft protective brace and early range of motion exercises"; wrong answer = "x-ray of the ankle to rule out fracture." - 40M + playing basketball + rolls ankle + pain anterior to lateral malleolus + swelling of ankle + no pain posterior to lateral malleolus + patient can bear weight; Q wants next best step in management → answer = rest, ice, compression, elevation (RICE); wrong answer is x-ray; the patient doesn't fulfill the Ottawa criteria for x-raying the ankle; he has pain in the malleolar zone but does not have pain posterior to the malleolus or on the tip of the malleolus, and he can bear weight. - 26F + went running and rolled her ankle + pain in lateral ankle + tenderness posterior to malleolus + can bear weight; Q wants next best step → answer = x-ray of ankle; patient fulfills Ottawa criteria for x-ray → she has pain in malleolar region + tenderness posterior to the malleolus; although she can bear weight, the former two findings satisfy the Ottawa criteria.

HY Pediatric hip disorders for Surg	
Primary hip dysplasia	- Aka developmental dysplasia of the hip. - Mechanism is "poorly developed acetabulum." - Initial diagnosis is with Ortolani and Barlow maneuvers, where a "clicking and clunking" is elicited on physical exam. - After the O&A maneuvers, next best step is ortho referral. Sounds wrong, but if it's listed, it's the answer before going to imaging. - Definitively diagnose with hip ultrasound if <6 months of age; hip x-ray if >6 months of age. - Treatment is "abduction harness," aka Pavlik harness, which positions the child's legs in a frog-leg-appearing manner.
Legg-Calve-Perthes	- Aka idiopathic avascular necrosis of the femoral head. - If the etiology for the avascular necrosis is known (i.e., Gaucher, sickle cell, corticosteroids), then the diagnosis is just "avascular necrosis," not LCP. - Vignette will be child 5-8 years old with hip pain. - First step in diagnosis is hip x-ray, which will show a "contracted" or flattened femoral head. The word "contracted" is HY and synonymous with avascular necrosis. - If x-ray is negative, diagnose with bone scan or MRI (on 2CK form). - Tx = hip replacement.

	 X-ray showing flattened/contracted femoral head (compare with the normal side that looks rounder).  On MRI, the right femoral head (left side of image) appears hypointense (more black). The necrotic / lack of bone in the black superior portion of the femoral head means the remaining white part of the femoral head is "flattened" or "contracted." The left femoral head (right side of image) shows a small area of necrosis as well (black medial portion).
Slipped capital femoral epiphysis (SCFE)	- Classic vignette is a 10-13-year-old (pre-adolescent) overweight boy with a painful limp. - NBME will write answer / mechanism as "displacement of the epiphysis of the femoral head." - Resources tend to emphasize obesity as the main risk factor, but maybe only ~1/2 of NBME Qs for SCFE give the child as overweight. This causes problems for students, where they rely on seeing high BMI to think SCFE. - This has led me to conclude that the **age** matters the most, since they will always give a kid who's about 10-13-ish. If they give you a kid who's younger, think LCP instead. - 2CK NBME Q gives 13M with painful gait + no mention of weight → answer is SCFE. - Another 2CK Q outright says BMI 20 in a 13-year-old who's 6 feet tall, and answer is SCFE. - X-ray shows "ice cream falling off the cone."

	 X-ray shows the “ice cream slipping off its cone.” - Offline 2CK gives Q where they say x-ray in 5-year-old shows “contracted capital femoral epiphysis” → answer is LCP, not SCFE. As I said above, “contracted” is HY for LCP. In this case, the younger age + the word “contracted” win over the words “femoral capital epiphysis.” Tx = surgical pinning.

Other Peds bone issues for Surg	
Rickets	- Rickets = vitamin D deficiency in children. Osteomalacia = vitamin D deficiency in adults. - Rickets = craniotabes (soft skull), rachitic rosary (bony knobs at costochondral junctions), genu varum (bowing of tibias). - Activated vitamin D (1,25) is necessary to convert unmineralized osteoid into mineralized hydroxyapatite, therefore hardening bones. - In both rickets and osteomalacia, patient will have “increased unmineralized osteoid,” or “deficient mineralization of osteoid.” - Important cause of Vit D deficiency on USMLE is impaired intestinal malabsorption (i.e., CF, Crohn). For CF, answer can be written as “exocrine pancreas insufficiency.” - “Pseudofracture” on x-ray is buzzy finding in osteomalacia/rickets. - Patients have ↓ Ca^{2+}, ↓ $PO4^{3-}$, ↑ PTH (due to ↓ negative feedback at parathyroid glands).  Normal Bow legs
Tibia vara	- Aka Blount disease.

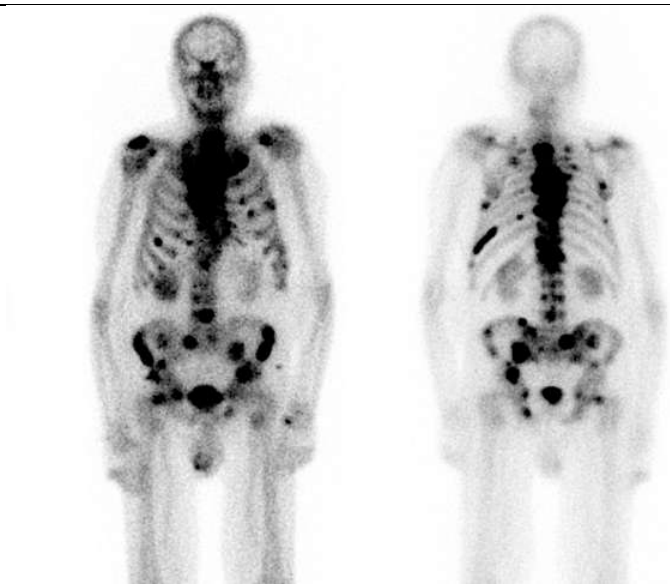
	 - Bowing of the tibias after the age of 2 years in a patient whom rickets has been ruled out. - Can be unilateral or bilateral. - Bowing of one or both tibias is sometimes **normal** until age 2 years. - Treatment is surgery (osteotomy).
Osteopetrosis	- Osteoclast dysfunction resulting in recurrent fractures in children due to bone density being **too high.** Sounds weird, but bone strength is based on balanced internal architecture of canals and networks, not just density alone. - HY DDx against osteogenesis imperfecta and child abuse. - Osteoclast dysfunction is due to deficiency of carbonic anhydrase II. This enzyme inside osteoclasts normally allows osteoclasts to form H^+ to resorb bone.
Growing pains	- No, this is not a joke. This is the answer straight-up on a 2CK NBME form. - Vignette is healthy child age 3-12 who awakens from sleep with throbbing pain in the legs. - No treatment necessary. You just need to know this Dx is exists and isn't a troll.
Talipes equinovarus	- Aka clubbed foot.  - USMLE just wants you to know that this is treated initially with **serial casting.** - Usually idiopathic; can be seen in Potter sequence. - Not the same as rocker-bottom foot (aka congenital vertical talus), seen in Edward syndrome.
Arthrogryposis	- You just need to know this is fancy name for a child born with multiple joint contractures. - If they give you a child with not just a clubbed foot, but also knee and/or elbow contractures, etc., the answer is arthrogryposis.

Paget disease
- Idiopathic disorder of increased bone turnover. Bone is described as having mixed osteoblastic and -clastic phases, where bone appears heterogenous on x-ray. - Buzzy vignette is male over 50 who's hat doesn't fit him anymore + has tinnitus (narrowing of acoustic foramina). - 19/20 questions will give isolated increase in serum ALP. You need to know Ca^{2+}, $PO4^{3-}$, and PTH are all normal in Paget disease. There is one Q on a 2CK CMS form where ALP is given as not elevated, but it's a one-off Q and rare. - High-output cardiac failure can occur due to intraosseous AV-fistulae, where patient has an S3 heart sound with high, rather than low, ejection fraction.

Osteoporosis
- Bone density >2.5 SD below mean compared to young adult woman. Osteopenia is 1.5-2.5 SD below mean. - Bone densitometry done at age 65. - If Q forces you to choose between female gender and age as most important risk factor, choose gender. - If Q gives you a female and forces you to choose between family history and age, choose family history. - Males are unlikely to develop osteoporosis, even with family history of females with the disorder. - If Q gives two women without family history and asks what is most protective against osteoporosis, answer is ethnicity. Black race is protective against osteoporosis. - If Q gives old woman who has femoral fracture + no mention of osteoporosis in the question, answer = "activity level before fracture" as most important predictor of success in the rehabilitation of the patient → weight-bearing exercise during life is protective against osteoporosis later. - USMLE loves corticosteroids and Cushing as causes of osteoporosis. - Compression fractures = osteoporosis on USMLE; e.g., patient with RA on steroids who has compression fracture → easy Dx of osteoporosis. - Low/low-normal BMI causing osteoporosis is HY; 2CK NBMEs have a couple of nonsense Qs where they give BMI of 19 and 20 in young woman, where they ask what patient is at increased risk of; answer = osteoporosis. Student says, "Wait, but isn't low BMI under 18.5?" I agree. But it's on NBME. - Metatarsal stress fracture HY in low-BMI young female runners who have low bone density. - USMLE also is known to assess low vitamin D in the setting of intestinal malabsorption (i.e., CF, Crohn) as cause of osteoporosis, even though that makes no sense, since low Vit D causes osteomalacia. - Serum calcium, phosphate, PTH, and ALP are all normal in osteoporosis. - Tx = weight-bearing exercise first (NBME has "go for a long walk outside daily" as correct; wrong answer = "increase participation in swimming pool-based exercise classes to at least three times weekly"). - Calcium and vitamin D are the first medical / pharmacologic treatment. - Bisphosphonates can be used after Ca^{2+}/VitD. - Teriparatide is an *N*-terminus PTH analogue that stimulates bone development. - Denosumab is a RANK-L monoclonal antibody.

HY bone tumor points for Surg	
- I only discuss the ones relevant to Surg shelf.	
Osteosarcoma	- Most common primary bone cancer; usually in patients age 10-30. - *Rb* mutations (congenital retinoblastoma) are associated with osteosarcoma (HY) – i.e., 1-year-old boy has enucleation of eye for retinoblastoma; what is he at risk of developing later in life? → answer = osteosarcoma. - Can also occur in Paget disease of bone patients (older age). - Buzzy findings are "Codman triangle" and "Sunburst appearance." - Codman triangle = periosteal reaction with lifting of periosteum off the bone. - Sunburst appearance = periosteal reaction described on NBME as "spiculated new born formation."

	 - The white arrow is the Codman triangle; the white star is the sunburst appearance. - NBME writes osteosarcoma description as “pleomorphic neoplastic cells producing new woven bone” as correct answer choice.
Unicameral bone cyst	- Underrated diagnosis for USMLE. Asked on 2CK exam. - Benign bone tumor that looks similar to osteoclastoma but age of onset usually birth to age 20, rather than 20-40. - Unicameral means “one chamber”; it is fluid-filled.  2CK wants you to know this image for unicameral bone cyst.
Metastases	- USMLE loves mets to the vertebrae, particularly from breast, prostate, and lung. - The exam will not show images of spinal cancer mets, but they will give vignette of either lytic lesions of vertebrae in patient with background of cancer, or will give neurologic syndrome (i.e., of cauda equina). - Diffuse bone pain in patient with background of cancer = mets.

	 - Above image shows Technetium-99 bone scan of cancer mets. Similar imaging is on 2CK forms for prostate mets.

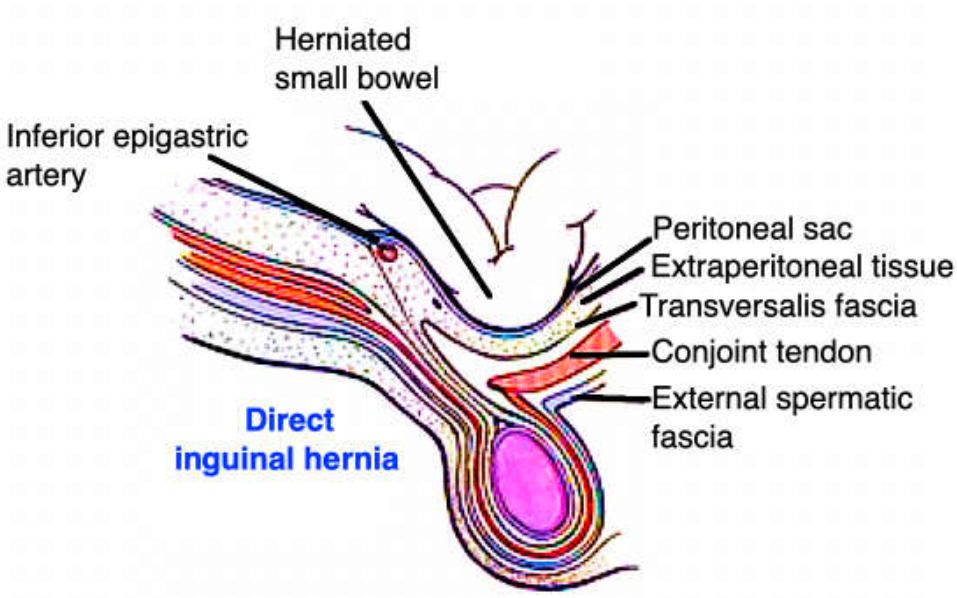
HY myopathies / muscular dystrophies for USMLE	
Duchenne / Becker muscular dystrophy	- XR disorder caused by mutation in dystrophin (*DMD*) gene. - Mutation results in disruption of α-/β-dystroglycan, which is required for proper internal cytoskeletal anchoring of the muscle cell to the extracellular matrix. - Presents with pseudohypertrophy, where muscles appear large but are replaced with fibroadipose tissue (connective tissue stromal cells). - Duchenne presents in a young boy who implements Gower maneuver to stand up (uses arms to walk up off the floor because leg muscles are weak). - Becker presents in adolescence or young adulthood (less severe form of Duchenne). - Duchenne is classically frameshift mutation; Becker is classically not frameshift.
Polymyalgia rheumatica (PMR)	- Usually patient over 50 with proximal muscle pain and stiffness. - No weakness on physical exam + creatine kinase (CK) levels are normal. If one or both of these findings is present, the answer is polymyositis, not PMR. - Can present with high ESR and low-grade fever (any autoimmune disease flare can present with low-grade fever). - PMR can present with or without temporal (giant cell) arteritis. Temporal arteritis can present bilaterally on NBME exams; do steroids first to prevent blindness, followed by biopsy second. - Temporal arteritis can cause jaw claudication (pain in the jaw during episodes). In contrast to temporomandibular joint dysfunction (a separate diagnosis), jaw claudication will not be precipitated by eating. - No specific diagnostic test; diagnosis is made clinically. - Tx = steroids. NSAIDs are wrong answer and are not proven to be effective.
Polymyositis / Dermatomyositis	- Usually patient over 50 with proximal muscle pain and stiffness. These findings are **not** unique to PMR. The USMLE will happily give pain and stiffness in polymyositis vignettes. - Key distinction between polymyositis and PMR is that polymyositis will present with **1) muscle weakness on physical exam, and/or 2) increased serum CK.** - The muscle weakness *must be on physical exam.* If the patient complains of "weakness" but there is no physical exam findings mentioned in vignette or

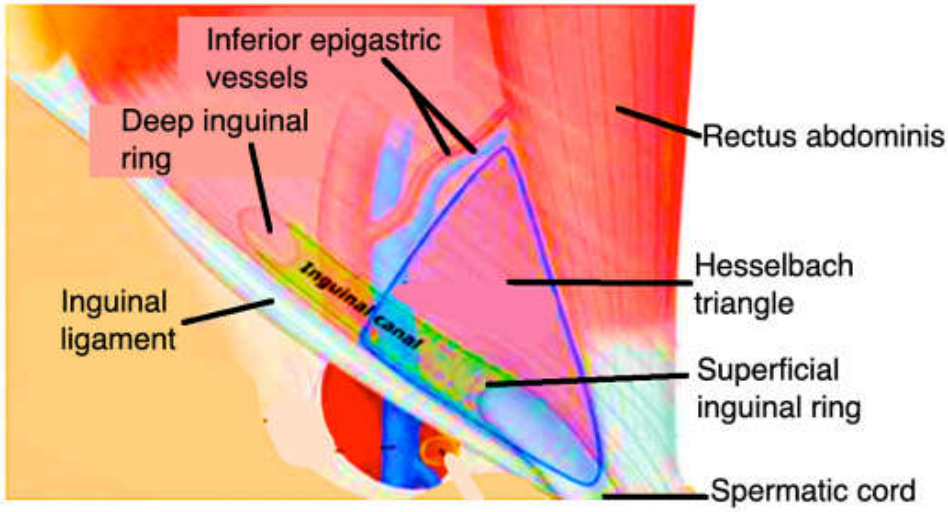
	physical exam shows 5/5 strength, there's no weakness. Patients will sometimes mention "weakness," even though they really just have pain and/or stiffness. - If polymyositis presents with skin findings, it is called dermatomyositis – i.e., Gottron papules (violaceous papules on the knuckles), mechanics' hands (rough-surfaced hands), shawl rash (body rash), heliotrope rash (violaceous eyelids / periorbital rash; don't confuse with malar rash of SLE). - Patients often positive for anti-Jo1 antibodies. - USMLE wants "electromyography and nerve conduction studies" as first step in management for polymyositis/dermatomyositis. This is what they ask on 2CK NBME forms. I have not seen them ask anti-Jo1 antibodies vs EMG+NCS as two separate answer choices. Usually anti-Jo1 antibodies are mentioned in the vignette rather than as the test you need to order. - Muscle biopsy is confirmatory, showing CD8+ T cell infiltration. The histo can be described as "CD8 + T cells and macrophages surrounding muscle fibers." - For whatever reason, dermatomyositis can be a paraneoplastic syndrome of ovarian cancer (shows up on NBME). Tx = steroids.  Gottron papules
Fibromyalgia	- This is a psych condition, not an actual muscle disorder, but is often confused with polymyositis and PMR. - Labs will be normal. ESR will not be elevated. Patient will not have fever. - Will be described as woman 20s-50s with multiple (and often symmetric) muscle tenderness points. - Treatment is SSRIs. USMLE can write this as "anti-depressant therapy." This confuses students ("But she doesn't have depression though.") → Right. But SSRIs are still anti-depressant medication.
Temporomandibular joint dysfunction	- The answer on USMLE if they give jaw pain that is **precipitated by eating.** - Often confused with jaw claudication seen in temporal arteritis. In the latter, however, the pain is not precipitated by eating.
Myotonic dystrophy	- Autosomal dominant, CTG trinucleotide repeat expansion disease. - Myotonia is inability to relax muscles. - The answer on USMLE if they say patient cannot relax grip on doorknob / handshake, or cannot let go of golf club. - Sometimes associated with early / frontal balding.
Hypothyroid myopathy	- Myopathy can occur in both hypo- and hyperthyroidism, yes, but this is especially HY for hypothyroidism on USMLE. - They will often sneak this in as proximal muscle weakness or increased CK in patient who has ongoing fatigue, dysthymia, menstrual irregularities, etc.
Drug-induced myopathy	- Classically seen when statins and fibrates are combined, but both drugs can cause myopathy independently. - Mild CK elevations are normal and expected in patients when commencing these agents. Dose does not need to be decreased for mild CK elevations. - USMLE wants "P450-mediated interaction" as the cause of the myopathy when statins and fibrates are combined.

Mitochondrial myopathy	- Broad term that can refer to numerous mitochondrial diseases. - USMLE wants you to know the patient has a mitochondrial disorder when he/she presents with **hypotonia, ear/eye problems, and lactic acidosis.** You want to memorize this tetrad as synonymous with mitochondrial disorders. - "Ragged red fibers" can be a buzzy descriptor in mitochondrial myopathy Qs. - Mitochondrial disorders are maternally inherited only. - Heteroplasmy refers to offspring having varying disease severity based on variation in allocation of diseased mitochondrial genes (I talk more about this stuff in my HY biochemistry PDF).
Inclusion body myositis	- The answer if they tell you patient 50 or older has months to years of progressive muscle weakness + biopsy of muscle shows **basophilic rimmed vacuoles.** - Shows up on a new 2CK form.

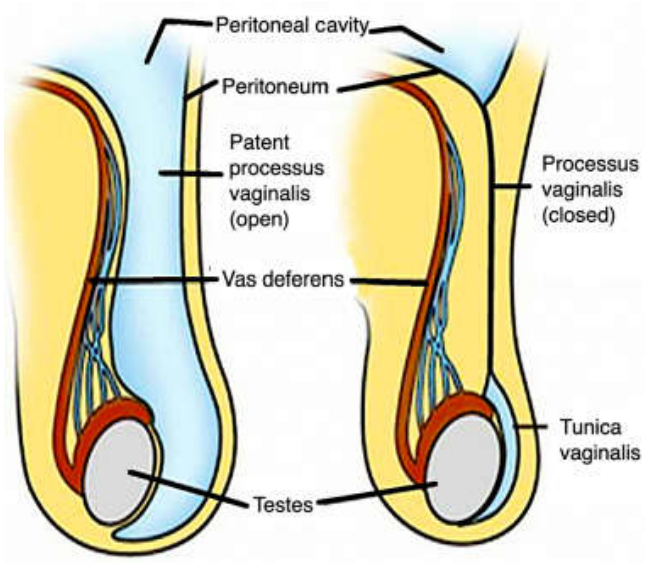
Connective tissue disorders	
Marfan	- Autosomal dominant; chromosome 15, *FBN1/2*. - Codes for fibrillin, which is a glycoprotein that forms a sheath around elastin. - Tall, lanky body habitus with flat feet, chest wall abnormalities (i.e., pectus excavatum or carinatum), flat feet (pes planus), scoliosis, mitral valve prolapse (mid-systolic click), increased risk for aortic dissection (can retrograde propagate toward aortic root, causing root dilatation and aortic regurgitation [decrescendo diastolic murmur]). - Is not associated with berry/saccular aneurysms (unlike Ehlers-Danlos and ADPKD).
Ehlers-Danlos	- Defect in collagen III synthesis; can be written as "Defect in synthesis of fibrillar collagen." - This can be confusing because fibrillin is completely unrelated and refers to Marfan syndrome. It is just coincidental that "fibrillar collagen" means type III collagen. - Wrong answer = "abnormal synthesis of extracellular glycoprotein" (refers to fibrillin for Marfan syndrome, which as mentioned above, is a glycoprotein that forms a sheath around elastin. - Presents as easy bruising + hyperextensible skin. 
Osteogenesis imperfecta	- Collagen I defect that results in recurrent fractures in a child; important DDx are child abuse and osteopetrosis. - Blue sclerae too buzzy and often not mentioned. - Conductive hearing loss due to defective ossicles (middle ear bones). - Many different types of OI, some resulting in miscarriage. Harder vignette can mention child with multiple fractures, where the mom has Hx of recurrent miscarriage.
Mixed connective tissue disease	- Answer on USMLE if they give you a patient who has anti-U1-ribonucleoprotein (U1-RNP) antibodies. - Patient presents as having combined features/symptoms from three different disorders → LPS → Lupus, Polymyositis, Scleroderma.

HY neck MSK masses for Surg	
Thyroglossal duct cyst	- The answer on USMLE if they say there's a painless, midline neck lump in a child that moves upward with swallowing or protrusion of the tongue. This buzzy description is seen for maybe only about half of Qs. - Can also be described as painless mass inferior to the hyoid bone that demonstrates uptake with a Technetium-99 scan. - USMLE wants "endoderm of foramen cecum" as the embryology.
Sternocleidomastoid injury	- The answer on USMLE if they say nodular mass in the lateral neck in an infant who had been born via forceps delivery (risk factor for damage to the muscle).
Branchial cleft cyst	- The answer on USMLE if they give idiopathic lateral neck mass in infant that may or may not have an opening to the skin.
Lingual thyroid	- Neonates or infants. - Answer for hypothyroidism + a midline neck lump located high in the neck. - They say nothing about protrusion of the tongue or uptake into the mass (of course this is thyroglossal duct cyst instead). - Can sometimes cause dysphagia (trouble swallowing), dysphonia (voice changes), or dyspnea (difficulty breathing).

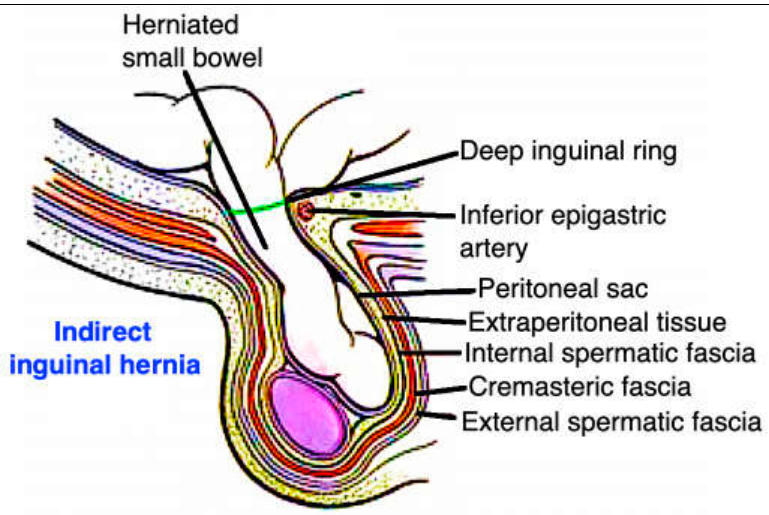
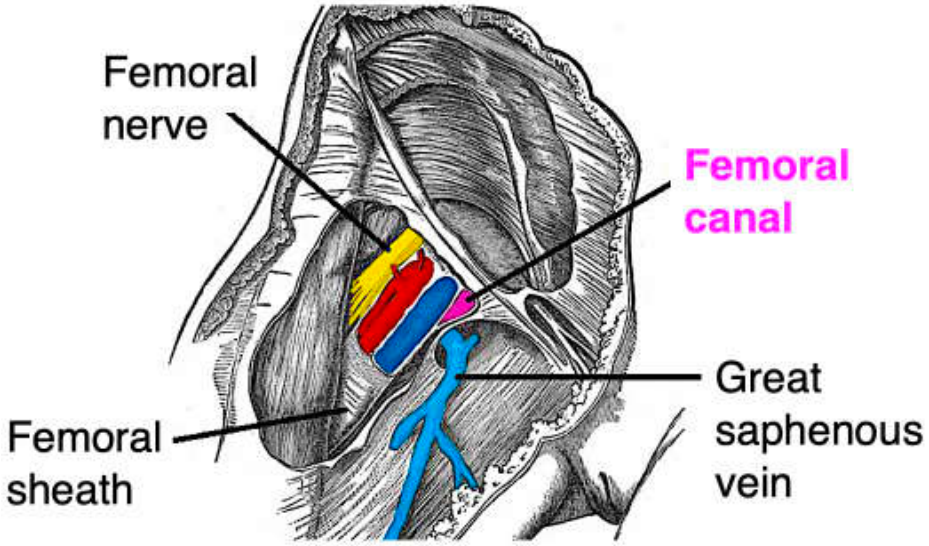
HY MSK hernia points for Surg	
Direct inguinal hernia	- Small bowel herniates **medial** to inferior epigastric vessels. - Occurs usually in older men. - Occurs due to weakness of abdominal wall musculature / transversalis fascia.  - The coverings of direct inguinal hernia = skin, superficial fascia, external spermatic fascia, cremasteric fascia, extraperitoneal tissue. - Hernia passes through Hesselbach triangle (inguinal triangle), which is an area of weakness in abdominal wall through which a direct inguinal hernia can protrude. - Boundaries of Hesselbach triangle: medial border = the lateral border of rectus abdominus muscle, aka linea semilunaris; lateral border = inferior epigastric vessels; inferior border = inguinal ligament.

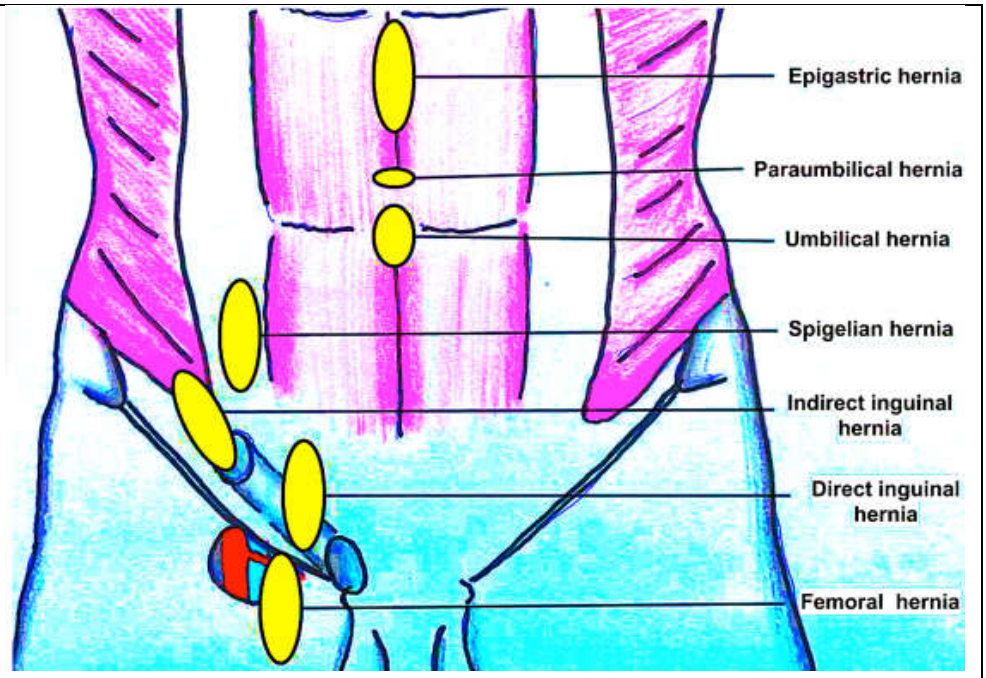
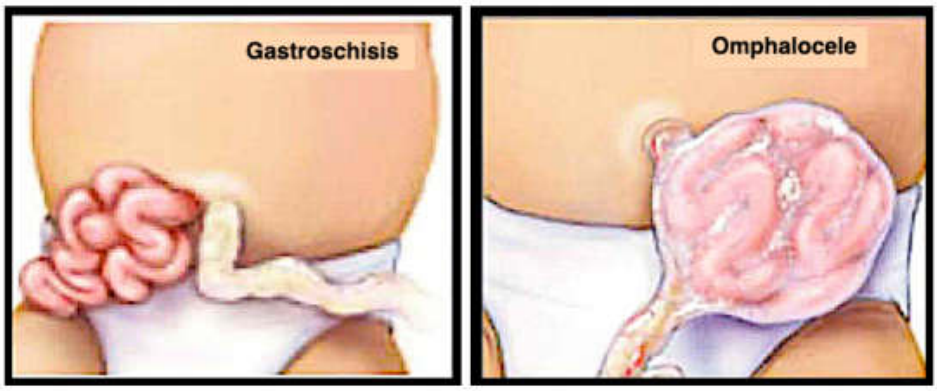
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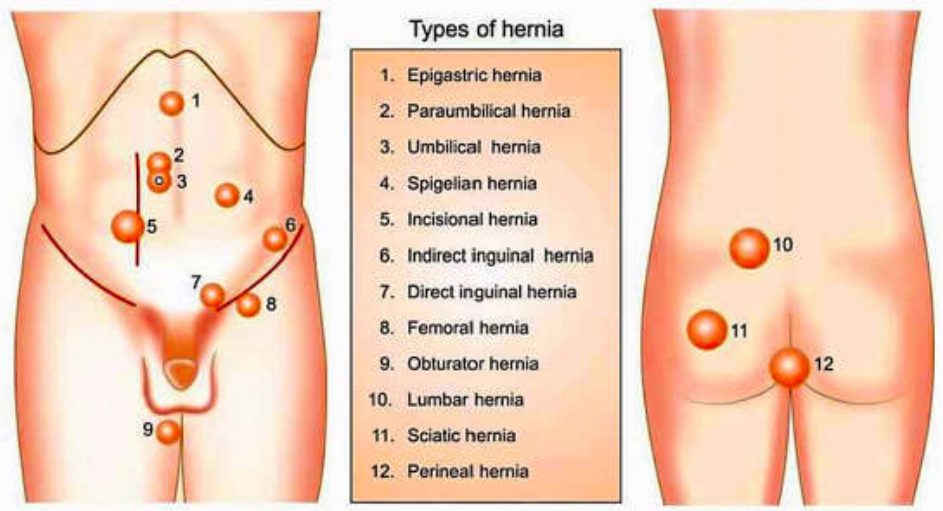
- NBME wants “medial to inferior epigastric artery; superior to inguinal ligament” as the answer for direct inguinal hernia.
- Q might say older patient has palpable mass in groin that reduces when he lies down, or worsens when he coughs.
Tx = “operative management” or “elective hernia repair,” since it is not an overt emergency but closure should be performed prior to any type of incarceration and strangulation (ischemia leading to pain, fever, and necrosis).
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<td>Indirect inguinal hernia</td>
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- Small bowel herniates lateral to inferior epigastric vessels, through deep inguinal ring.
- Seen classically in male infants, but can occur any age.
- Mechanism is patent processus vaginalis. This is also the mechanism for hydrocele (asked on NBME).
- NBME wants “lateral to inferior epigastric artery; superior to inguinal ligament” as the answer for indirect inguinal hernia.



- Since it passes through deep inguinal ring, it can be reduced with pressure / a finger placed over deep inguinal ring. Direct inguinal hernia, since it does not pass through deep inguinal ring, will not reduce with pressure applied here.
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	 - Tx = elective hernia repair. - For hydroceles, observation is the answer under the age of 1 (HY on Peds forms).
Femoral hernia	- Small bowel protrudes through femoral canal, inferior to inguinal ligament. - Account for 5% of hernias and more common in women (3:1) because of the wider anatomy of the pelvis in women.  - Boundaries of femoral canal (and hernia) are: lateral border = femoral vein; medial border = lacunar ligament; posterior border = pectineal ligament; anterosuperior border = inguinal ligament. - Tx = surgery.
Spigelian hernia	- Rare abdominal hernia (<2%) that occurs lateral to the linea semilunaris of the rectus abdominis, but NBME asks it. You should also know this for 2CK Surg. - Usually in older adults.

	 - Highest yield point is that the medial border of the hernia is the rectus abdominis.
Umbilical hernia	- Small bowel herniation through the umbilicus. - Can occur at any age, but I've seen this show up on NBME in **congenital hypothyroidism** (i.e., cretinism). - Not to be confused with omphalocele in neonates. An umbilical hernia is completely covered by skin. In contrast, an omphalocele is a herniation merely covered by a thin, translucent layer of peritoneal membrane. It can be idiopathic, but is often associated with the Trisomy 13 and 18, and Beckwith-Wiedemann syndrome. 
Paraumbilical hernia	- Small bowel herniation through the linea alba (tendinous, fibrous raphe that runs vertically down the abdomen). - Not to be confused with gastroschisis in neonates. A paraumblical hernia is completely covered by skin and protrudes usually superior to the umbilicus. Gastroschisis is not covered by anything (not even a layer of peritoneal membrane as with omphalocele) and protrudes to the right of the umbilucus; gastroschisis is seen sometimes in Trisomy 13 and 18.
Epigastric hernia	- Small bowel protrusion in epigastrium through the linea alba, similar to paraumbilical hernia. - The difference is that epigastric hernia is further up in the abdomen in the epigastrium, whereas paraumbilical hernia is literally adjacent the umbilicus.
Incisional hernia	- Can occur at the site of any abdominal incision. Just know it's possible.

Obturator hernia	- Rare hernia of the pelvic floor. Small bowel contents protrude through obturator foramen.  - Despite being rare, obturator hernia is HY for 2CK Surg and shows up on 2CK exam. - You need to know that the **Howship-Romberg sign** is used to diagnosis obturator hernia. In this test, thigh extension, medial rotation, and abduction cause lancinating pain in the medial thigh / obturator distribution due to compression of obturator nerve.
Sciatic hernia	- Rare hernia of the pelvis with protrusion of small bowel through either the greater or lesser sciatic foramen. - Can mimic sciatica due to compression of sciatic nerve. - Just know this type of hernia exists. As I mentioned above, obturator hernia is HY for 2CK. They will usually list sciatic hernia as incorrect answer choice alongside it.
Lumbar hernia	- Rare hernia that presents with pain and herniation in the lower back.
Perineal hernia	- Rare hernia of pelvic floor. Can occur due to atrophy of the levator ani muscle.

Highest yield "MSK pharm" for Surg	
Alendronate	- Bisphosphonate; inhibits osteoclasts. This MOA is HY. - Used for osteoporosis after Ca^{2+}/VitD. - I've seen pamidronate (not alendronate) show up on 2CK forms for Tx of hypercalcemia (after normal saline is given). - Students get fanatical about bisphosphonates causing osteonecrosis of the jaw. The yieldness of this adverse effect is basically non-existent on NBME exams. USMLE wants you to know bisphosphonates cause **pill-induced esophagitis.** This is very HY for 2CK forms (K^+ supplements also cause esophagitis).
Teriparatide	- *N*-terminus PTH analogue that can induce bone formation. Even though PTH causes bone resorption, this agent stimulates osteoblast-mediated bone formation more than it induces RANK-L-mediated activation of osteoclasts. - Can be used for severe/advanced osteoporosis.
Denosumab	- Monoclonal antibody against RANK-L. - Can be used for severe/advanced osteoporosis.
Baclofen	- Agonizes $GABA_B$. - Used for spasticity, classically in multiple sclerosis, but I've seen one NBME Q where it's used for random spasticity in an older dude. - Students frequently remember $GABA_B$ for this drug, but often say "antagonist" when I probe them further. So remember: it's an agonist, not an antagonist, at $GABA_B$.
Cyclobenzaprine	- Muscle relaxant used for spasms. - Structurally similar to TCAs; helps modulate pain sensation at brain stem.
Benztropine	- Muscarinic (cholinergic) receptor antagonist.

	- Used to treat acute dystonia due to anti-psychotics. - If patient starts anti-psychotic and then gets stiff neck, oculogyric crisis (abnormal eye movements), or muscle rigidity **without** fever, the answer = benztropine.
Diphenhydramine / Chlorpheniramine	- First-generation histamine-1 (H1) antagonists. - Diphenhydramine is quite possibly the highest-yield drug on USMLE. - Used to treat acute dystonia, similar to benztropine, as well as motion sickness. - H1 blockers can treat allergies in theory, but they have **nasty anti-cholinergic (anti-muscarinic) side-effects.** - The anti-cholinergic side-effects are interestingly a *good* thing, however, when we want to treat acute dystonia. Psych Qs will either list benztropine or diphenhydramine (or chlorpheniramine) as the answer, but not both at the same time. - For whatever reason, anti-cholinergic effects treat motion sickness. Scopolamine is an anti-cholinergic used to treat motion sickness classically. But I've seen NBME ask diphenhydramine straight-up for this as well – i.e., the nasty anti-cholinergic side-effects are, once again, a good thing if the aim is Tx of motion sickness. - 1st-gen H1 blockers can cause **cognitive dysfunction** (delirium, as well as worsening of dementia) and **drowsiness.** Therefore avoid in elderly and locomotive/machine operators if at all possible. - 1st-gen H1 blockers can also cause anti-α1-adrenergic effects (orthostatic hypotension). - I talk about all of the pharm-related stuff in a lot more detail in my free pharm modules on the website.
Dantrolene	- Blocks ryanodine Ca^{2+} channel. - Tx for neuroleptic malignant syndrome (NMS) and malignant hyperthermia (MH). - If patient gets muscle rigidity and fever following commencement of anti-psychotic, or following administration of succinylcholine during surgery, answer = dantrolene. (Bromocriptine for NMS is low-yield and rarely seen on NBME). - NBME will sometimes give vignette of NMS or MH, and then the answer for Tx is "decreases sarcoplasmic calcium release." - In NMS and MH, the ryanodine channel, which allows calcium to move from the sarcoplasmic reticulum into the cytosol, gets stuck open, so high amounts of calcium moves into the cytoplasm. The cell then needs to use a lot of ATP to pump the calcium back into the sarcoplasmic reticulum. This generates heat → fever. Dantrolene closes this channel.

Surg Endocrine

Acromegaly
- Caused by excess growth hormone secretion (usually by a tumor) following closure of the growth plates (↑ GH secretion prior to growth plate closure causes gigantism). - As mentioned earlier, GH goes to the liver, which causes it to secrete IGF-1. It is then IGF-1 that induces growth effects at tissues. An NBME Q gives easy vignette of acromegaly and then asks what you check; serum GH is wrong; answer is serum IGF-1. - Acromegaly causes prognathism (lantern jaw), large ears/nose, arthritis and carpal tunnel syndrome (due to ↑ joint and tendon growth), enlargement of hands/feet, hypertension, type II diabetes (GH causes insulin resistance), and ↑ risk of arrhythmia and cardiomyopathy. - Treatment is somatostatin analogue (octreotide). Somatostatin is aka growth hormone-inhibiting hormone (GHIH) and shuts off its secretion from the anterior pituitary. It also ↓ secretion of insulin and glucagon from the pancreas.

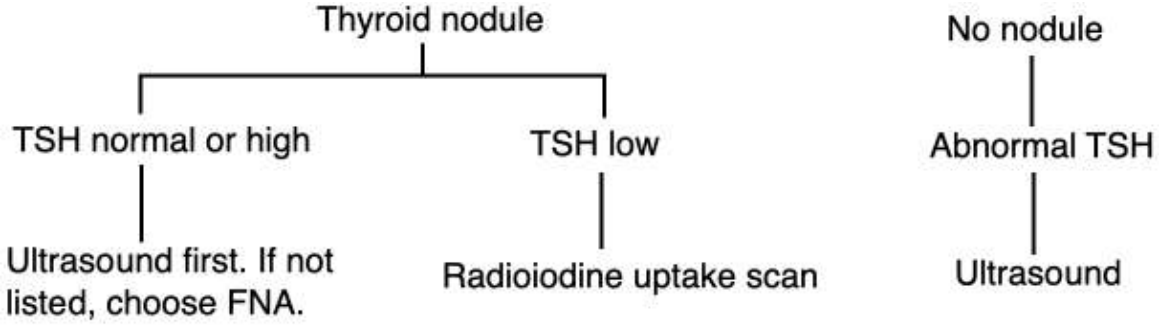
Vasopressin conditions	
- Vasopressin (aka anti-diuretic hormone; ADH). - ADH is produced by both the supraoptic nucleus and paraventricular nucleus of the hypothalamus → stored in posterior pituitary. - ADH ↑ free water reabsorption by the medullary collecting duct (MCD) of the kidney by causing aquaporin insertion. - If ADH is ↑, we get ↑ free water reabsorption and dilution of our serum → serum sodium and osmolality ↓. Likewise, our urine becomes more concentrated, meaning urine osmolality and specific gravity ↑. - If ADH is ↓, we get ↓ free water reabsorption and concentration of our serum → serum sodium and osmolality ↑. Likewise, our urine becomes more dilute, meaning urine osmolality and specific gravity ↓.	
SIADH	- Syndrome of Inappropriate Anti-Diuretic Hormone secretion → means too much ADH (vasopressin secretion). - Central SIADH → follows head trauma, meningitis, brain cancer, and pain (latter on 2CK Surg). - Ectopic SIADH → small cell lung cancer secreting ADH. - Drug-induced ADH → ultra-rare on USMLE, but carbamazepine can do it. - Patient will have dilute serum and concentrated urine: - ↓ serum sodium, ↓ serum osmolality, ↓ serum specific gravity. - ↑ urinary osmolality, ↑ urinary specific gravity. - You must know serum sodium is normally 135-145 mEq/L. So in SIADH, it's <135. - Serum vs urinary osmolalities will be all over the place and you do *not* need to memorize values. They might say serum osmolality is 250 and urinary is 750, and then you say, "Well I can tell serum is dilute compared to urine, which sounds like SIADH." - Specific gravity will be 1.000-1.030 on USMLE. Sounds obscure, but it shows up quite a bit, particularly on 2CK Qs. It's to my observation that values 1.000-1.006ish are "dilute"; 1.024-1.030 are "concentrated." For 9/10 Qs, values will be what you expect. - "Fluid restriction" is first answer in diagnosis on 2CK. We want to see how serum/urinary values change in response. - Demeclocycline is answer on 2CK offline NBME 8 for treatment of SIADH. Demeclocycine is technically a tetracycline antibiotic but isn't used because it can cause insensitivity to ADH at the kidney (i.e., nephrogenic diabetes insipidus). So we essentially cause a 2nd problem that cancels out the 1st problem.
DI	- Central diabetes insipidus → means not enough ADH secretion by hypothalamus, or the posterior pituitary is unable to release it properly. - Nephrogenic DI → insensitivity to ADH at the kidney (serum ADH is ↑).

	- Similar to central SIADH, central DI can be caused by head trauma, meningitis, and cancer. - Nephrogenic DI is caused by **lithium, demeclocycline, hypercalcemia,** and **NSAIDs.** - 2CK Q gives patient with primary hyperparathyroidism and ↑↑ serum calcium + nephrogenic DI; Q asks cause of the DI → answer = hypercalcemia. High calcium can cause renal insensitivity to vasopressin. - Serum vs urinary values are the opposite of SIADH: - ↑ serum sodium (>145 mEq/L), ↑ serum osmolality, ↑ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - When we are trying to first diagnose DI, the first thing we do is fluid restriction, same as with SIADH. We want to see how serum/urinary values change first. - After we determine that the urine is staying dilute + the serum is staying concentrated, the next best step is desmopressin (analogue of vasopressin). If the urine gets more concentrated, (i.e., if the drug works), we know central DI is the diagnosis and we're merely deficient in ADH. - If desmopressin doesn't work, we know we have nephrogenic DI. I should point out that even in nephrogenic DI, desmopressin might work *but only very little,* whereas with central DI, administration will ↑↑ urinary osmolality robustly. It will be obvious on USMLE. But my point is, don't say, "Oh well desmopressin worked like 5% so we can't have nephrogenic DI here." - Treatment for central DI is therefore desmopressin. - Treatment for nephrogenic DI is NSAID + a thiazide. Sounds weird, but ↓ Na^+ reabsorption induced by thiazides in the early-DCT promote compensatory ↑ Na^+ reabsorption in the PCT, where water follows Na^+ and our net loss of fluid is less than without the thiazide. In healthy individuals, however, they *will* lose more net fluid with the thiazide. The NSAID presumably ↓ renal blood flow, which will ↓ GFR and ↓ net fluid loss. - There is difficult 2CK NBME Q where they say patient is on lithium + has ↑ urinary output, and fluid restriction is wrong answer to this question (I say hard because 9/10 times, fluid restriction is correct when it's listed); answer = NSAID + thiazide diuretic. The implication is, if it's obvious what the patient's diagnosis is already nephrogenic DI, going straight to Tx is acceptable.
PP	- Psychogenic polydipsia means the patient is simply drinking too much. - Both the urine and serum will be dilute. - Serum vs urinary values are the opposite of SIADH: - ↓ serum sodium (<135 mEq/L), ↓ serum osmolality, ↓ serum specific gravity. - ↓ urinary osmolality, ↓ urinary specific gravity. - I'd say 3/4 Qs on USMLE are obvious and will say some psych patient is drinking lots to "clear himself from evil spirits," etc. - Probably 1/4 Qs won't be an obvious psych vignette, but will just show you the lab values where you have to say, "The urine and serum are *both* dilute, so this is psychogenic polydipsia." - First step in diagnosis is fluid restriction in order to see how urinary/serum values change.

HY Thyroid diagnoses	
Hashimoto thyroiditis	- Autoimmune destruction of thyroid gland. - Lymphocytic infiltrate seen on biopsy (asked on NBME). - Anti-microsomal (aka anti-thyroperoxidase) + anti-thyroglobulin antibodies. - ↓ T3, ↓ T4, ↑ TSH. - ^{131}I uptake scan is ↓ or patchy (i.e., the gland is not producing hormone). - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol,

	proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy); transaminitis (sounds weird, but LFTs can be ↑). - Associated with other autoimmune diseases (e.g., type I diabetes, pernicious anemia, vitiligo) and immunodeficiencies (e.g., IgA deficiency). For USMLE, a HY big-picture concept is that "Autoimmune diseases go together," where if you have one autoimmune disease, there's ↑ risk of other autoimmune diseases. The strict HLA associations aren't important (outside of HLA-B27). In addition, "Autoimmune diseases and immunodeficiencies to together," where having autoimmune disease in general ↑ risk of immunodeficiencies, and vice versa. - ↑ risk of non-Hodgkin lymphoma (e.g., primary CNS lymphoma). - Tx = levothyroxine or triiodothyronine.
Graves disease	- Activating autoantibody against TSH receptor. This antibody is called thyroid-stimulating immunoglobulin (TSI). - ↑ T3, ↑ T4, ↓ TSH. - Lymphocytic infiltrate + hyperplastic thyroid follicles seen on biopsy. - ^{131}I uptake scan shows diffusely ↑ uptake. - Exophthalmos (aka proptosis; protrusion of eyes) and pretibial myxedema (non-pitting edema of shins due to GAG deposition) are specific findings in Graves. The vignette need not mention either of these findings, but if the Q specifically tells you "there's no exophthalmos," they're saying it's not Graves. - Buzzy findings are tachycardia, heat intolerance, sweating, and palpitations. - Myopathy can also occur with ↑ CK, but USMLE tends to give this finding in hypothyroidism. - Can cause atrial fibrillation in younger patients (normally AF is elderly). - "Thyroid storm" is an acute exacerbation of Graves, often precipitated by a stress factor such as surgery, trauma, or infection, where thyroid hormone production goes ↑↑↑, causing dangerous physiologic responses, such as hyperthermia, tachycardia, and altered mental status. - Thyroid storm can also cause adrenal crisis (i.e., acute ↓ in blood pressure due to rapid consumption of cortisol), especially in those with concurrent adrenal insufficiency taking exogenous steroids. Tx for the low BP is IV glucocorticoid (i.e., hydrocortisone or methylprednisolone). - Tx for general Graves is thionamides (propylthiouracil [PTU] and methimazole). - Both PTU and methimazole can cause neutropenia/agranulocytosis. - Propranolol is used for tachycardia in hyperthyroidism. - Oral prednisone is specifically used for Tx of exophthalmos, not thionamides. - Tx of thyroid storm on USMLE is tetrad of: 1) PTU, 2) propranolol, 3) steroids, and 4) potassium iodide.
deQuervain thyroiditis	- Aka subacute granulomatous thyroiditis, or just simply subacute thyroiditis. - Mechanism is viral infection followed by a **painful/tender** thyroid. There is inflammation of the thyroid gland, which causes the spacing between the cells to increase slightly, allowing for the release of pre-formed thyroid hormone into the blood. Therefore we have ↓ TSH, ↑ T3, ↑ T4, ↓ ^{131}I uptake. - The gland is not over-producing thyroid hormone. This is why uptake is not increased. - Subacute granulomatous thyroiditis can be either hypo- or hyperthyroid. The key detail you need to know is that **uptake is always decreased *even if the patient is hyper-.*** - DeQuervain vignettes will almost always be given as hyperthyroidism because the USMLE wants to specifically assess that you know uptake is decreased. If they give you hypo-, of course you'll select decreased for uptake. - Decreased uptake applies to all thyroiditis conditions (i.e, deQuervain, drug-induced, and postpartum).

	- Viral infections are often asymptomatic, so most deQuervain vignettes will not mention the viral infection.
Drug-induced thyroiditis	- Caused by lithium or amiodarone on USMLE. - Can in theory be hypo-, eu-, or hyperthyroid. - Will usually present on USMLE as painless hypothyroidism in patient being treated for bipolar disorder (lithium) or started on new anti-arrhythmic (amiodarone). - Will present with ↓ ^{131}I uptake, where inflammation of the gland can sometimes cause leakage of thyroid hormone into the blood, but the gland itself is not demonstrating increased production of hormone.
Subclinical hypothyroidism	- ↑ TSH, ↔ T3, ↔ T4. - In subclinical hypothyroidism, the patient will be asymptomatic (hence subclinical) and will have normal T3 and T4, despite an elevated TSH. - Most patients with subclinical do not need to be treated. - Don't treat unless TSH >10, patient is pregger, or anti-Hashimoto Abs are +.
Euthyroid sick syndrome	- Vignette will give patient who's had recent major trauma or surgery, or for patients in hospital on ventilators. - ↔ TSH, ↔ T4, ↓ T3, ↑ rT3. - Etiologies are manifold, but one proposed mechanism is that in times of stress (usually acute), spikes in cortisol and other inflammatory mediators can inhibit peripheral conversion of T4 to T3. This causes T3 to go down. - T4, however, rather than being detectable in excess, is converted to an inactive form of thyroid hormone called reverse T3 (rT3), so rT3 is high. - T4 is therefore still in the normal range, as is TSH (presumably because T4 is normal). - Call it weird all you want, but USMLE asks this. This is one of the most "underrated" thyroid diagnoses, as students will often have heard of it but then disregard it, thinking it's minutiae and probably won't show up.
Toxic adenoma	- Autonomously secreting thyroid nodule (aka hot nodule). - ^{131}I scan shows uptake onto into the area of a single nodule. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential).
Toxic multinodular goiter	- Autonomously secreting thyroid nodules (i.e., many hot nodules). - ^{131}I scan shows multinodular uptake. - ↑ T3, ↑ T4, ↓ TSH. - Not malignant (i.e., no metastatic potential). - I don't think I've ever seen this as a correct answer on USMLE, but it shows up quite a bit as a distractor, e.g., for deQuervain Qs. - Occurs in elderly. You might get a younger patient with hyperthyroidism due to deQuervain, and you can eliminate toxic multinodular goiter because 1) the patient is young, and 2) deQuervain is painful/tender, whereas TMG is not.
Riedel thyroiditis	- Obscure fibrosing disorder characterized by replacement of the thyroid gland with dense, fibrous tissue (sometimes called "woody thyroid" for this reason). - Can sometimes resemble anaplastic thyroid carcinoma, where it extends into adjacent structures such as the esophagus or trachea.
Factitious thyrotoxicosis	- Aka surreptitious thyrotoxicosis. - USMLE need not mention the patient is a pharmacist (implying access to the hormone). - Q will give hyperthyroid patient with small, non-palpable thyroid gland (atrophic due to suppressed TSH). - As mentioned earlier, T4 is converted to T3, but T3 isn't converted to T4. - So if thyroxine is administered, we will have ↑ T3, ↑ T4, ↓ TSH. - If triiodothyronine is administered, we will have ↑ T3, **↓ T4,** ↓ TSH.
Struma ovarii	- Abstruse ovarian tumor that for some magical reason secretes T3 and T4.

Thyroid nodule evaluation
- When evaluating for thyroid cancer, the first step is "palpation of thyroid gland" before "check serum TSH." There is an NBME Q for 2CK where both are listed and palpation is correct (i.e., do a history and exam before jumping into tests). - Once a nodule is palpated, *then* serum TSH is the next best step. - If patient with nodule has normal or high TSH (i.e., is euthyroid or hypothyroid), do ultrasound before fine-needle aspiration (FNA). This is assessed on the new 2CK Free 120, where ultrasound is correct over FNA for evaluation of thyroid nodule. It had long been pushed in resources that ultrasound is always wrong. But this is on Free 120. - 2CK IM CMS Q gives euthyroid patient with nodule where FNA is the answer, but ultrasound isn't listed. - So my conclusion (based on NBME/Free 120 content; not my opinion) is: for eu- or hypothyroid patient with thyroid nodule, ultrasound is correct over FNA if both are listed. If ultrasound isn't listed, choose FNA. - If patient has low TSH (i.e., is hyperthyroid), do ^{131}I uptake scan, not ultrasound. - Since carcinomas are non-secretory of thyroid hormone, if a patient is hyperthyroid, we're not concerned about carcinoma, which is why we don't go the ultrasound then FNA route. We just do uptake to better see if the patient's etiology for hyperthyroidism is Graves (diffuse), toxic adenoma (single nodular uptake), or toxic multinodular goiter (multifocal nodular uptake). - A caveat about the above algorithm for nodule evaluation is that if a patient has hyperthyroidism (i.e., low TSH) *and does not have a nodule*, then you will choose ultrasound over uptake scan. I know this sounds annoying, but USMLE assesses this. There is a new NBME Q where they give low TSH in young patient with no mention of nodule, and the answer is ultrasound over uptake scan. But in Qs where TSH is low and the patient has a nodule, the answer is uptake scan first over ultrasound. If you want a high score, then know the difference.    - TSH is screening for everyone not pregnant. - T3+T4 can be ordered after TSH, but Q often omits this. - Screening in pregnancy is Free-T4. - Most diagnostic/accurate for everyone is Free-T4. - Algorithms that are more detailed as far as size of nodule, etc., are basically nonsense for USMLE.

Hyperparathyroidism conditions	
Primary	- ↑ Ca^{2+}, ↓ $PO4^{3-}$, ↑ PTH. - Primary hyperparathyroidism is usually due to a PTH-secreting adenoma, but can also be due to diffuse four-gland hyperplasia. - Can be part of MEN-1 or -2A (discussed later).
Secondary	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH, ↓ 1,25-D3. - Secondary hyperparathyroidism is due to renal failure. - There are two reasons why calcium is low: 1) The kidney cannot reabsorb it in the late-DCT like it's supposed to. 2) The kidney cannot synthesize activated 1,25-D3 in the PCT like it's supposed to. Since 1,25-D3 absorbs calcium in the small bowel, and 1,25-D3 levels are lower, serum calcium will also be lower.

	- The low serum calcium will cause PTH to go up (deceased negative-feedback at the calcium-sensing receptors on the parathyroid glands). - Since the etiology for the increased PTH secretion is not the parathyroid glands themselves, we call this secondary hyperparathyroidism. If the etiology is due to the parathyroid glands themselves (i.e., adenoma or diffuse hyperplasia), we call that primary hyperparathyroidism. - The calcium and phosphate arrows for secondary hyperPTH are the opposite of primary hyperPTH. - Phosphate is high in renal failure, despite the high PTH, because the kidney is not able to downregulate the PCT phosphate transporters, so there is too much phosphate reabsorption.
Tertiary	- ↑ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH. - In order to understand tertiary hyperparathyroidism, let's first review secondary hyperparathyroidism: - Patients with renal failure will initially develop secondary hyperparathyroidism, where calcium is low, phosphate is high, and PTH is high. Calcium is low due to failure of reabsorption at the late-DCT of the kidney and because of decreased synthesis of activated vitamin D3 (leading to decreased small bowel absorption). Phosphate is high due to failure to downregulate PCT reabsorption pumps (i.e., more pumps reabsorb phosphate). PTH is high because calcium is low (less negative feedback). - Tertiary hyperparathyroidism results from hyperplasia of the parathyroid glands in patients with long-standing secondary hyperparathyroidism, such that even if the renal failure is brought under control and serum calcium is brought back into the normal range, PTH continues being autonomously secreted at higher basal levels than prior to the renal disease, effectively resetting the body's setpoint for calcium homeostasis. - This causes a rise in serum calcium in a patient with renal failure. This should immediately raise a red flag for tertiary hyperPTH, since renal failure patients will almost always have low, not high, calcium. - Even though the parathyroid glands will be hyper-secreting PTH, this is not primary hyperparathyroidism, since the etiology for the high PTH was not idiopathic adenoma or hyperplasia. - Phosphate is high, not low, because the patient has renal failure. Phosphate is always high in renal failure. Even though PTH is high, remember that the kidney can't downregulate the PCT reabsorption pumps the way they're supposed to.

Hypoparathyroidism conditions	
DiGeorge	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH, ↓ T cell levels, ↔ B cell levels. - Mechanism is most frequently 22q11 deletion, resulting in agenesis of the 3rd and 4th pharyngeal pouches. - Tetralogy of Fallot and truncus arteriosus are common heart defects. - Agenesis of the parathyroid glands → primary hypoparathyroidism → low calcium + high phosphate. - DiGeorge is characterized by T cell deficiency. Recurrent viral, fungal, and protozoal infections are characteristic. - Absent thymic shadow = T cell deficiency → DiGeorge. - Scanty lymph nodes/tonsils = B cell deficiency → Bruton agammaglobulinemia. - If Q says kid has both absent thymic shadow *and* scanty lymph nodes/tonsils → answer = SCID.
Post-thyroidectomy	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH. - Thyroidectomy can result in removal of or damage to the parathyroid glands, resulting in decreased PTH secretion and, in turn, hypocalcemia. Phosphate goes up because there is increased renal reabsorption (PTH normally downregulates apical PCT phosphate transporters, thereby promoting excretion). - Twitching of the masseter with stimulation is called Chvostek sign of hypocalcemia.

	- Hypocalcemia causes “up” findings – i.e., muscle tetany and hyperreflexia. - Trousseau sign of hypocalcemia is carpopedal spasm with blood pressure cuff inflation (i.e., twitching of the hand/wrist). The USMLE can give you either Chvostek or Trousseau sign when Ca^{2+} is low. - Hypercalcemia, in contrast, causes “down” findings – i.e., muscle flaccidity and hyporeflexia. - Do not confuse Trousseau sign of hypocalcemia with Trousseau and Troisier signs of malignancy. - Troisier sign of malignancy is a palpable left supraclavicular lymph node sometimes seen in visceral malignancy (aka Virchow node). - Trousseau sign of malignancy is migratory thrombophlebitis classically seen in head of pancreas adenocarcinoma (but can be other adenocarcinomas, such as bronchogenic).
Hypomagnesemia	- ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↓ PTH, ↓ Mg^{2+}. - Alcoholics are susceptible to hypomagnesemia as a result of dietary deficiency (EtOH is 7kcal/g, so they fill up on alcohol). - Low magnesium can cause hypocalcemia and hypokalemia nonresponsive to supplementation (i.e., you give calcium or potassium for low serum levels, but the serum levels do not appreciably rise). - Basal levels of magnesium are required for proper functioning of the parathyroid gland, so low magnesium can cause low PTH secretion. - Patients with hypomagnesemia may indeed have completely normal serum levels for calcium, phosphate, and PTH, but if we are forced to choose arrows, we would choose a Ca^{2+}/$PO4^{3-}$/PTH combo that reflects low PTH secretion. - The Step 2 exam will often just say an alcoholic has low calcium or potassium nonresponsive to supplementation, and then the next best step in management is “check serum magnesium levels.” This isn’t hard at all. But some students will be like “Oh wow.”

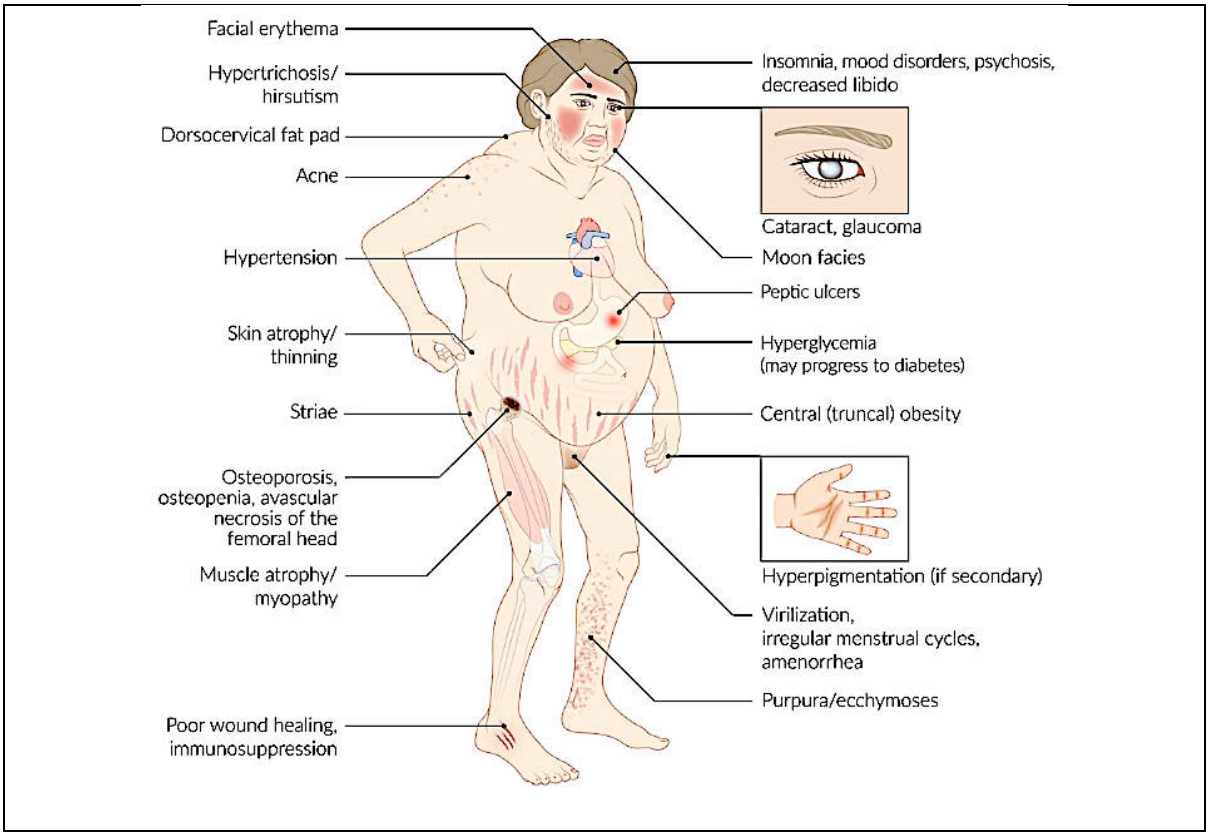
Other calcium-related conditions	
Paraneoplastic PTHrp	- Squamous cell carcinoma of the lung and renal cell carcinoma both secrete parathyroid hormone-related peptide (PTHrp). *This is not the same as endogenous PTH.* - USMLE Qs for renal cell carcinoma and squamous cell of the lung will have: o ↑ Serum Ca^{2+}, ↓ serum $PO4^{3-}$, ↓ PTH. - PTHrp exerts very similar physiologic effects as PTH, causing high serum calcium and low phosphate. - PTH secretion by the parathyroid glands will be suppressed in this setting due to the high serum calcium. - NBME will give you lung cancer + high serum calcium + low serum PTH, with the answer being squamous cell carcinoma of the lung. The low PTH will throw some people off. But this is not weird. PTH is not the same as PTHrp (which will be up-arrow).
Metastases	- ↑ Serum Ca^{2+}, ↓ PTH. - Once metastases seed at bony locations, cytokine activity causes lysis of the bone and release of calcium into the blood. Hypercalcemia is common in the setting of metastatic malignancies. - PTH is suppressed due to the high calcium.
Paget disease	- ↔ Serum Ca^{2+}, ↔ serum $PO4^{3-}$, ↔ PTH, ↑ ALP. - Idiopathic disorder of increased bone turnover. Bone is described as having mixed osteoblastic and -clastic phases, appearing heterogenous on x-ray. - Buzzy vignette is male over 50 who’s hat doesn’t fit him anymore + has tinnitus (narrowing of acoustic foramina).

	- 19/20 questions will give isolated increase in serum ALP. You need to know Ca^{2+}, $PO4^{3-}$, and PTH are all normal in Paget disease. There is one Q on a 2CK CMS form where ALP is given as not elevated, but it's a one-off Q and rare. - High-output cardiac failure can occur due to intraosseous AV-fistulae, where patient has an S3 heart sound with high, rather than low, ejection fraction.
Milk-alkali syndrome	- Diagnosis of exclusion on USMLE, meaning we eliminate to get there. - Textbook scenario is **hypercalcemia and metabolic alkalosis** (high bicarb) in someone taking too many calcium carbonate antacid tablets for GERD or dyspepsia (indigestion/upset stomach). This can lead to **urolithiasis.** - What the USMLE vignette will do, however, is not mention the patient's Hx, but will simply give you a kidney stone + hypercalcemia + bicarb that is elevated (or toward the upper limit of normal). Then you eliminate to get there, where you say, "Even though they don't mention the textbook presentation, no other answers explain the high calcium + high bicarb." - Normal bicarb is 22-28 mEq/L, but an NBME Q for milk-alkali syndrome gives a bicarb of 27, where this throw students off. Acid-base disorders can occasionally have bicarb in the normal range, albeit in the direction you expect.

Primary adrenal DDx affecting aldosterone	
Conn syndrome	- Aldosterone-secreting tumor. - Causes ↑ Na^+, ↓ K^+, ↑ $HCO3^-$, ↑ pH, ↓ renin. - In ~50% of aldosterone-derangement Qs, Na^+ will be normal. This confuses students, but it's because the body has strong ability to maintain Na^+ (i.e., also via ADH). - Tx = spironolactone (aldosterone receptor antagonist) prior to surgery.
Familial hyperaldosteronism	- There are gene mutations that can cause ↑ primary adrenal production of aldo. - If the USMLE gives you Q where various family members have ↑ BP + ↑ aldosterone and they don't tell you any other info, including the renin level, the answer is primary hyperaldosteronism, not renovascular hypertension (i.e., FMD or RAS). - There is a Q on a 2CK NBME form where they do this, and the student says, "But wait, how are we supposed to know it's primary and not FMD/RAS?" It's because if ↑ aldo is familial, it's likely due to gene mutations causing ↑ primary production from the zona glomerulosa.
Addison disease	- Autoimmune destruction of the adrenal cortex, leading to ↓ aldosterone *and* ↓ cortisol. - I haven't seen the USMLE ever care about "decreased androgens," so the effect is predominantly on the zona glomerulosa and fasciculata. - Called "primary adrenal insufficiency" because the adrenal gland itself is under-producing hormones. - Addison is idiopathic / can be seen in patients with Hx of other autoimmune diseases, either in themselves or family members (i.e., the vignette throws in the side-detail that the brother has RA, or the sister has IBD, or the patient has Hx of thyroiditis). - ↓ Na^+, ↑ K^+, ↓ pH, ↓ bicarb. - Blood pressure will either be low-normal or low, due to both ↓ aldosterone as well as ↓ cortisol. I explicate this because students tend to just mention the ↓ aldosterone as the reason BP is ↓, but the effect of ↓ cortisol is HY for USMLE. - ACTH and α-MSH are both ↑ due to ↓ cortisol → hyperpigmentation. Although I'd say they only mention this in ~50% of vignettes. - Patients can present with eosinophilia. This is because cortisol normally plays a role in helping to sequester eosinophils within the spleen + facilitates eosinophil apoptosis, so ↓ cortisol means ↑ eosinophils. Don't go chasing stool ova and parasites. Qs can give you eosinophils of 15-25%, where you're like wtf? They should normally be <5%.

	- Classic Addison vignette is patient (young or old, I've seen both) who has ongoing fatigue + hyperpigmentation, and then the answer is just "ACTH stimulation test." - Exogenous ACTH should normally cause a robust ↑↑ in cortisol from the adrenal fasciculata. In the setting of primary adrenal insufficiency, since the adrenal gland is fucked up, giving exogenous ACTH won't do jack shit. - ↓ cortisol causes chronic fatigue syndrome. This is one of the most over-looked details in NBME Qs and in real life. Any time a patient has ongoing chronic fatigue that is unexplained, cortisol levels should be checked. If they are ↓, hydrocortisone will immediately boost the patient. - Tx for Addison = fludrocortisone +/- hydrocortisone. Fludrocortisone is a mineralocorticoid similar to aldosterone. Hydrocortisone is a glucocorticoid similar to cortisol.
Waterhouse-Friderichsen	- Meningococcal septicemia causing bilateral hemorrhagic necrosis of the adrenal glands, resulting in ↓ aldosterone and ↓ cortisol. - Vignette will be ↓ BP + ↓ Na^+ + ↑ K^+ + ↓ bicarb in patient with meningococcal infection (e.g., recent meningitis + non-blanching rash), where answer = ACTH stimulation test.

Cushing syndrome vs disease
- Cushing syndrome = how the patient looks/presents; can refer to any cause of Cushingoid presentation (i.e., exogenous glucocorticoids, ACTH-secreting tumor of anterior pituitary, cortisol-secreting tumor of adrenal cortex, small cell bronchogenic carcinoma secreting ACTH). **- Cushing disease** = only ACTH-secreting tumor of anterior pituitary. - If patient walks through the door and appears Cushingoid due to an ACTH-secreting tumor of the anterior pituitary, we would say, "The patient has Cushing syndrome caused by Cushing disease." - Highest yield findings for Cushingoid appearance on USMLE, as per my observation, are: **purple striae** on the abdomen / stretch marks, **osteoporosis** (compression fractures of vertebrae)**,** and **avascular necrosis of the femoral head** (hip pain). Findings such as buffalo hump, moon facies, central obesity, peripheral muscle wasting, hypertension, and diabetes type II are important and pass-level, but you really need to know those 3 bold findings I write above. - In Peds, Cushings can present as precocious puberty without all of the classic findings. - Chronically elevated cortisol can cause **hypokalemia** similar to high aldosterone. Sometimes NBME will give you a vignette of Cushings + mention potassium is 3.0 mEq/L (NR 3.5-5.0), and you're like, "why the fuck is the potassium low. Is aldosterone high?" No. All glucocorticoids have some degree of mineralocorticoid effect at the kidney similar to aldosterone. In patients who have high cortisol, the effect longer term can occasionally cause hypokalemia. This detail is particularly important for 2CK vignettes.



Dexamethasone suppression test	
- Dexamethasone is a cortisol analogue that is capable of exerting negative-feedback at the hypothalamus and anterior pituitary. Given to healthy individuals, the result is suppression of ACTH and cortisol levels. - The dexamethasone suppression test is done to determine the etiology of *endogenous* Cushing syndrome (i.e., the patient isn't taking exogenous steroids). This is because if the patient is taking steroids, clearly we know the cause of the Cushings already. - Low-dose dex will suppress ACTH and cortisol in healthy patients. It is effectively a "yes or no" test for Cushing syndrome, where if there's suppression of cortisol with low-dose dex, then we can say, "Patient doesn't have Cushings"; if cortisol doesn't suppress, then we know the patient has Cushings, but we just don't know the cause yet. - High-dose dex helps us differentiate Cushing disease from other causes of Cushing syndrome. This is because only Cushing disease will have suppressed cortisol levels. - Therefore, if cortisol goes ↓ with high-dose dex, we know the diagnosis is Cushing disease. - If cortisol does not go ↓ with high-dose dex, we look at ACTH level. - If ACTH is ↓, we say, the only way that possible to have ↑ cortisol with ↓ ACTH is if the diagnosis is an adrenal cortical adenoma of the zona fasciculata secreting cortisol. - If ACTH is ↑, we know the cause is ectopic ACTH secretion from small cell bronchogenic carcinoma. The reason we know the ACTH isn't from the pituitary is because high-dose Dex didn't suppress cortisol.	
Cushing disease	- ↑ ACTH + ↑ cortisol. - Suppresses with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "↓ ACTH, ↓ cortisol."
Small cell lung cancer	- ↑ ACTH + ↑ cortisol. - Does not suppress with high-dose dex. - If high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "no change ACTH, no change cortisol."
Adrenal adenoma	- ↓ ACTH + ↑ cortisol.

	- Does not suppress with high-dose dex. - Same as with small cell lung cancer, if high-dose dex is given and they ask for the arrows for how ACTH and cortisol *will change*, the answer is "no change ACTH, no change cortisol."
Exogenous steroids	- ↓ ACTH + **↓ cortisol (holy shit).** - Exogenous steroids (i.e., prednisone, hydrocortisone) are not the same thing as cortisol. The result is suppression of endogenous cortisol. - Students get this wrong all of the time.

Miscellaneous	
Glucocorticoid psychosis	- Glucocorticoids can cause **"glucocorticoid psychosis,"** which can present as an overt psychosis, delirium-like presentation, or even just as general mood disturbance, i.e., depression. - The vignette can give you Cushing syndrome + depression, or **recent IV methylprednisolone injection** + delirium-like presentation 24-48 hours later.
Licorice	- Licorice contains glycyrrhizic acid, which leads to ↑ mineralocorticoid effect, at the kidney (i.e., similar to aldosterone). - Can cause hypokalemia in those who drink excessive licorice tea.

Catecholamine-secreting tumors	
Pheochromocytoma	- Catecholamine-secreting tumor of adrenal medulla. - Occurs in adults; can calcify. - Causes paroxysmal (i.e., comes and goes) ↑↑ BP, which presents as periodic palpitations and pounding headaches. - Because the HTN is paroxysmal, the patient can have BP measured in the office at 120/80, so do not exclude pheo in this case. - Stressors (e.g., sitting up on the examination table) can trigger ↑ NE + E release from the tumor, where BP shoots up to >180/100. This is different from whitecoat hypertension, where young patient has 140/90 measured due to being slightly nervous. - Diagnose with urinary/serum metanephrines. If positive, then do CT abdomen. - Tx = phenoxybenzamine (irreversible α1-blocker). If you give beta-blocker first, this will kill the patient. This is because if you block beta receptors, the NE and E have little to bind to now except α receptors (mainly α1), so we get ↑↑ arteriolar constriction and BP shoots up. This is called "unopposed alpha." - Some students ask about giving propranolol after phenoxybenzamine. The answer is, yes, in theory this can be done. But USMLE doesn't assess this. They want straight-up alpha-blockade. - Pheo can be idiopathic or part of condition such as MEN 2A/B or NF1. - If a patient has medullary thyroid cancer, you must check for pheochromocytoma before doing thyroidectomy. This is because if the patient has MEN 2 and you don't check for pheo first, doing surgery will cause ↑↑ BP due to catecholamine release and kill the patient.
Neuroblastoma	- Essentially the "pediatric pheo." - Secretes catecholamines; can calcify (same as pheo in adults). - Grows from the median (midline) sympathetic chain. Usually intra-abdominal but can sometimes be in the posterior mediastinum. Sounds odd, but the latter shows up on an 2CK NBME form. - Causes ↑ BP and findings such as opsoclonus-myoclonus syndrome (dancing eyes) and violaceous eyelids (sounds like heliotrope rash of dermatomyositis but unrelated).

	- Diagnose with urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA), which are metabolites of catecholamines. - Metaiodobenzylguanidine scan can be done after HVA and VMA detected.

Carcinoid tumor
- Neuroendocrine tumor of small bowel, appendix, or lung that is S-100 (+) and consists of small blue cells. - Secretes serotonin. - Presents as flushing, tachycardia, diaphoresis, and diarrhea. - Tricuspid regurg common (holosystolic murmur that increases with inspiration) due to tricuspid lesions. - Diagnose with urinary 5-hydroxyindole acetic acid (5-HIAA). - Don't confuse with serotonin syndrome, which is a drug interaction (e.g., starting a monoamine oxidase inhibitor too soon after discontinuing an SSRI, or if a patient takes St John wort with an SSRI).

Polyglandular syndromes	
- All MEN syndromes and Carney complex are autosomal dominant.	
MEN 1	- *MEN1* gene mutation. - Parathyroid adenoma (or hyperplasia), pituitary adenoma, pancreatic adenoma. ○ Zollinger-Ellison syndrome (gastrinoma) is highest yield pancreatic tumor in MEN 1. Will present as recurrent peptic ulcers. I discuss this stuff in detail in the HY Gastro PDF. ○ Parathyroid adenoma often presents with **calcium urolithiasis,** where they will mention flank or groin pain in patient with Hx of ulcers (Zollinger-Ellison). ○ Prolactinoma is highest yield pituitary tumor. ○ USMLE need not give the full constellation of findings for the MEN syndromes. There is an NBME Q floating around where the patient has a parathyroid adenoma – nothing more – and the answer is just *MEN1* as the gene that's mutated.
MEN 2A	- *RET* proto-oncogene mutation. - Parathyroid adenoma (or hyperplasia), pheochromocytoma, medullary thyroid carcinoma. ○ As mentioned earlier, *RET* mutations need not cause the full constellation of findings. NBME Qs can give isolated medullary thyroid carcinoma in a family, and the answer is just *RET.* ○ In theory, isolated parathyroid adenoma could also be *RET*, but with the NBME example I mentioned above they listed *MEN1* without also listing *RET*.
MEN 2B	- *RET* proto-oncogene mutation. - Marfanoid body habitus, mucosal neuromas, pheochromocytoma, medullary thyroid carcinoma. ○ -Oid means looks like but ain't. So the patient might look like he/she has Marfan's (i.e., tall, lanky) but doesn't, because this isn't *FBN1/2* mutation. ○ As mentioned earlier, before doing thyroidectomy for medullary thyroid carcinoma, you must check for pheochromocytoma (because stress of surgery will ↑↑ catecholamines from the tumor and kill the patient) by first doing urinary metanephrines. If positive, then do CT of the abdomen.
Carney complex	- Hyperpigmentation, atrial myxoma, endocrine hypersecretion. ○ The hyperpigmentation can present as perioral melanosis (i.e., hyperpigmentation around the lips/mouth) that sounds similar to Peutz-Jegher syndrome, but instead of the Q focusing on hamartomatous colonic polyps, they mention a cardiac tumor and another endocrine organ hyper-secreting. And you're like huh?

	○ Atrial myxoma normally shows up in adults; cardiac rhabdomyoma normally shows up in children with tuberous sclerosis. In Carney, the Q can say atrial myxoma *in a kid*. So it's the only example on USMLE that breaks the rule. ○ Endocrine hypersecretion is classically Cushing syndrome (granular hyperpigmentation of the zona fasciculata) or hyperthyroidism. ○ The reason you likely haven't heard of Carney complex before is because students coming out of the USMLE didn't even realize this is what they saw.

Diabetes mellitus (DM)	
- The word diabetes means polydipsia + polyuria (i.e., drinking a lot and peeing a lot). - I see students frequently misconstrue diabetes mellitus with diabetes insipidus, since both vignettes will give patients who have increased thirst and urination. - A key detail that distinguishes acute diabetes mellitus from diabetes insipidus is that DM vignettes can often mention weight loss due to ↑ catabolism. Weight loss is also seen in acute type II, even though the patient is usually overweight. Vomiting is another HY finding in acute DM. - So if vignette says ↑ thirst + ↑ urination + weight loss +/- vomiting, this is DM, not DI. - For extensive info on the effects of diabetes on different organ systems (i.e., cardio, renal, ophthal, GIT, pregnancy, etc.), use those corresponding HY PDFs I've made. I will keep the focus here on endocrine and electrolytes / acid-base. - Diagnosis of DM is made by measuring two fasting glucoses >126 mg/dL, any one random fasting glucose >200 mg/dL, or an HbA1c of >6.5%.	
Type I	- Absence of insulin; autoimmune-mediated; no one specific gene, although HLA-DR3/4 are associated. - Often described as "genetic susceptibility with environmental trigger" (i.e., viral infection, usually Coxsackie B). - Occurs almost always in children or teenagers of normal BMI. - Pathogenesis is usually combo of an antibody- (anti-GAD65 or -ZnT8) and T-cell-mediated process that leads to destruction of β-islet cells in the tail of the pancreas → ↓ insulin secretion → inability to recruit GLUT4 to cell surface of skeletal muscle and adipose tissue → inability to take up glucose into cells → cells are starved of energy → ↑ gluconeogenesis + ketogenesis. - Hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Biopsy of pancreas shows inflammatory infiltrate in acute type I DM; in late type I (i.e., years later), it shows atrophy and fibrosis. - It is the absence of insulin in type I DM that enables ketogenesis. In type II DM, ketones are low, not high, because insulin is present. - Ketone bodies (USMLE likes β-hydroxybutyrate) are highly acidic, so will drop the blood pH (high anion-gap metabolic acidosis). - Vignette will give kid or teenager with polyuria, polydipsia, and sometimes vomiting, who has severe metabolic acidosis (bicarb ↓), which is due to diabetic ketoacidosis. Breath can be fruity in odor (smell of ketones). DKA will have: - **↓ pH, ↓ $HCO3^-$, ↓ CO2.** - Ketone bodies are acidic → causes metabolic acidosis (decreased bicarbonate). - CO2 is acidic, so we blow that off to compensate. Kussmaul breathing is deep, labored breathing seen in DKA. - **↑ β-hydroxybutyrate, ↑ serum osmolality, ↑ anion gap.** - β-hydroxybutyrate is a ketone body. Since we have ketosis (ketone production), it will be increased. I have seen this specific ketone body as an arrow on an offline NBME. - Serum osmolality is increased because serum glucose is markedly increased (insulin isn't present to drive glucose into cells). Even though serum sodium is low (so you might think osmolality is low similar to SIADH), the ↑ glucose makes serum osmolality high. - DKA is the D in MUDPILES, so anion gap is up. - **↓ Serum Na^+, ↑ serum K^+, ↓ total body K^+, ↑ GFR.**

	- Serum sodium is decreased mostly due to dilutional hyponatremia. As mentioned earlier, serum osmolality is high, which causes fluid retention intravascularly → dilutes out serum sodium. In addition, low insulin means less glucose enters cells → less ATP production → less cellular Na^+/K^+-ATPase activation → sodium not pumped out of cells. - We describe the state of K^+ as: hyperkalemia (high serum K^+) despite a low total body potassium. - Serum potassium is high for two main reasons: 1) **Potassium-proton shift:** Acidosis (due to ketosis) causes protons in the blood to exchange with potassium in the cell via the H^+/K^+-antiporter (i.e., H^+ goes into cell; K^+ moves out). 2) **Insulin drives potassium into cells (and we don't have insulin in DKA):** Insulin normally leads to the upregulation of a Na^+/K^+-ATPase antiporter that moves $3Na^+$ out for every $2K^+$ in. If insulin is low, then less K^+ will be moved into the cell → hyperkalemia (high potassium in blood). - Total body potassium is low because of increased losses at the kidney. Since serum potassium is high, more potassium is filtered through the glomerulus, so more is excreted (i.e., the kidney thinks there's too much potassium in the body since more is being filtered, so it tries to get rid of more of it). However, this causes bodily depletion of K^+ because the processes that cause hyperkalemia are unmitigated, so the kidney continues to excrete potassium. - GFR is increased because of hyperfiltration secondary to hyperglycemia. That is, since serum glucose is high, more glucose is filtered through the glomerulus, which pulls water with it → hyperfiltration. - Tx for DKA is normal saline first (i.e., 0.9% NaCl), prior to giving insulin. - Serum potassium will drop as insulin is given. If the Q tells you patient has DKA + shows you normal potassium (even though it's usually high) + insulin is given + patient now has arrhythmia, the answer is potassium for which electrolyte is disturbed (i.e., potassium was normal prior to giving insulin, but now it's been driven into cells and the patient's developed hypokalemia). - Bicarb is always a wrong answer for low pH in DKA. - Don't confuse DKA with a hypoglycemic episode. The latter will be a type I diabetic who went out to do exercise or who was cleaning the roof of his house, who is now found unconscious → if insulin dose is not decreased prior to exercise, patient can get hypoglycemia. The Tx is intramuscular glucagon (2CK likes this) or dextrose in water. - Sometimes the USMLE can give a bit of a weird Q where they say a patient is given insulin for diabetes but has prolonged hypoglycemia + they ask why → answer = "deficiency of counter-regulatory glucagon." You need to know that in advanced diabetes, the glucagon-producing α-islet cells of the pancreas can also get fucked up. I've also seen this Q asked for chronic alcoholism and pancreatectomy; in both cases, if insulin is given, prolonged hypoglycemia can result from "deficiency of counter-regulatory glucagon."
Type II	- Insulin resistance; seen usually in high-BMI adults, but can rarely show up in Peds (2CK NBME form gives type II in a 17-yr-old). - High BMI causes insulin resistance because adipose tissue secretes adipokines that impair insulin signaling. - Insulin is *high* initially (hyperinsulinemia) due to β-cell hypertrophy in order to compensate for peripheral insulin resistance. - Ketones are absent/low because the presence of insulin inhibits ketogenesis. - The two above points are the highest yield pieces of info for USMLE. - Biopsy of the pancreas initially shows β-cell hypertrophy, but years later will show amyloid deposition and cell shrinkage. - Same as with type I, hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Hyperosmolar hyperglycemic state (HHS) is seen in type II; DKA is strictly type I on USMLE.

	- In HHS, glucose is usually >600 mg/dL (in DKA, it need not be this high) and bicarb is usually not below 20 mEq/L (NR 22-28); in DKA, bicarb can be as low as it wants (I've seen 5 mEq/L on NBME). - The USMLE vignette will give massive paragraph Q where they mention glucose is 700 and then ask for the cause of mental status change → answer = "hyperosmolality," or simply "hyperglycemia," or "osmotic diuresis" (causes severe dehydration). I point this out because this is particularly prevalent on 2CK NBMEs, where they'll give a 15-line bullshit paragraph with tons of lab values that the student spends 9 minutes reading, but meanwhile you can see glucose is 650 and then the answer is just "osmotic diuresis" or "hyperosmolality"; should take two seconds to answer. - Mental status change is technically due to hyperosmolality in the ECF of the brain pulling water out of the ICF (i.e., the cells). - Treat HHS with normal saline first prior to insulin. - I discuss the various meds for type II diabetes later in this PDF.
Other causes	- **Hereditary hemochromatosis** ("bronze diabetes") → iron deposition in tail of the pancreas, leading to impaired fasting glucose / overt diabetes. Hyperpigmentation is due to hemosiderin deposition. Patients can also get cardiomyopathy, infertility, and pseudogout. - **Cushing syndrome** (as discussed earlier) can cause diabetes due to the insulin resistance caused by glucocorticoids. - **Acromegaly** (growth hormone causes insulin resistance). - **Alcoholism** can cause chronic pancreatitis that not only leads to exocrine pancreatic burnout and steatorrhea, but can also cause diabetes due to loss of islet cells. - **Pancreatectomy** is self-explanatory. If you remove some of the tail of the pancreas, this can lead to diabetes. - **Tacrolimus** is a HY drug that can cause diabetes. Other drugs can do this, but tacrolimus is the notable one for USMLE. Tacrolimus is an immunosuppressant that ↓ IL-2 synthesis and T cell stimulation.

Electrolytes	
Calcium	- 8.4-10.2 mg/dL. - Low Ca causes "up" presentation of tetany and hyperreflexia; high Ca causes "down" presentation of muscle flaccidity and hyporeflexia; high Ca also associated with delirium/confusion (hypercalcemic crisis). - High calcium HY causes are sarcoidosis (due to ↑ vitamin D), primary hyperparathyroidism (adenoma, hyperplasia, MEN 1/2A), metastatic malignancy, multiple myeloma, and thiazides (cause - Ca reabsorption in DCT). - Low calcium HY causes are rickets/osteomalacia (↓ vitamin D), secondary hyperparathyroidism (renal failure), post-thyroidectomy (concomitant loss of parathyroids), DiGeorge syndrome (agenesis of 3rd and 4th pharyngeal pouches), and loop diuretics (↓ paracellular reabsorption), hypomagnesemia (hypo-Mg can cause hypo-Ca and hypo-K non-responsive to supplementation; usually seen in alcoholics).
Sodium	- 135-145 mEq/L. - Sodium derangement (high and low) causes CNS dysfunction – i.e., confusion, stupor, or coma. - Na can often be normal in USMLE vignettes despite your expectation that it might be characteristically deranged – e.g., patient has Conn syndrome and yet the sodium is normal, and you're like wtf? (because you expect it to be elevated); this is typical for USMLE vignettes. - High Na HY causes are primary hyperaldosteronism, renal artery stenosis, fibromuscular dysplasia, dehydration, diabetes insipidus. - Low Na HY causes are Addison disease, psychogenic polydipsia, SIADH.
Potassium	- 3.5-5.0 mEq/L. - Potassium derangement (high and low) causes cardiac dysfunction → arrhythmia (contrast this with Na, which causes CNS dysfunction).

	- High K HY causes are Addison disease, renal failure, polypharmacy with agents such as potassium-sparing diuretics and digoxin. - Low K HY causes are primary hyperaldosteronism, renal artery stenosis, fibromuscular dysplasia, Cushing disease (chronically high glucocorticoid levels can cause hypo-K due to distal renal secretion similar to mineralocorticoids), vomiting, diarrhea, loop diuretics and thiazides, and hypomagnesemia (hypo-Mg can cause hypo-K and hypo-Ca non-responsive to supplementation; usually seen in alcoholics).
Phosphate	- 2.5-4.5 mg/dL. - High phosphorus HY causes are hypoparathyroidism, secondary hyperparathyroidism, renal failure, tumor lysis syndrome, and sarcoidosis (↑ vitamin D). - Low phosphorus HY causes are primary hyperparathyroidism, rickets / osteomalacia, and refeeding syndrome.
Magnesium	- 1.7-2.2 mg/dL. - Magnesium derangement causes effects similar to Ca – i.e., low Mg presents with an "up" state of hyperreflexia and increased muscle tone; high Mg presents with a "down" state of hyporeflexia and muscle flaccidity. - Low Mg HY cause is alcoholism (decreased dietary intake); leads to hypocalcemia and hypokalemia non-response to supplementation.

Acid/base	
(If you want practice with acid/base USMLE Qs and/or are confused by certain things I list in this table, go through my HY Arrows PDF. I discuss this stuff in comprehensive detail).	
- Normal blood pH: 7.35-7.45. **- Normal blood $HCO3^-$: 22-28 mEq/L.** **- Normal blood pCO2: 33-44 mmHg.**	
Metabolic acidosis	- Low pH caused by low HCO3; pCO2 will go down to compensate (CO2 is acidic).
High anion-gap metabolic acidoses	- MUDPILES is mnemonic for high-anion-gap metabolic acidoses → Methanol, Uremia (renal failure), DKA, Phenformin (weird drug you don't need to worry about), Iron tablets / Isoniazid, Lactic acidosis, Ethylene glycol (antifreeze), Salicylates (aspirin). - If the patient has metabolic acidosis and the anion gap is normal in the Q, you can eliminate the MUDPILES answers and choose an answer like renal tubular acidosis (RTA). - Anion gap is calculated as $Na^+ - (HCO3^- + Cl^-)$. - Anion gap of 8-12 is normal. So 13 or greater is high. I specifically write "13 or greater" because there is an NBME Q where anion gap comes out to 12, and they want renal tubular acidosis. I talk in detail in the HY Renal PDF about RTA.
Normal anion-gap metabolic acidoses	- Diarrhea, RTA, adrenal insufficiency, and CAH (21-deficiency) are most important.
Metabolic alkalosis	- High pH caused by high HCO3; pCO2 will go up to compensate. - Conn syndrome, renal artery stenosis, fibromuscular dysplasia, vomiting, loops and thiazides (promote RAS activation due to volume depletion).
Respiratory acidosis	- Low pH caused by high CO2; HCO3 goes up *if chronic* to compensate (bicarb is basic). - Acute (**normal** bicarb): hypoventilation due to opioids, benzos, barbiturates. - Chronic (high bicarb): COPD, obstructive sleep apnea, obesity, ankylosing spondylitis, severe kyphoscoliosis.
Respiratory alkalosis	- High pH caused by low CO2; HCO3 goes down *if chronic* to compensate. - Acute (**normal** bicarb): asthma attack, panic attack, most pulmonary emboli, altitude sickness (first day at high altitude). - Chronic (**low** bicarb): living at high altitude, pregnancy (progesterone upregulates respiratory center).
- Should be noted that primary metabolic acid/base derangements are compensated for instantly by changes in respiration (i.e., it's easy to retain or breathe off CO2 merely by changing respiratory rate),	

whereas primary respiratory derangements are compensated for **slowly** because it takes time for the kidney to alter bicarb excretion.
- Winter formula is used to calculate predicted pCO2 based on any change in bicarb, where: Predicted pCO2 = (1.5xHCO3) + 8 +/- 2. For example, if a patient has DKA and bicarb is 14 mEq/L, we expect pCO2 to be 27-31 mmHg. If pCO2 is lower than 27, then the patient has a concurrent respiratory alkalosis; if higher than 31, then the patient has a concurrent respiratory acidosis.

Salicylate toxicity

- Diagnosis is salicylate (aspirin) toxicity. Tinnitus (ear-ringing) is the most common first symptom.
- This patient has mixed metabolic acidosis-respiratory alkalosis (i.e., both a metabolic acidosis and respiratory alkalosis at the same time).
- Aspirin initially causes an isolated respiratory alkalosis (within the first 20 minutes) due to direct upregulation of respiratory centers in the brain.
- However, aspirin itself is an acid (salicylic acid), so it will ultimately cause a metabolic acidosis that wins over the respiratory alkalosis (i.e., pH will ultimately go low).
- Anion gap is high. Salicylates are the S in MUDPILES.
- Sodium and potassium are normal in salicylate toxicity. I have seen these arrows listed before, which causes students confusion, but they are unchanged. Sometimes the Q will give you numerical values where you need to calculate anion gap – i.e., Na^+ - ($HCO3^-$ + Cl^-). The value must be 13 or greater since it is one of the MUDPILES. Normal range is 8-12.
- In summary, once again, for aspirin:
    - First 20 mins: acute respiratory alkalosis (**↑ pH, ↔ $HCO3^-$, ↓ CO2).**
    - After 20 mins: mixed metabolic acidosis-respiratory alkalosis (**↓ pH, ↓ $HCO3^-$, ↓ CO2, ↑ anion-gap).**
- The low bicarbonate we see after 20 minutes **is not compensation.** It coincidentally goes low (as we'd expect for renal compensation in the setting of respiratory alkalosis). But remember that the kidney cannot induce this change until a minimum of 12-24 hours later. The reason bicarb is low after 20 minutes is because aspirin itself is an acid that is causing a metabolic acidosis and driving the bicarb low.
- A 2CK-level point is that Na^+ and Cl^- can be abnormal insofar as the bicarb is low and anion gap is high.
- Potassium must always be normal.
- You might get an aspirin Q where you're forced to choose either an up or down arrow for chloride and you say, "I haven't heard of that before. I thought it would just be normal." But if you're forced to choose, clearly you'd go with the answer where chloride is low, since anion gap is calculated as Na^+ - ($HCO3^-$ + Cl^-), and a low chloride would make the anion gap higher than the answer choice with high chloride.

Renal failure key disturbances

- As discussed earlier, renal failure causes secondary hyperparathyroidism, where we have:
    - ↓ Ca^{2+}, ↑ $PO4^{3-}$, ↑ PTH, ↓ 1,25-D3.
- However, it is also pass-level you know that in renal failure we also have:
    - ↑ K^+, ↓ $HCO3^-$, ↓ pH, ↑ anion gap (U in MUDPILES = uremia = renal failure), with Na^+ variable.
    - The kidney normally secretes potassium distally in the cortical collecting duct under the action of aldosterone. So if the kidney is fucked up, we can't do that, so K^+ is ↑.
    - Likewise, we normally have $HCO3^-$ reabsorbed in the PCT and H^+ secreted in the cortical collecting duct, so if we can't do either of those things, blood $HCO3^-$ and pH ↓.

HY Pulmonary acid/base findings

- Acute asthma attack and PE: **↓ pCO2, no change $HCO3^-$, ↑ pH,**
    - The key here is that these conditions cause acute respiratory alkalosis, not acidosis, as I mentioned in above table. Students normally fuck this up and think pCO2 is high. pCO2 only goes ↑ if the patient becomes tired as an asthma attack prolongs. But initially, for both of these conditions, we

have low pCO2 (holy shit), not high. Bicarb is unchanged because it takes a minimum of 12-24 hours for bicarb excretion to be sufficiently modulated by the kidney.
  - Tangentially, metabolic acid-base disturbances are instantly compensated for by changes in pCO2 by simply breathing more or less.
- Patients with COPD and obstructive sleep apnea are **chronic CO2 retainers,** where we have:
  - **↑ pCO2, ↑ HCO_3^-, ↓ pH.**
  - Bicarb goes up to compensate.

Miscellaneous HY acid/base	
Vomiting	**- ↓ Serum K^+, ↓ Cl^-, ↑ $HCO3^-$, ↑ CO2, ↑ pH.** - Very high-yield you know that vomiting causes a hypokalemic, hypochloremic, metabolic alkalosis. - Vomitus contains lots of potassium, chloride, and protons. In addition, volume depletion can promote distal renal losses of K^+ and H^+ secondary to RAAS upregulation. - CO2 is acidic, so we want to retain it to compensate.
Loop/thiazide diuretic abuse	**- ↓ Serum K^+, ↓ Cl^-, ↑ $HCO3^-$, ↑ CO2, ↑ pH.** **-** Furosemide is a loop diuretic. It will block the apical $2Cl^-/K^+/Na^+$ symporter in the thick ascending limb. Since water follows ions, and ion reabsorption is impaired, increased diuresis (urination) occurs. Volume loss causes upregulation of RAAS and distal secretion of protons under the control of aldosterone, leading to metabolic alkalosis. This is referred to as "contraction alkalosis," where volume contraction induced by diuretics promotes metabolic alkalosis. In addition, additional distal secretion of K^+ occurs. **-** Thiazide diuretics will also follow this arrow pattern. They block the Na^+/Cl^- symporter on the apical membrane of the early-DCT, leading to volume depletion and ↑ RAAS, thereby ↑ H^+ and K^+ secretion distally. **-** CO2 is acidic and retained because it compensates for our metabolic alkalosis. **-** It should be noted that these arrows are the same as for vomiting.
Diarrhea	**- ↓ Serum K^+, ↑ Cl^-, ↓ $HCO3^-$, ↓ pH, ↓ CO2, ↔ anion gap.** - Stool is rich in potassium and bicarb. Diarrhea and laxative abuse can cause a normal anion-gap metabolic acidosis. Since CO2 is acidic, we blow this off to compensate. - Serum chloride is high due to cellular shifts. - Anion gap is normal because diarrhea is not part of MUDPILES.
Laxative abuse	- ↓ **Serum K^+, ↑ Cl^-, ↓ $HCO3^-$, ↓ CO2, ↓ pH, ↔ anion gap.** - Stool is rich in potassium and bicarb. Diarrhea and laxative abuse can cause a normal anion-gap metabolic acidosis. Since CO2 is acidic, we blow this off to compensate. - Serum chloride is high due to cellular shifts.

Endocrine pharm for USMLE (keeping things HY)	
Metformin	- MOA = "↑ Glycolysis + ↓ gluconeogenesis." - Only used for type II DM. - Causes lactic acidosis. - Discontinue if bicarb is low (means lactic acidosis). - Discontinue (or don't commence) if creatinine is 1.5 or greater, as reduced renal function ↑ risk of lactic acidosis. - Choose insulin over metformin for regulation of glycemic control in pregnancy / intrapartum, as it allows for optimal fine-tuning / control of glucose. - USMLE can give you patient on metformin who had CT with contrast + gets high creatinine as a result + they ask you what would have prevented the increase in creatinine prior to the CT → answer = "infusion of 0.9% saline" or "adequate intravenous hydration"; the wrong answer is "discontinuation of metformin." This one is a little tricky, where you need to know that adequate hydration saline hydration is the #1 way to prevent contrast nephropathy. There is ↑ risk of lactic

	acidosis due to metformin in patients who have renal insufficiency, but metformin itself doesn't cause renal insufficiency. In other words, ↑ creatinine is a reason to stop metformin, but ↑ creatinine doesn't result from metformin itself.
Insulin	- USMLE doesn't give a fuck that you know about all of the different insulin types and their durations of action. - What USMLE might do on 2CK is give you a patient with diabetes who wakes up with severe hyperglycemia + ask you what needs to be done. The answer = **decrease, not increase (holy shit),** dose of evening intermediate-acting insulin. This is called **Somogyi phenomenon,** where it appears as though because morning glucose is elevated, this means evening/night insulin is insufficient. But it's actually the opposite, where the evening/night dose is too high, which pushes the patient into overnight hypoglycemia, causing a compensatory sympathetic response that results in a robust rebound hyperglycemia by morning. - The other Q they will give is a patient with type II DM who is already on metformin + glyburide who has uncontrolled HbA1c at 12% + give you either low bicarb or high creatinine → they ask what needs to be done → answer = "discontinue metformin and glyburide and commence intermediate-acting insulin." I've made YouTube clips on this Q and have talked about it many times, as it gets asked in our Telegram group probably once a month. The point is that 1) if a patient has low bicarb while on metformin, this means lactic acidosis due to metformin, so you must choose an answer choice where metformin is discontinued; and 2) a patient who has a creatinine of 1.5 or greater needs to be off metformin (I've seen max 1.4 on NBME where metformin is acceptable), since renal insufficiency is a risk factor for lactic acidosis due to metformin.
Sulfonylureas	- Glyburide, glipizide, glibenclamide. - Insulin secretagogues – i.e., ↑ insulin secretion by blocking the ATP-gated K^+ channel on β-islet cells without the need for ATP. - Glyburide and glipizide tend to be the ones that shows up repeatedly on NBME. I don't think I've ever seen other sulfonylurea on NBME for that matter. - Glibenclamide was known for causing severe hypoglycemia. You could be aware of this in theory.
PTU/methimazole	- Propylthiouracil (PTU) and methimazole are both used for Graves. - Both inhibit thyroperoxidase; PTU has an additional MOA of inhibiting 5-deiodonise. - Both PTU and methimazole can cause **neutropenia/agranulocytosis.** You need to know this often presents as mouth ulcers on USMLE. For example, they'll say patient was recently treated in hospital for thyroid storm + now has mouth ulcers and fever → answer = "drug adverse effect," or "drug-induced neutropenia." - Propranolol is used for tachycardia in hyperthyroidism (beta-blockade inhibits 5-deiodonase). Steroids are used for exophthalmos. - Methimazole is teratogenic in first-trimester.
Demeclocycline	- Shows up as Tx for SIADH on NBME. - Tetracycline antibiotic that causes nephrogenic DI as an adverse effect, but this is actually a "good thing" if we want to treat SIADH (i.e., let's just treat one problem by causing another problem, and they somehow cancel out).
Desmopressin	- Vasopressin (ADH) analogue used as treatment for central DI. - Also used after fluid restriction in DI cases for differentiating central from nephrogenic.
Octreotide	- Somatostatin analogue used for acromegaly and esophageal varices.
Cinacalcet	- Increases sensitivity of calcium-sensing receptors on parathyroid glands. - Helps to suppress PTH in secondary hyperparathyroidism.

Surg Bullet points

- When is sitz bath the answer? → Tx of anal fissure → Q will say "in addition to stool softeners, what's the next best step in Mx?" → answer = sitz bath.
- When is anal fissure the answer? → Q will say physical exam shows of the perineum shows an ulcer in the posterior midline of the anal canal +/- an adjacent edematous skin tag at the anal verge → highest yield descriptor is if they say the **rectal exam cannot be performed** because of the exquisite pain and increased anal sphincter tone.
- When is hydrocortisone enema the answer? → sometimes for IBD → bear in mind first-line Tx for IBD is 5-ASA compounds (mesalamine), but I mention this because "hydrocortisone enema" is a common answer choice on surg forms and students ask me when that's the answer.
- When is anal manometry the answer? → to Dx Hirschsprung → they'll say a child passes one string-like stool every 4 days despite maximum laxative therapy + there is no stool in the rectal vault; then they'll ask for next best step in Dx? → anal manometry (pressure study) → and of course confirmatory biopsy is most accurate, but anal manometry is done first.
- When is "placement of elastic bands" the answer? → Tx for hemorrhoids on USMLE.
- Cystic mass at top of gluteal cleft (by the tailbone); Dx + Tx? → pilonidal cyst (aka pilonidal abscess according to the literature) → Tx = surgical drainage of abscess (incision + drainage just off the midline).
- How do you Dx GERD? → answer = trial of PPIs (diagnostic + therapeutic).
- When is 24-hour pH monitoring the answer? → if patients have failed PPI trial but have suggestive symptomatology, or if initial Sx of GERD are equivocal (i.e., no burning but chronic/nocturnal cough).
- When is endoscopic ultrasound the answer? → it almost never is, but it's an occasional answer choice on surg forms, so I'm mentioning it; may have utility in Dx pancreatic cancer if CT is negative.
- When is esophageal manometry the answer? → USMLE wants this for achalasia **after** a barium or gastrografin (water-soluble contrast) swallow has been performed showing the bird's beak appearance.
- 42M + 3-year-Hx of GERD + CXR shows air-fluid level posterior to the cardiac silhouette; Dx? → hiatal hernia.

- Above 42M; next best step in diagnosis? → barium swallow → will show proximal stomach herniating through the esophageal hiatus.
- 42M w/ hiatal hernia suggested on barium swallow; next best step? → esophagogastroduodenoscopy (answer on the NBME).
- High ALP + high direct bilirubin + high amylase or lipase → gallstone pancreatitis = choledocholithiasis
- High ALP + high direct bilirubin + high amylase or lipase + remote Hx of cholecystectomy → sphincter of Oddi dysfunction (can't be a stone cuz the gallbladder was removed ages ago)
- High ALP + high direct bilirubin + normal amylase or lipase in someone with recent cholecystectomy → choledocholithiasis (retained stone in cystic duct that descended, but not distal to pancreatic duct entry point)
- Dx and Tx of choledocholithiasis → ERCP
- High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy → pancreatic cancer
- Dx of pancreatic cancer → CT abdo with contrast
- High ALP + high direct bilirubin + normal amylase or lipase in someone with remote cholecystectomy + CT is negative → cholangiocarcinoma
- High ALP + high direct bilirubin + normal amylase or lipase + diffuse pruritis + high cholesterol → primary biliary cirrhosis (PBC)
- High ALP + high direct bilirubin + normal amylase or lipase + autoimmune disease (in pt or family) → PBC
- Dx of PBC → anti-mitochondrial Abx next best step; liver biopsy is confirmatory
- Recent cholecystectomy + fever + abdo pain → post-op bile leak
- High ALP + high direct bilirubin + normal amylase or lipase + CT shows cystic lesion in bile duct → choledochal cyst → do simple excision of cyst (cholangiocarcinoma not cystic + CT can be negative)
- Imaging to view liver or pancreas → CT with contrast
- Imaging to view gallbladder → Ultrasound
- Imaging to view gallbladder in suspected cholecystitis only if USS negative → HIDA scan
- Imaging to view bile ducts → ERCP or MRCP (choose ERCP > MRCP if both listed)
- UC + high bilirubin + high ALP → primary sclerosing cholangitis

- 8-day-old neonate + jaundice + direct bilirubin 14 mg/dL + total bilirubin 15 mg/dL; Dx? → answer = biliary atresia → super HY Dx.
- 8-day-old neonate + jaundice + direct bilirubin 14 mg/dL + total bilirubin 15 mg/dL; next best step in diagnosis? → answer = straight-up liver biopsy → if positive, must do liver transplant.
- 79M + Hx of atrial fibrillation + severe, acute, diffuse abdo pain; Dx? → acute mesenteric ischemia caused by mural thrombus embolizing to SMA or IMA.
- Above 79M; next best step in Mx? → mesenteric arteriography.
- Above 79M; Tx? → antibiotics (for necrotic bowel) then laparotomy (to remove necrotic bowel) → they will tell you in last line of vignette that IV Abx are administered and then ask for the next step, which is just laparotomy. It should be noted that the literature mentions various Txs like embolectomy, but the USMLE wants resection of nonviable bowel as the answer.
- 52F + short episode of ventricular fibrillation + defibrillated + now has severe abdo pain; Dx? → acute mesenteric ischemia due to ischemia caused by VF, not an embolus → antibiotics; CT if stable; if unstable go straight to laparotomy.
- 55F diabetic + Hx of intermittent claudication + Hx of abdo pain 1-2 hours after eating meals; Dx? → **chronic** mesenteric ischemia (CMI) caused by severe atherosclerosis of SMA or IMA (essentially angina of the bowel).
- 55F diabetic + Hx of CABG + Hx of abdo pain 1-2 hours after eating meals; next best step in Dx? → mesenteric arteriography (CMI).
- 55F diabetic + Hx renal artery stenosis + Hx of abdo pain 1-2 hours after eating meals; Tx? → angioplasty + stenting (CMI) to restore blood flow.
- Patient with CMI who has a 2-day Hx of severe abdo pain + fever; Dx? → acute mesenteric ischemia (acute on chronic due to a thrombosis; essentially akin to an "MI" of the bowel) → do mesenteric arteriography to Dx; Tx with Abx + laparotomy to remove necrotic bowel.
- What is pectinate line? → separates upper 2/3 of the anal canal (part of hindgut; endoderm-derived) from the lower 1/3 of anal canal (aka proctodeum, which is ectodermal).
- Lymphatic drainage above/below pectinate line? → above: internal iliac; below: superficial inguinal.
- Arterial supply above/below pectinate line? → above: superior rectal artery; below: middle/inferior rectal arteries.

- Venous drainage above/below pectinate line? → above: superior rectal vein; below: middle/inferior rectal veins.
- How does pectinate line relate to hemorrhoids? → above: internal hemorrhoids (painless); below: external hemorrhoids (painful).
- Tx for hemorrhoids? → conservative first, i.e., fiber + exercise; if they want intervention, banding (rubber band ligation) is the answer.
- How do you Dx congenital midgut volvulus? → upper-GI series (AXR + contrast follow-through of esophagus, stomach, and duodenum with barium or gastrografin).
- 87F + coffee bean sign on AXR + obstipated; Dx? → sigmoid volvulus
- Tx for sigmoid volvulus? → answer on surgery NBME = "sigmoidoscopy-guided placement of rectal tube"
- 22M + stressed studying for exams + yellow eyes + has had a few similar episodes in the past + is otherwise healthy; Dx? → answer = Gilbert syndrome (proncouned jeel-BEAR).
- Mechanism for Gilbert? → decreased expression of glucuronosyltransferase (bilirubin conjugating enzyme) in the liver → decreased ability to take up unconjugated bilirubin at the liver → jaundice.
- Tx for Gilbert? → reassurance; benign condition.
- Criteria for pathologic jaundice? → student says "I really need to know that?" Absolutely. HY on peds shelves and 2CK. If any one or more of the following is positive, the etiology of the kid's jaundice is considered pathologic:
    - Any jaundice present in first 24 hours of life, period.
    - Any jaundice present after one week if term or two weeks if preterm.
    - Total bilirubin >15 mg/dL.
    - Direct bilirubin >10% of total bilirubin, *even if total is under 15 mg/dL.*
    - (The one everyone forgets) Rate of change of increase of total bilirubin >0.5 mg/dL/hour.
- 16F + fever + high leukocytes + RLQ pain that migrated from epigastrium; Dx? → appendicitis (easy, but so HY how can I not at least mention it classically) → USMLE wants you to know that migration is because, initially, epigastric pain = visceral pain; RLQ pain = inflammation of parietal peritoneum. Must do a pregnancy test if female + adnexal ultrasound to look for gyn causes, i.e., ruptured cyst, etc. If male, go straight to laparoscopy. If rule out gyn cause in female, do laparoscopic removal.

Ultrasound + CT can be done, but false-negatives have led to rupture + death, so they don't change management if clinical suspicion is high, which is why pt goes straight to laparoscopy if under high suspicion for appendicitis → if during surgery the appendix is normal, **answer = still remove it.**

- Mallory-Weiss tear vs esophageal varices HY point → MWT usually caused by vomiting/retching in alcoholic + **presents with a little bit** of blood in the vomitus; varices present with **LOTS of blood** → about half of patients with ruptured varix die.
- Mallory-Weiss tear vs Boerhaave → MWT is not transmural; Boerhaave is transmural → causes subcutaneous emphysema (crepitus due to air under the skin).
- Tx of varix? → banding + octreotide.
- Prophylaxis for varix? → propranolol.
- Person who's vomiting; what's the biochemical disturbance? → hypokalemic hypochloremic metabolic alkalosis → low K, low Cl, high pH, high bicarb, low H, anion gap normal (even though it's alkalosis, not acidosis, the USMLE will still ask an arrow for the anion gap here).
- 2-week-old male + forceful non-bilious vomiting; Dx? → hypertrophic pyloric stenosis.
- Dx of pyloric stenosis? → abdominal ultrasound to show olive-shaped hypertrophied pylorus.
- Tx of pyloric stenosis? → myomectomy
- Who gets pyloric stenosis? → first-born males (weird, but it's on an old NBME) + neonates taking oral erythromycin for chlamydial ophthalmia neonatorum (erythromycin is a motilin-receptor agonist).
- 2-week-old male + bilious vomiting; Dx? → duodenal atresia, annular pancreas, congenital midgut volvulus, or Hirschsprung (correct, Hirschsprung can present with bilious vomiting).
- 2-week-old + Down syndrome + bilious vomiting + passed meconium ok; Dx? → duodenal atresia
- 2-week-old + Down syndrome + bilious vomiting + slow to pass meconium; Dx? → Hirschsprung
- How do you Dx duodenal atresia? → abdominal x-ray (AXR) showing double-bubble sign (very HY).
- How do you Dx Hirschsprung? → rectal manometry, followed by confirmatory rectal biopsy showing absence of ganglion cells.
- Mechanism for Hirschsprung? → failure of migration of neural crest cells distally to the rectum.
- Failure to pass meconium at birth. Most likely cause overall? → cystic fibrosis.
- 18-month-old + intermittent abdominal pain + crying + blood in stool; Dx? → intussusception.
- 18-month-old + intermittent squatting + crying + FOBT positive; Dx? → intussusception.

- 18-month-old + occasionally brings legs to chest + vomits + FOBT positive; Dx? → intussusception.
- 18-month-old + occasionally brings legs to chest + vomits + FOBT negative; Dx? → volvulus → this is congenital midgut volvulus.
- 18-month-old + occasionally brings legs to chest + vomits + FOBT negative
- Presentation sounds like intussusception but no blood per rectum → answer = congenital midgut volvulus.
- Cause of intussusception? → >99% are in kids under age 2; caused by lymphoid hyperplasia due to viral infection (e.g., rotavirus) or recent vaccination; if in adult (usually elderly), it is caused by colorectal cancer.
- Dx and Tx of intussusception? → USMLE wants enema as the answer. Even though ultrasound can be done which shows a target sign, the USMLE always wants enema. And it can be any type. I've seen "air contrast enema", "air enema," "contrast enema" all as answers. I also had a student simply get "water-soluble contrast enema" on the exam, which means gastrografin. Barium would refer to regular contrast.
- For contrast swallows, when to do barium vs water-soluble (gastrografin)? → barium most of the time; if at risk of aspiration, must do barium because aspiration of gastrografin will cause pneumonitis. If patient has suspected esophageal perforation, do not do barium, as that will cause mediastinitis; must do gastrografin in this case.
- Level of celiac trunk and main branches + what's it supply? → T12; splenic artery, common hepatic artery, left gastric artery; supplies foregut (mouth to duodenum at ampulla of Vater).
- Level of SMA + what's it supply? → L1; supplies midgut (duodenum at ampulla of Vater until 2/3 distal transverse colon); so for instance, the right colic and middle colic arteries come off SMA.
- Level of IMA + what's it supply? → L3; supplies hindgut (2/3 distal transverse colon until the pectinate line 2/3 distal on the anal canal); left colic artery comes off IMA.
- Renal + gonadal (testicular in men; ovarian in women) arteries come off of L2 most often.
- Abdominal aorta bifurcates into common iliacs at L4.
- Abdominal aortic aneurysm occurs in males over 55 who are ever-smokers → a one-off abdo ultrasound is done in this patient group.

- Most common locations for atherosclerosis (in descending order) → abdominal aorta, coronary arteries, popliteal arteries, carotid arteries.
- USMLE favorite question → "Which of the following is supplied by an artery of the foregut but is not itself derived from the foregut?" → answer = spleen.
- Short gastric arteries come off which artery? → splenic.
- What's the main arterial supply to the pancreas? → Arteria pancreatica magna (greater pancreatic artery) → **a branch of the splenic.**
- Which nerve must be severed to remove cancer at gastroesophageal junction → answer = vagus (just memorize it; it's on a retired NBME).
- HY structures passing through diaphragm? → "I Ate 10 Eggs At 12." → IVC T8; T10 Esophagus + thoracic duct; Aortic hiatus (aorta, azygous vein, thoracic duct) at T12.
- Celiac disease important points? → gluten intolerance; gluten found in wheat, oats, rye, and barley, so therefore will get Sx after eating, e.g., pasta (too easy for the vignette to mention though); causes flattening of intestinal villi on biopsy (image HY); patients often present with **iron deficiency anemia (HY way to differentiate from lactose intolerance);** Dx with Abs: anti-endomysial IgA (anti-gliadin IgA), anti-tissue transglutaminase IgA → after you get positive Abs in Dx of Celiac, USMLE wants **duodenal biopsy** to confirm ("no further studies indicated" is the wrong answer) → Tx = dietary changes to avoid gluten.
- Dx of C. difficile? → answer = stool AB toxin test, **not** stool culture (exceedingly HY).
- Tx of C. difficile? → **guidelines as of Feb 2018 say oral vancomycin first-line, not metronidazole** → apparently UW is updated on this too now → note that vanc is given orally → apart from C. diff, it's always given IV because it has terrible oral bioavailability, but in the case of C. diff, where we want the drug confined to the lumen of the colon, that makes sense.
- Mechanism of colonic necrosis in C. diff colonic necrosis? → answer = "cytoskeletal disruption."
- Patient is treated with vanc for C. diff but gets recurrence weeks later; why? → answer = "regermination of spores."
- C. diff + fever of 104F + tachy + diffuse abdominal pain; next best step in Mx? → AXR → look for toxic megacolon → Tx w/ NPO (nothing by mouth), NG decompression + rectal tube (decompression) + Abx (vancomycin or fidaxomicin) + steroids (if UC) + correct any electrolyte imbalances (sometimes low K)

→ if patient doesn't improve with conservative therapy, must do surgery (subtotal colectomy + ileostomy); do **not** do a colonoscopy on a patient with toxic megacolon as this will cause perforation.

- Damage to which nerves can cause constipation? → answer = pelvic splanchnic (because these are parasympathetic, which drive bowel function) → in contrast, hypogastric nerves = sympathetic (internal anal sphincter); pudendal nerve = somatic (external anal sphincter).
- 45M + cirrhosis + fluid wave + fever + abdo pain; Dx? → spontaneous bacterial peritonitis (SBP).
- 69F diabetic + undergoing peritoneal dialysis + fever + abdo pain; Dx? → SBP.
- 8M + viral infection + pedal/periorbital edema + fluid wave + fever + abdo pain; Dx? → SBP → minimal change disease (nephrotic syndrome) causing ascites.
- SBP Dx → paracentesis (peritoneal aspiration; don't confuse with periocardiocentesis) + do gram-stain + look for >250 WBCs per HPF.
- Tx for SBP → ceftriaxone.
- "What do I need to know about serum-ascites albumin gradient (SAAB)?"
    - Equal to the (serum albumin) – (ascitic fluid albumin).
    - If SAAB >1.1, Dx = transudative process (i.e., congestive heart failure, Budd-Chiari, cirrhosis with portal hypertension) → all cause increased hydrostatic pressure in portal vein, so fluid leakage into peritoneal cavity carries less protein.
    - If SAAB <1.1, Dx = exudative process (i.e., infection [e.g., TB, SBP]), pancreatitis, malignancy), **or nephrotic syndrome (HY);** if the serum protein is very low because of nephrotic syndrome, even though the fluid leakage into the peritoneal cavity also carries very low protein, the result is still a low SAAB.
- Woman 20s-50s + high cholesterol + diffuse pruritis + sister has rheumatoid arthritis; Dx? → primary biliary cirrhosis → USMLE likes "autoimmune diseases go together" in patient (or family).
- How to Dx PBC? → anti-mitochondrial Abs first; if positive, liver biopsy is confirmatory.
- Neonate born 32 weeks gestation + fever + abdominal distension + feeding intolerance + hematochezia; Dx? → necrotizing enterocolitis → bowel infection in premature neonates usually <32 weeks gestation.
- Dx of NE? → abdominal x-ray (AXR) visualizing **pneumatosis intestinalis** (air in the bowel wall), **air in the portal vein,** or free air under the diaphragm.

- Tx of NE? → NPO (nil per os; nothing by mouth), NG decompression, broad-spectrum Abx; if necrotic bowel, must do surgery.
- 49M + Hx of abdo pain after meal; now presents with sepsis + diffuse, acute abdo pain; Dx? → ruptured viscus (duodenum).
- Above 49M; next best step in Dx? → answer = "x-ray of chest + abdomen" to look for air under the diaphragm (confirms diagnosis); USMLE will never give you choice A) CXR; B) AXR, etc.; they'll either give you CXR alone, AXR alone, or both; one of the 2CK surgery NBMEs has both as the answer.
- 42M + dysphagia to solids + liquids + no other Hx; Dx? → achalasia → inability to swallow solids + liquids together implies neurogenic etiology.
- 42M + EtOH Hx + dysphagia to solids that progresses to include liquids; Dx? → esophageal cancer (SCC) → dysphagia to solids **that progresses to** solids + liquids = cancer until proven otherwise.
- HY points about esophageal cancer? → adenocarcinoma is distal 1/3 and is caused by GERD (obesity → low LES tone → GERD → Barrett → adenocarcinoma); SCC is upper 2/3 of esophagus and is caused by smoking + alcohol; can also be caused by webs, burns, chemicals, and achalasia (difficult, because achalasia is LES so your thought is, "how can that cause SCC of upper 2/3?" → probably the dysphagia causes increased esophageal irritation, which then becomes the risk factor for SCC).
- Mechanism for achalasia? → loss of NO-secreting neurons in myenteric plexus of LES → increased LES tone → bird's beak appearance on contrast swallow + increased tone on esophageal manometry; cause is often idiopathic, but Chagas disease (Trypansoma cruzi) is a known infective cause (rare).
- How do you Dx achalasia? → **USMLE wants barium (or gastrografin) swallow, then manometry, then confirmatory biopsy, in that order.** There is a Q on an NBME for 2CK where both barium and manometry were listed, and the answer was barium swallow, not manometry.
- So when is manometry the answer for achalasia? → when they show you a pic of the bird's beak from the barium swallow already performed, so clearly the next best step is manometry. The USMLE will sometimes show a graph of a manometry that's been performed, and you'll simply see that the pressure is high at the LES → hence Dx = achalasia. The confirmatory / most accurate test is biopsy of LES showing loss of neurons.
- Tx for achalasia → surgery first-line (endoscopic myotomy is superior to pneumatic dilatation); if patient has high surgical risk, can use botulinum toxin as first-line therapy → if fails, dCCB or nitrates.

- 42M + overweight + halitosis + gurgling sound when drinking fluids + occasionally regurgitates undigested food; next best step in Mx? → barium (or gastrografin) swallow; Dx = Zenker.
- Above 42M + Hx of GERD → go straight to endoscopy as the answer (cancer, not Zenker).
- Location + mechanism of Zenker? → false diverticulum just superior to the cricopharyngeus on the posterior pharyngeal wall → USMLE answers are "increased oropharyngeal pressure" and "cricopharyngeal muscle spasm." They want you to know it is **not** a congenital weakness. Dysphagia is a risk factor because this increases oropharyngeal pressure. I've noticed Zenker vignettes often mention the patient is overweight, although the association is non-specific.
- What is Whipple triad? → means insulinoma → 1) hypoglycemia, 2) Sx of hypoglycemia (diaphoresis, tachycardia, tremulousness), 3) relief of Sx with meals (or they'll say it gets worse between meals).
- If patient has Whipple triad, what's next best step in Mx? → check serum C-peptide levels → now this is where I get you a point: if C-peptide is high, the wrong answer is CT abdo to look for insulinoma because **the Dx is not automatically insulinoma.** Now you're probably like, "Really? Wait, why? I'm not following." → if C-peptide high, answer = **first check serum hypoglycemic levels** → meaning, some patients can surreptitiously take sulfonylureas (i.e., glyburide, etc.), which are insulin secretagogues, **so they're C-peptide levels will be high.** Only after the serum hypoglycemic screen is negative do you consider CT abdo.
- 52F + 2 kids + BMI 28 + recurrent colicky epigastric pain; next best step in Dx? → ultrasound (cholelithiasis).
- 45F + BMI 29 + BP 140/90 + high TGAs + low HDL + elevated fasting glucose + slightly elevated AST and ALT; Dx? → NASH (non-alcoholic steatohepatitis) → caused by metabolic syndrome → Tx is lifestyle modification to eradicate the metabolic syndrome.
- Above 45F + completely normal liver enzymes; Dx? → NASH → I've seen plenty of Qs where all labs values they show you are completely normal + the only thing you're left with is, "well she's fat / has metabolic syndrome," and you eliminate the others to just say, "well, this is NASH."
- First Tx for pancreatitis in general → USMLE wants triad of NPO (nil per os; nothing by mouth) + fluids + NG tube → if USMLE asks for imaging, do CT with contrast to look for fluid collections + degree of necrosis → **drain fluid collections percutaneously;** if pseudoabscess forms and doesn't regress, answer = **internal drainage via ERCP or EUS (endoscopic ultrasound)**; if frank pus (fat enzymatic

necrotic pancreas) >30%, must do necrosectomy (excision of necrotic pancreas) or aggressive percutaneous drainage.

- 69M + LLQ pain + fever = diverticulitis → Dx with CT with contrast of abdomen → Tx w/ Abx (metronidazole, PLUS fluoroquinolone or Augmentin; USMLE won't ask you the exact Abx, but you should be aware that metro covers anaerobes below the diaphragm) → **never** do a colonoscopy on someone with suspected diverticulitis, as you may cause perforation. However, after the diverticulitis is fully treated + cleared, patient will need a follow-up colonoscopy to rule out malignancy.
- Crohn GI findings? → mouth to anus; classically terminal ileum; frequently intermittent bloody diarrhea if colonic involvement; skip lesions causing "string sign" on contrast studies; "cobblestone mucosa"; transmural inflammation with non-caseating granulomas; perianal fistulae; B12 + fat-soluble vitamin malabsorption.
- 25M + oral ulcer + bloody diarrhea; Dx? → Crohn
- Extra-intestinal manifestations of Crohn? → classically erythema nodosum (red shins; not a rash; this is panniculitis, which is inflammation of subcutaneous fat); anterior uveitis (red eyes); oxalate nephro-/ ureterolithiasis (decreased fat absorption → more binding of fat to calcium [saponification] → therefore less calcium binds to oxalate → more oxalate absorbed → oxalate stones).
- Any weird factoid about Crohn? → sometimes patients (+) for anti-Saccharomyces cerevisiae Abs
- How to Dx Crohn? → USMLE wants colonoscopy
- Tx for Crohn? → USMLE wants NSAIDs (either sulfasalazine or mesalamine [5-ASA] will be listed) before steroids → use immunosuppressants late (i.e., infliximab, azathioprine, etc.).
- Does Crohn share anything with UC? → Yes, bear in mind in real life, there is overlap between the two diseases, so don't pigeonhole things; think of these disease-associations as propensities rather than as intransigent rules.
- 28M + lower back pain worse in morning and gets better throughout the day + mouth ulcer; Dx? → Crohn disease (oral involvement only) + sacroiliitis (back pain Sx of ankylosing spondylitis).
- 20F + bloody diarrhea + sore joints + eczematoid plaque on forehead → IBD + psoriatic arthritis
- UC GI findings? → rectum-ascending (meaning, starts at rectum and ascends; does not involve anus; Crohn of course is mouth to anus); bloody diarrhea; **not** transmural (mucosa + submucosa involved only; unlike Crohn); **no** granulomas (unlike Crohn); **lead pipe** appearance of colon on contrast studies

(unlike “string sign” of Crohn); crypt abscesses (just memorize) → lead pipe means **loss of haustra** (so the colon looks smooth from the outside; this is really HY!) → USMLE might also there are “microabscesses within the crypts” on colonoscopy (meaning crypt abscesses in UC).

- Extra-intestinal manifestations of UC? → primary sclerosing cholangitis (beaded appearance of common bile duct; can be p-ANCA positive); pyoderma gangrenosum (crater on the forearm with necrotic debris); like Crohn, is associated with anterior uveitis + HLA-B27 associations.
- How to Dx UC? → USMLE wants colonoscopy.
- Tx for UC? → same as Crohn for USMLE purposes, but just be aware in severe cases colectomy is performed.
- 65M + intermittent bloody diarrhea + now has fever of 104F + abdominal pain + high leukocytes; Dx? → answer = toxic megacolon → **Dx with AXR, not colonoscopy!** → if you scope, patient will perforate and die → AXR will show dilated bowel (e.g., one NBME Q says “12-cm cecum”); in general, know that AXR is done when you’re looking for gas.
- Where do most colonic ischemic ulcers occur? → watershed areas → splenic flexure (watershed of SMA and IMA) + sigmoidal-rectal junction (watershed of IMA and hypogastric artery).
- 72M + advanced CVD + bloody stool; Dx? → ischemic colitis (due to ischemic ulcer).
- 55M + BMI of 33 + vignette doesn’t mention diabetes + 3 months burning in throat → Dx = GERD → Tx? = **trial of PPIs** (i.e., **trial of omeprazole)** for two weeks → relief of Sx is consistent with GERD as the correct Dx.
- 55M + BMI of 33 + poorly controlled diabetes (type I or II) + 3 months of burning in throat → Dx = **diabetic gastroparesis, NOT GERD** (woahhh crazy) → first pharm **Tx = metoclopramide, not PPIs.** If metoclopramide not listed, choose erythromycin (motilin receptor agonist).
- Regarding gastroparesis, the USMLE vignette will make an explicit point about bad diabetic disease, i.e., peripheral edema (renal insufficiency due to decreased oncotic pressure from albuminuria) +/- cataracts (osmotic damage from intracellular sorbitol) +/- urinary retention (neurogenic bladder due to osmotic denervation leading to hypocontractile detrusor) +/- they simply say HbA1C of 12% (diabetes is 6.5% or greater; prediabetic is 6-6.49). There will be no question as to whether they want gastroparesis vs simple GERD.

- If the vignette (more 2CK here) doesn't ask straight-up which drug you choose and they ask for next best step in Mx for suspected gastroparesis → first do endoscopy to rule out physical obstruction → if negative, then do gastric-emptying scintigraphy (scintigraphic gastric-emptying assay) → if delayed gastric emptying, first Tx = smaller meals; if insufficient, then do metoclopramide, then add erythromycin.
- Ursodeoxycholic acid (Ursodiol) → naturally occurring bile acid given to patients with cholesterol cholelithiasis as an alternative to cholecystectomy. On USMLE, Dx of cholelithiasis = abdominal ultrasound, followed by Tx = cholecystectomy. Ursodiol can be given to select patients but is not considered the universal next best step.
- What is Dumping syndrome? → caused by gastric bypass surgery, diabetes, or malfunctioning pyloric sphincter, in which stomach contents following a meal enter the duodenum too quickly; there are two types: early vs late → both show up in vignettes (without people even realizing they're seeing a Dumping syndrome Q).
- Early Dumping syndrome → 10-30 minutes after a meal → rapid entry of hyperosmolar gastric contents into duodenum → osmotic expansion of small bowel lumen → diarrhea + bloating → on USMLE, answer = "rapid emptying of hyperosomlar chyme into small bowel."
- Late Dumping syndrome → 1-2 hours after meal → rapid absorption of carbohydrates through small bowel wall → hyperglycemia → pancreas secretes lots of insulin → **rebound hypoglycemia** → USMLE merely wants you to identify this in a vignette as Dumping syndrome. They might say Hx of gastric bypass + now there's a meal + patient gets diarrhea +/- hypoglycemia → Dx simply = Dumping syndrome.
- What is Blind loop syndrome? → disturbance of normal floral balance in the small bowel due to disruption of peristalsis (i.e., surgery / post-surgical ileus), but may also be caused by conditions like IBD and scleroderma → leads to steatorrhea + B12 def + fat-soluble vitamin deficiencies → USMLE merely wants you to be able to make the diagnosis from a vignette → Tx is with antibiotics (doxycycline or fidaxomicin).
- 8M + bloody stool + perfectly healthy otherwise; Dx? → Meckel diverticulum; student says, "huh, I thought that was age 2." I agree with you. But there's an NBME Q where the kid was 8, and the answer was Meckel → bleeding due to "heterotopic gastric / pancreatic tissue."

- How to Dx Meckel diverticulum? → Meckel scan (Tc99 uptake scan that localizes to diverticulum).
- Tx for Meckel → if asymptomatic, can leave alone; if symptomatic, surgical removal.
- 16F + fever + high leukocytes + RLQ pain that migrated from epigastrium; Dx? → appendicitis (easy, but so HY how can I not at least mention it classically) → USMLE wants you to know that migration is because, initially, epigastric pain = visceral pain; RLQ pain = inflammation of parietal peritoneum. Must do a pregnancy test if female + adnexal ultrasound to look for gyn causes, i.e., ruptured cyst, etc. If male, go straight to laparoscopy. If rule out gyn cause in female, do laparoscopic removal. Ultrasound + CT can be done, but false-negatives have led to rupture + death, so they don't change management if clinical suspicion is high, which is why pt goes straight to laparoscopy if under high suspicion for appendicitis → if during surgery the appendix is normal, **answer = still remove it.**
- Mallory-Weiss tear vs esophageal varices HY point → MWT usually caused by vomiting/retching in alcoholic + **presents with a little bit** of blood in the vomitus; varices present with **LOTS of blood** → about half of patients with ruptured varix die.
- Mallory-Weiss tear vs Boerhaave → MWT is not transmural; Boerhaave is transmural → causes subcutaneous emphysema (crepitus due to air under the skin).
- Tx of varix? → banding + octreotide.
- Prophylaxis for varix? → propranolol.
- Most common cause of death due to fall or MVA? → traumatic rupture of the aorta (thoracic).
- Where does rupture of the aorta occur? → where the ligamentum arteriosum wraps around the top of the descending arch → ligament is taut but arch is more mobile → leads to shearing.
- What will the NBME/USMLE Q say for traumatic rupture → MVA or fall followed by "widening of the mediastinum on CXR."
- 32M + MVA + widening of mediastinum on CXR; next best step in Dx? → aortic arteriography (aka aortography).
- Tx for traumatic rupture? → if ascending arch: labetalol + surgery; if descending arch: labetalol only. Sodium nitroprusside is the **wrong** answer.
- Traumatic rupture + low BP; next best step? → labetalol (decreases shearing forces, even with low BP it's the answer on the NBME).

- If patient is diagnosed with bicuspid aortic valve, next best step in Mx? → annual transthoracic echos → if valve cross-sectional area falls below 1.0 cm^2 then do aortic valve replacement; there's a surgery NBME Q where they say cross-sectional area is 0.8 cm^2 and the answer is straight-up "aortic valve replacement."
- Most common cause of carotid plaques? → HTN → the strong systolic impulse from the heart pounds the carotids --> endothelial damage --> atherosclerosis.
- 55M + BP 150/90 + TIA; next best step in Mx? → carotid duplex USS → the first thing you want to think about is, "does this guy have a carotid plaque that has resulted in a clot embolizing to his brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA; next best step in Mx? → ECG → you want to think, "Does he have atrial fibrillation with a LA mural thrombus that's now embolized to the brain."
- 80M + good blood pressure (e.g., 110/70) + stroke or TIA + ECG shows sinus rhythm with no abnormalities; next best step in Mx? → Holter monitor → when you first see this scenario you're probably like, "Wait, the ECG is normal, so it's not AF?" → No, *it is* likely AF, but AF is often paroxysmal, so in order to detect it in this scenario, **the next best step is a Holter monitor (24-hour wearable ECG).** This means that later in the day when he sits down to have dinner and then pops into AF, the Holter monitor will pick it up.
- What % of people over age 80 have AF? → 8% of people over age 80 have AF, which is why age is a huge risk factor. In other words, if the vignette says the guy is 58, AF is probably less likely just based on shear probability, regardless of hypertensive status." And, once again, knowing that AF is often paroxysmal is really important.
- Age 50s-60s + high BP + TIA/stroke/retinal artery occlusion; next best step in Dx? → answer = carotid duplex ultrasound to look for carotid plaques.
- Age >75 + good BP + TIA/stroke/retinal artery occlusion; answer = ECG to look for AF → if normal, do Holter monitor to pick up paroxysmal AF.
- 55M + good BP + carotid bruit heard on auscultation; next best step in Mx? → answer = carotid duplex ultrasound to look for carotid plaques → in this case, if they are obvious and explicit about the suspected etiology of the stroke, TIA, or retinal artery occlusion, then you can just do the carotid duplex ultrasound.

- How to Mx carotid plaques? → first we have to ask whether the patient is symptomatic or asymptomatic. A bruit does not count as symptoms (that's a sign). Symptomatic means stroke, TIA, or retinal artery occlusion. According to recent guidelines: carotid occlusion >70% if symptomatic, or >80% if asymptomatic → answer = do carotid endarterectomy. Below these thresholds → answer = medical management = statin, PLUS clopidogrel OR dipyridamole + aspirin. The USMLE will actually *not* be hyper-pedantic about the occlusion %s (that's Qbank). They'll make it obvious for you which answer they want. They'll say either 90% → answer certainly = carotid endarterectomy, or they'll say 50% → answer = medical management only. There's one NBME Q where they say a guy has a bruit but is asymptomatic, and has 10 and 30% occlusion in the left vs right carotids, respectively, and he's already on aspirin + statin, and the answer is "maintain current regimen" → if he were symptomatic, even with low occlusion, he'd certainly need statin, PLUS clopidogrel OR dipyridamole + aspirin.
- 68F + diabetic + diffuse, dull abdo pain 1-2 hours after meals; Dx? → chronic mesenteric ischemia due to atherosclerosis of SMA or IMA, **not** duodenal ulcer (if they want the latter, they'll say 29M from Indonesia) → essentially stable angina of the bowel.
- 68F + Hx of intermittent claudication + CABG + abdo pain 1-2 hours after eating meals; Dx? → chronic mesenteric ischemia.
- 78M + Hx of AF + acute-onset severe abdo pain "out of proportion to physical exam"; Dx? → acute mesenteric ischemia due to embolus.
- 16F + Hx of severe anorexia + BMI of 14 + has episode of ventricular fibrillation due to hypokalemia + now has severe abdo pain; Dx? → acute mesenteric ischemia due to episode of decreased blood flow (should be noted that hypokalemia causing arrhythmia is most common cause of death in anorexia).
- 68F + diabetic + Hx of diffuse, dull abdo pain 1-2 hours after meals + now has 2-day Hx of severe abdo pain out of proportion to physical exam; Dx? → acute on chronic mesenteric ischemia due to ruptured atherosclerotic plaque (akin to an "MI" of the bowel).
- Dx of acute + chronic mesenteric ischemia? → USMLE answer = mesenteric angiography.
- Tx of acute mesenteric ischemia? → endarterectomy might be able to restore blood flow if caught in time, but on the USMLE, they will say "IV antibiotics are administered; what's the next best step in Mx?" and the answer is just "laparotomy."
- Tx of chronic mesenteric ischemia → endarterectomy to clear vessel.

- 32M + pneumothorax that does not resolve following placement of a chest tube; Dx? –> **ruptured bronchus or rupture of intrathoracic trachea.**
- 32M + pneumothorax with a persistent air leak despite chest tube placement; Dx? –> **ruptured bronchus or rupture of intrathoracic trachea.**
- 32M + contralateral tracheal deviation + low BP; Dx? –> tension pneumothorax; Tx = needle decompression followed by chest tube. If out in the field (i.e., not in hospital), if there is penetrating chest trauma, tape over the wound on three sides only, so air can exit but not enter. Tension pneumothorax need not be associated with penetrating chest trauma, but this is one of the most common etiologies.
- Mechanism of low BP in tension pneumothorax? –> answer = compression of venous structures (IVC).
- 25M + tall +/- uses cocaine + has dyspnea + air in pleural space; mechanism? –> rupture of subapical bleb causing spontaneous pneumothorax; Tx = can technically observe if very small and patient stable; however answer on USMLE will still be needle decompression followed by chest tube.
- 35F + C-section 24-48 hours ago + crackles at both lung bases; Dx? –> **atelectasis;** normally the fever is within 24 hours, but a Q on one of the forms says “two days after surgery.”
- 48M + motor vehicle accident (MVA) + rib fractures + paradoxical breathing (i.e., chest wall moves outward with exhalation and inward with inhalation); Dx? –> **flail chest.**
- 48M + MVA + has severe pain and/or bruising *over the sternum*; Dx? –> **myocardial contusion** –> do an ECG + monitor in hospital due to high risk of arrhythmia. Get troponins.
- 72F + 6-month Hx of small painless papule from a chickenpox scar on her chin; Dx? → answer = Marjolin ulcer (squamous cell carcinoma) → SCC growing from previous scar or burn site.
- 34F + painless lump on dorsal aspect of proximal hand; Dx? → ganglion cyst; Tx = needle drainage.
- 44F + frequently wears high-heel shoes + painful lump on the underside of her foot between her third and fourth toes; Dx? → answer = Morton neuroma → benign growth of nerve tissue between the 2nd and 3rd, or 3rd and 4th, metatarsal heads; usually from chronic irritation from high-heel shoes; Mulder sign is replication of Sx when the metatarsal heads are compressed together.



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